



CAPSTONE PROJECT REPORT

Report 4 – Software Design Document

Table of Contents

I. Project Report	6
1. Status Report	6
2. Team Involvements	6
3. Issues/Suggestions	6
II. Software Design Document	7
1. System Design	7
1.1 System Architecture	7
1.2 Package Diagram	8
2. Database Design	12
2.1 Auth Service Database	12
2.2 Account Service Database	15
2.3 Core Clinic Service Database	20
2.4 Core Medical Service Database	29
2.5 Payment Service Database	40
3. Detailed Design	42
3.1 Register	42
3.2 Login	44
3.3 Logout	46
3.4 Login With Google	47
3.5 Reset Password	48
3.6 Forgot Password	50
3.7 Change Password	51
3.8 View Profile	52
3.9 Update Profile	53
3.10 Send OTP	54
3.11 View Role	56
3.12 Update Role	57
3.13 Create Permission	58
3.14 View Permission	59
3.15 Update Role Permission	60
3.16 Assign Role	61

3.17 Revoke Role.....	63
3.18 Access Audit Log.....	64
3.19 Verify Identity KYC	65
3.20 View User List.....	66
3.21 Lock Ban User Account.....	67
3.22 Unlock Unban User Account	68
3.23 View Clinic Transform	69
3.24 View Clinic Details	70
3.25 Update Clinic Information.....	71
3.26 Add Treatment Room.....	72
3.27 View Treatment Room	73
3.28 Update Treatment Room	74
3.29 Delete Treatment Room	75
3.30 Create Work Schedule	76
3.31 Manage Working Schedule	78
3.32 View Personal Schedule	80
3.33 Request Personal Schedule Change.....	81
3.34 Update Personal Schedule	83
3.35 Notify Schedule Change	84
3.36 Notify Shift Transfer	85
3.37 Add Patient Profile	86
3.38 View Patient Profile	87
3.39 Update Patient Profile.....	88
3.40 Add Medical Record.....	89
3.41 View Medical Record	91
3.42 Update Medical Record	92
3.43 Delete Medical Record.....	93
3.44 Add Treatment Profile	94
3.45 View Treatment Profile	95
3.46 Update Treatment Profile	96
3.47 Export Medical Record.....	97
3.48 Create Appointment At Facility.....	98
3.49 Create Appointment By Specialty	99

3.50 Create Appointment By Doctor	101
3.51 Create Appointment Outside Working Hours.....	102
3.52 View Appointment	103
3.53 Edit Appointment.....	104
3.54 Cancel Appointment	105
3.55 Confirm Appointment	106
3.56 Send Appointment Confirmation.....	107
3.57 Send Appointment Reminder	108
3.58 Initiate Payment.....	109
3.59 View Payment History.....	110
3.60 Refund Cancel Payment	111
3.61 Chatbot Support For Booking	112
3.62 View Specialty	113
3.63 Add Specialty.....	114
3.64 Update Specialty	115
3.65 Delete Specialty	116
3.66 Enter Symptoms.....	117
3.67 View Symptoms.....	118
3.68 Edit Symptoms	119
3.69 Delete Symptoms.....	120
3.70 Create Treatment Plan	121
3.71 View Treatment Plan.....	122
3.72 Edit Treatment Plan	123
3.73 Send Treatment Plan.....	124
3.74 Delete Treatment Plan	125
3.75 Create Electronic Prescription	126
3.76 Order Xray CBCT.....	127
3.77 Order Laboratory Test Service	128
3.78 Order Clinical Test.....	129
3.79 Upload Endodontic Image.....	130
3.80 Upload Xray CBCT Image.....	131
3.81 View Dental Image Library	132
3.82 Attach Image To Treatment Profile	133

3.83 Edit Image By Treatment Profile 134

3.84 View Doctor Performance Report..... 135

3.85 View Dashboard By Doctor 136

3.86 View Dashboard By Customer 137

3.87 Revenue Financial Report 138

I. Project Report

1. Status Report

#	Work Item	Status	Notes (Work Item in Details)
1		Pending	
2		In Progress	
3		Completed	

2. Team Involvements

#	Task	Member	Notes (Task Details, etc.)
1		KienNT	
2		TuanTV	
3		AnhLM	

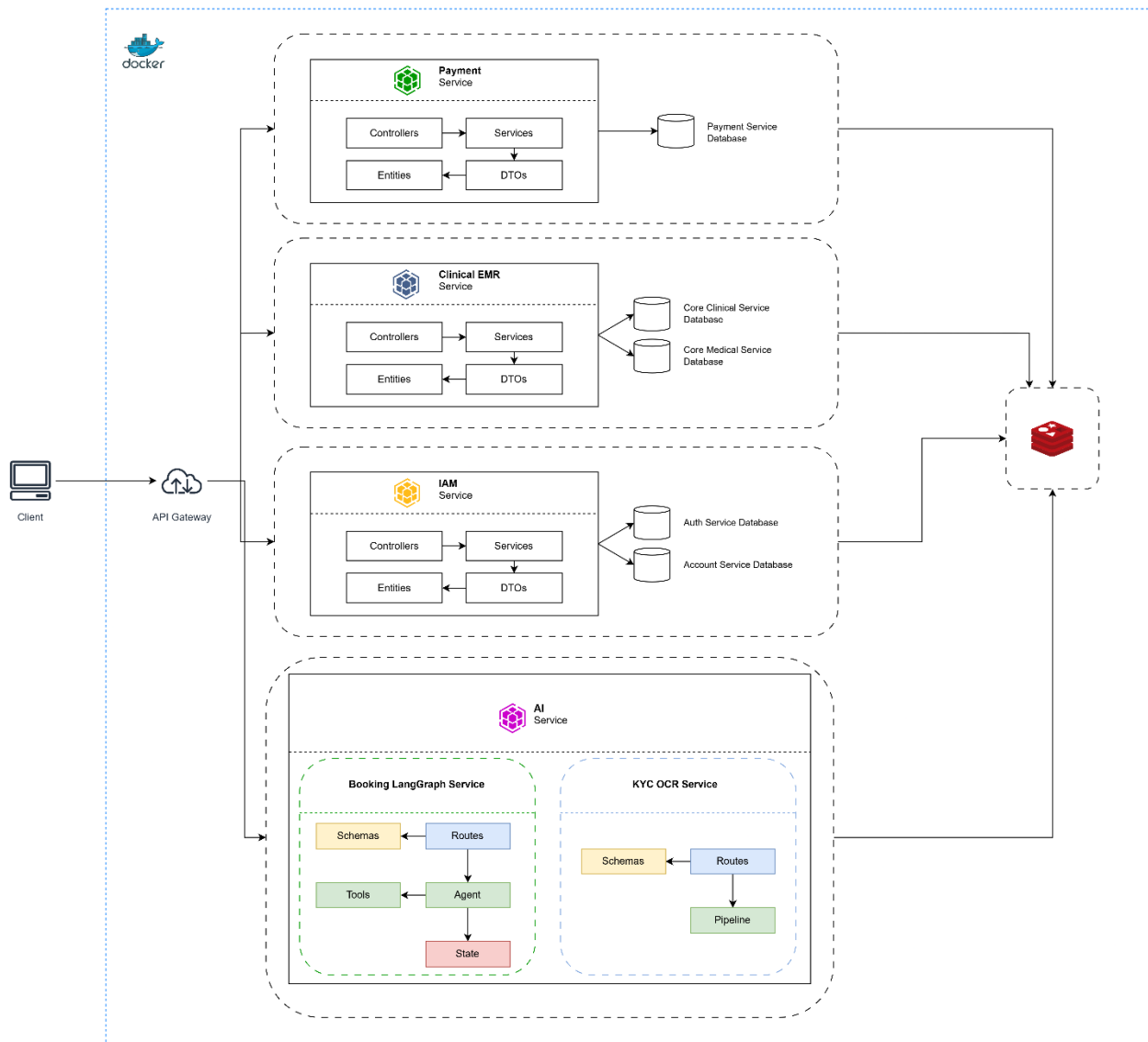
3. Issues/Suggestions

#	Issue	Status	Notes (Solution, Suggestion, etc.)
1		Pending	
2		In Progress	
3		Completed	

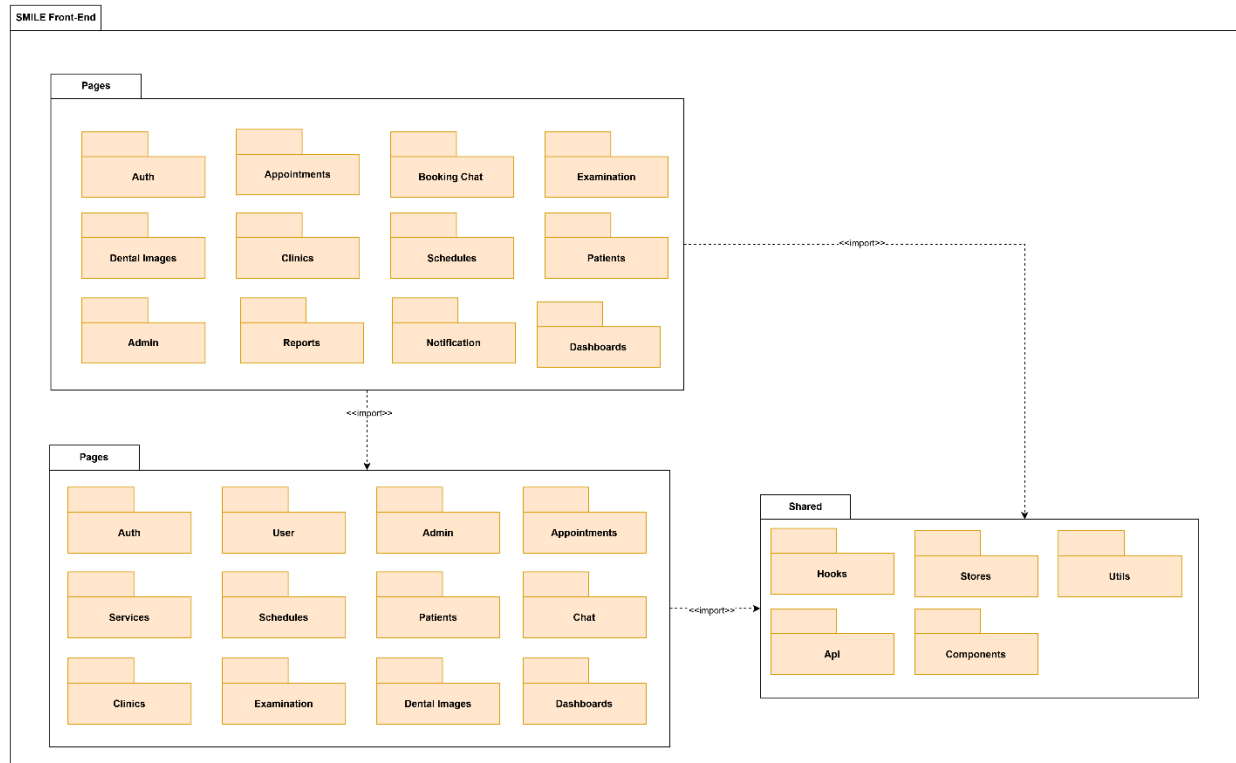
II. Software Design Document

1. System Design

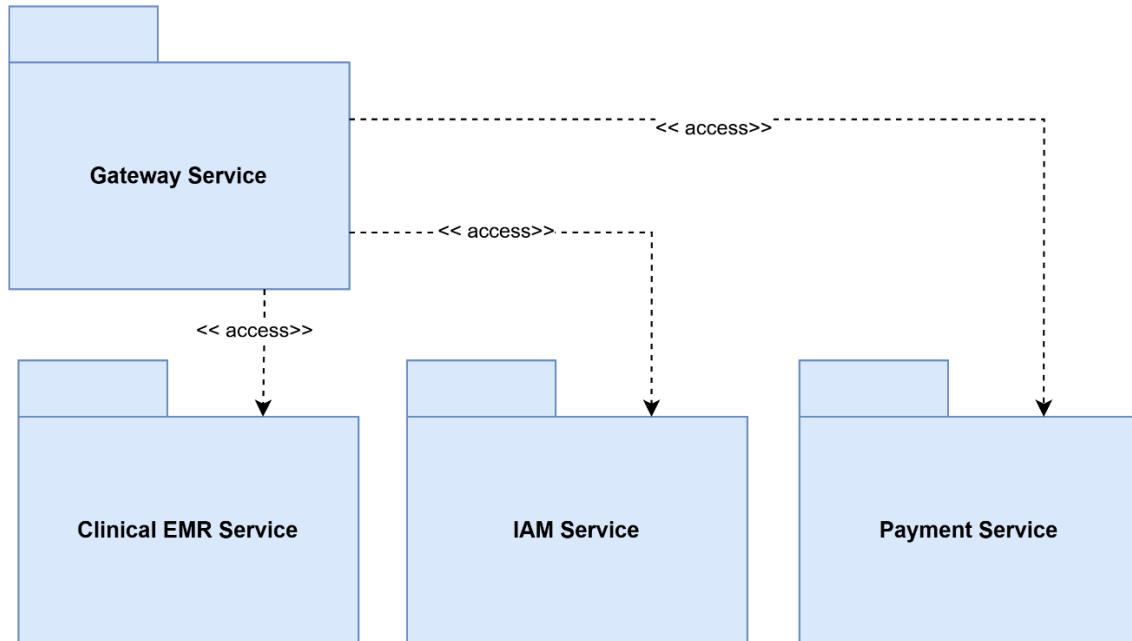
1.1 System Architecture

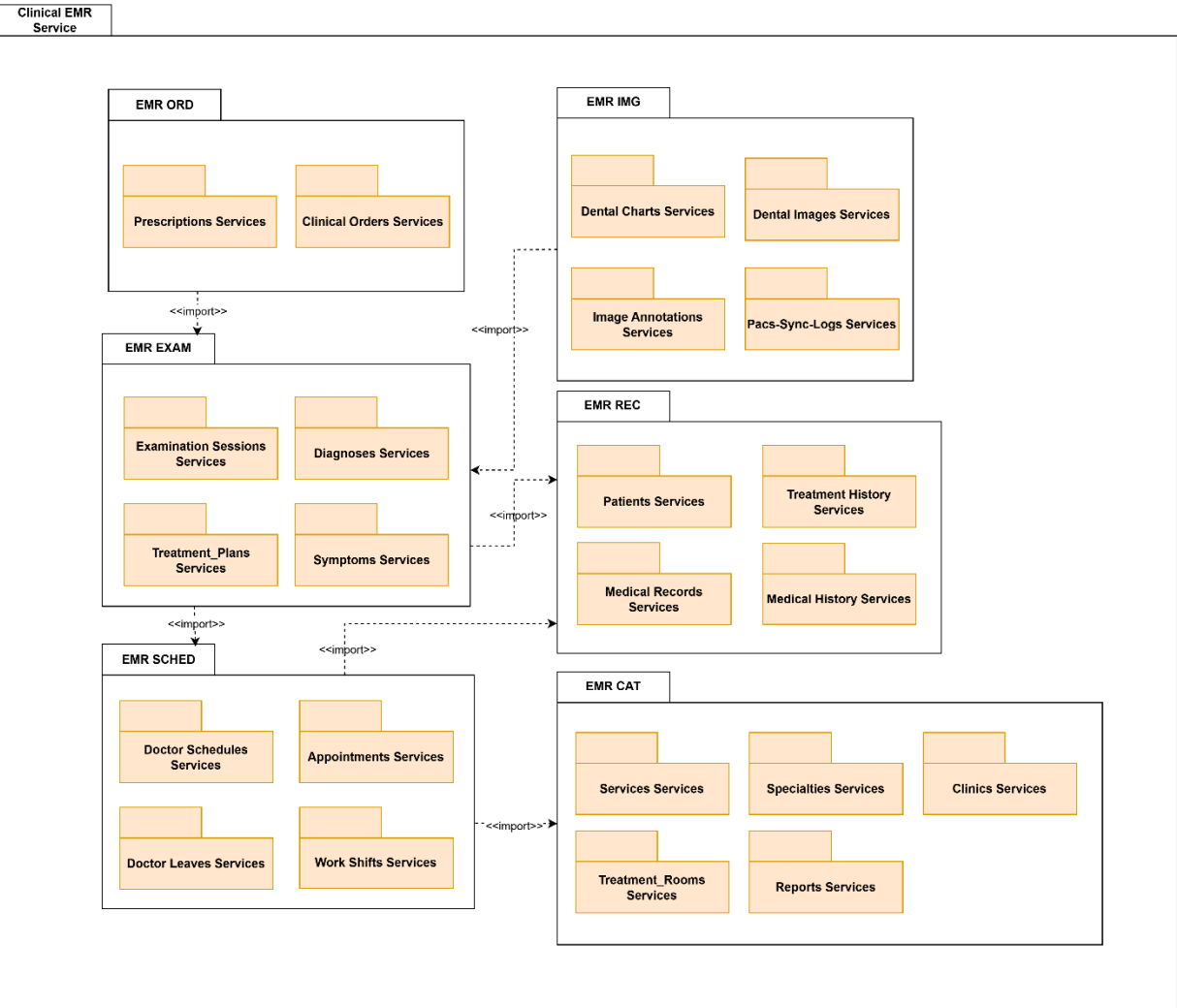


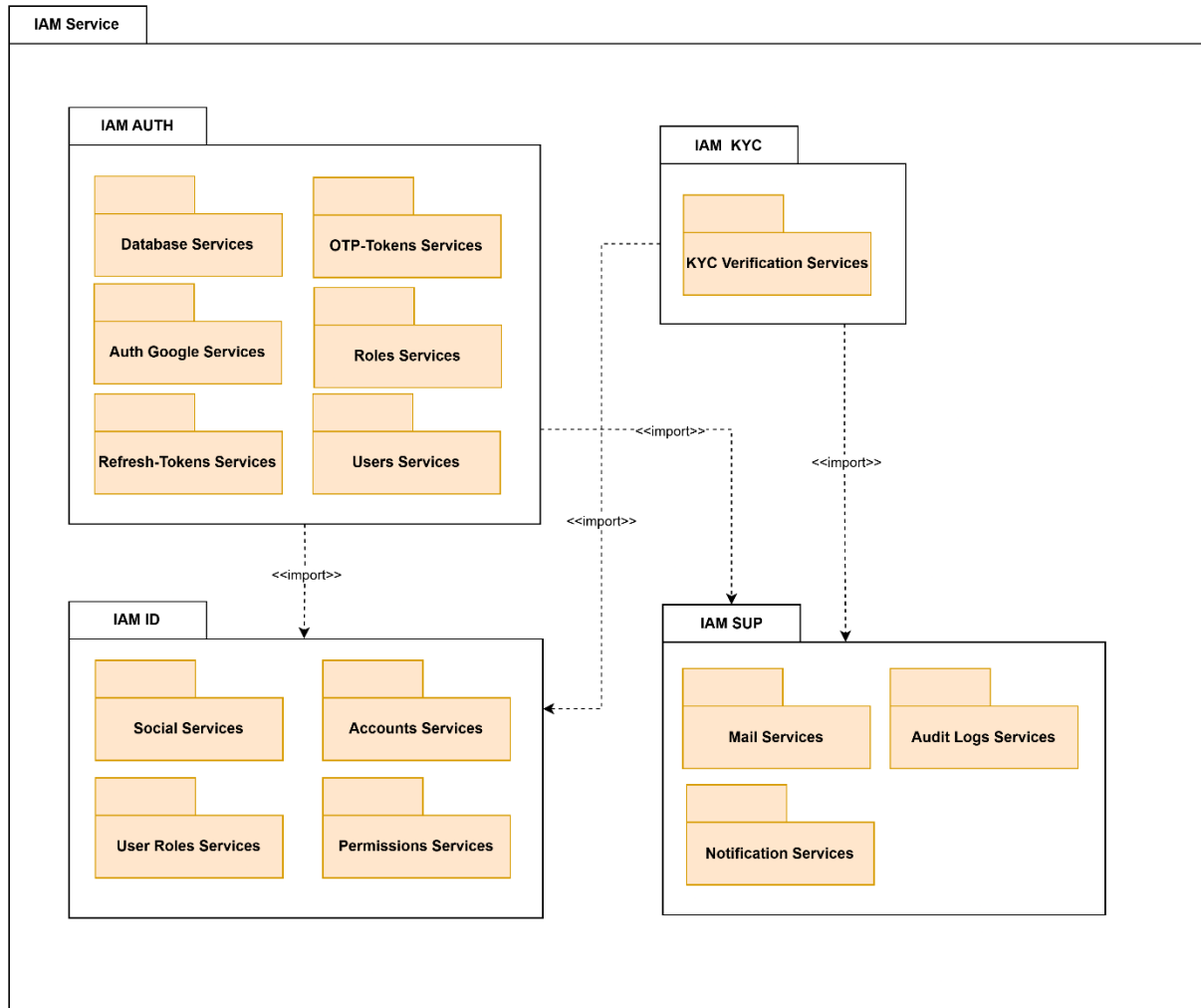
1.2 Package Diagram



SMILE Back-end







Code Packages

No	Package	Description
01	SMILE Front-End	Containing the user interface built with Next.js 15 App Router and React 19, including all pages, features, shared components, API client, and state management.
02	SMILE Back-end	Containing all three NestJS microservices that handle business logic, data access, and API responses for the S.M.I.L.E platform.
03	IAM Service	Identity and Access Management microservice handling authentication, authorization, account management, OAuth2 social login, RBAC, and user profile management.

No	Package	Description
04	Clinical EMR Service	Electronic Medical Records microservice handling all clinical operations: appointments, patient records, examinations, prescriptions, dental imaging, and reporting.

2. Database Design

2.1 Auth Service Database

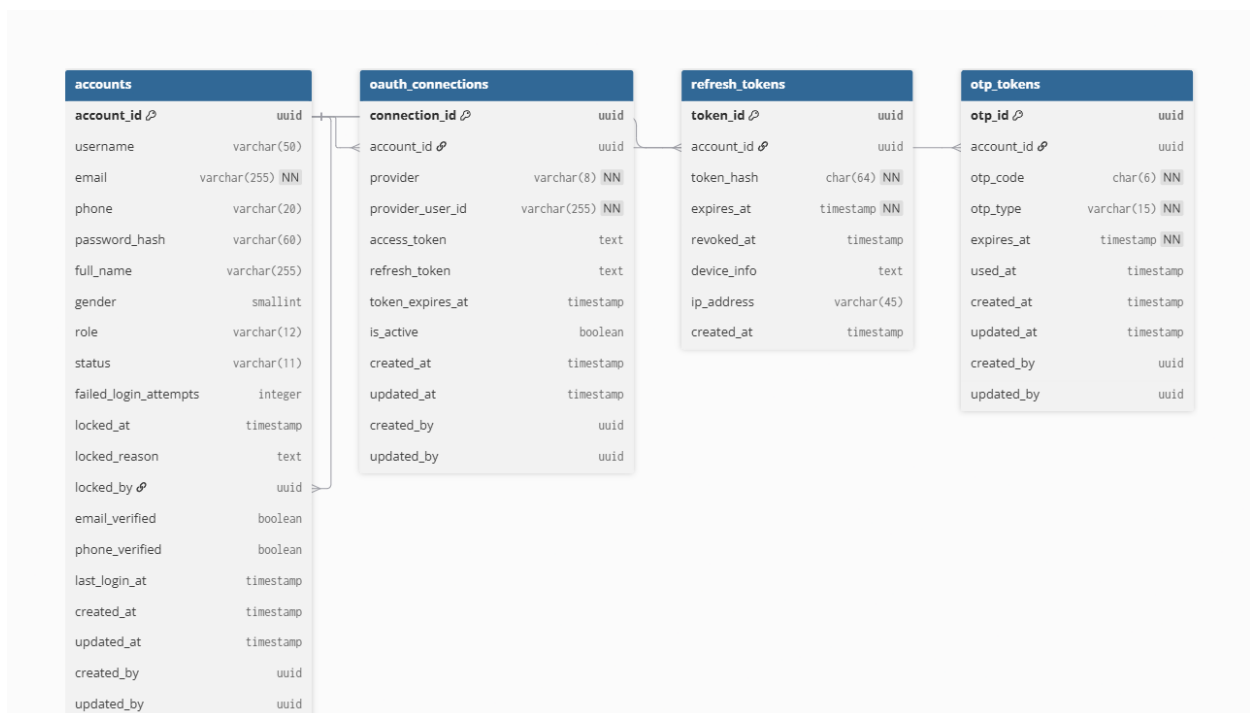


Figure 1. Auth Service Database

Table Descriptions

No	Table	Description
01	accounts	Primary key: - account_id: UUID Foreign keys: - account_id → account_service_db.users.user_id (1:1 identity link) (<i>logical, cross-DB</i>)

		Attributes: • username: VARCHAR (unique, nullable) • email: VARCHAR (unique, nullable) • phone: VARCHAR (unique, nullable) • password_hash: VARCHAR (nullable) • status: VARCHAR (default: ACTIVE) • failed_login_attempts: INT (default: 0) • locked_at: TIMESTAMP (nullable) • locked_reason: TEXT (nullable) • locked_by: UUID (nullable) • email_verified: BOOLEAN (default: false) • phone_verified: BOOLEAN (default: false) • last_login_at: TIMESTAMP (nullable) • created_at: TIMESTAMP • updated_at: TIMESTAMP • created_by: UUID (nullable) • updated_by: UUID (nullable)
02	oauth_connections	Primary key: • connection_id: UUID Foreign keys: • account_id → accounts(account_id) Attributes: • provider: VARCHAR(50) • provider_user_id: VARCHAR(255) • access_token: TEXT (nullable) • refresh_token: TEXT (nullable) • token_expires_at: TIMESTAMP (nullable) • is_active: BOOLEAN (default: true) • created_at: TIMESTAMP • updated_at: TIMESTAMP • created_by: UUID (nullable) • updated_by: UUID (nullable)
03	otp_tokens	Primary key: • otp_id: UUID Foreign keys: • account_id → accounts(account_id) Attributes: • otp_code: VARCHAR(10) • otp_type: VARCHAR(20) • expires_at: TIMESTAMP • used_at: TIMESTAMP (nullable) • created_at: TIMESTAMP • updated_at: TIMESTAMP

		• created_by: UUID (nullable) • updated_by: UUID (nullable)
04	refresh_tokens	Primary key: • token_id: UUID Foreign keys: • account_id → accounts(account_id) Attributes: • token_hash: VARCHAR(255) • expires_at: TIMESTAMP • revoked_at: TIMESTAMP (nullable) • device_info: TEXT (nullable) • ip_address: VARCHAR(45) (nullable) • created_at: TIMESTAMP

2.2 Account Service Database

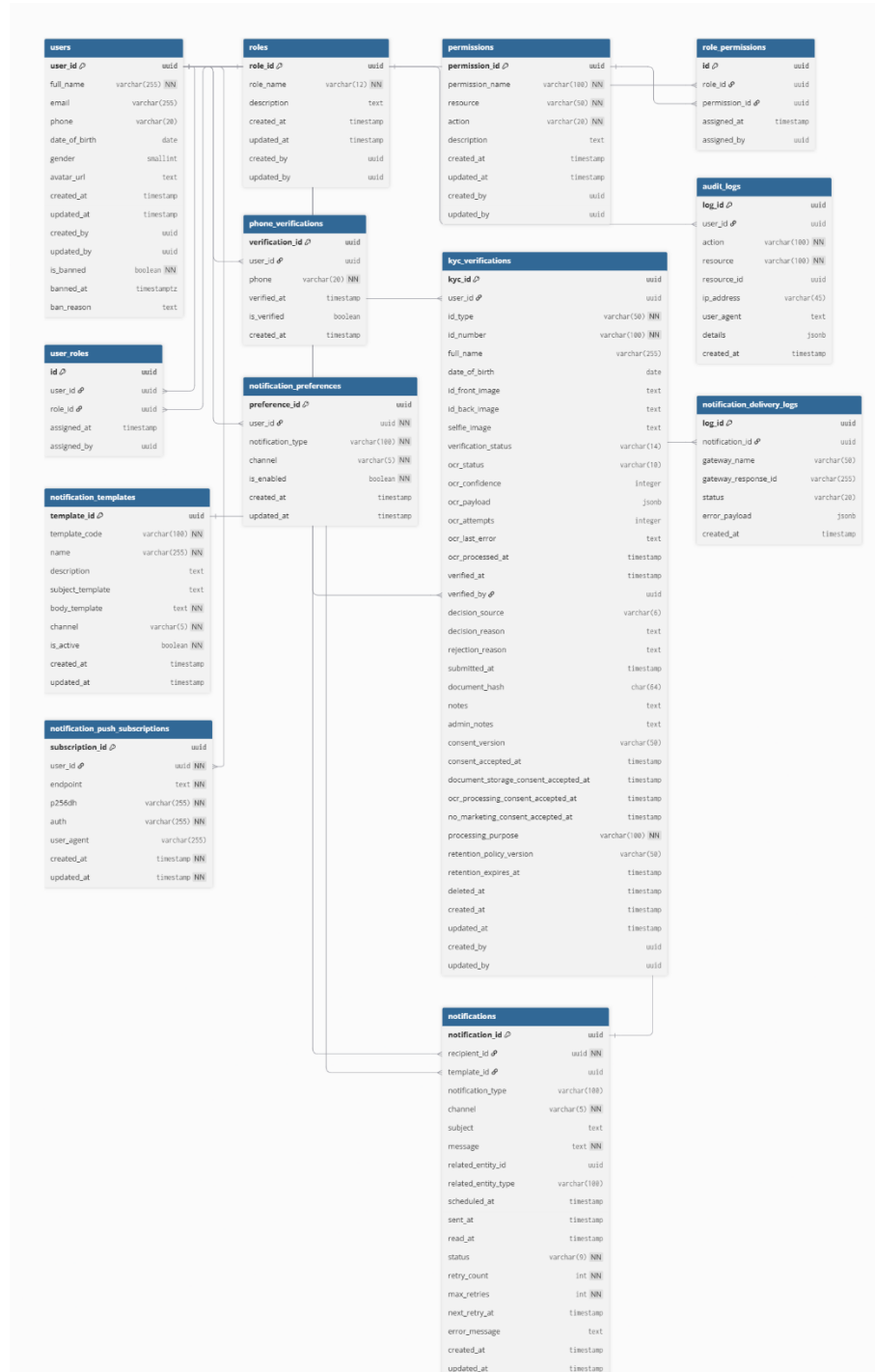


Figure 2. Account Service Database

Table Descriptions

No	Table	Description
01	users	Primary key: • user_id: UUID Foreign keys: • user_id → auth_service_db.accounts.account_id (logical, cross-DB) Attributes: • full_name: VARCHAR(255) • email: VARCHAR(255) (nullable) • phone: VARCHAR(20) (nullable) • date_of_birth: DATE (nullable) • gender: SMALLINT (nullable) • avatar_url: TEXT (nullable) • created_at: TIMESTAMP (default: now, nullable) • updated_at: TIMESTAMP (default: now, nullable) • created_by: UUID (nullable) • updated_by: UUID (nullable) • is_banned: BOOLEAN (default: FALSE) • banned_at: TIMESTAMPTZ (nullable) • ban_reason: TEXT (nullable)
02	roles	Primary key: • role_id: UUID Foreign keys: — none Attributes: • role_name: VARCHAR(12) (unique) • description: TEXT (nullable) • created_at: TIMESTAMP (default: now, nullable) • updated_at: TIMESTAMP (default: now, nullable) • created_by: UUID (nullable) • updated_by: UUID (nullable)
03	permissions	Primary key: • permission_id: UUID Foreign keys: — none Attributes: • permission_name: VARCHAR(100) (unique) • resource: VARCHAR(50) • action: VARCHAR(20)

		• description: TEXT (nullable) • created_at: TIMESTAMP (default: now, nullable) • updated_at: TIMESTAMP (default: now, nullable) • created_by: UUID (nullable) • updated_by: UUID (nullable)
04	role_permissions	Primary key: • id: UUID Foreign keys: • role_id → roles(role_id) (nullable) • permission_id → permissions(permission_id) (nullable) Attributes: • assigned_at: TIMESTAMP (default: now, nullable) • assigned_by: UUID (nullable)
05	user_roles	Primary key: • id: UUID Foreign keys: • user_id → users(user_id) (nullable) • role_id → roles(role_id) (nullable) Attributes: • assigned_at: TIMESTAMP (default: now, nullable) • assigned_by: UUID (nullable)
06	phone_verifications	Primary key: • verification_id: UUID Foreign keys: • user_id → users(user_id) (nullable) Attributes: • phone: VARCHAR(20) • verified_at: TIMESTAMP (nullable) • is_verified: BOOLEAN (default: FALSE, nullable) • created_at: TIMESTAMP (default: now, nullable)
07	kyc_verifications	Primary key: • kyc_id: UUID Foreign keys: • user_id → users(user_id) (nullable) • verified_by → users(user_id) (nullable) Attributes: • id_type: VARCHAR(50) • id_number: VARCHAR(100) • full_name: VARCHAR(255) (nullable) • date_of_birth: DATE (nullable) • id_front_image: TEXT (nullable) • id_back_image: TEXT (nullable)

		• selfie_image: TEXT (nullable) • verification_status: VARCHAR(14) (default: pending, nullable) • ocr_status: VARCHAR(10) (default: PENDING, nullable) • ocr_confidence: INTEGER (nullable) • ocr_payload: JSONB (nullable) • ocr_attempts: INTEGER (default: 0, nullable) • ocr_last_error: TEXT (nullable) • ocr_processed_at: TIMESTAMP (nullable) • verified_at: TIMESTAMP (nullable) • decision_source: VARCHAR(6) (nullable) • decision_reason: TEXT (nullable) • rejection_reason: TEXT (nullable) • submitted_at: TIMESTAMP (nullable) • document_hash: CHAR(64) (nullable) • notes: TEXT (nullable) • admin_notes: TEXT (nullable) • consent_version: VARCHAR(50) (nullable) • consent_accepted_at: TIMESTAMP (nullable) • document_storage_consent_accepted_at: TIMESTAMP (nullable) • ocr_processing_consent_accepted_at: TIMESTAMP (nullable) • no_marketing_consent_accepted_at: TIMESTAMP (nullable) • processing_purpose: VARCHAR(100) (default: identity_verification_and_booking_safety) • retention_policy_version: VARCHAR(50) (nullable) • retention_expires_at: TIMESTAMP (nullable) • deleted_at: TIMESTAMP (nullable) • created_at: TIMESTAMP (default: now, nullable) • updated_at: TIMESTAMP (default: now, nullable) • created_by: UUID (nullable) • updated_by: UUID (nullable)
08	audit_logs	Primary key: • log_id: UUID Foreign keys: • user_id → users(user_id) (nullable) Attributes: • action: VARCHAR(100) • resource: VARCHAR(100) • resource_id: UUID (nullable) • ip_address: VARCHAR(45) (nullable) • user_agent: TEXT (nullable)

		• details: JSONB (nullable) • created_at: TIMESTAMP (default: now, nullable)
09	notification_templates	Primary key: • template_id: UUID Foreign keys: — none Attributes: • template_code: VARCHAR(100) (unique) • name: VARCHAR(255) • description: TEXT (nullable) • subject_template: TEXT (nullable) • body_template: TEXT • channel: VARCHAR(5) • is_active: BOOLEAN (default: TRUE) • created_at: TIMESTAMP (default: now, nullable) • updated_at: TIMESTAMP (default: now, nullable)
10	notification_preferences	Primary key: • preference_id: UUID Foreign keys: • user_id → users(user_id) Attributes: • notification_type: VARCHAR(100) • channel: VARCHAR(5) • is_enabled: BOOLEAN (default: TRUE) • created_at: TIMESTAMP (default: now, nullable) • updated_at: TIMESTAMP (default: now, nullable)
11	notifications	Primary key: • notification_id: UUID Foreign keys: • recipient_id → users(user_id) • template_id → notification_templates(template_id) (nullable) Attributes: • notification_type: VARCHAR(100) (nullable) • channel: VARCHAR(5) • subject: TEXT (nullable) • message: TEXT • related_entity_id: UUID (nullable) • related_entity_type: VARCHAR(100) (nullable) • scheduled_at: TIMESTAMP (nullable) • sent_at: TIMESTAMP (nullable) • read_at: TIMESTAMP (nullable) • status: VARCHAR(9) (default: PENDING)

		• retry_count: INT (default: 0) • max_retries: INT (default: 3) • next_retry_at: TIMESTAMP (nullable) • error_message: TEXT (nullable) • created_at: TIMESTAMP (default: now, nullable) • updated_at: TIMESTAMP (default: now, nullable)
12	notification_delivery_logs	Primary key: • log_id: UUID Foreign keys: • notification_id → notifications(notification_id) (nullable) Attributes: • gateway_name: VARCHAR(50) (nullable) • gateway_response_id: VARCHAR(255) (nullable) • status: VARCHAR(20) (nullable) • error_payload: JSONB (nullable) • created_at: TIMESTAMP (default: now, nullable)
13	notification_push_subscriptions	Primary key: • subscription_id: UUID Foreign keys: • user_id → users(user_id) Attributes: • endpoint: TEXT • p256dh: VARCHAR(255) • auth: VARCHAR(255) • user_agent: VARCHAR(255) (nullable) • created_at: TIMESTAMP (default: now) • updated_at: TIMESTAMP (default: now)

2.3 Core Clinic Service Database

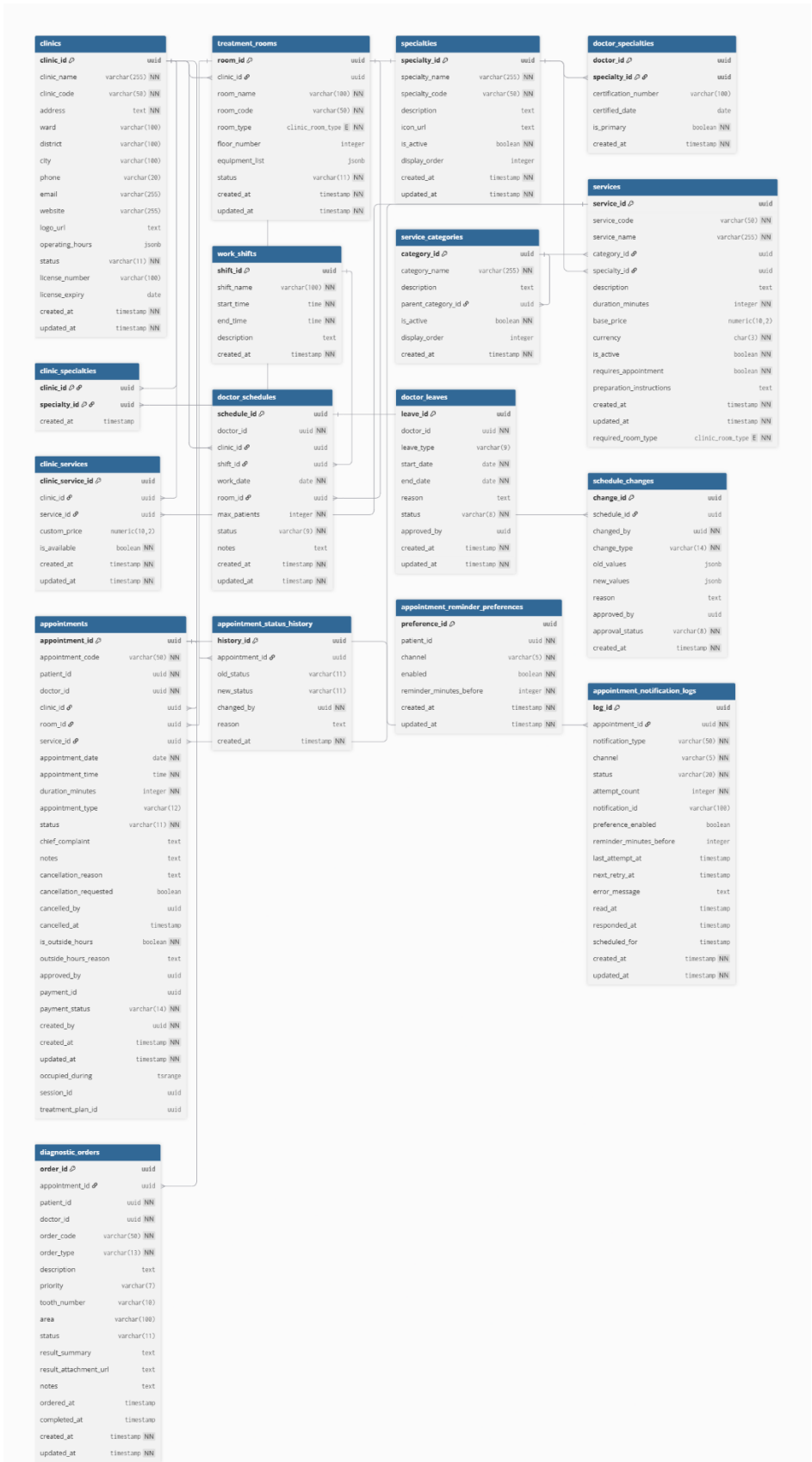


Table Descriptions

No	Table	Description
01	clinics	Primary key: • clinic_id: UUID Foreign keys: — none Attributes: • clinic_name: VARCHAR(255) • clinic_code: VARCHAR(50) (unique) • address: TEXT • ward: VARCHAR(100) (nullable) • district: VARCHAR(100) (nullable) • city: VARCHAR(100) (nullable) • phone: VARCHAR(20) (nullable) • email: VARCHAR(255) (nullable) • website: VARCHAR(255) (nullable) • logo_url: TEXT (nullable) • operating_hours: JSONB (nullable) • status: VARCHAR(20) (default: ACTIVE) • license_number: VARCHAR(100) (nullable) • license_expiry: DATE (nullable) • created_at: TIMESTAMP • updated_at: TIMESTAMP
02	treatment_rooms	Primary key: • room_id: UUID Foreign keys: • clinic_id → clinics(clinic_id) Attributes: • room_name: VARCHAR(100) • room_code: VARCHAR(50) • room_type: VARCHAR (enum) • floor_number: INT (nullable) • equipment_list: JSONB (nullable) • status: VARCHAR(20) (default: AVAILABLE) • created_at: TIMESTAMP • updated_at: TIMESTAMP
03	work_shifts	Primary key: • shift_id: UUID Foreign keys: — none Attributes:

No	Table	Description
		• shift_name: VARCHAR(100) • start_time: TIME • end_time: TIME • description: TEXT (nullable) • created_at: TIMESTAMP
04	specialties	Primary key: • specialty_id: UUID Foreign keys: — none Attributes: • specialty_name: VARCHAR(255) • specialty_code: VARCHAR(50) (unique) • description: TEXT (nullable) • icon_url: TEXT (nullable) • is_active: BOOLEAN (default: true) • display_order: INT (nullable) • created_at: TIMESTAMP • updated_at: TIMESTAMP
05	doctor_specialties	Primary key: • doctor_id: UUID • specialty_id: UUID Foreign keys: • specialty_id → specialties(specialty_id) • doctor_id → account_service_db.users.user_id (<i>logical, cross-DB</i>) Attributes: • certification_number: VARCHAR(100) (nullable) • certified_date: DATE (nullable) • is_primary: BOOLEAN (default: false) • created_at: TIMESTAMP
06	doctor_leaves	Primary key: • leave_id: UUID Foreign keys: • doctor_id → account_service_db.users.user_id (<i>logical, cross-DB</i>) Attributes: • leave_type: VARCHAR(50) (nullable) • start_date: DATE

No	Table	Description
		• end_date: DATE • reason: TEXT (nullable) • status: VARCHAR(20) (default: pending) • approved_by: UUID (nullable) • created_at: TIMESTAMP • updated_at: TIMESTAMP
07	doctor_schedules	Primary key: • schedule_id: UUID Foreign keys: • clinic_id → clinics(clinic_id) • shift_id → work_shifts(shift_id) • room_id → treatment_rooms(room_id) • doctor_id → account_service_db.users.user_id (logical, cross-DB) Attributes: • work_date: DATE • max_patients: INT (default: 20) • status: VARCHAR(20) (default: scheduled) • notes: TEXT (nullable) • created_at: TIMESTAMP • updated_at: TIMESTAMP
08	schedule_changes	Primary key: • change_id: UUID Foreign keys: • schedule_id → doctor_schedules(schedule_id) Attributes: • changed_by: UUID • change_type: VARCHAR(50) • old_values: JSONB (nullable) • new_values: JSONB (nullable) • reason: TEXT (nullable) • approved_by: UUID (nullable) • approval_status: VARCHAR(20) (default: pending) • created_at: TIMESTAMP
09	service_categories	Primary key: • category_id: UUID Foreign keys: • parent_category_id →

No	Table	Description
		service_categories(category_id) Attributes: • category_name: VARCHAR(255) • description: TEXT (nullable) • is_active: BOOLEAN (default: true) • display_order: INT (nullable) • created_at: TIMESTAMP
10	services	Primary key: • service_id: UUID Foreign keys: • category_id → service_categories(category_id) • specialty_id → specialties(specialty_id) Attributes: • service_code: VARCHAR(50) (unique) • service_name: VARCHAR(255) • description: TEXT (nullable) • duration_minutes: INT (default: 30) • required_room_type: VARCHAR (enum) • base_price: DECIMAL(10,2) (nullable) • currency: VARCHAR(10) (default: VND) • is_active: BOOLEAN (default: true) • requires_appointment: BOOLEAN (default: true) • preparation_instructions: TEXT (nullable) • created_at: TIMESTAMP • updated_at: TIMESTAMP
11	clinic_services	Primary key: • clinic_service_id: UUID Foreign keys: • clinic_id → clinics(clinic_id) • service_id → services(service_id) Attributes: • custom_price: DECIMAL(10,2) (nullable) • is_available: BOOLEAN (default: true) • created_at: TIMESTAMP • updated_at: TIMESTAMP
12	appointments	Primary key: • appointment_id: UUID Foreign keys: • clinic_id → clinics(clinic_id)

No	Table	Description
		• room_id → treatment_rooms(room_id) • service_id → services(service_id) • patient_id → core_medical_service_db.patients.patient_id (<i>logical, cross-DB</i>) • doctor_id → account_service_db.users.user_id (<i>logical, cross-DB</i>) • payment_id → payment_service_db.payments.payment_id (<i>logical, cross-DB</i>) Attributes: • appointment_code: VARCHAR(50) (unique) • appointment_date: DATE • appointment_time: TIME • duration_minutes: INT (default: 30) • appointment_type: VARCHAR(50) (nullable) • status: VARCHAR(20) (default: scheduled) • chief_complaint: TEXT (nullable) • notes: TEXT (nullable) • session_id: UUID (nullable) • treatment_plan_id: UUID (nullable) • cancellation_reason: TEXT (nullable) • cancelled_by: UUID (nullable) • cancelled_at: TIMESTAMP (nullable) • is_outside_hours: BOOLEAN (default: false) • outside_hours_reason: TEXT (nullable) • approved_by: UUID (nullable) • payment_status: VARCHAR(20) (default: unpaid) • created_by: UUID • created_at: TIMESTAMP • updated_at: TIMESTAMP • occupied_during: TSRANGE (generated)
13	appointment_status_history	Primary key: • history_id: UUID Foreign keys: • appointment_id → appointments(appointment_id) Attributes: • old_status: VARCHAR(20) (nullable)

No	Table	Description
		• new_status: VARCHAR(20) (nullable) • changed_by: UUID • reason: TEXT (nullable) • created_at: TIMESTAMP
14	appointment_reminder_preferences	Primary key: • preference_id: UUID Foreign keys: • patient_id → core_medical_service_db.patients.patient_id (<i>logical, cross-DB</i>) Attributes: • channel: VARCHAR(20) (default: APP) • enabled: BOOLEAN (default: true) • reminder_minutes_before: INT (default: 1440) • created_at: TIMESTAMP • updated_at: TIMESTAMP
15	appointment_notification_logs	Primary key: • log_id: UUID Foreign keys: • appointment_id → appointments(appointment_id) Attributes: • notification_type: VARCHAR(50) • channel: VARCHAR(20) (default: APP) • status: VARCHAR(20) • attempt_count: INT (default: 0) • notification_id: VARCHAR(100) (nullable) • preference_enabled: BOOLEAN (nullable) • reminder_minutes_before: INT (nullable) • last_attempt_at: TIMESTAMP (nullable) • next_retry_at: TIMESTAMP (nullable) • error_message: TEXT (nullable) • read_at: TIMESTAMP (nullable) • responded_at: TIMESTAMP (nullable) • created_at: TIMESTAMP • updated_at: TIMESTAMP
16	diagnostic_orders	Primary key: • order_id: UUID Foreign keys: • appointment_id →

No	Table	Description
		appointments(appointment_id) • patient_id → core_medical_service_db.patients.patient_id (<i>logical, cross-DB</i>) • doctor_id → account_service_db.users.user_id (<i>logical, cross-DB</i>) Attributes: • order_code: VARCHAR(50) (unique) • order_type: VARCHAR(50) • description: TEXT (nullable) • priority: VARCHAR(20) (default: routine) • tooth_number: VARCHAR(10) (nullable) • area: VARCHAR(100) (nullable) • status: VARCHAR(20) (default: ordered) • result_summary: TEXT (nullable) • result_attachment_url: TEXT (nullable) • notes: TEXT (nullable) • ordered_at: TIMESTAMP (nullable) • completed_at: TIMESTAMP (nullable) • created_at: TIMESTAMP • updated_at: TIMESTAMP
17	idempotency_keys	Primary key: • idempotency_key: VARCHAR(255) Foreign keys: — none Attributes: • method: VARCHAR(10) • path: VARCHAR(512) • status: VARCHAR(20) (default: in_progress) • response_status: INT (nullable) • response_body: JSONB (nullable) • created_at: TIMESTAMP • expires_at: TIMESTAMP

2.4 Core Medical Service Database

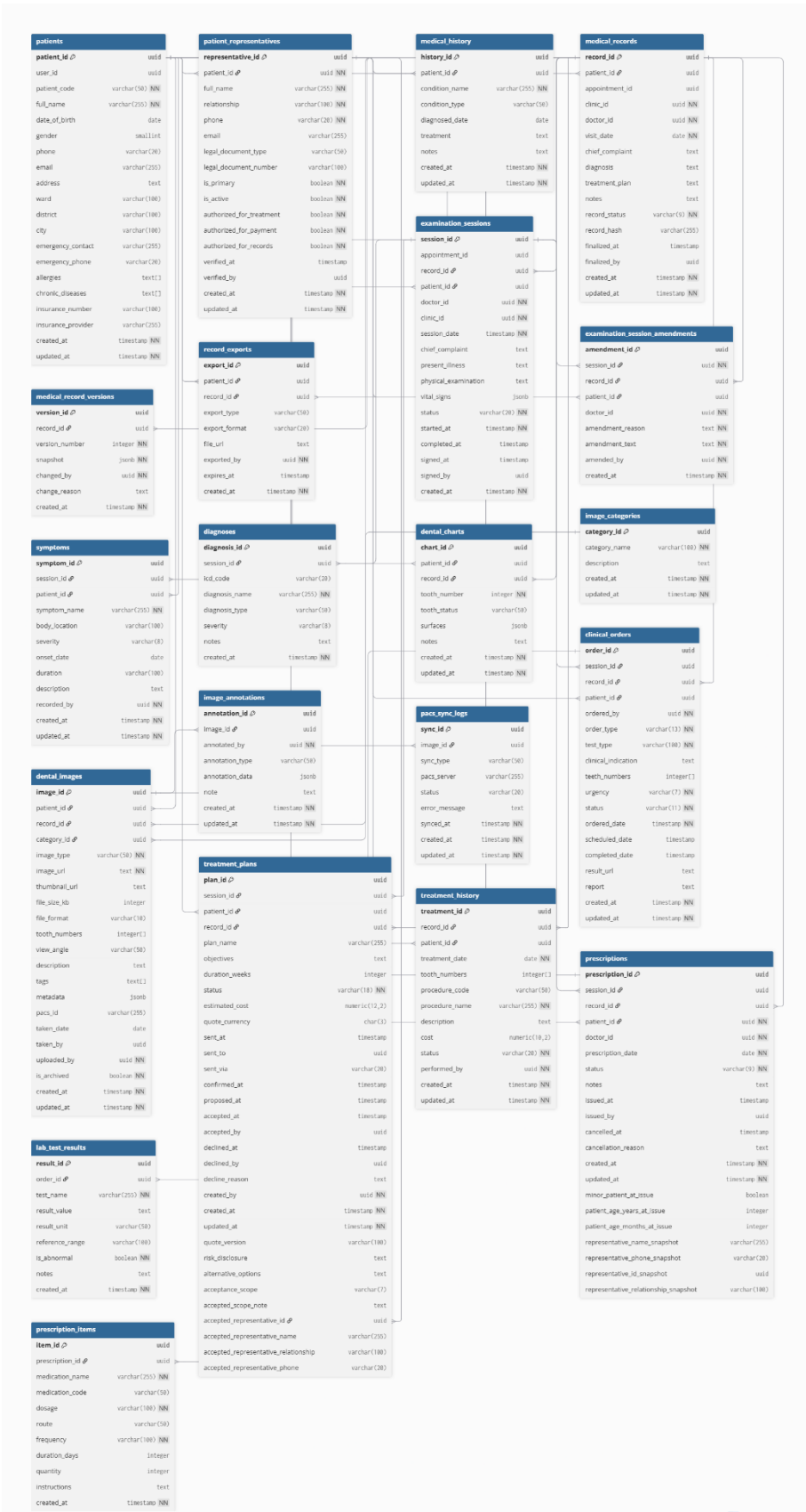


Table Descriptions

No	Table	Description
01	patients	Primary key: • patient_id: UUID Foreign keys: • user_id → account_service_db.users.user_id (<i>logical, cross-DB</i>) Attributes: • patient_code: VARCHAR(50) (unique) • full_name: VARCHAR(255) • date_of_birth: DATE (nullable) • gender: VARCHAR(10) (nullable) • phone: VARCHAR(20) (nullable) • email: VARCHAR(255) (nullable) • address: TEXT (nullable) • ward: VARCHAR(100) (nullable) • district: VARCHAR(100) (nullable) • city: VARCHAR(100) (nullable) • emergency_contact: VARCHAR(255) (nullable) • emergency_phone: VARCHAR(20) (nullable) • blood_type: VARCHAR(10) (nullable) • allergies: TEXT[] (nullable) • chronic_diseases: TEXT[] (nullable) • insurance_number: VARCHAR(100) (nullable) • insurance_provider: VARCHAR(255) (nullable) • created_at: TIMESTAMP • updated_at: TIMESTAMP
02	patient_representatives	Primary key: • representative_id: UUID Foreign keys: • patient_id → patients(patient_id) Attributes: • full_name: VARCHAR(255) • relationship: VARCHAR(100) • phone: VARCHAR(20) • email: VARCHAR(255) (nullable) • legal_document_type: VARCHAR(50) (nullable) • legal_document_number: VARCHAR(100) (nullable) • is_primary: BOOLEAN (default: false) • is_active: BOOLEAN (default: true) • authorized_for_treatment: BOOLEAN (default: false) • authorized_for_payment: BOOLEAN (default: false)

No	Table	Description
		• authorized_for_records: BOOLEAN (default: false) • verified_at: TIMESTAMP (nullable) • verified_by: UUID (nullable) • created_at: TIMESTAMP • updated_at: TIMESTAMP
03	medical_history	Primary key: • history_id: UUID Foreign keys: • patient_id → patients(patient_id) Attributes: • condition_name: VARCHAR(255) • condition_type: VARCHAR(50) (nullable) • diagnosed_date: DATE (nullable) • treatment: TEXT (nullable) • notes: TEXT (nullable) • created_at: TIMESTAMP • updated_at: TIMESTAMP
04	medical_records	Primary key: • record_id: UUID Foreign keys: • patient_id → patients(patient_id) • appointment_id → core_clinic_service_db.appointments.appointment_id (<i>logical, cross-DB</i>) • clinic_id → core_clinic_service_db.clinics.clinic_id (<i>logical, cross-DB</i>) • doctor_id → account_service_db.users.user_id (<i>logical, cross-DB</i>) Attributes: • visit_date: DATE • chief_complaint: TEXT (nullable) • diagnosis: TEXT (nullable) • treatment_plan: TEXT (nullable) • notes: TEXT (nullable) • record_status: VARCHAR(20) (default: draft) • record_hash: VARCHAR(255) (nullable) • finalized_at: TIMESTAMP (nullable) • finalized_by: UUID (nullable) • created_at: TIMESTAMP • updated_at: TIMESTAMP

No	Table	Description
05	medical_record_versions	Primary key: • version_id: UUID Foreign keys: • record_id → medical_records(record_id) Attributes: • version_number: INT • snapshot: JSONB • changed_by: UUID • change_reason: TEXT (nullable) • created_at: TIMESTAMP
06	examination_sessions	Primary key: • session_id: UUID Foreign keys: • record_id → medical_records(record_id) • patient_id → patients(patient_id) • doctor_id → account_service_db.users.user_id (<i>logical, cross-DB</i>) • clinic_id → core_clinic_service_db.clinics.clinic_id (<i>logical, cross-DB</i>) Attributes: • appointment_id: UUID (nullable) • session_date: TIMESTAMP (default:) • chief_complaint: TEXT (nullable) • present_illness: TEXT (nullable) • physical_examination: TEXT (nullable) • vital_signs: JSONB (nullable) • status: VARCHAR(20) (default: in_progress) • started_at: TIMESTAMP (default:) • completed_at: TIMESTAMP (nullable) • signed_at: TIMESTAMP (nullable) • signed_by: UUID (nullable) • created_at: TIMESTAMP
07	examination_session_amendments	Primary key: • amendment_id: UUID Foreign keys: • session_id → examination_sessions(session_id) • record_id → medical_records(record_id) Attributes: • patient_id: UUID (nullable) • doctor_id: UUID • amendment_reason: TEXT

No	Table	Description
		• amendment_text: TEXT • amended_by: UUID • created_at: TIMESTAMP
08	diagnoses	Primary key: • diagnosis_id: UUID Foreign keys: • session_id → examination_sessions(session_id) Attributes: • icd_code: VARCHAR(20) (nullable) • diagnosis_name: VARCHAR(255) • diagnosis_type: VARCHAR(50) (nullable) • severity: VARCHAR(20) (nullable) • notes: TEXT (nullable) • created_at: TIMESTAMP
09	symptoms	Primary key: • symptom_id: UUID Foreign keys: • session_id → examination_sessions(session_id) • patient_id → patients(patient_id) Attributes: • symptom_name: VARCHAR(255) • body_location: VARCHAR(100) (nullable) • severity: VARCHAR(20) (nullable) • onset_date: DATE (nullable) • duration: VARCHAR(100) (nullable) • description: TEXT (nullable) • recorded_by: UUID • created_at: TIMESTAMP • updated_at: TIMESTAMP
10	clinical_orders	Primary key: • order_id: UUID Foreign keys: • session_id → examination_sessions(session_id) • record_id → medical_records(record_id) • patient_id → patients(patient_id) Attributes: • ordered_by: UUID • order_type: VARCHAR(50) • test_type: VARCHAR(100) • clinical_indication: TEXT (nullable)

No	Table	Description
		- teeth_numbers: INT[] (nullable) - urgency: VARCHAR(20) (default: routine) - status: VARCHAR(20) (default: ordered) - ordered_date: TIMESTAMP (default:) - scheduled_date: TIMESTAMP (nullable) - completed_date: TIMESTAMP (nullable) - result_url: TEXT (nullable) - report: TEXT (nullable) - created_at: TIMESTAMP - updated_at: TIMESTAMP
11	lab_test_results	Primary key: - result_id: UUID Foreign keys: - order_id → clinical_orders(order_id) Attributes: - test_name: VARCHAR(255) - result_value: TEXT (nullable) - result_unit: VARCHAR(50) (nullable) - reference_range: VARCHAR(100) (nullable) - is_abnormal: BOOLEAN (default: false) - notes: TEXT (nullable) - created_at: TIMESTAMP
12	prescriptions	Primary key: - prescription_id: UUID Foreign keys: - session_id → examination_sessions(session_id) - record_id → medical_records(record_id) - patient_id → patients(patient_id) - doctor_id → account_service_db.users.user_id (logical, cross-DB) Attributes: - prescription_date: DATE (default:) - status: VARCHAR(20) (default: draft) - notes: TEXT (nullable) - digital_signature_id: UUID (nullable) - issued_at: TIMESTAMP (nullable) - issued_by: UUID (nullable) - minor_patient_at_issue: BOOLEAN (nullable) - patient_age_years_at_issue: INT (nullable) - patient_age_months_at_issue: INT (nullable) - representative_id_snapshot: UUID (nullable)

No	Table	Description
		• representative_name_snapshot: VARCHAR(255) (nullable) • representative_relationship_snapshot: VARCHAR(100) (nullable) • representative_phone_snapshot: VARCHAR(20) (nullable) • cancelled_at: TIMESTAMP (nullable) • cancellation_reason: TEXT (nullable) • created_at: TIMESTAMP • updated_at: TIMESTAMP
13	prescription_items	Primary key: • item_id: UUID Foreign keys: • prescription_id → prescriptions(prescription_id) Attributes: • medication_name: VARCHAR(255) • medication_code: VARCHAR(50) (nullable) • dosage: VARCHAR(100) • route: VARCHAR(50) (nullable) • frequency: VARCHAR(100) • duration_days: INT (nullable) • quantity: INT (nullable) • instructions: TEXT (nullable) • created_at: TIMESTAMP
14	treatment_plans	Primary key: • plan_id: UUID Foreign keys: • session_id → examination_sessions(session_id) • patient_id → patients(patient_id) • record_id → medical_records(record_id) Attributes: • plan_name: VARCHAR(255) (nullable) • objectives: TEXT (nullable) • duration_weeks: INT (nullable) • status: VARCHAR(20) (default: draft) • estimated_cost: DECIMAL(12,2) (nullable) • quote_currency: VARCHAR(3) (nullable) • quote_version: VARCHAR(100) (nullable) • risk_disclosure: TEXT (nullable) • alternative_options: TEXT (nullable) • sent_at: TIMESTAMP (nullable) • sent_to: UUID (nullable) • sent_via: VARCHAR(20) (nullable)

No	Table	Description
		• confirmed_at: TIMESTAMP (nullable) • proposed_at: TIMESTAMP (nullable) • accepted_at: TIMESTAMP (nullable) • accepted_by: UUID (nullable) • declined_at: TIMESTAMP (nullable) • declined_by: UUID (nullable) • decline_reason: TEXT (nullable) • acceptance_scope: VARCHAR(20) (nullable) • accepted_scope_note: TEXT (nullable) • accepted_representative_id: UUID (nullable) • accepted_representative_name: VARCHAR(255) (nullable) • accepted_representative_relationship: VARCHAR(100) (nullable) • accepted_representative_phone: VARCHAR(20) (nullable) • created_by: UUID • created_at: TIMESTAMP • updated_at: TIMESTAMP
15	treatment_history	Primary key: • treatment_id: UUID Foreign keys: • record_id → medical_records(record_id) • patient_id → patients(patient_id) Attributes: • treatment_date: DATE • tooth_numbers: INT[] (nullable) • procedure_code: VARCHAR(50) (nullable) • procedure_name: VARCHAR(255) • description: TEXT (nullable) • cost: DECIMAL(10,2) (nullable) • status: VARCHAR(20) (default: completed) • performed_by: UUID • created_at: TIMESTAMP • updated_at: TIMESTAMP
16	dental_charts	Primary key: • chart_id: UUID Foreign keys: • patient_id → patients(patient_id) • record_id → medical_records(record_id) Attributes: • tooth_number: INT • tooth_status: VARCHAR(50) (nullable)

No	Table	Description
		• surfaces: JSONB (nullable) • notes: TEXT (nullable) • created_at: TIMESTAMP • updated_at: TIMESTAMP
17	image_categories	Primary key: • category_id: UUID Foreign keys: — none Attributes: • category_name: VARCHAR(100) • description: TEXT (nullable) • created_at: TIMESTAMP • updated_at: TIMESTAMP
18	dental_images	Primary key: • image_id: UUID Foreign keys: • patient_id → patients(patient_id) • record_id → medical_records(record_id) • category_id → image_categories(category_id) Attributes: • image_type: VARCHAR(50) • image_url: TEXT • thumbnail_url: TEXT (nullable) • file_size_kb: INT (nullable) • file_format: VARCHAR(10) (nullable) • tooth_numbers: INT[] (nullable) • view_angle: VARCHAR(50) (nullable) • description: TEXT (nullable) • tags: TEXT[] (nullable) • metadata: JSONB (nullable) • pacs_id: VARCHAR(255) (nullable) • taken_date: DATE (nullable) • taken_by: UUID (nullable) • uploaded_by: UUID • is_archived: BOOLEAN (default: false) • created_at: TIMESTAMP • updated_at: TIMESTAMP
19	image_annotations	Primary key: • annotation_id: UUID Foreign keys:

No	Table	Description
		• image_id → dental_images(image_id) Attributes: • annotated_by: UUID • annotation_type: VARCHAR(50) (nullable) • annotation_data: JSONB (nullable) • note: TEXT (nullable) • created_at: TIMESTAMP • updated_at: TIMESTAMP
20	pacs_sync_logs	Primary key: • sync_id: UUID Foreign keys: • image_id → dental_images(image_id) Attributes: • sync_type: VARCHAR(50) (nullable) • pacs_server: VARCHAR(255) (nullable) • status: VARCHAR(20) (nullable) • error_message: TEXT (nullable) • synced_at: TIMESTAMP (default:) • created_at: TIMESTAMP • updated_at: TIMESTAMP
21	record_exports	Primary key: • export_id: UUID Foreign keys: • patient_id → patients(patient_id) • record_id → medical_records(record_id) Attributes: • export_type: VARCHAR(50) (nullable) • export_format: VARCHAR(20) (nullable) • file_url: TEXT (nullable) • exported_by: UUID • expires_at: TIMESTAMP (nullable) • created_at: TIMESTAMP

2.5 Payment Service Database

payments	
payment_id	uuid
appointment_id	uuid NN
amount	numeric(12,2) NN
currency	char(3) NN
status	varchar(8) NN
provider	varchar(30) NN
provider_txn_ref	varchar(100)
order_info	text
refund_amount	numeric(12,2)
refunded_at	timestamp
refund_status	varchar(12)
refund_reason	text
refund_requested_by	uuid
refund_requested_at	timestamp
refund_reviewed_by	uuid
refund_reviewed_at	timestamp
created_at	timestamp
updated_at	timestamp

Table Descriptions

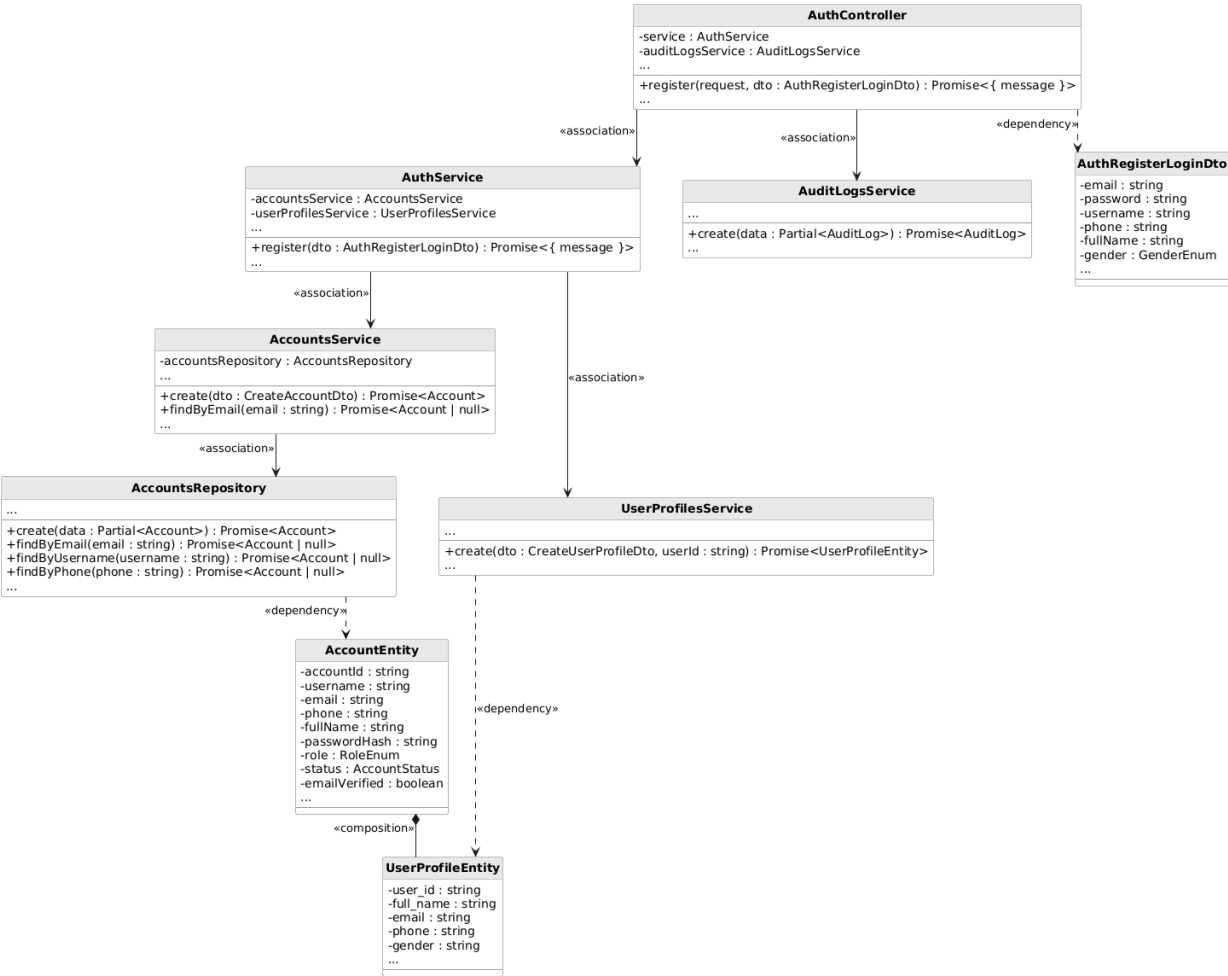
No	Table	Description
01	payments	Primary key: • payment_id: UUID Foreign keys: • appointment_id → core_clinic_service_db.appointments.appointment_id (<i>logical, cross-DB</i>) Attributes: • amount: NUMERIC(12,2) • currency: VARCHAR(10) (default: VND) • status: VARCHAR(20) (default: pending) • provider: VARCHAR(30) (default: vnpay) • provider_txn_ref: VARCHAR(100) (nullable) • order_info: TEXT (nullable)

No	Table	Description
		• refund_amount: NUMERIC(12,2) (nullable) • refunded_at: TIMESTAMP (nullable) • refund_status: VARCHAR(20) (nullable) • refund_reason: TEXT (nullable) • refund_requested_by: UUID (nullable) • refund_requested_at: TIMESTAMP (nullable) • refund_reviewed_by: UUID (nullable) • refund_reviewed_at: TIMESTAMP (nullable) • created_at: TIMESTAMP • updated_at: TIMESTAMP

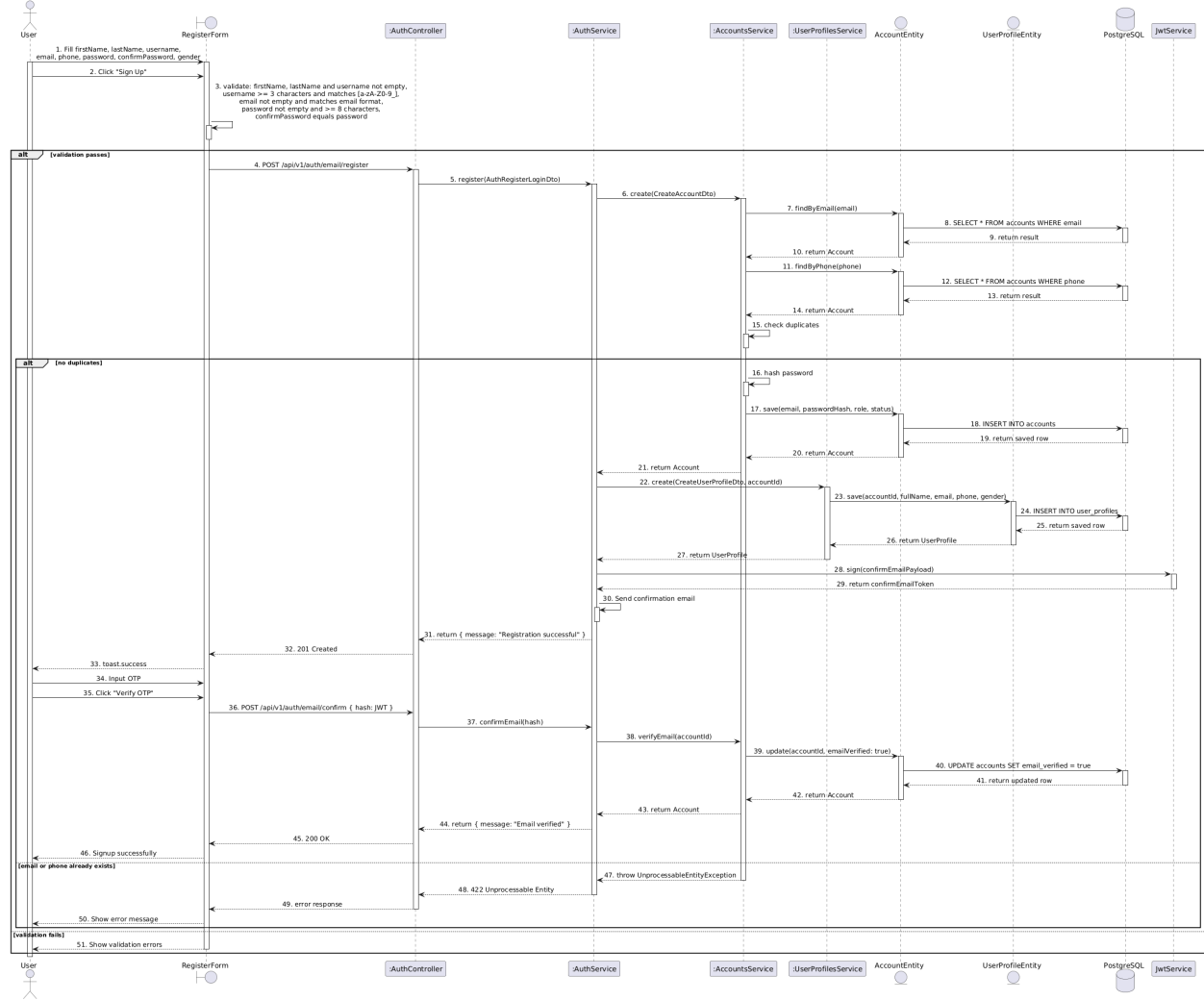
3. Detailed Design

3.1 Register

a. Class Diagram

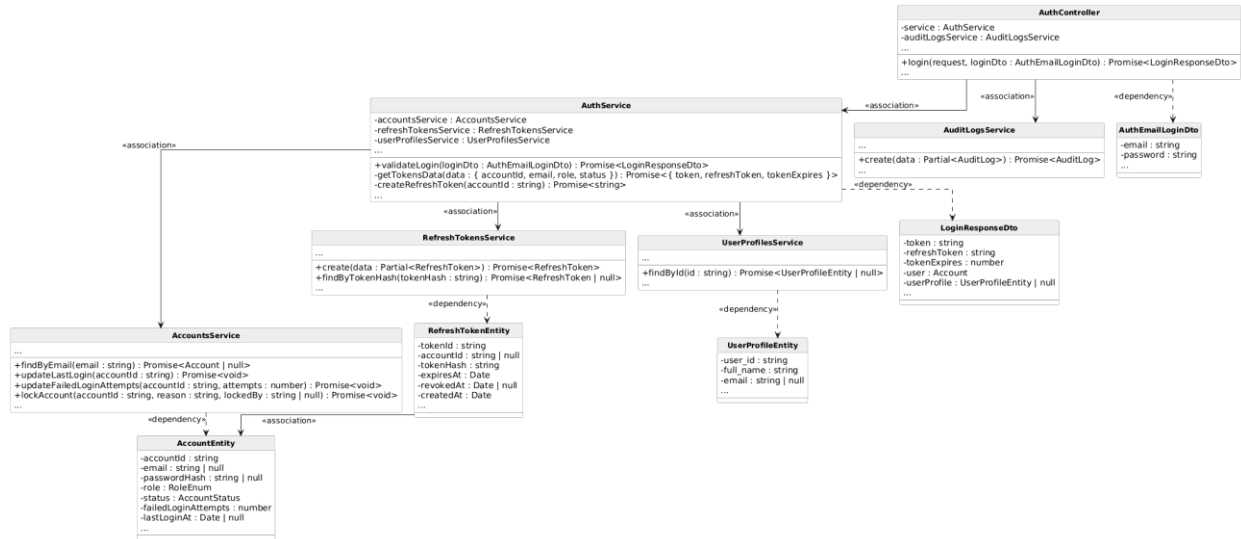


b. Sequence Diagram(s)

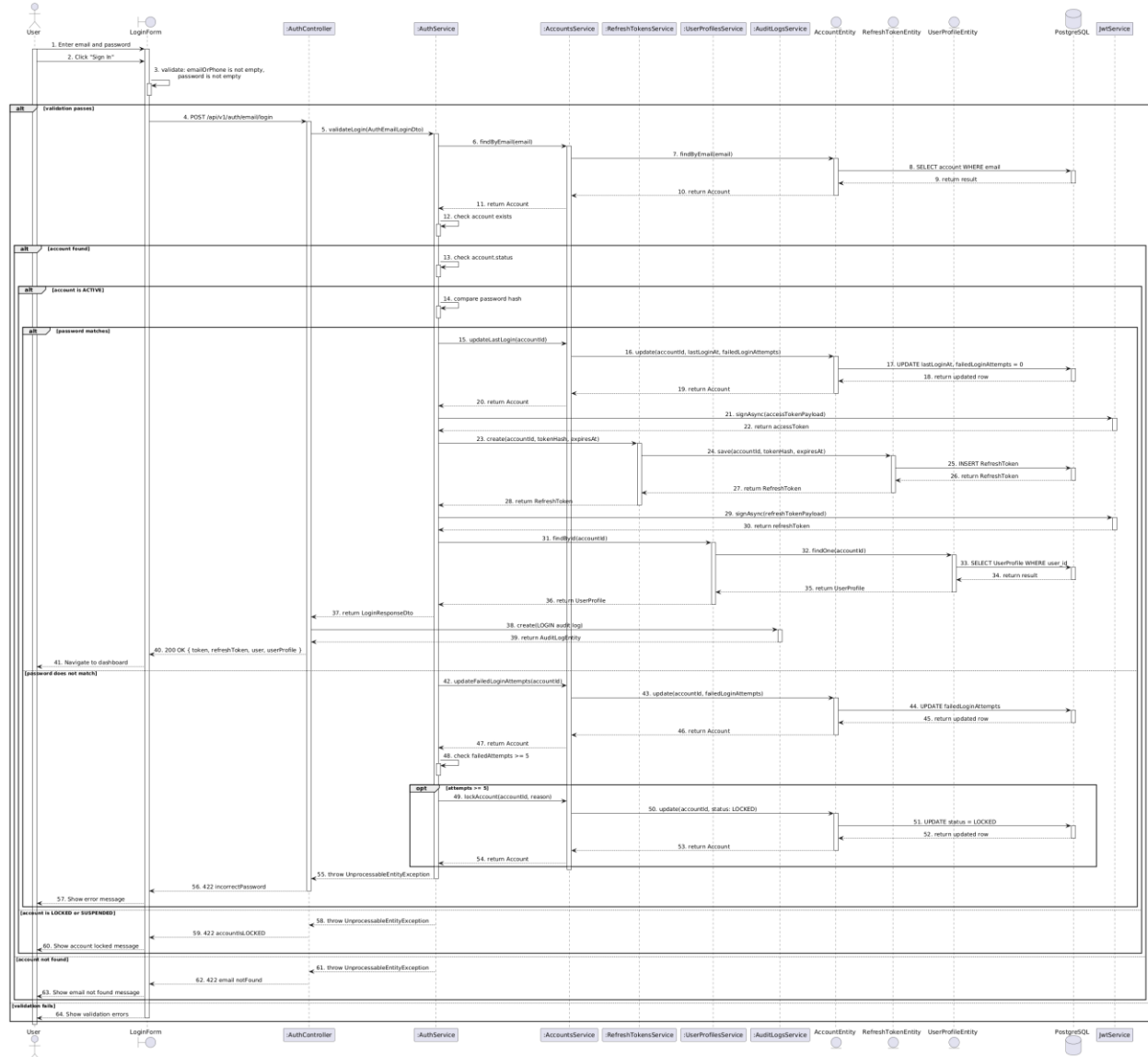


3.2 Login

a. Class Diagram

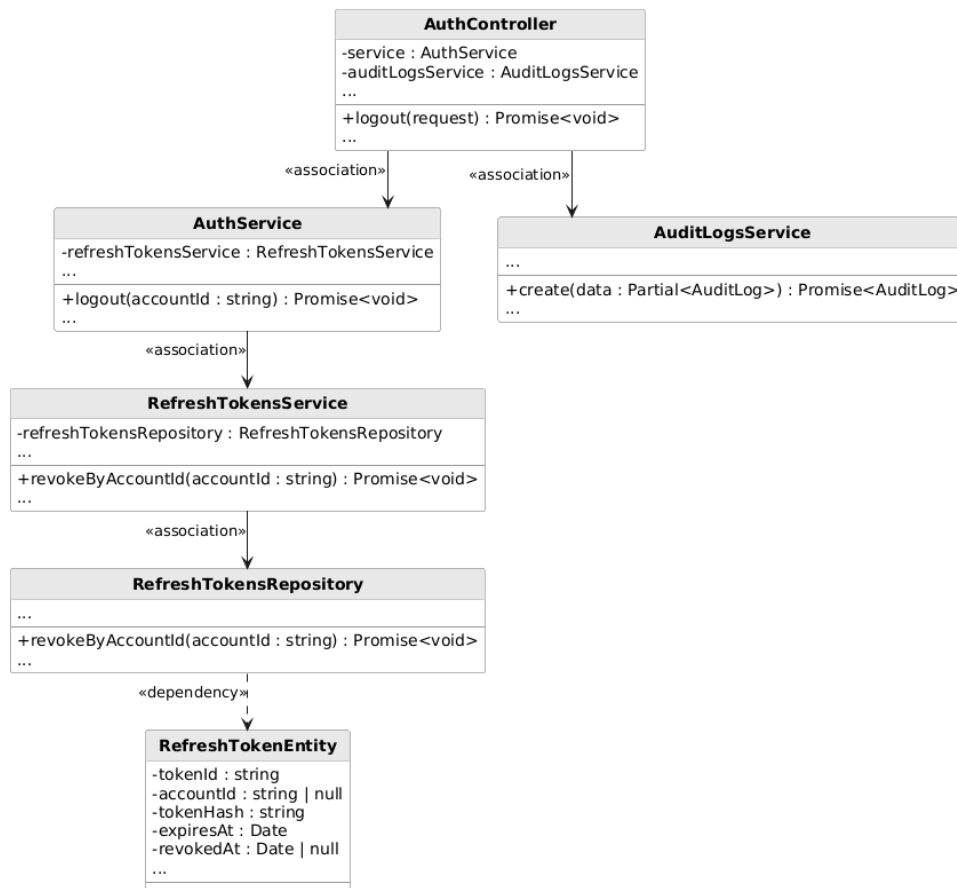


b. Sequence Diagram(s)

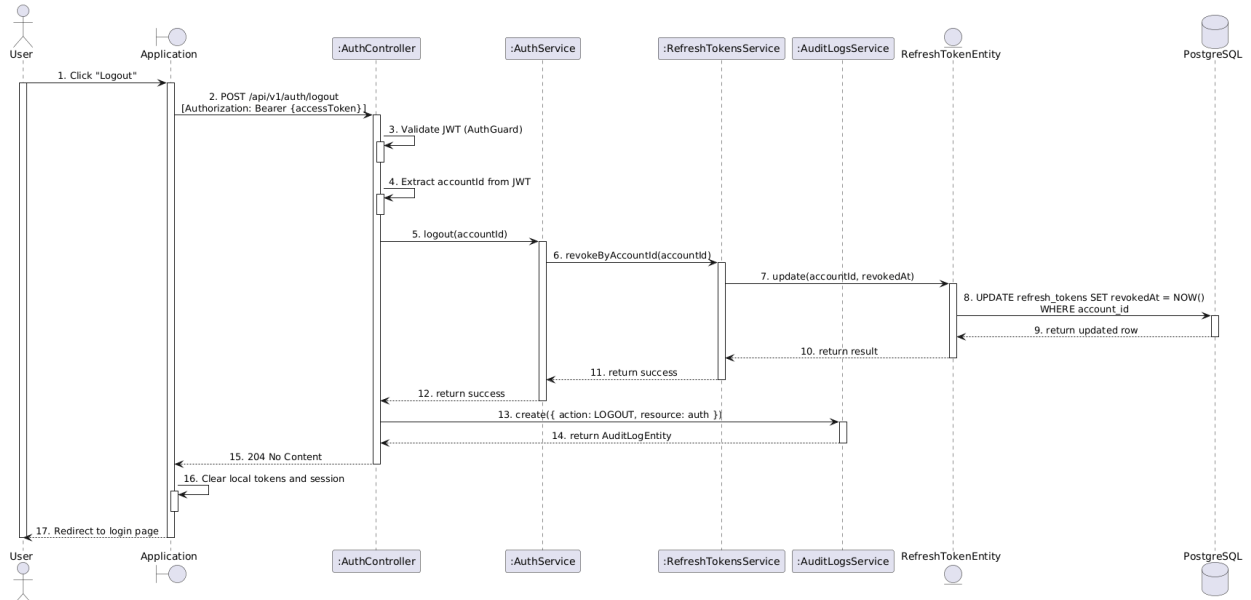


3.3 Logout

a. Class Diagram

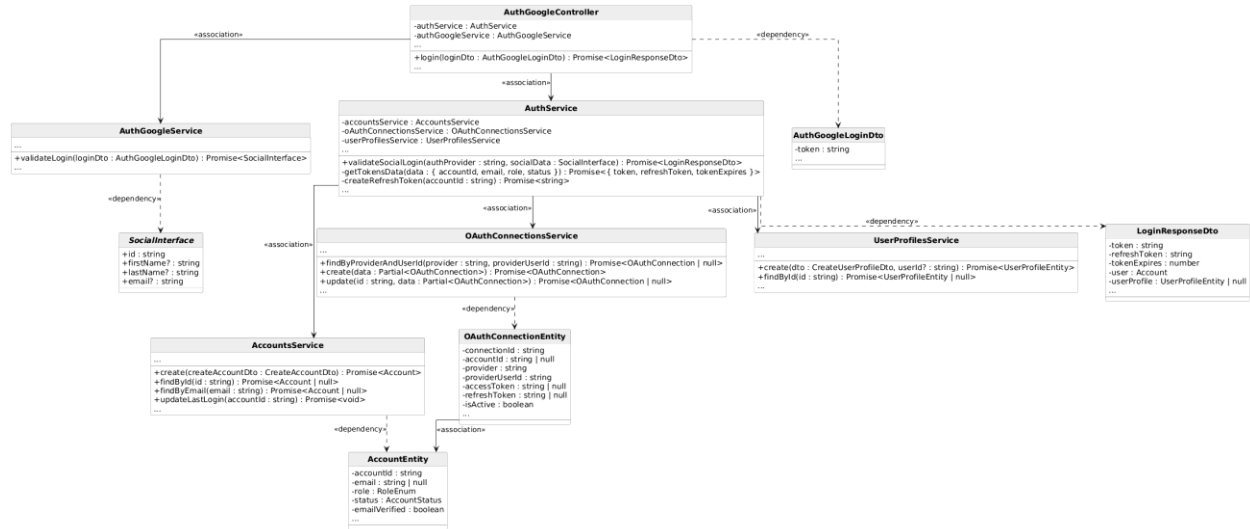


b. Sequence Diagram(s)

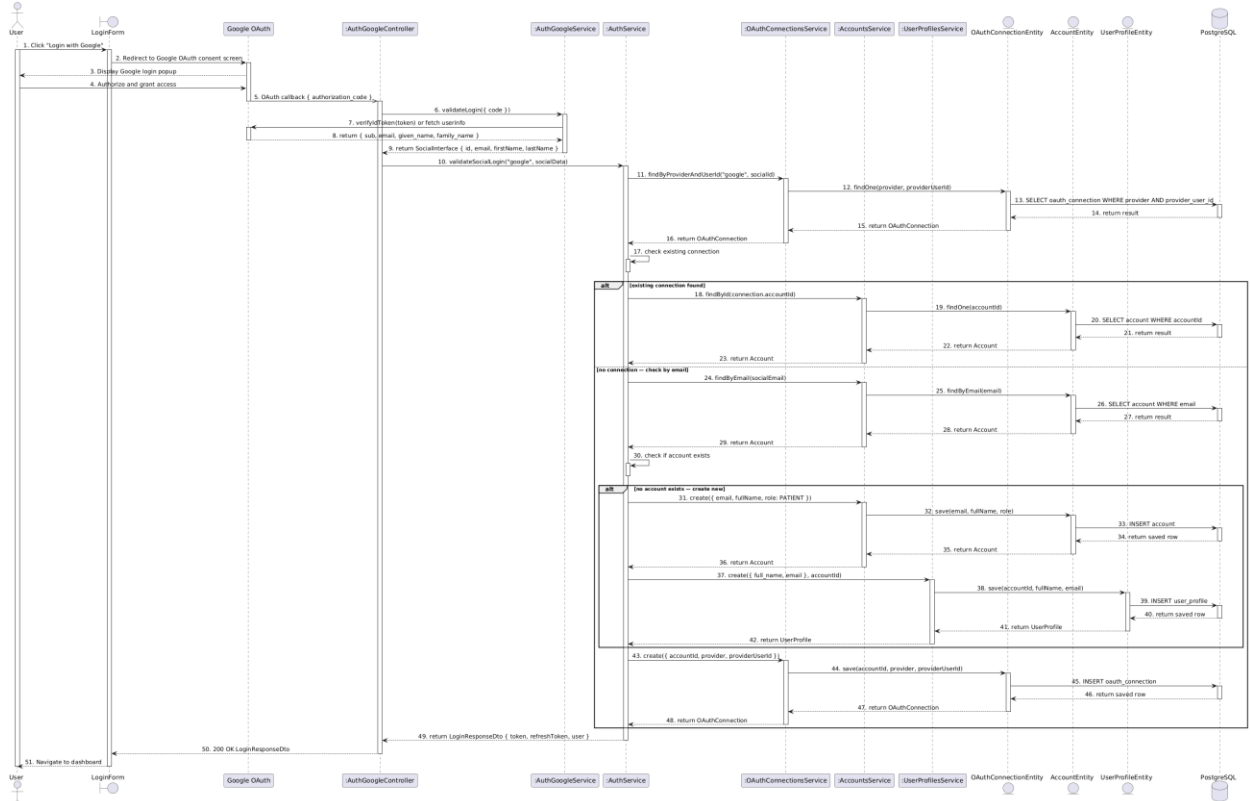


3.4 Login With Google

a. Class Diagram

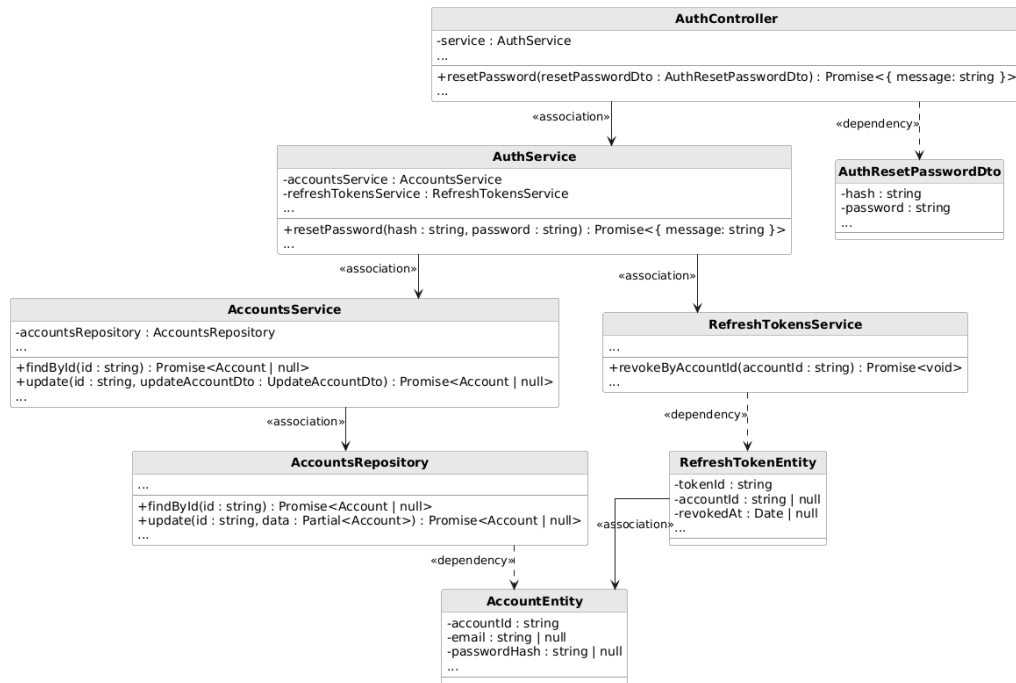


b. Sequence Diagram(s)

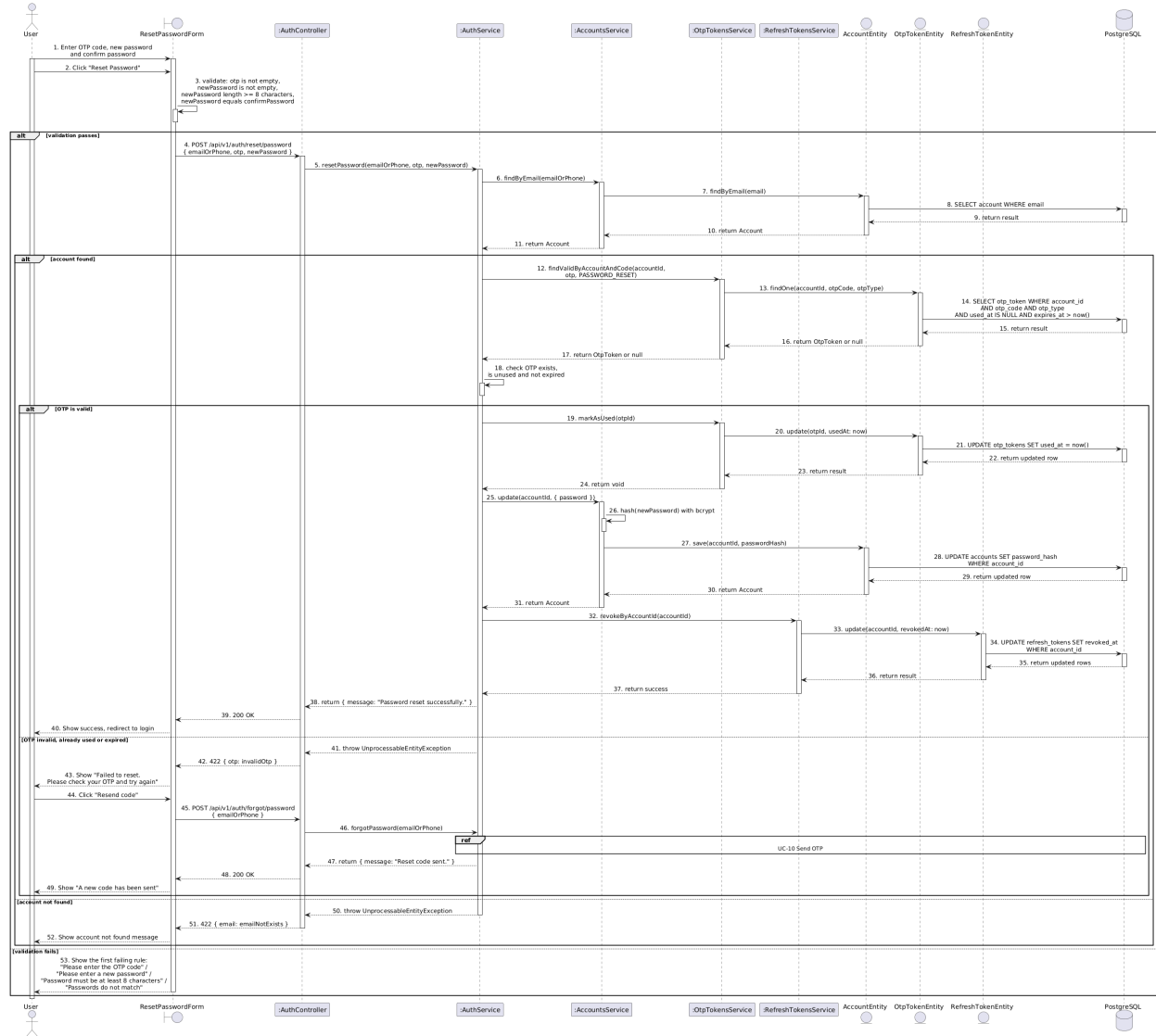


3.5 Reset Password

a. Class Diagram

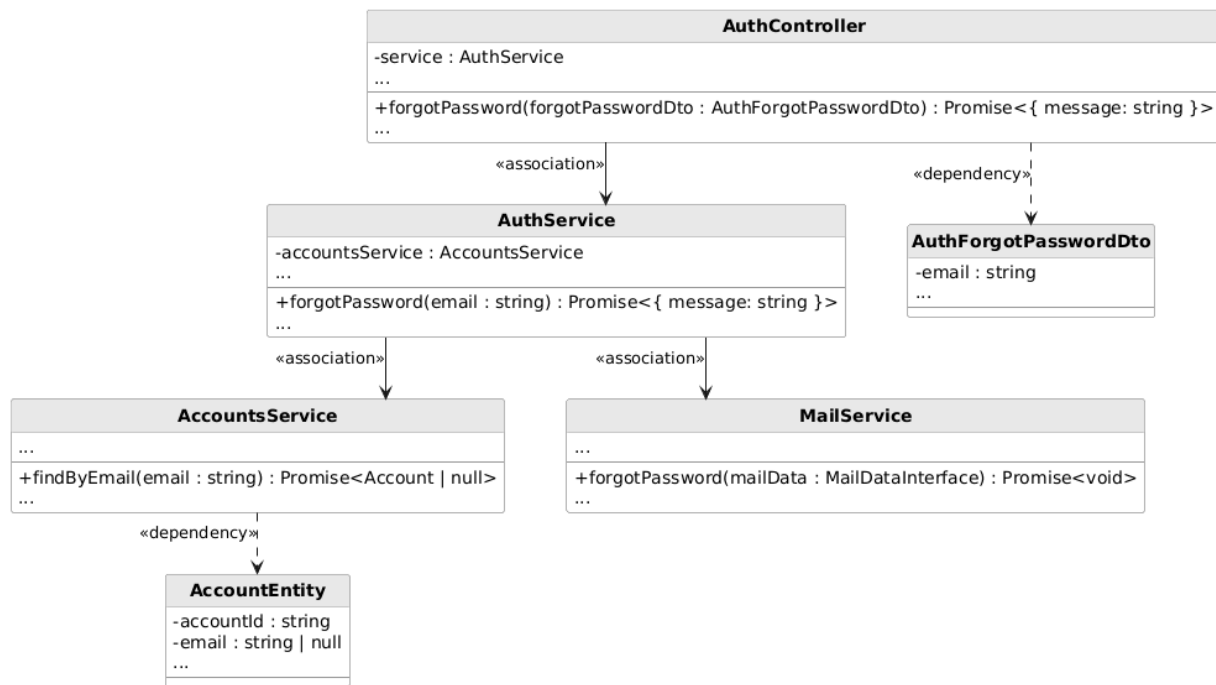


b. Sequence Diagram(s)

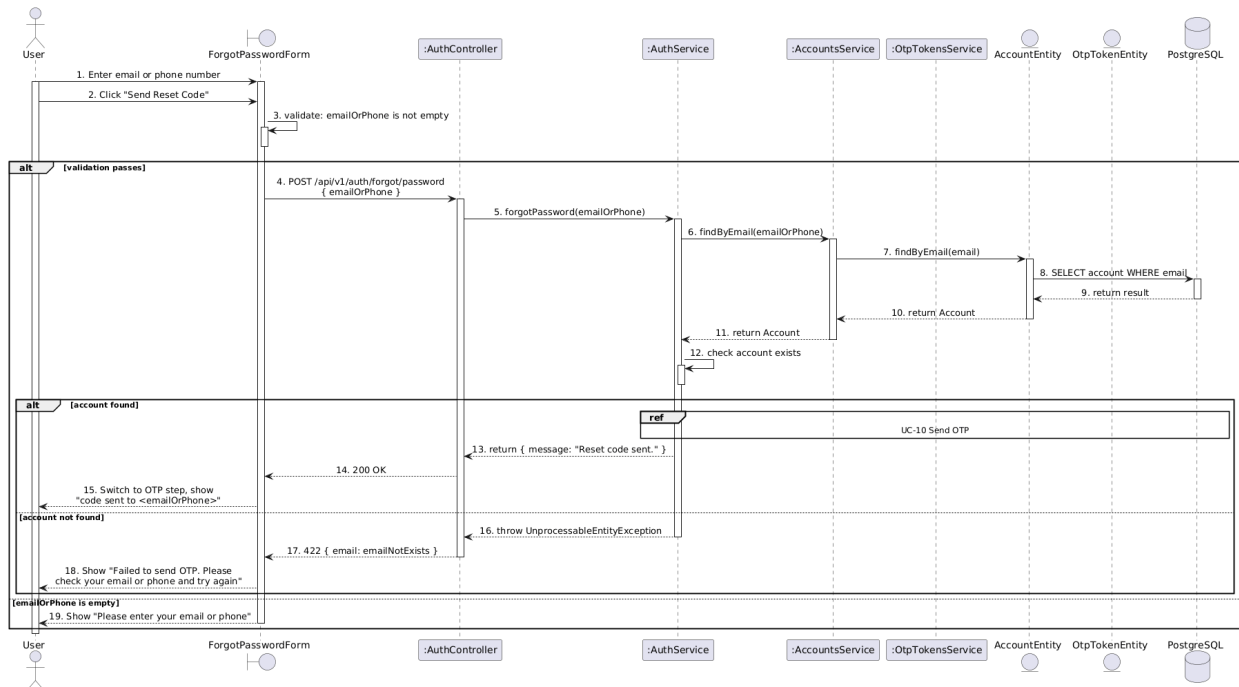


3.6 Forgot Password

a. Class Diagram

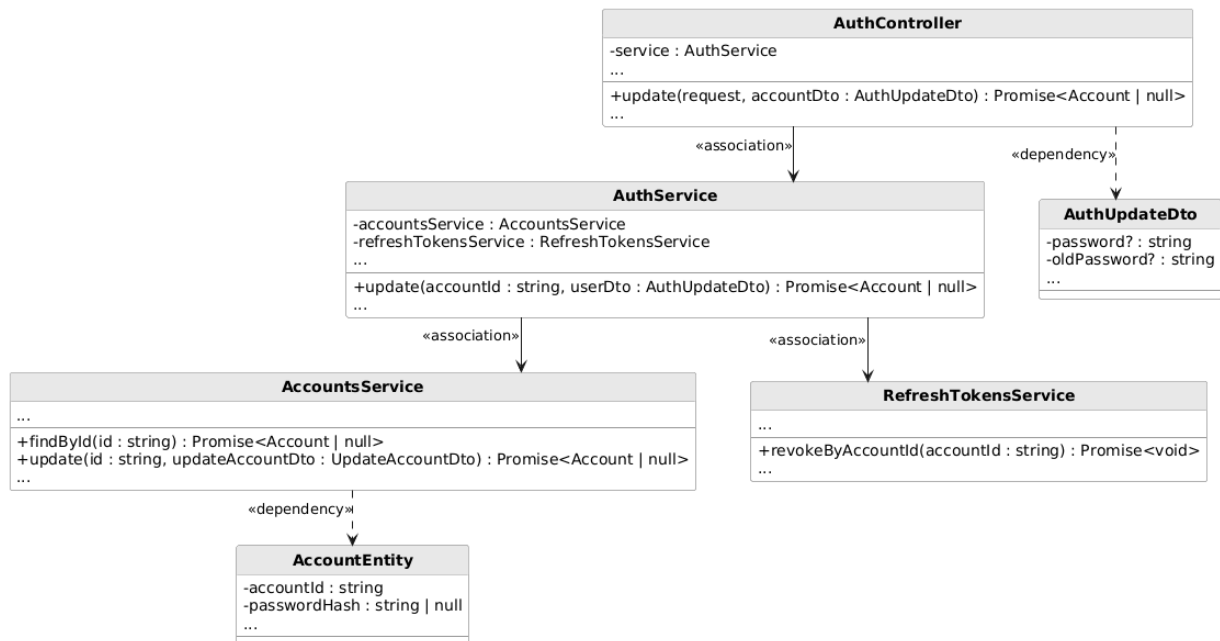


b. Sequence Diagram(s)

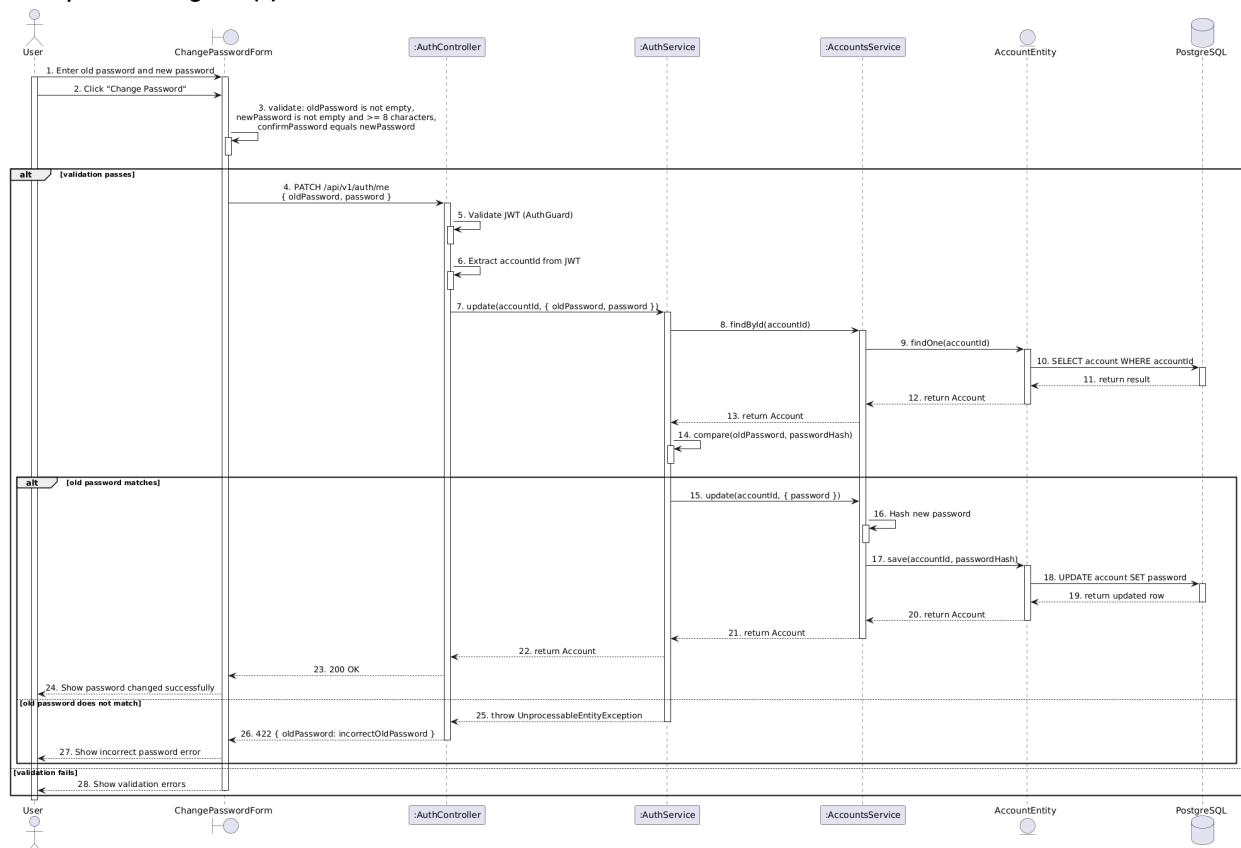


3.7 Change Password

a. Class Diagram

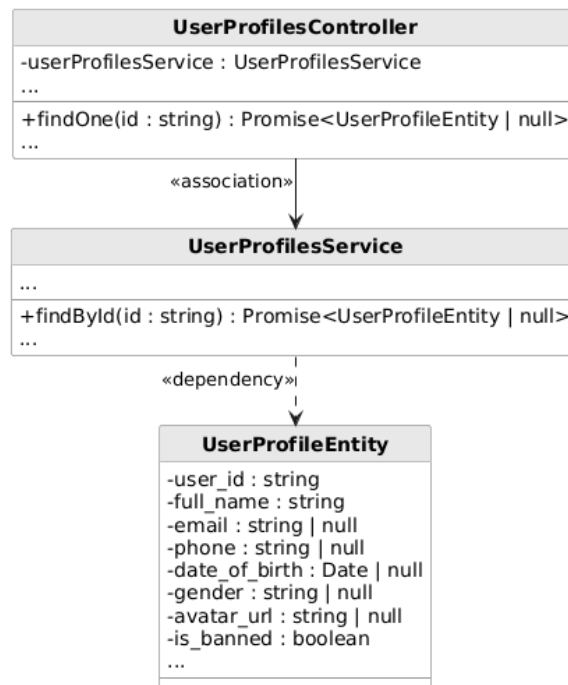


b. Sequence Diagram(s)

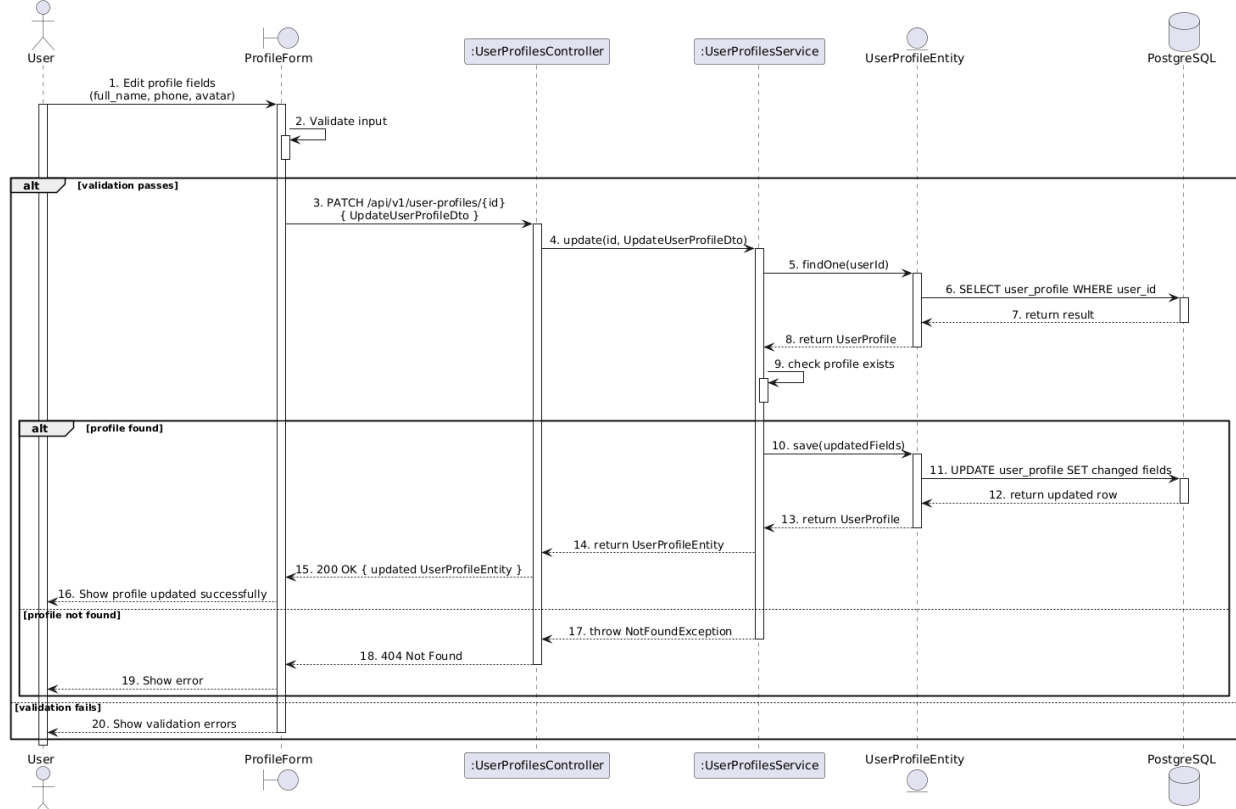


3.8 View Profile

a. Class Diagram

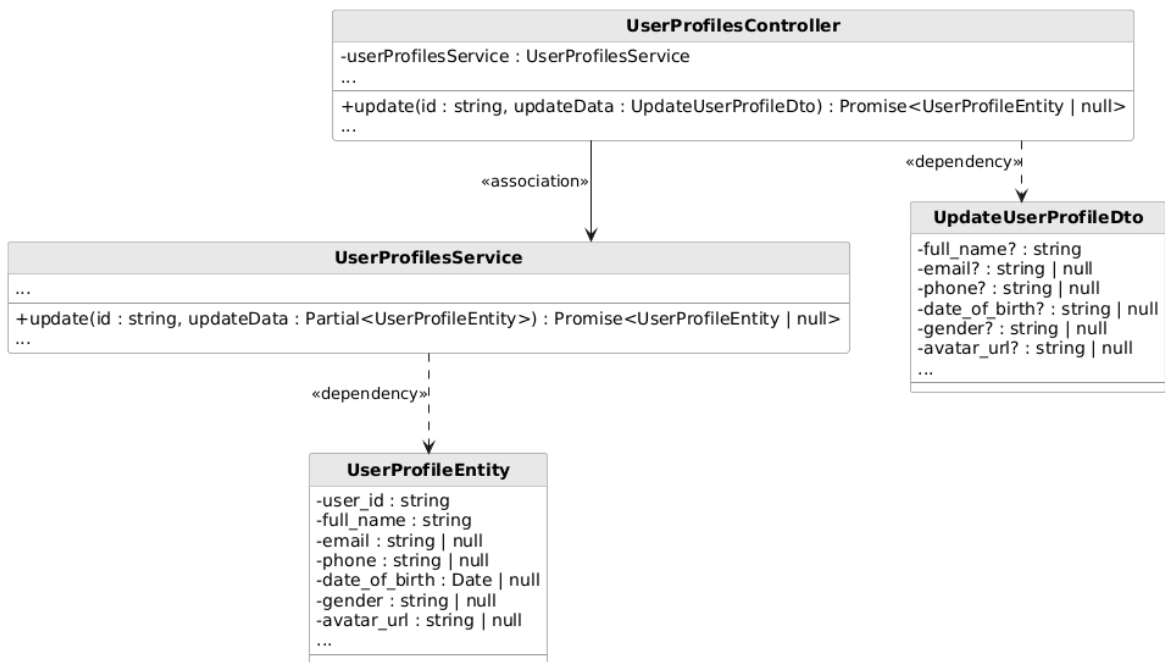


b. Sequence Diagram(s)

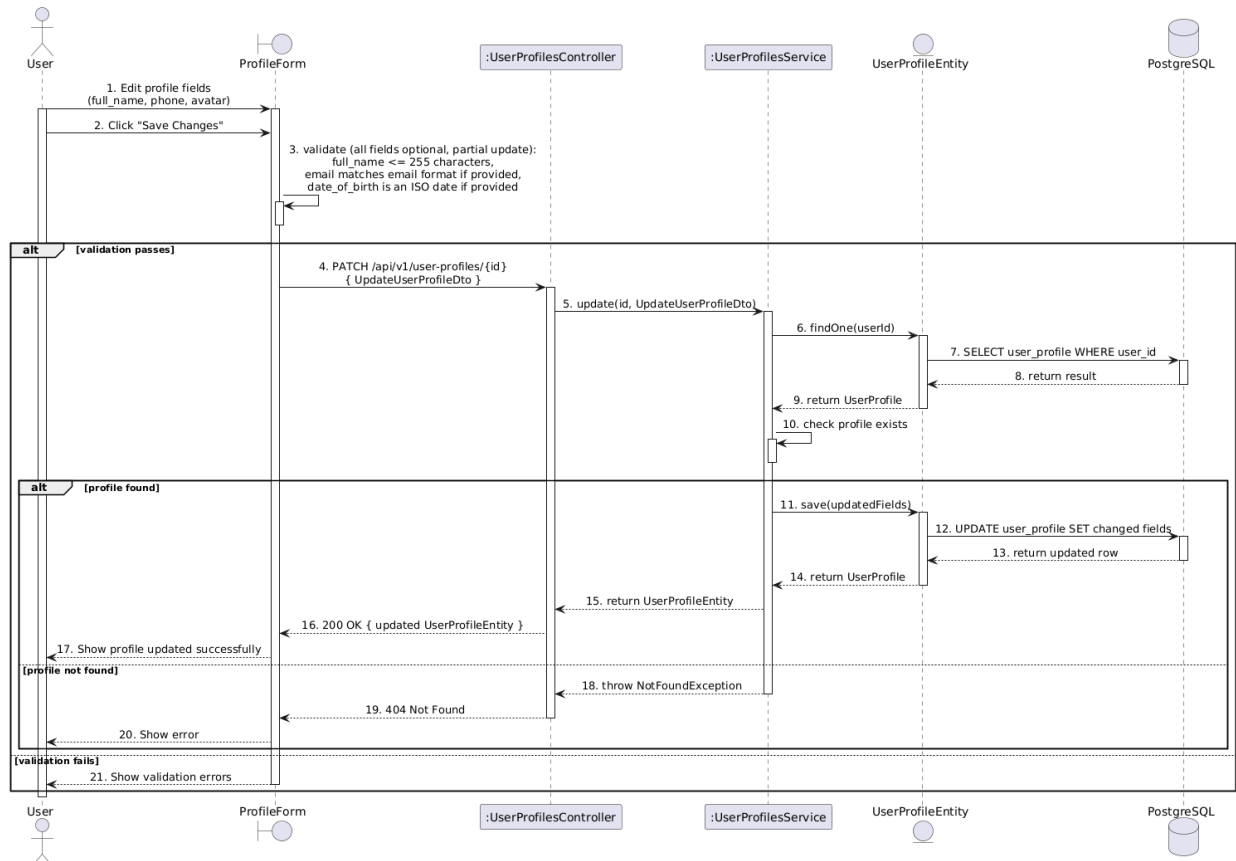


3.9 Update Profile

a. Class Diagram

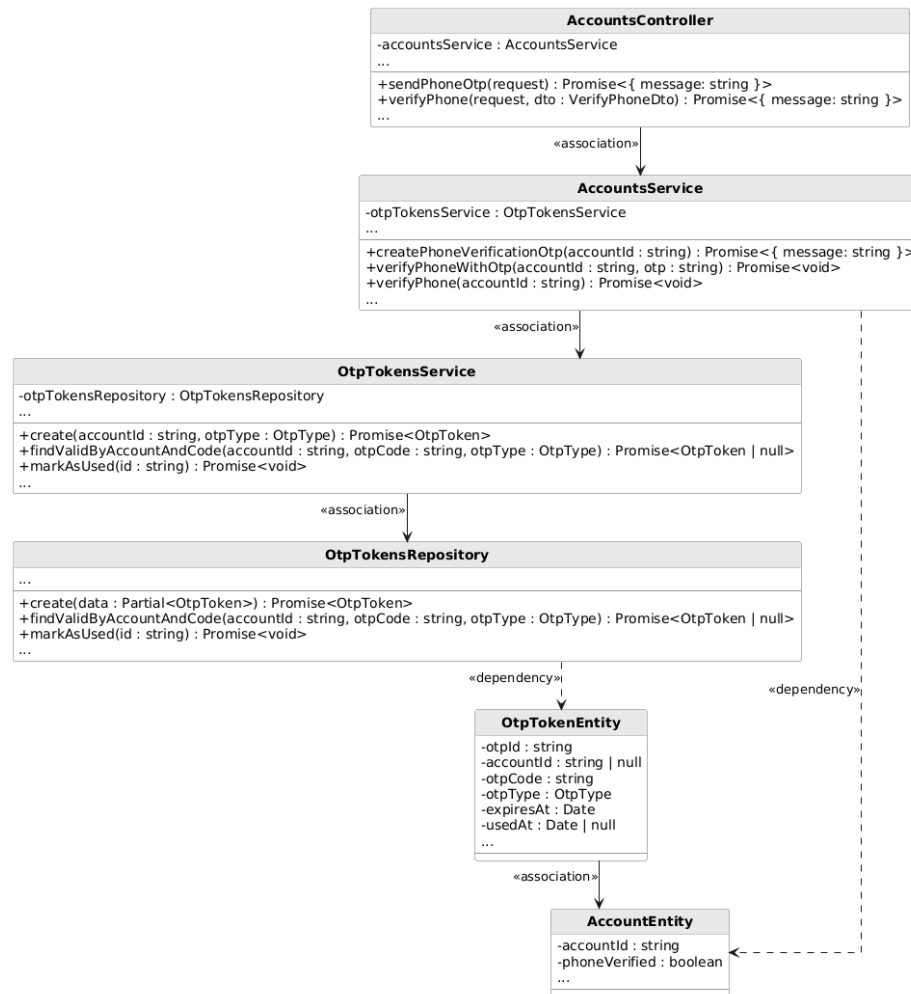


b. Sequence Diagram(s)

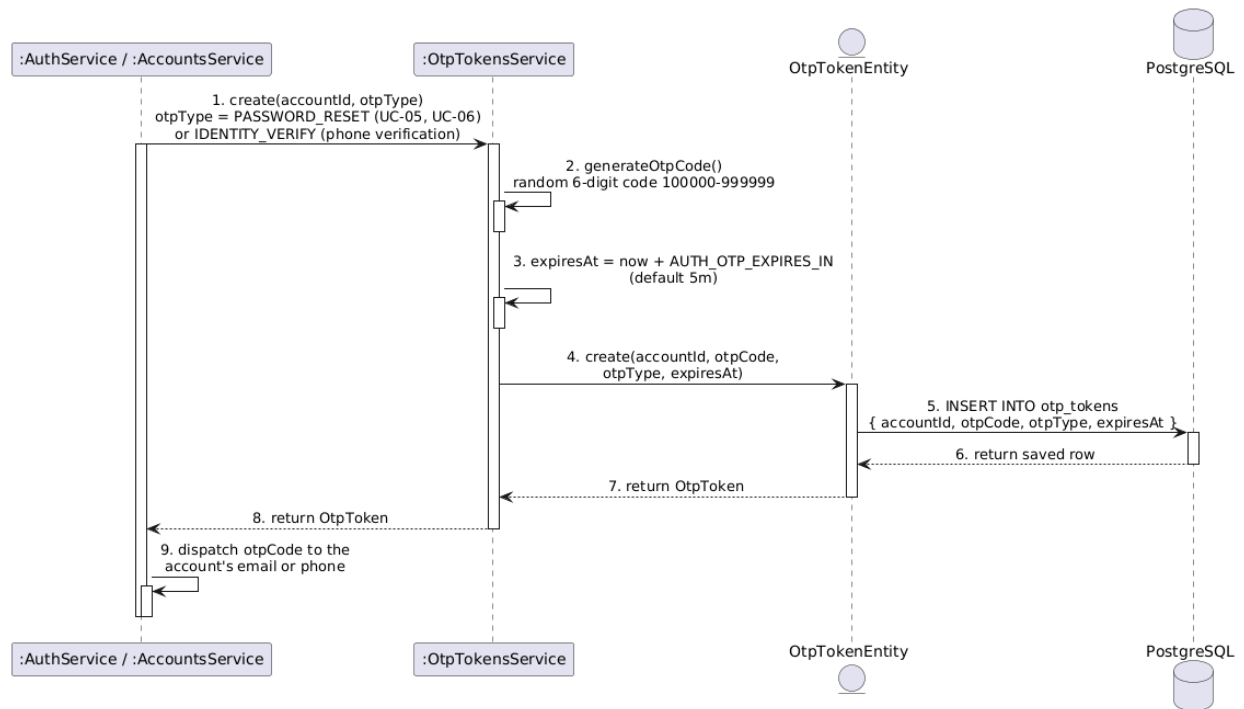


3.10 Send OTP

a. Class Diagram

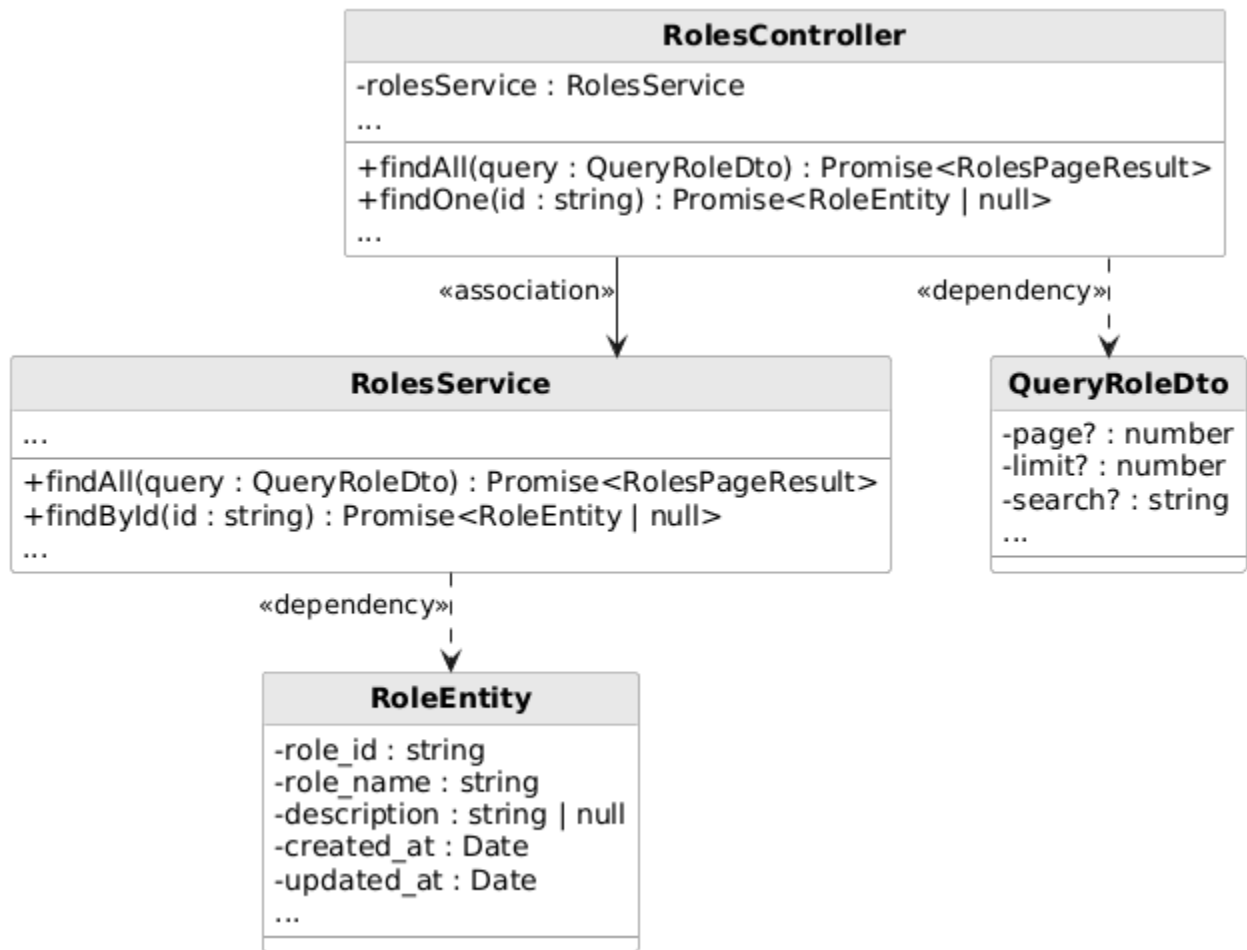


b. Sequence Diagram(s)

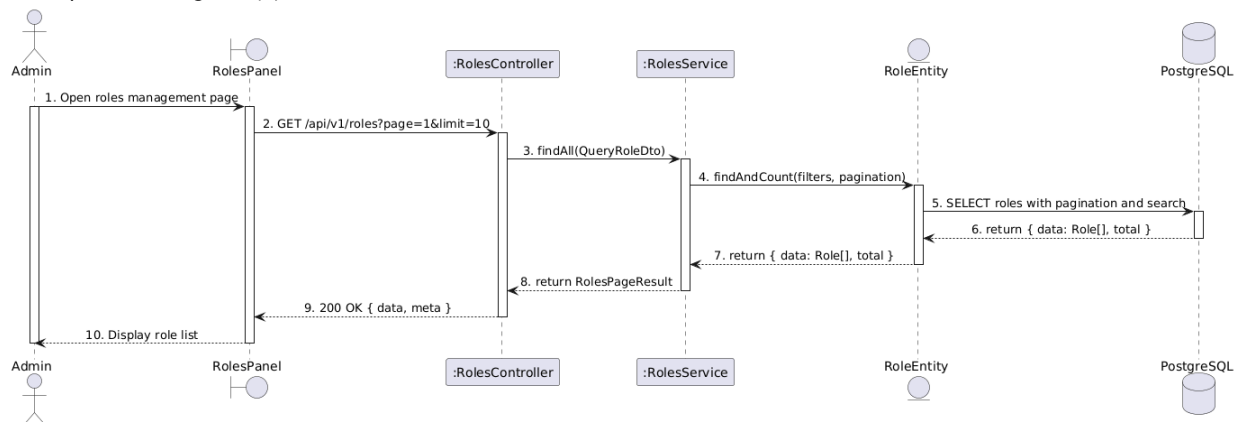


3.11 View Role

a. Class Diagram

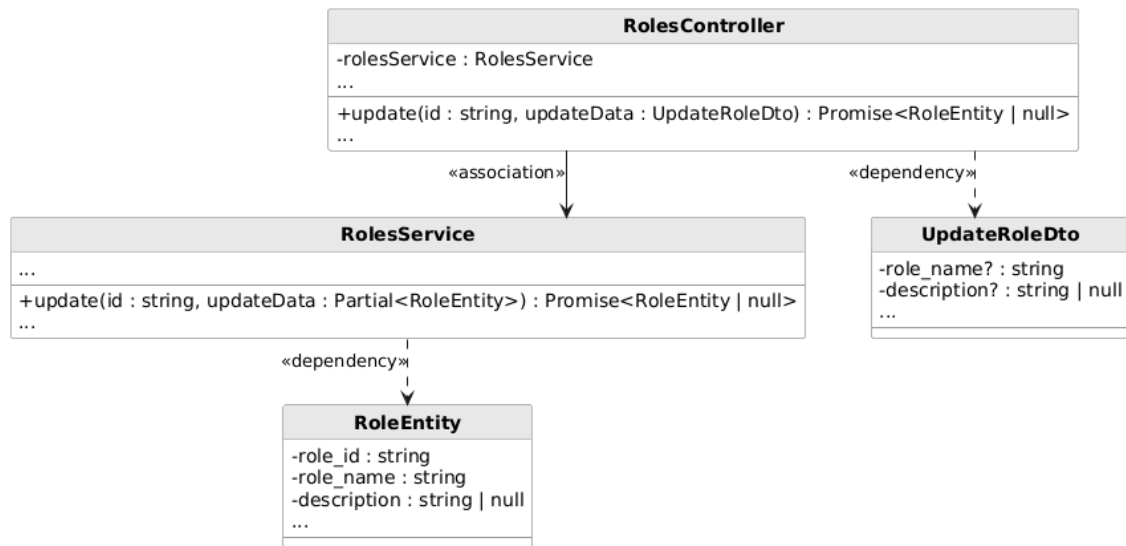


b. Sequence Diagram(s)

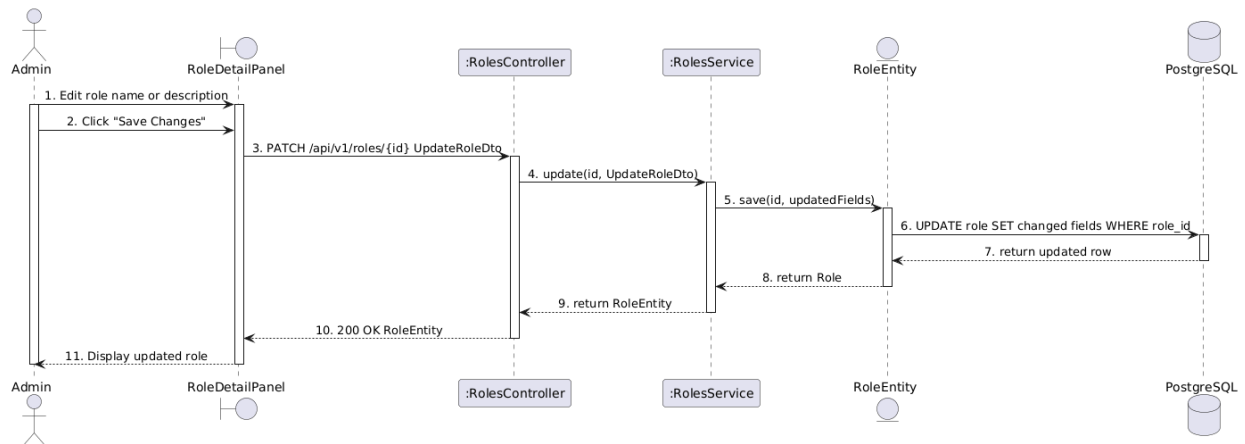


3.12 Update Role

a. Class Diagram

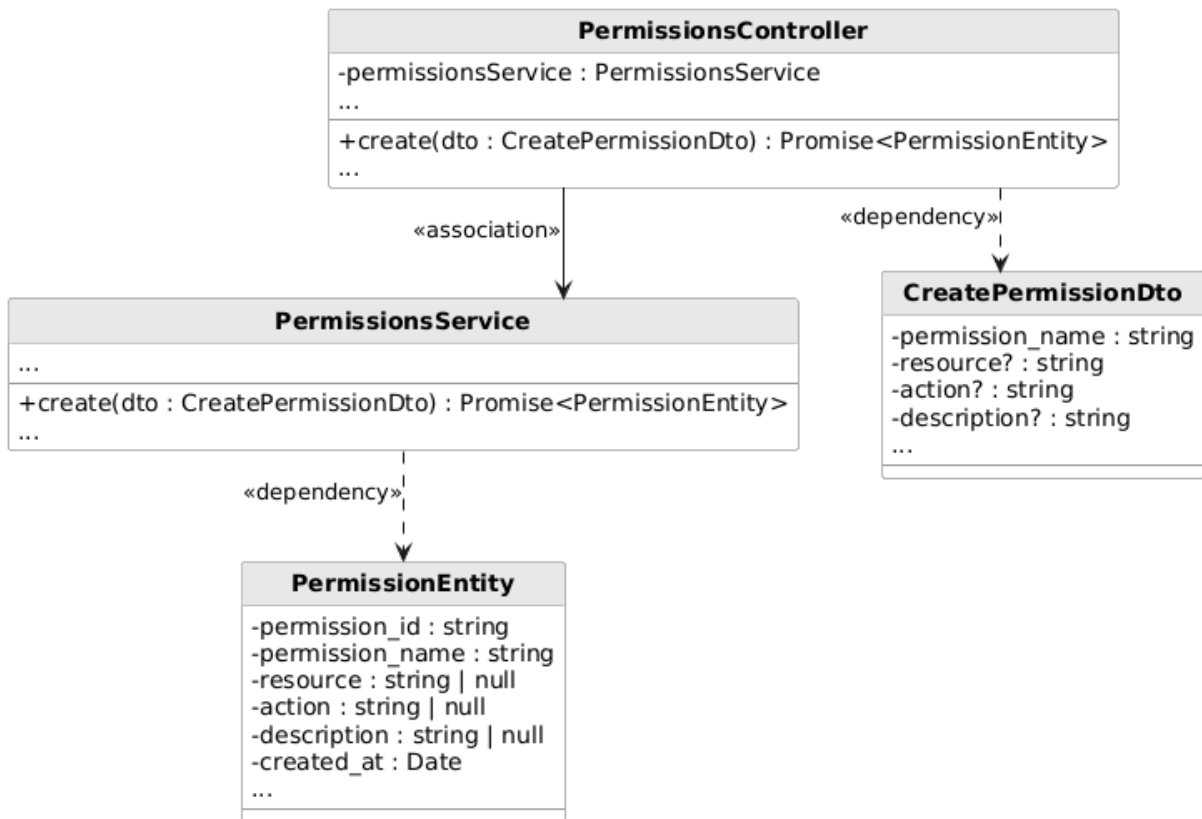


b. Sequence Diagram(s)

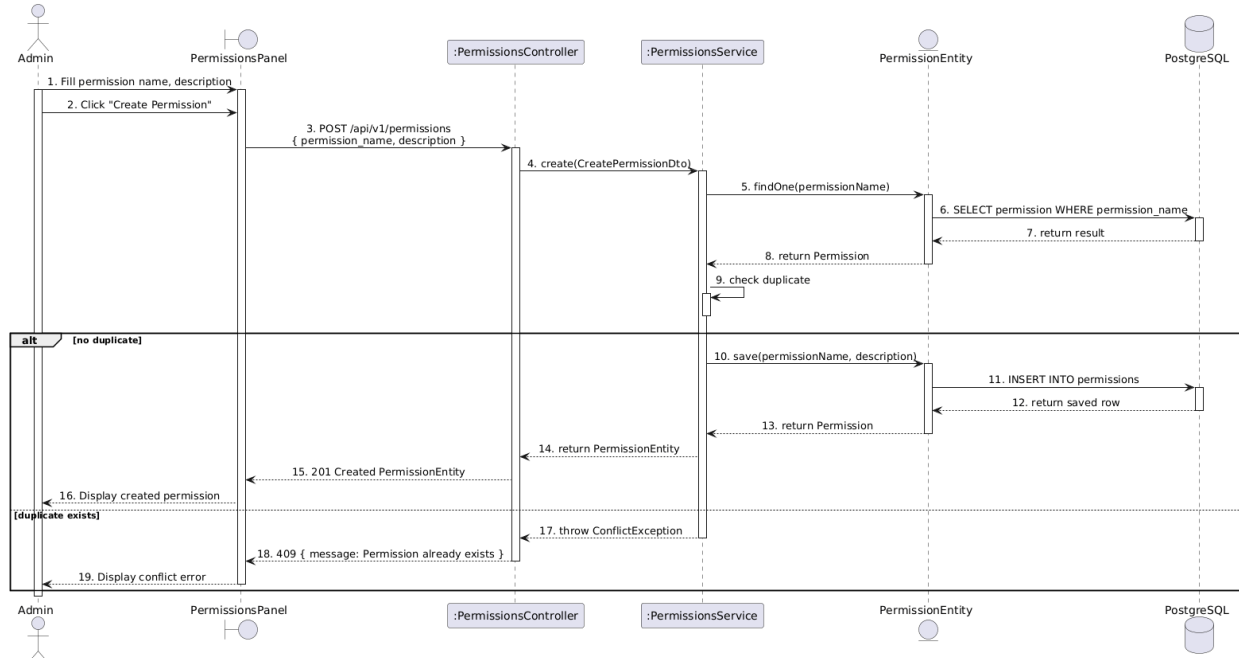


3.13 Create Permission

a. Class Diagram

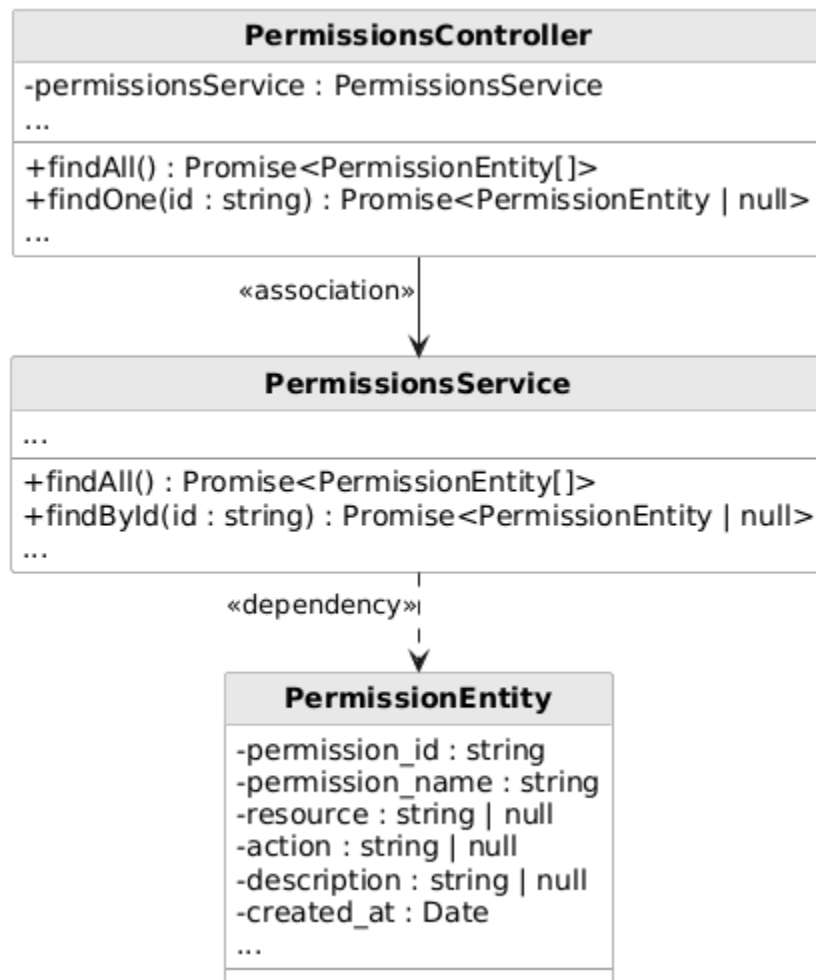


b. Sequence Diagram(s)

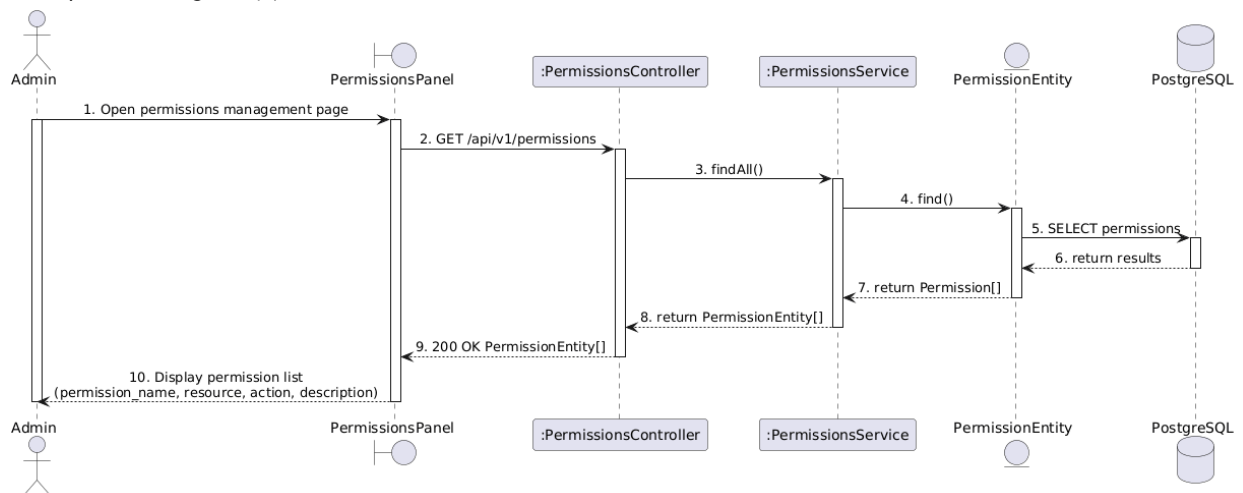


3.14 View Permission

a. Class Diagram

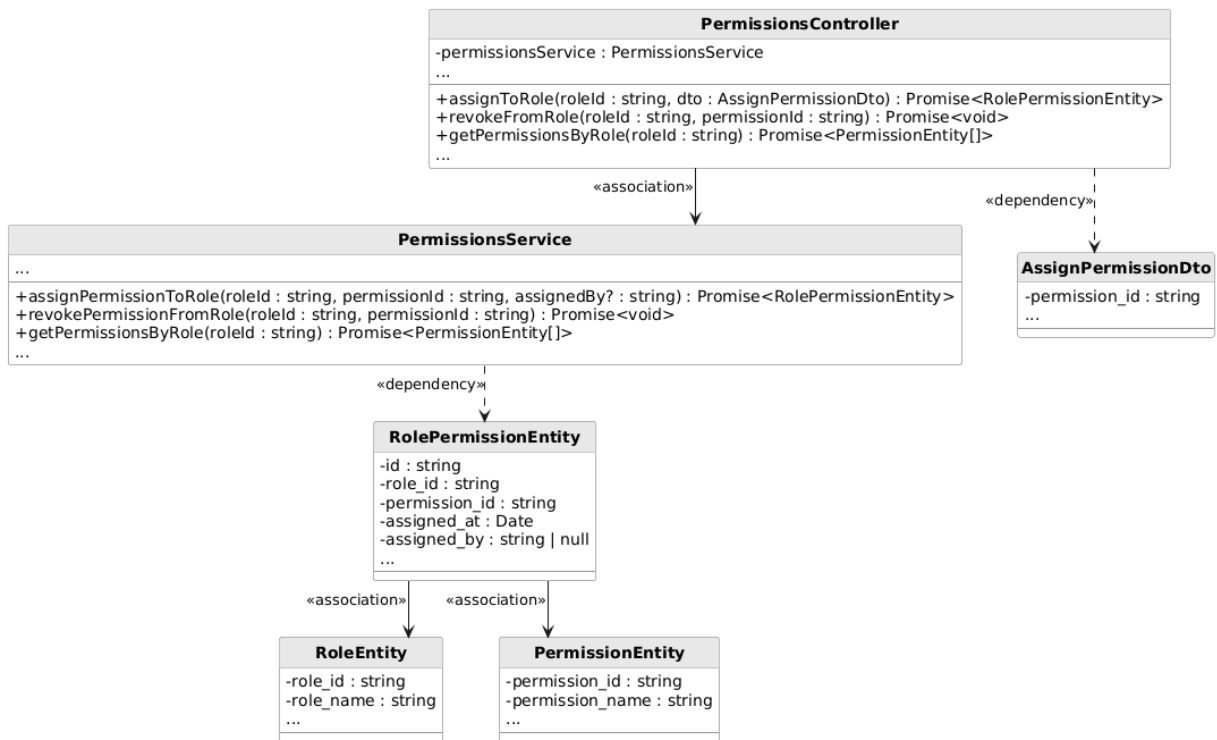


b. Sequence Diagram(s)

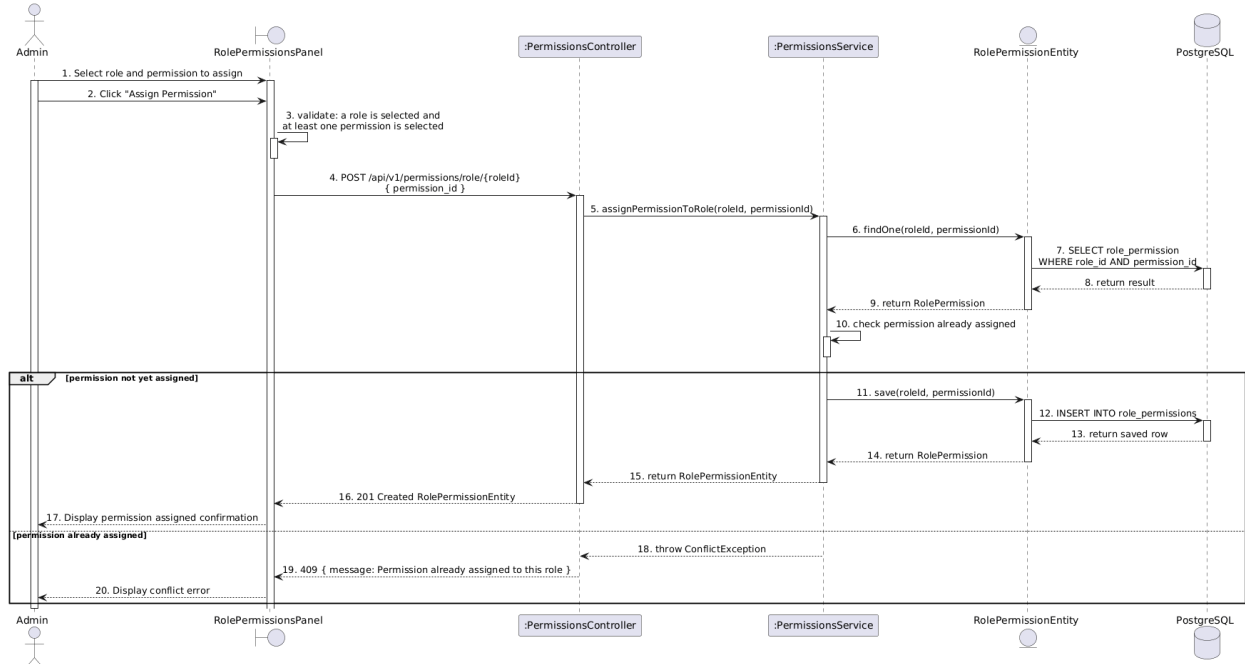


3.15 Update Role Permission

a. Class Diagram

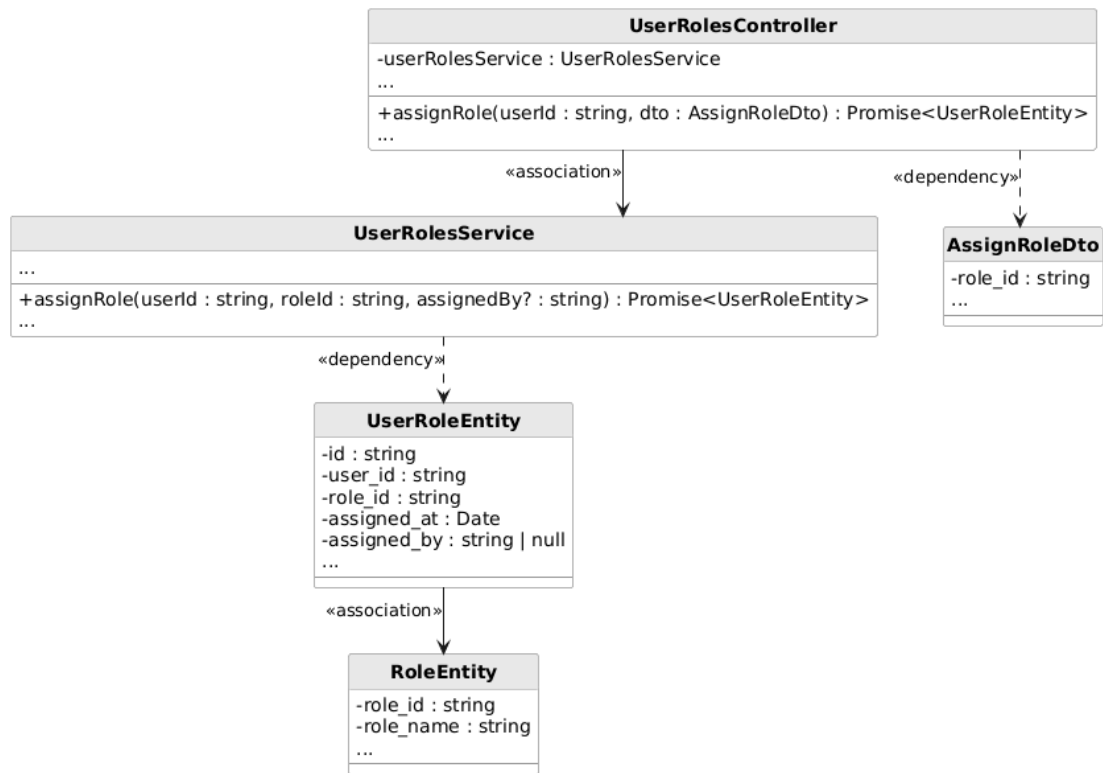


b. Sequence Diagram(s)

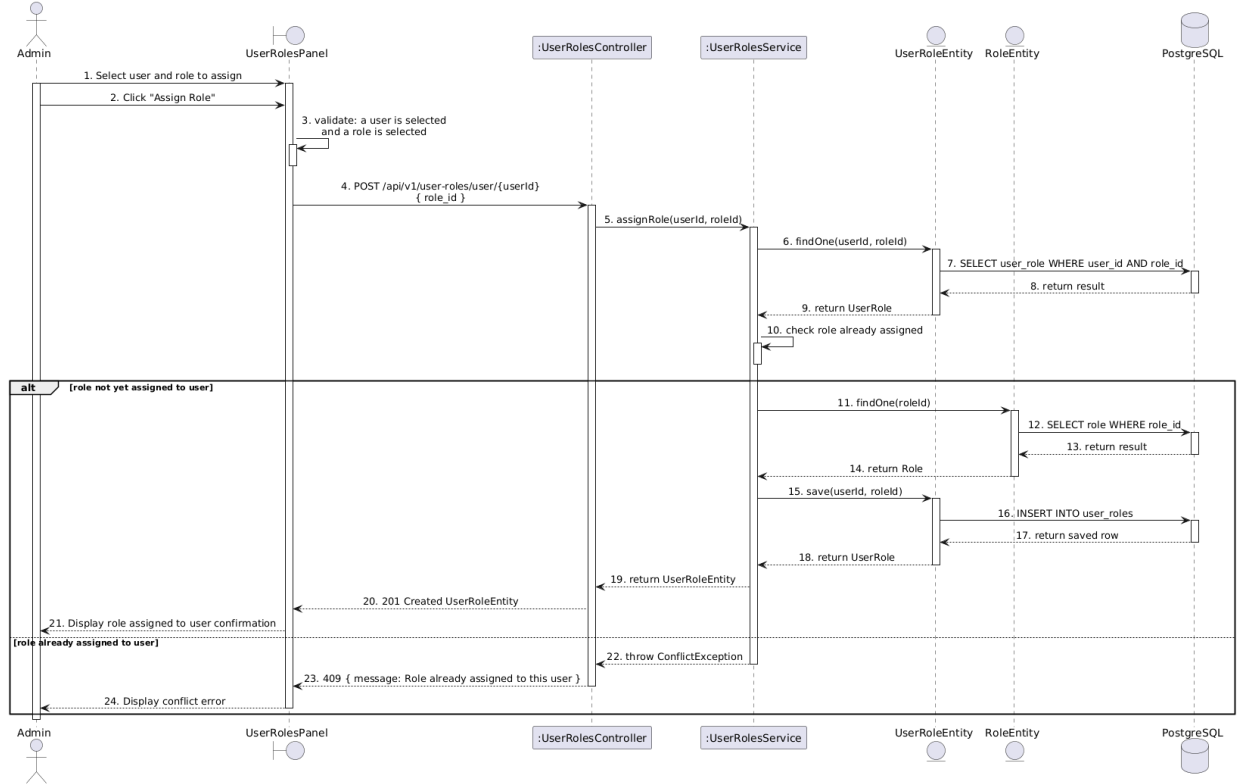


3.16 Assign Role

a. Class Diagram

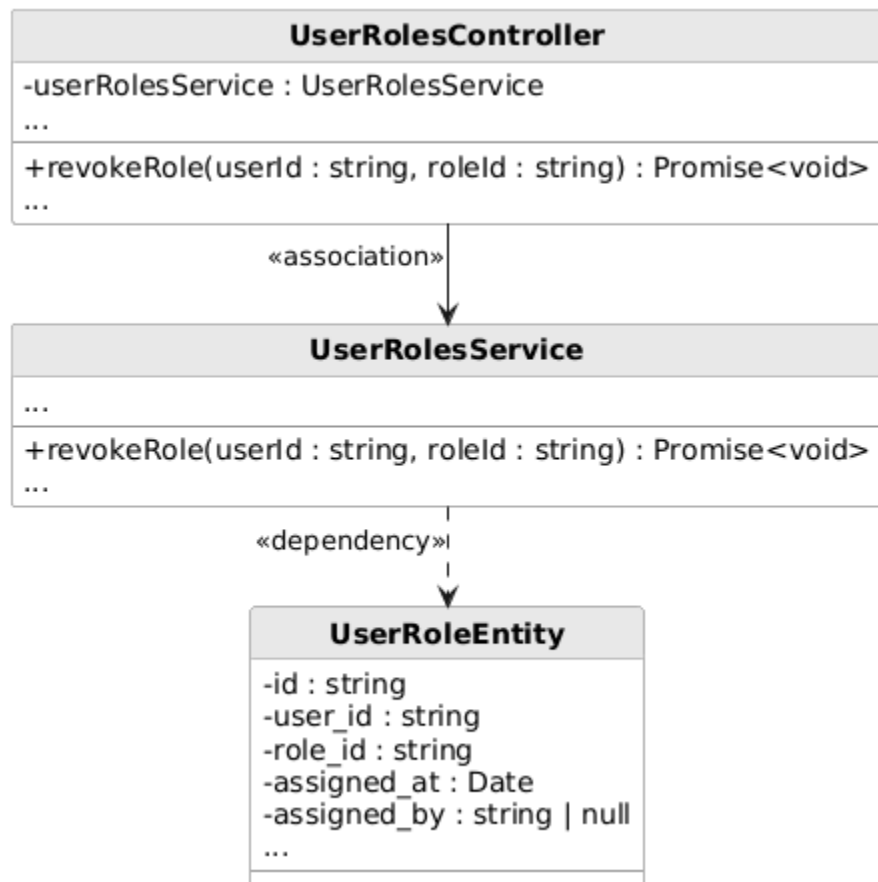


b. Sequence Diagram(s)

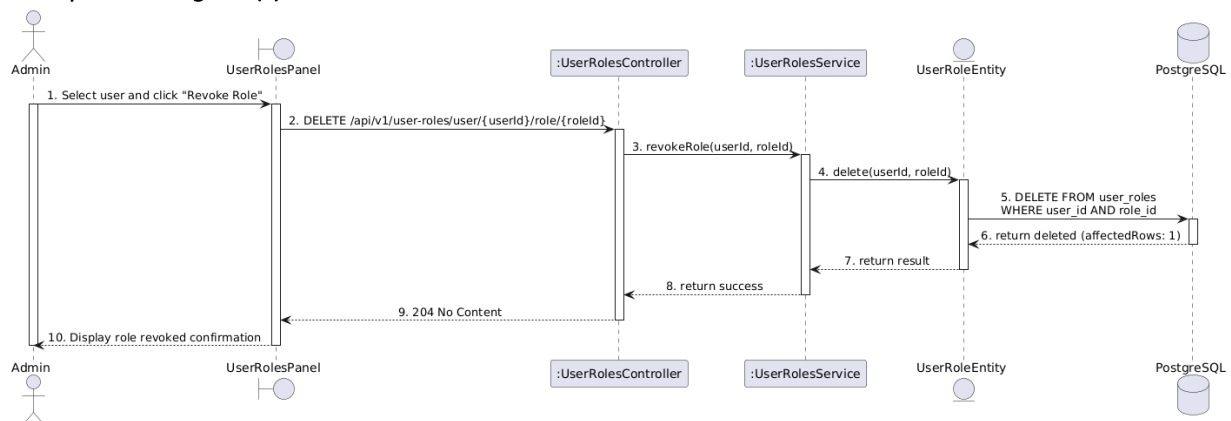


3.17 Revoke Role

a. Class Diagram

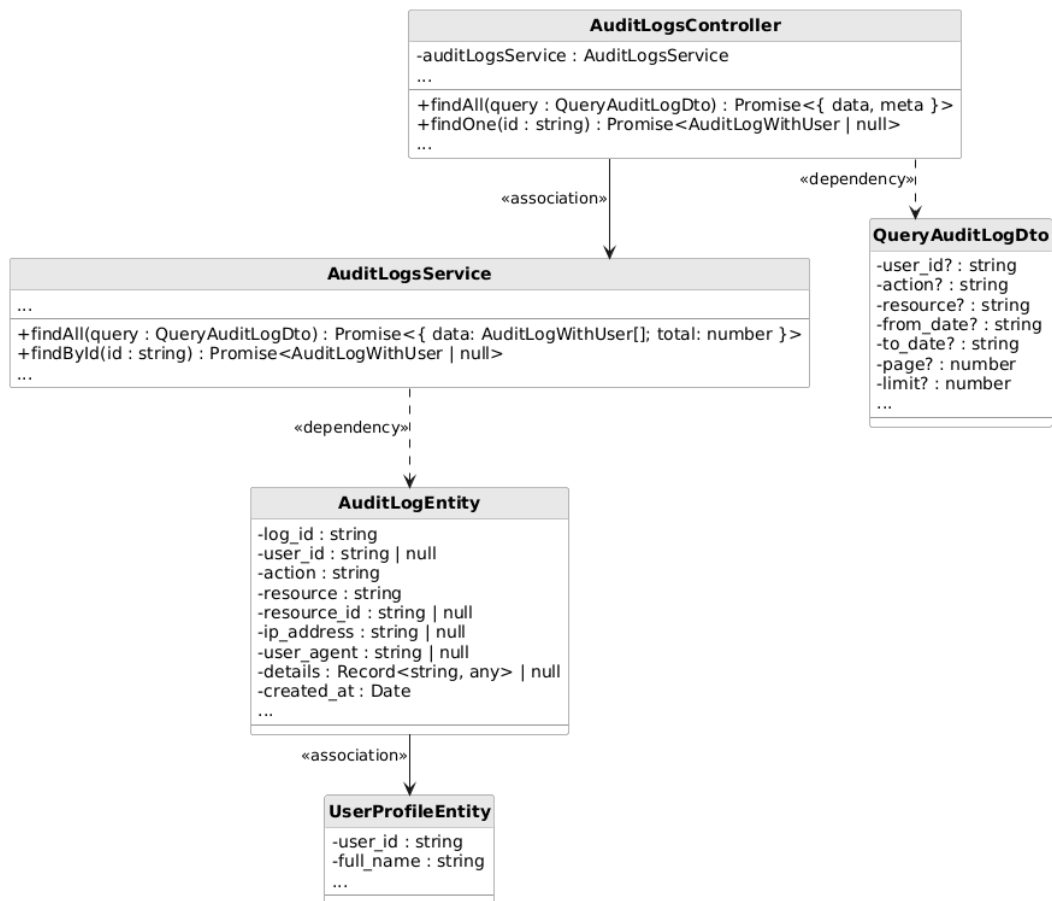


b. Sequence Diagram(s)

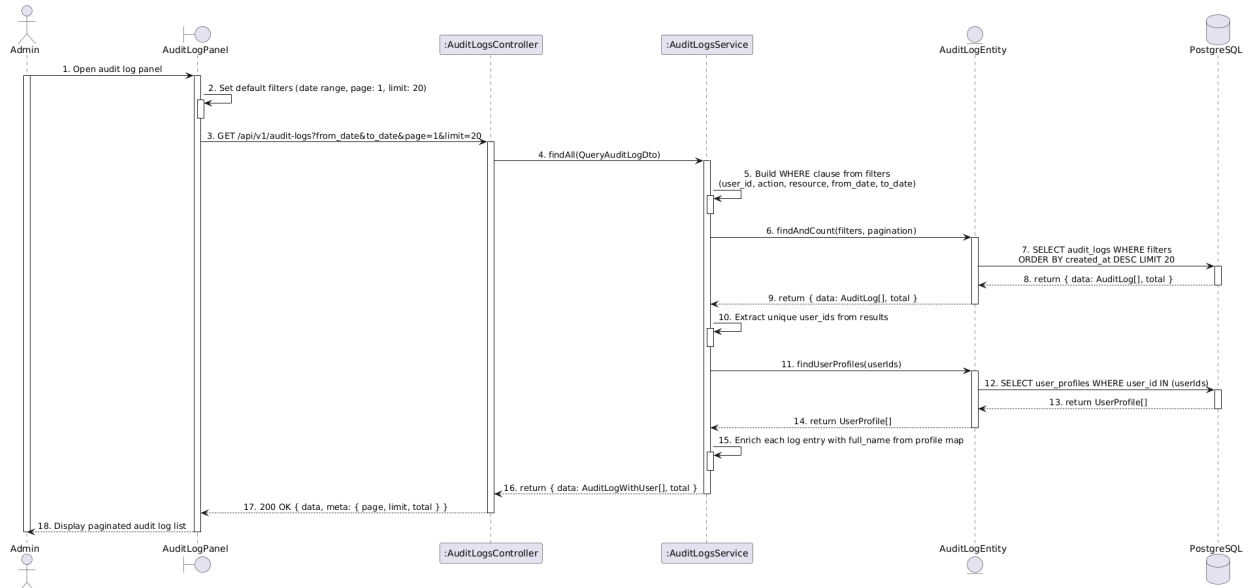


3.18 Access Audit Log

a. Class Diagram

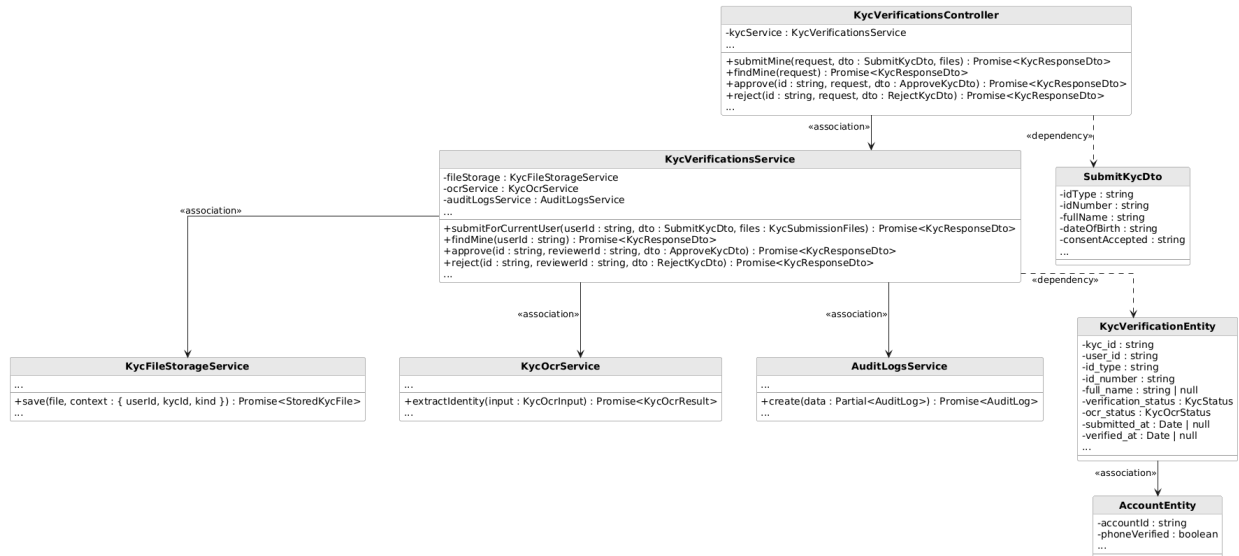


b. Sequence Diagram(s)

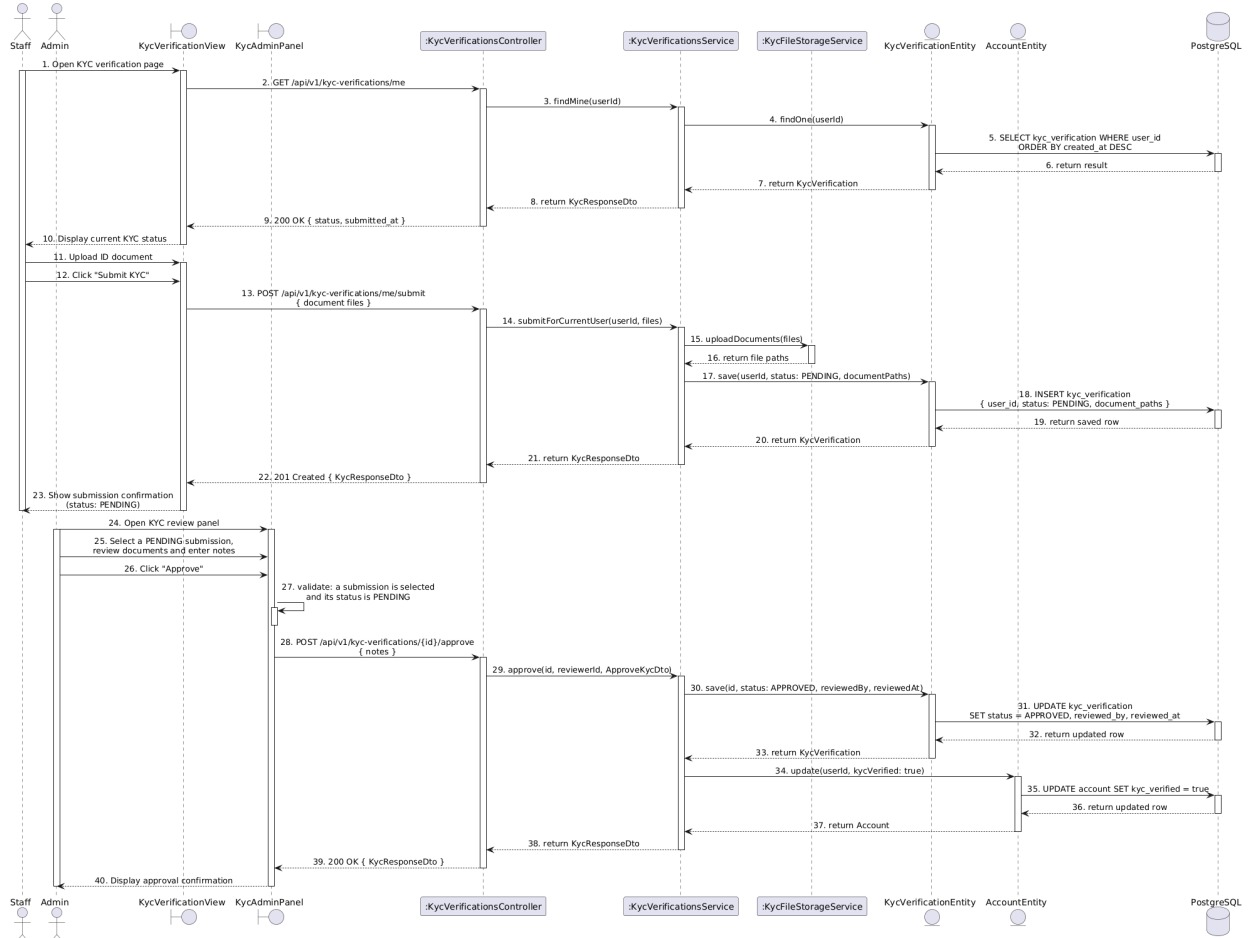


3.19 Verify Identity KYC

a. Class Diagram

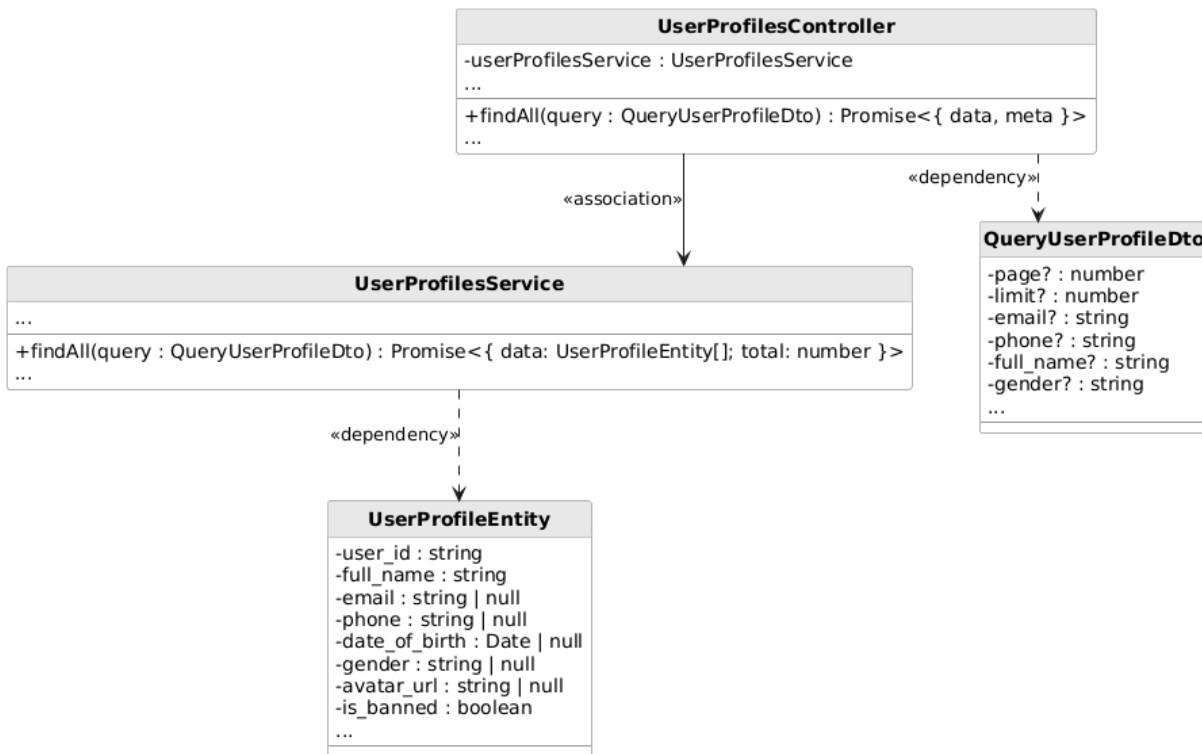


b. Sequence Diagram(s)

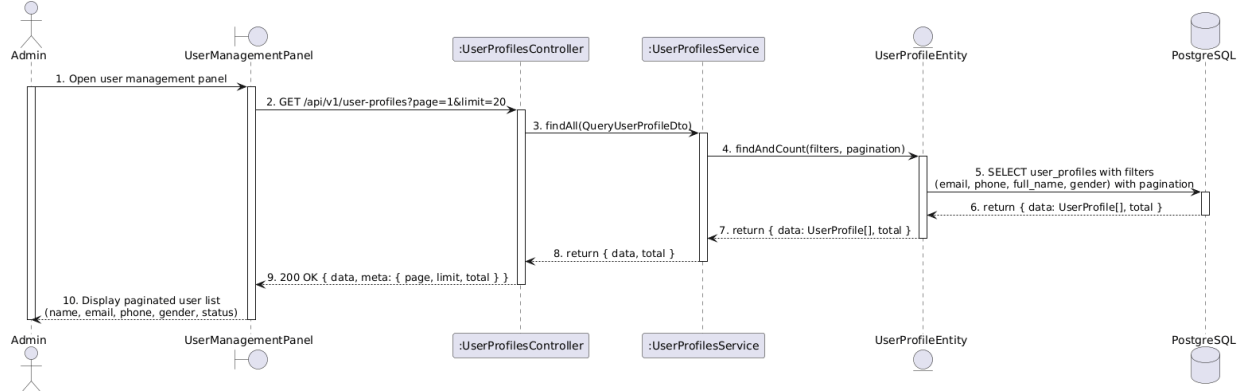


3.20 View User List

a. Class Diagram

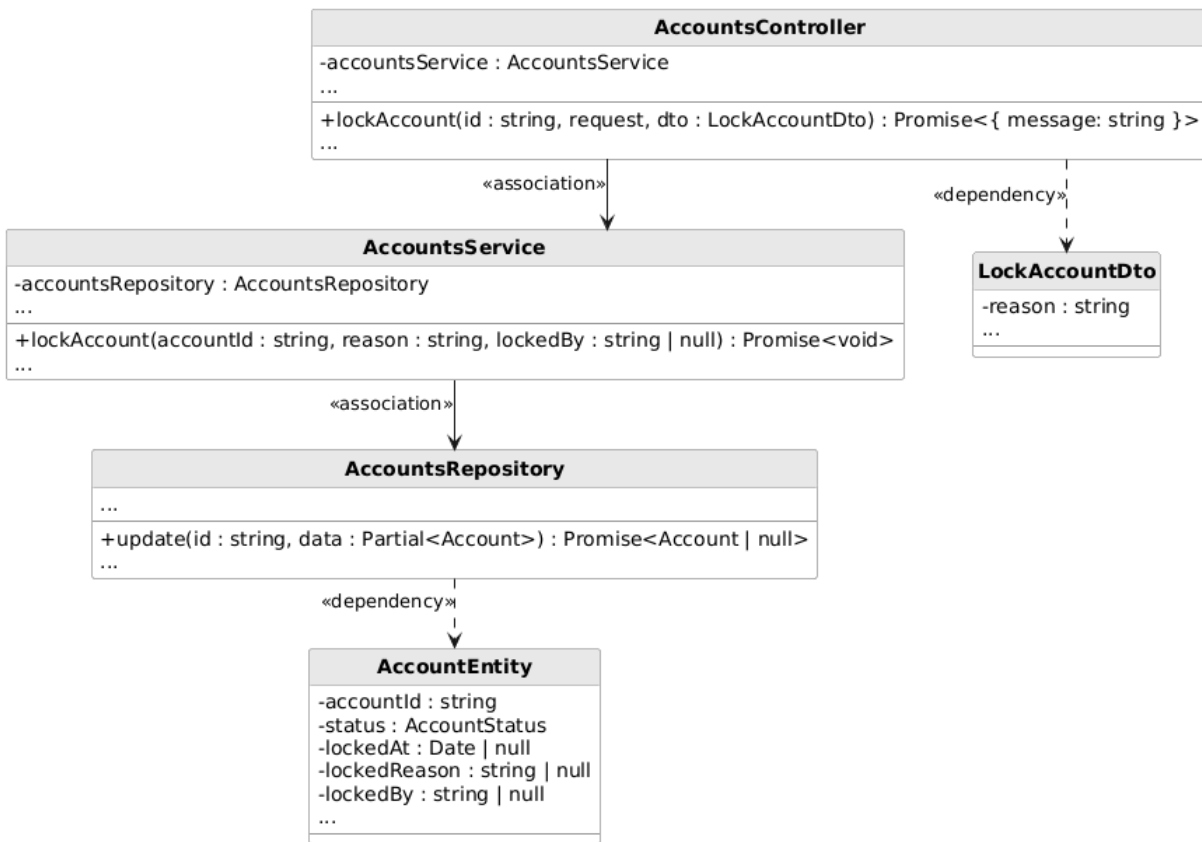


b. Sequence Diagram(s)

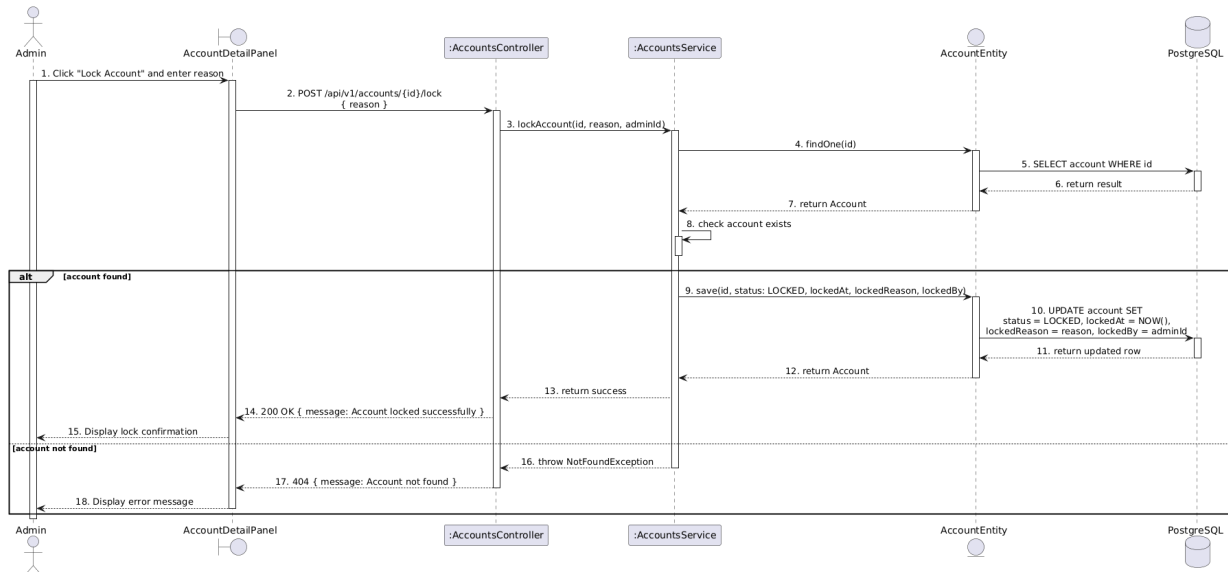


3.21 Lock Ban User Account

a. Class Diagram

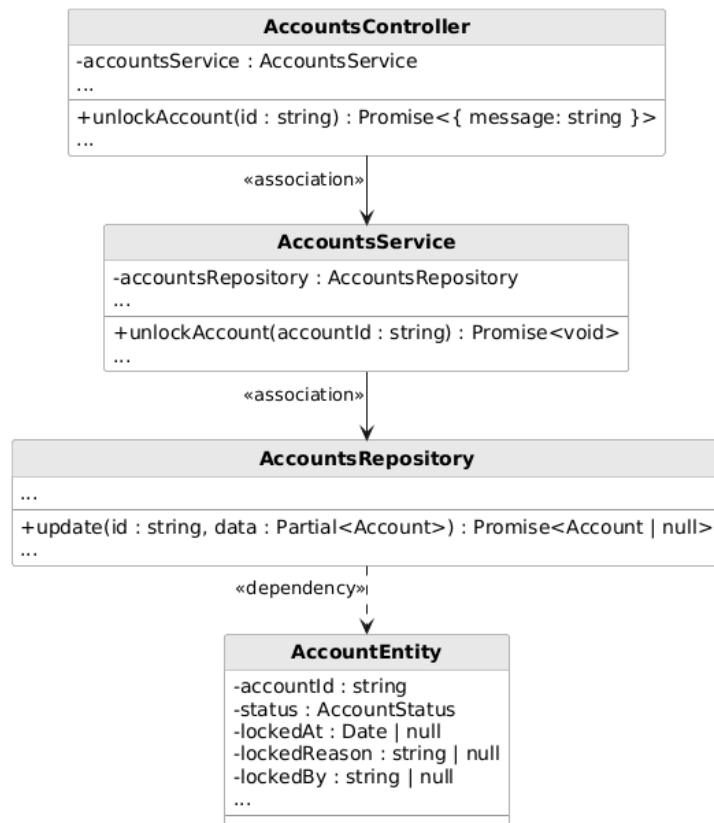


b. Sequence Diagram(s)

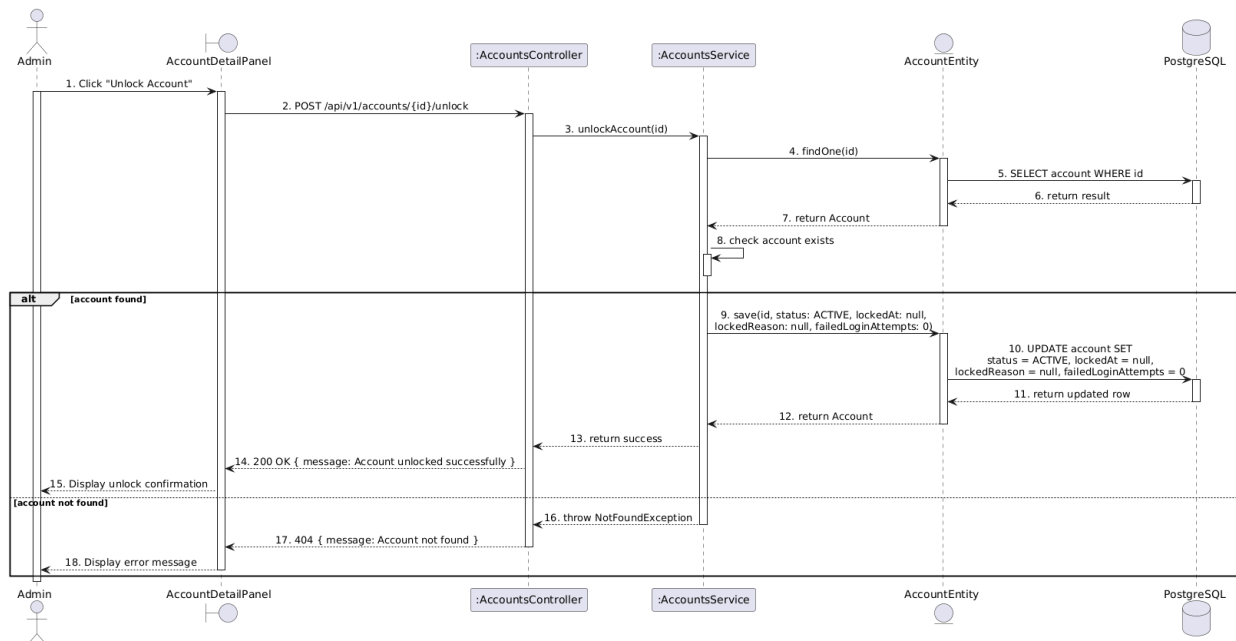


3.22 Unlock Unban User Account

a. Class Diagram

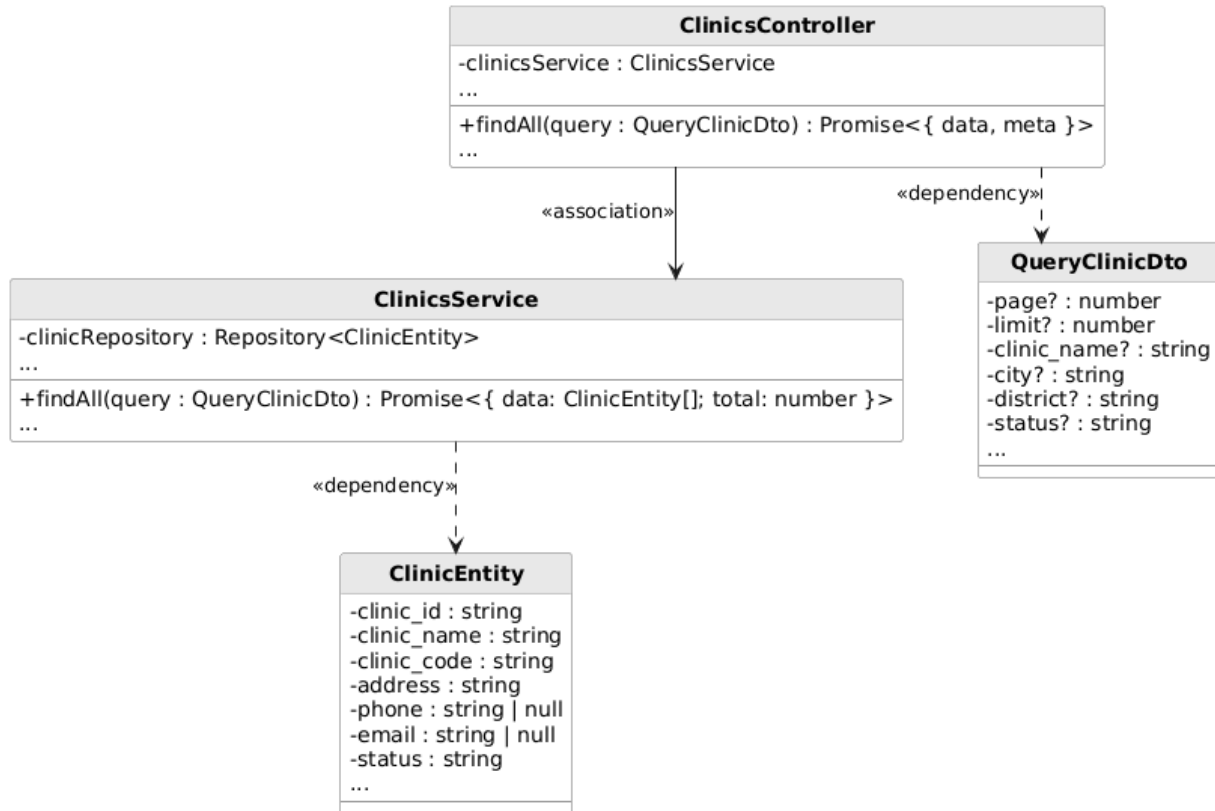


b. Sequence Diagram(s)

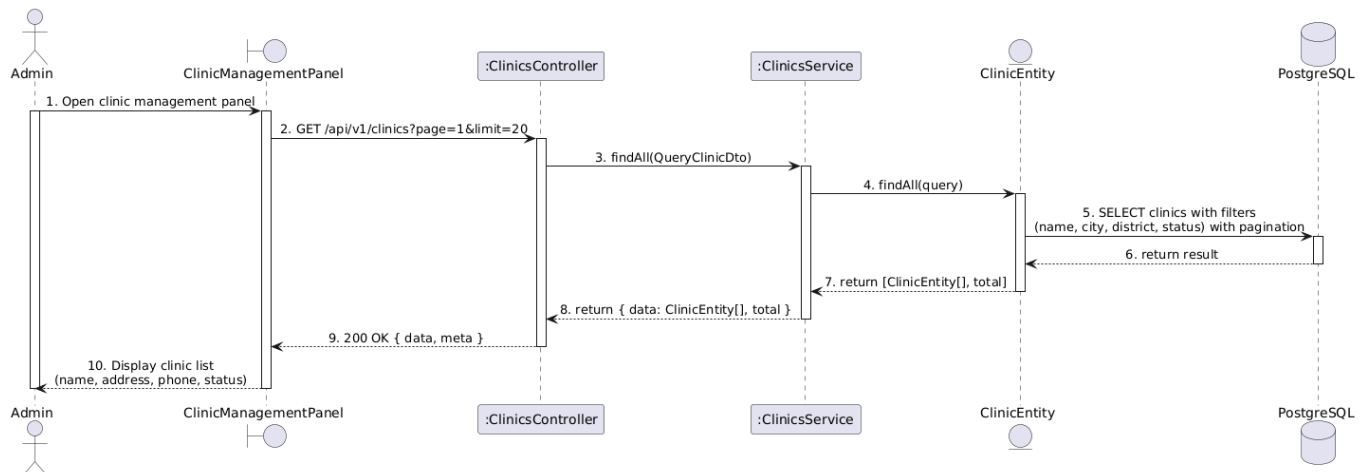


3.23 View Clinic Transform

a. Class Diagram

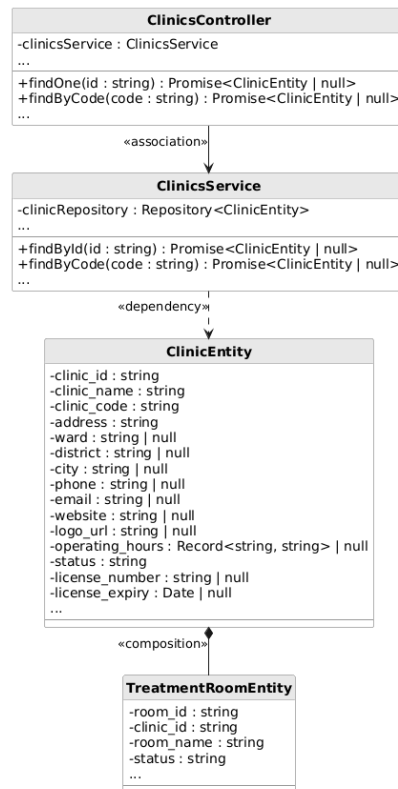


b. Sequence Diagram(s)

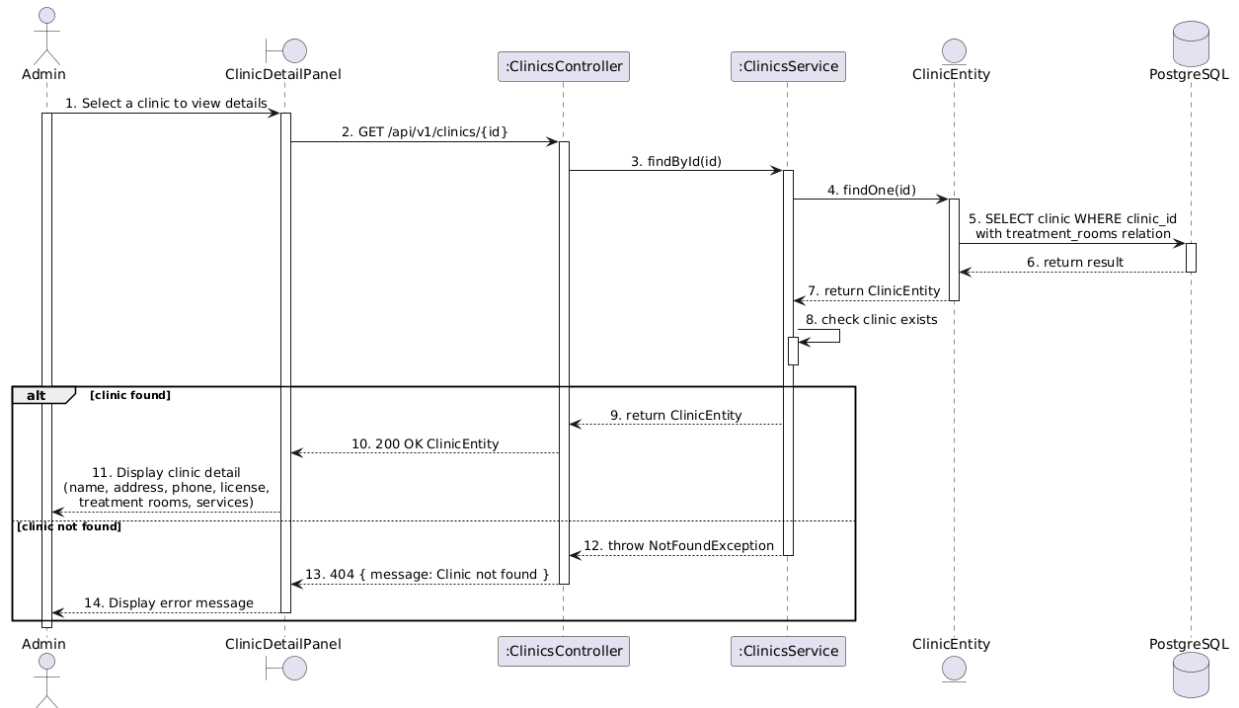


3.24 View Clinic Details

a. Class Diagram

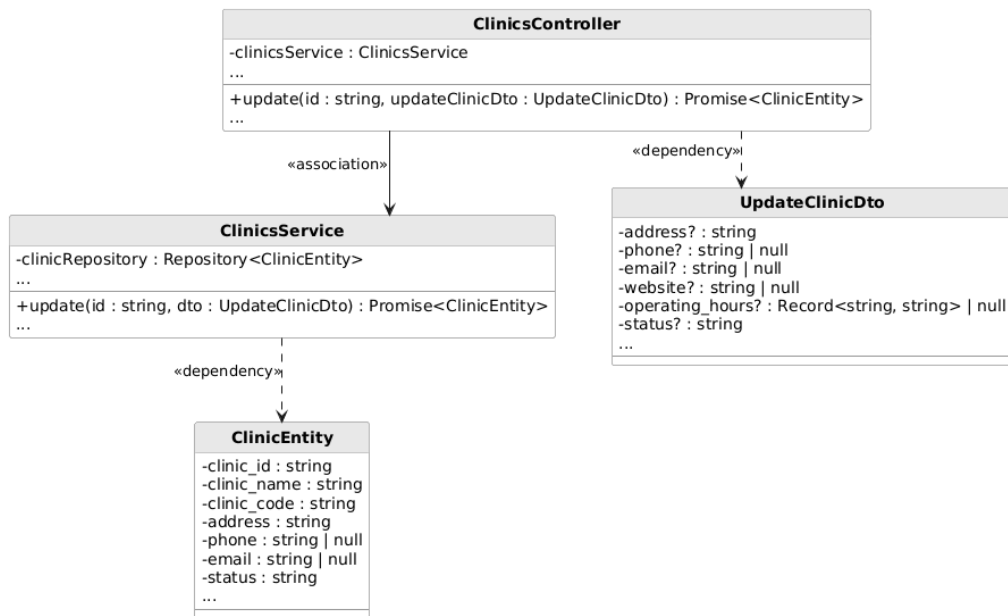


b. Sequence Diagram(s)

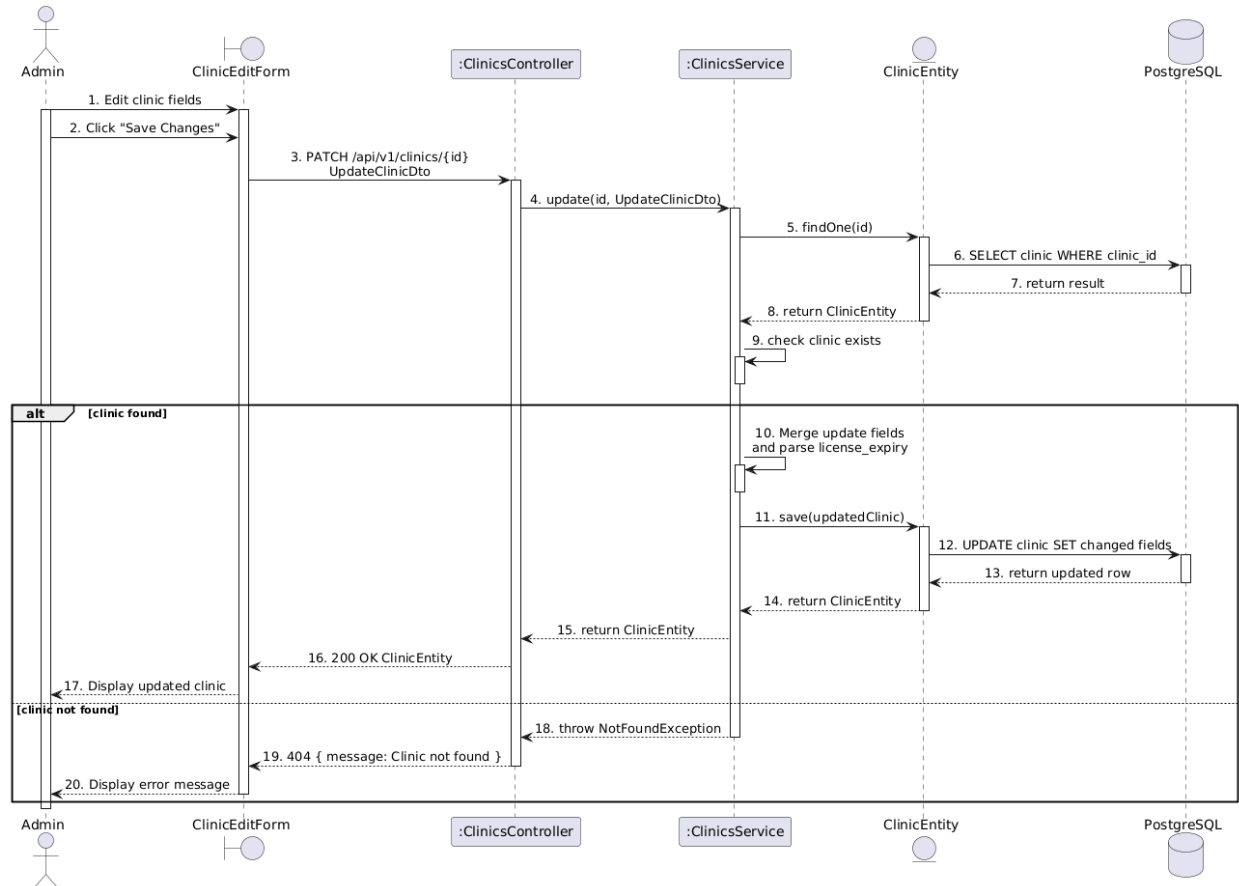


3.25 Update Clinic Information

a. Class Diagram

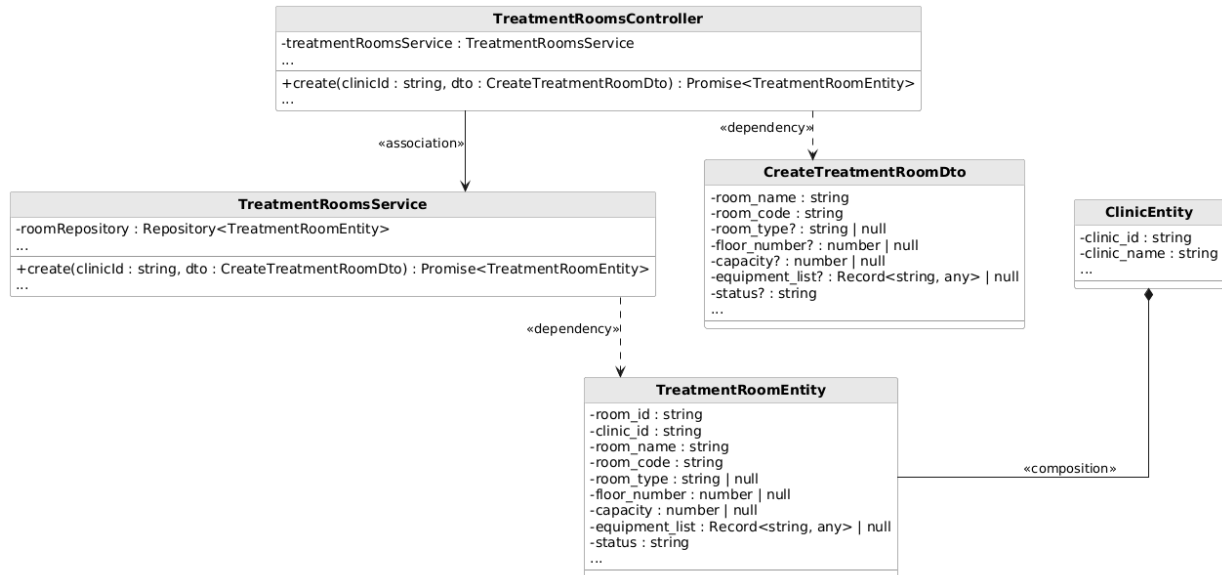


b. Sequence Diagram(s)

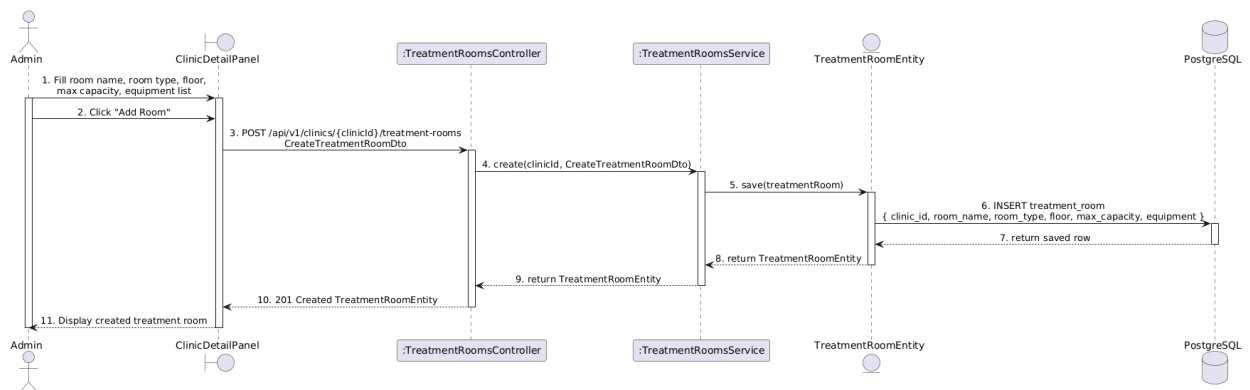


3.26 Add Treatment Room

a. Class Diagram

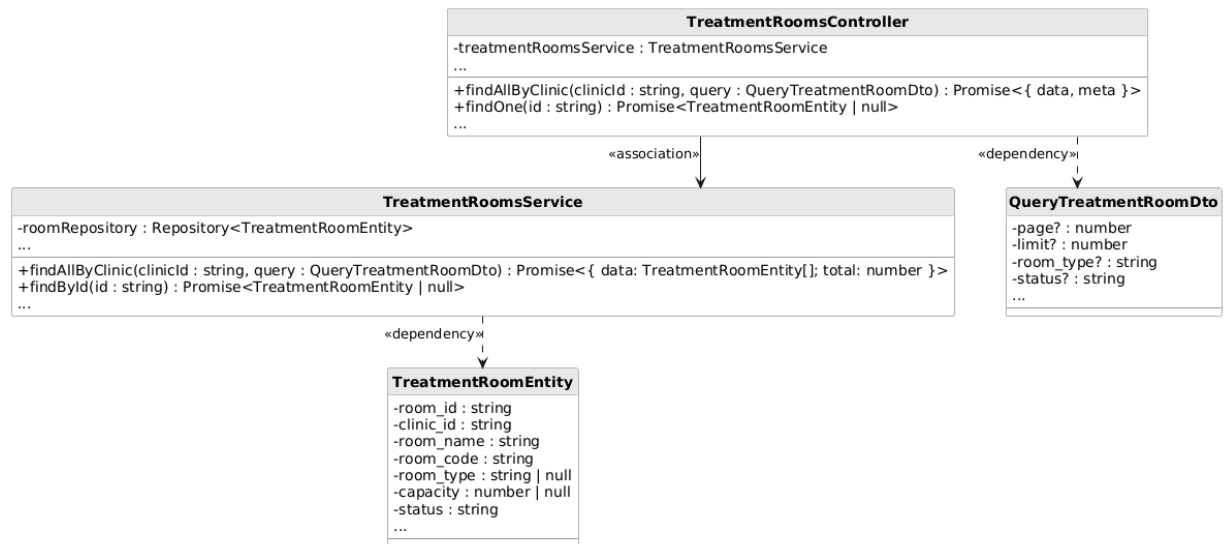


b. Sequence Diagram(s)

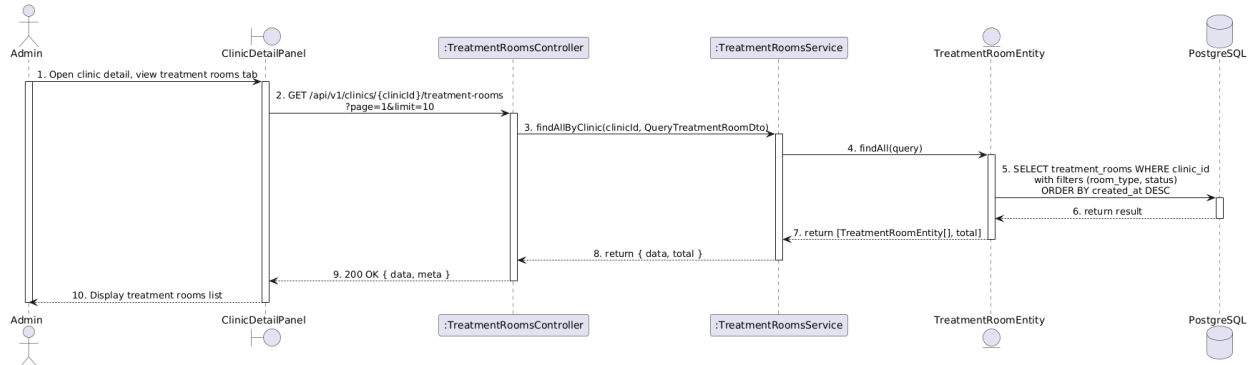


3.27 View Treatment Room

a. Class Diagram

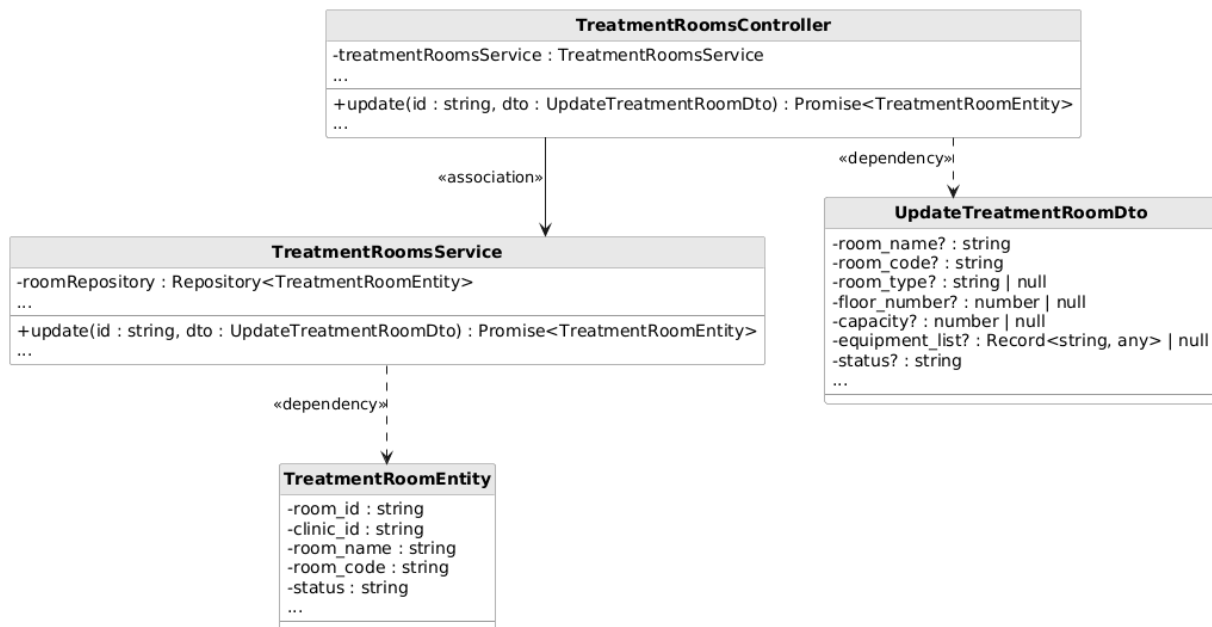


b. Sequence Diagram(s)

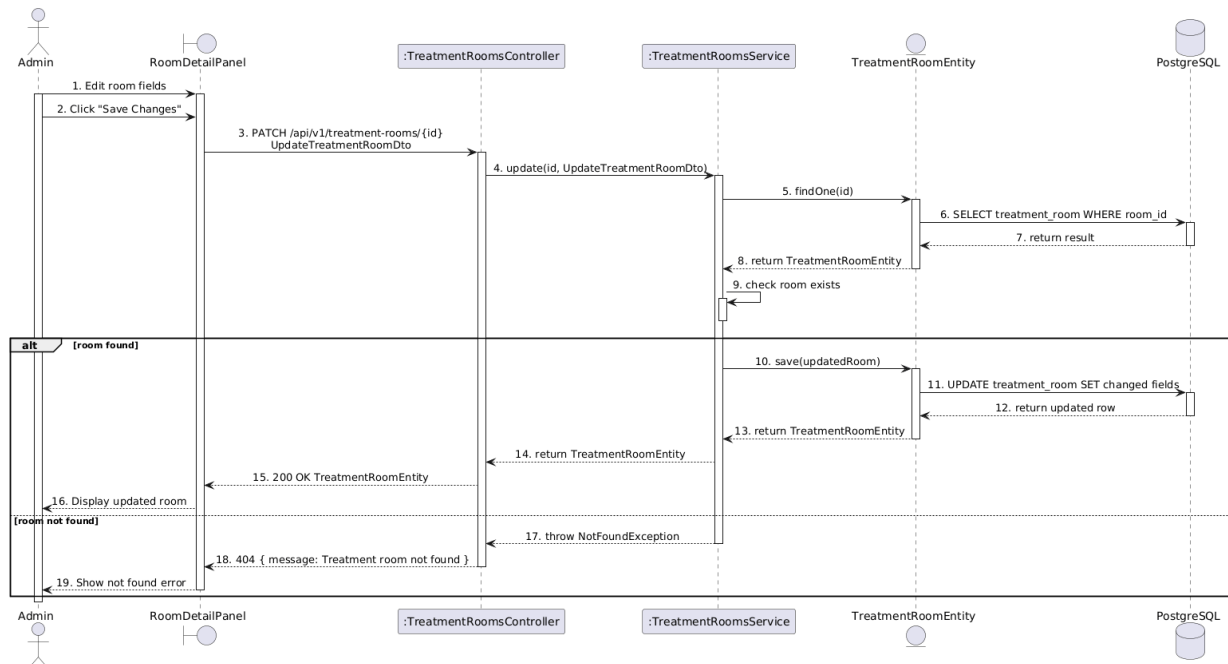


3.28 Update Treatment Room

a. Class Diagram

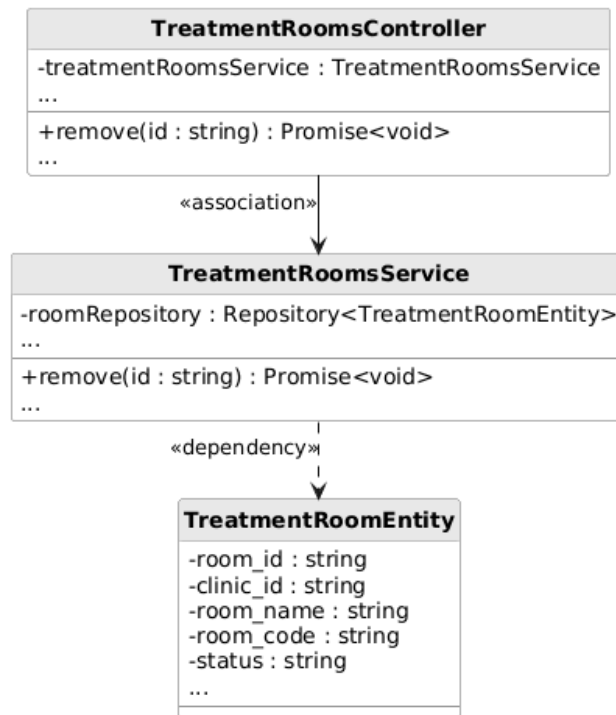


b. Sequence Diagram(s)

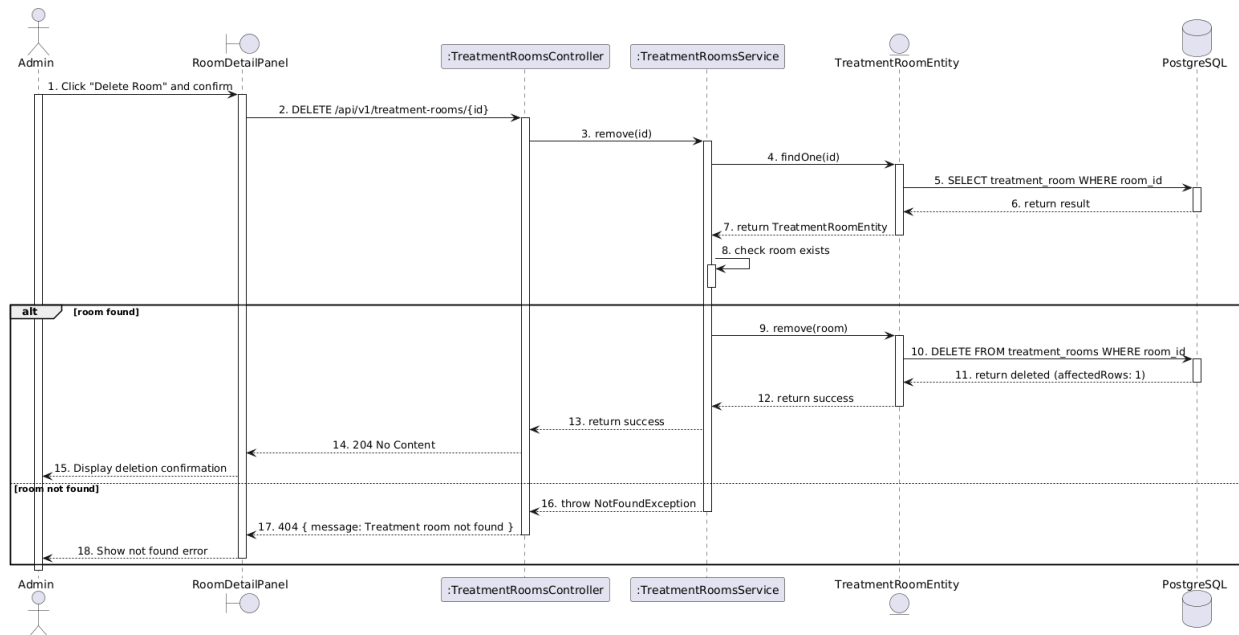


3.29 Delete Treatment Room

a. Class Diagram

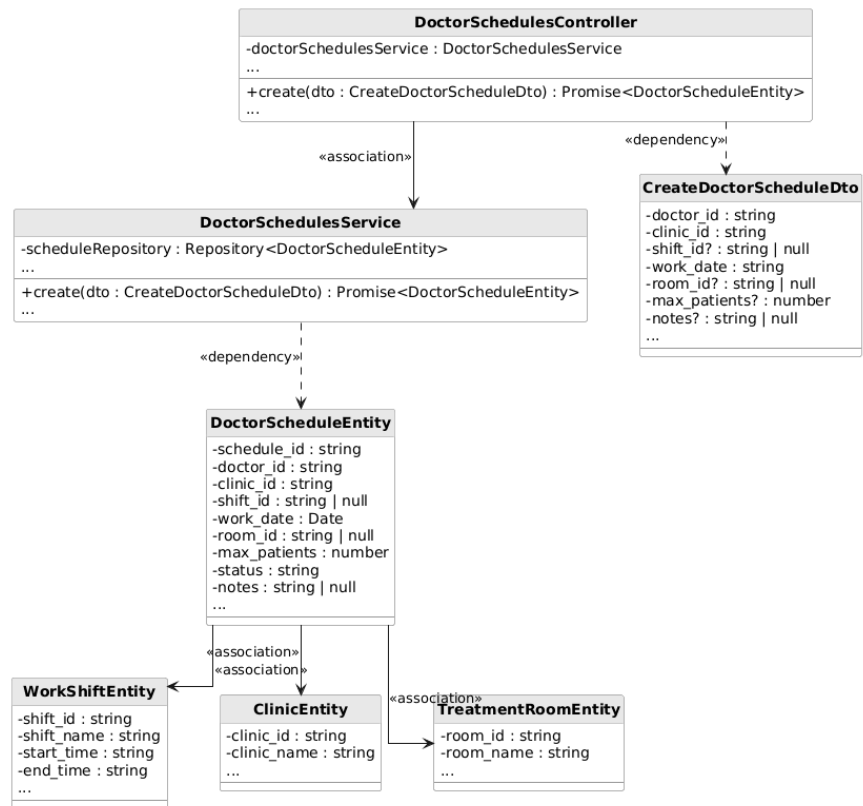


b. Sequence Diagram(s)

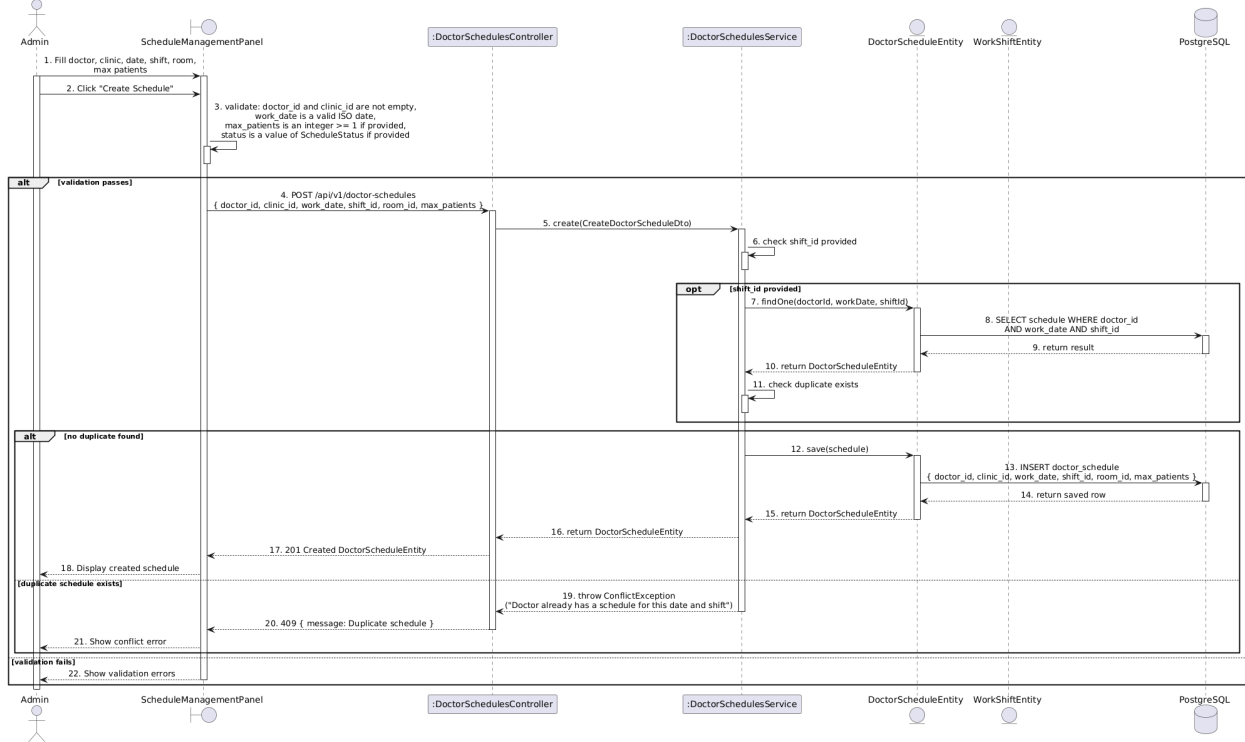


3.30 Create Work Schedule

a. Class Diagram

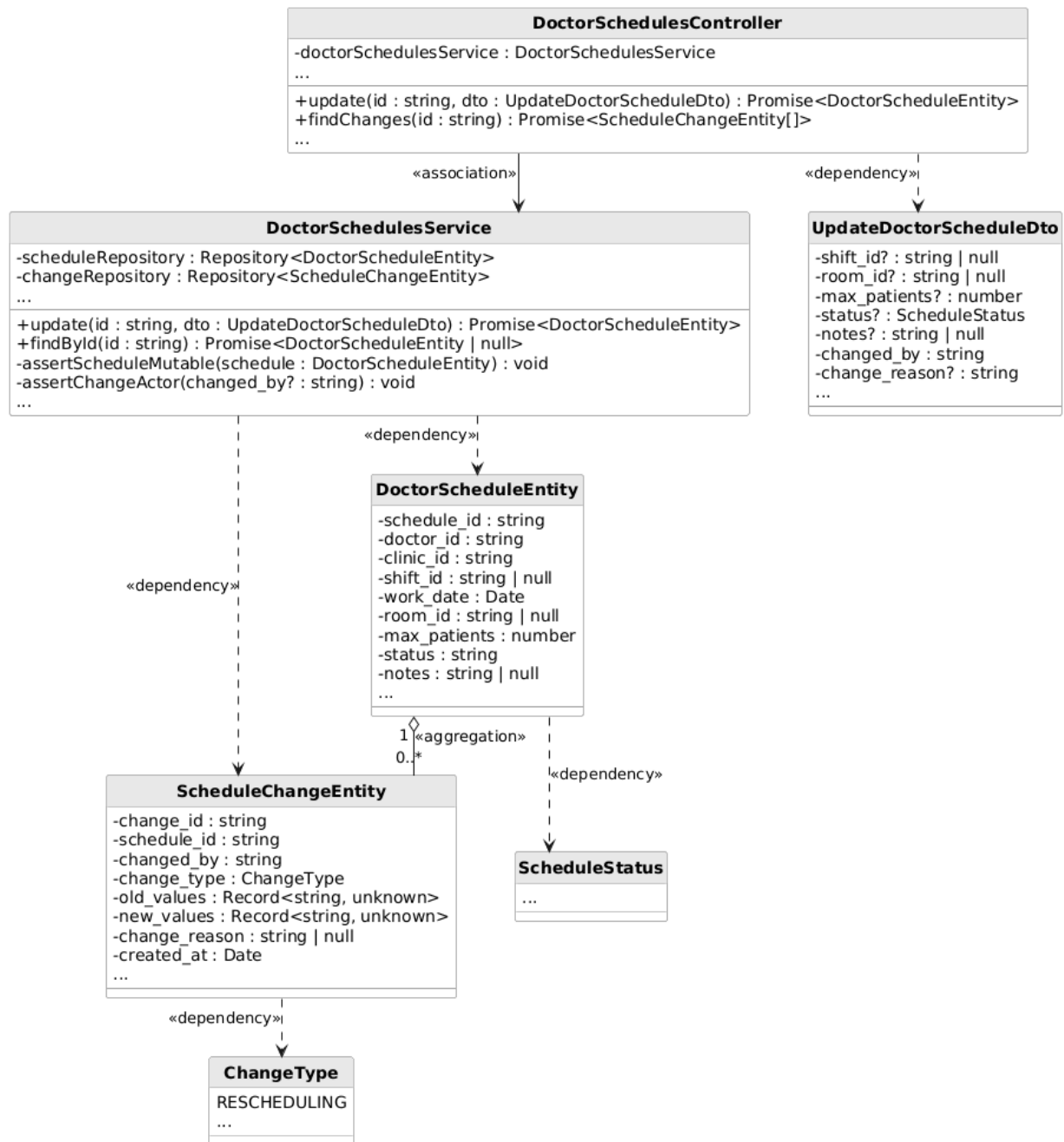


b. Sequence Diagram(s)

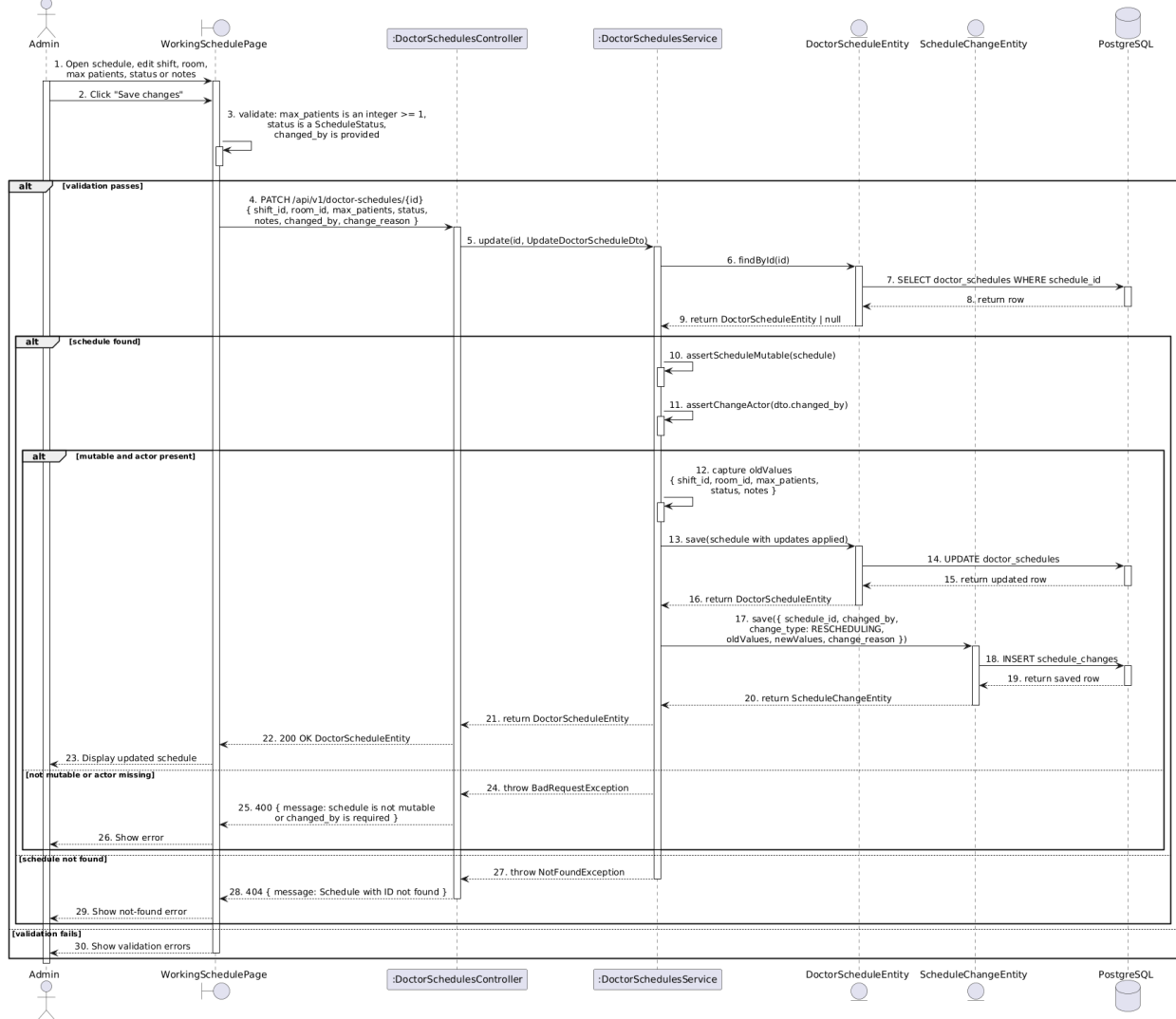


3.31 Manage Working Schedule

a. Class Diagram

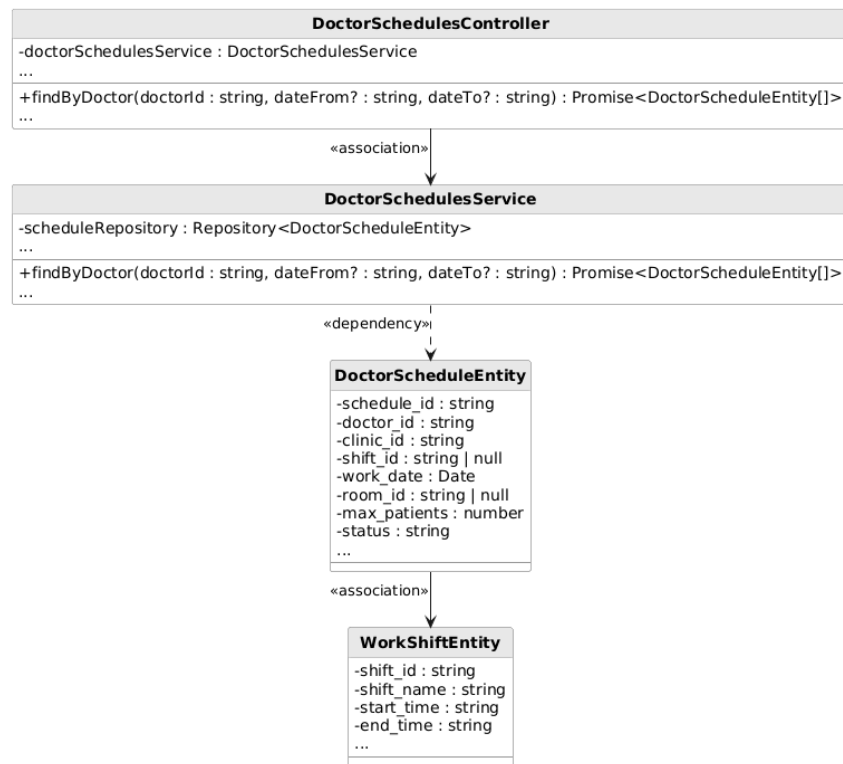


b. Sequence Diagram(s)

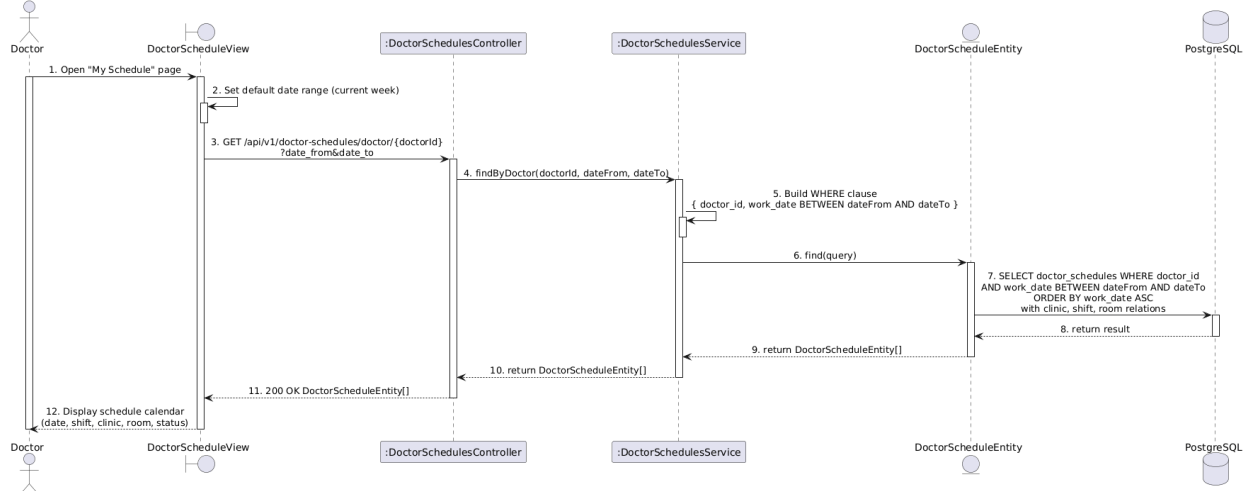


3.32 View Personal Schedule

a. Class Diagram

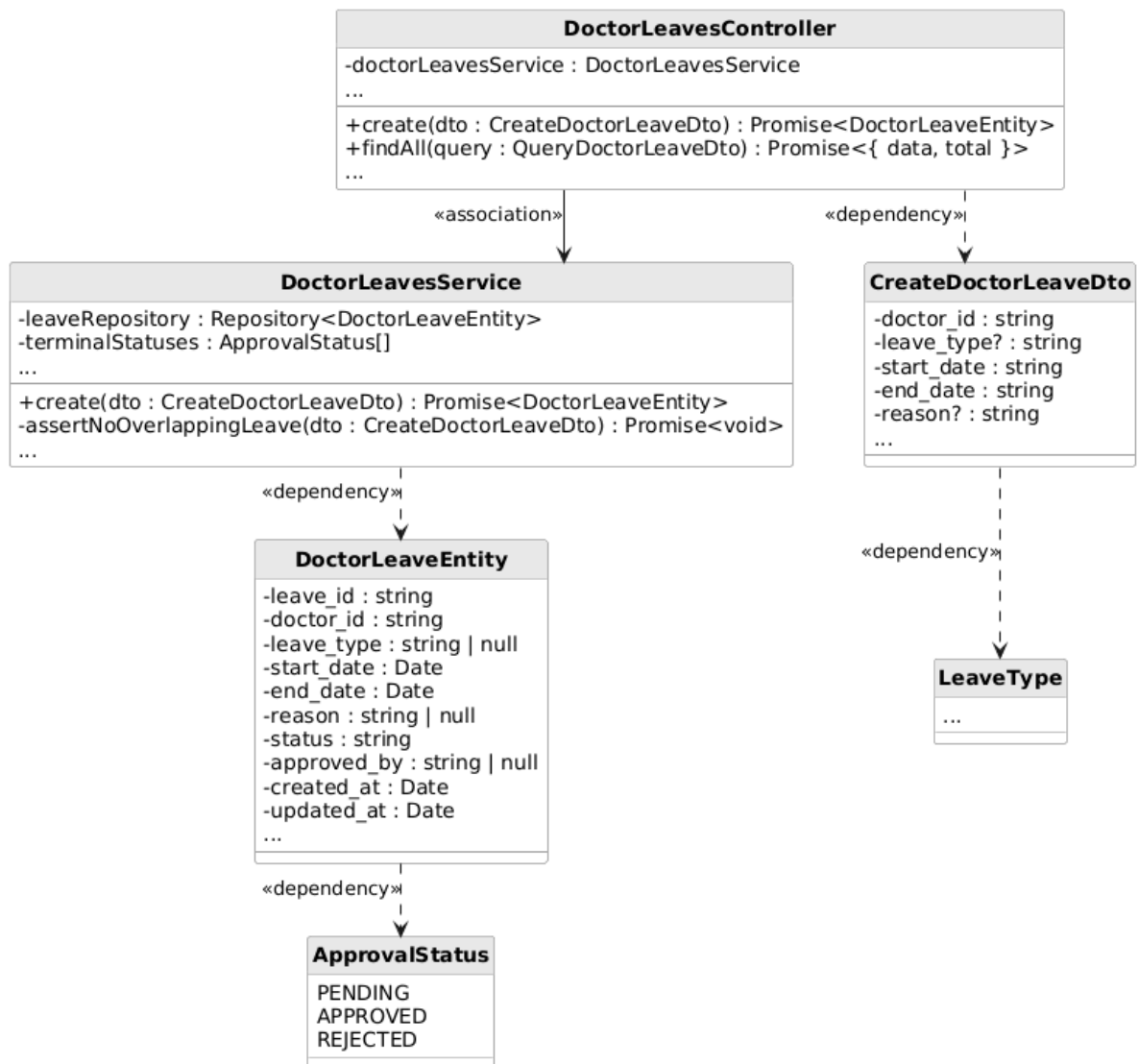


b. Sequence Diagram(s)

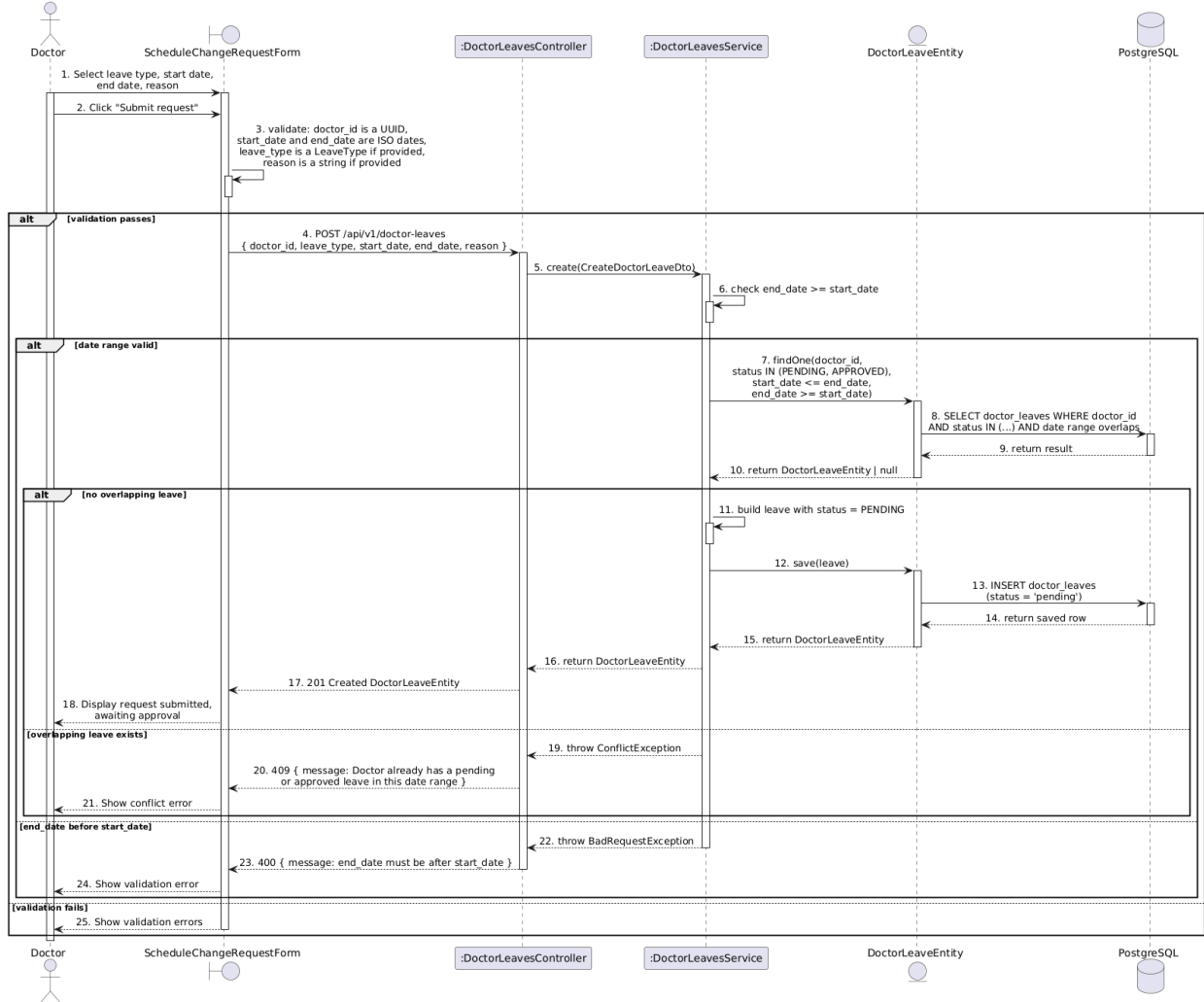


3.33 Request Personal Schedule Change

a. Class Diagram

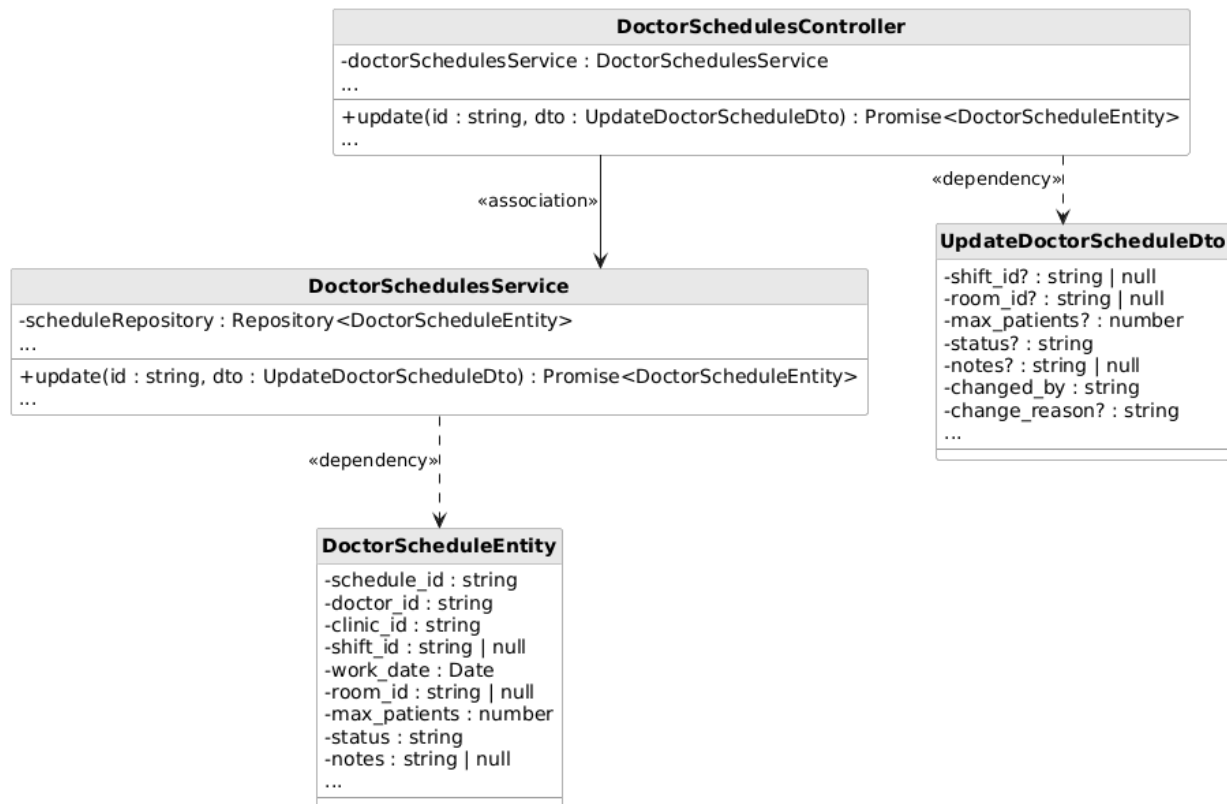


b. Sequence Diagram(s)

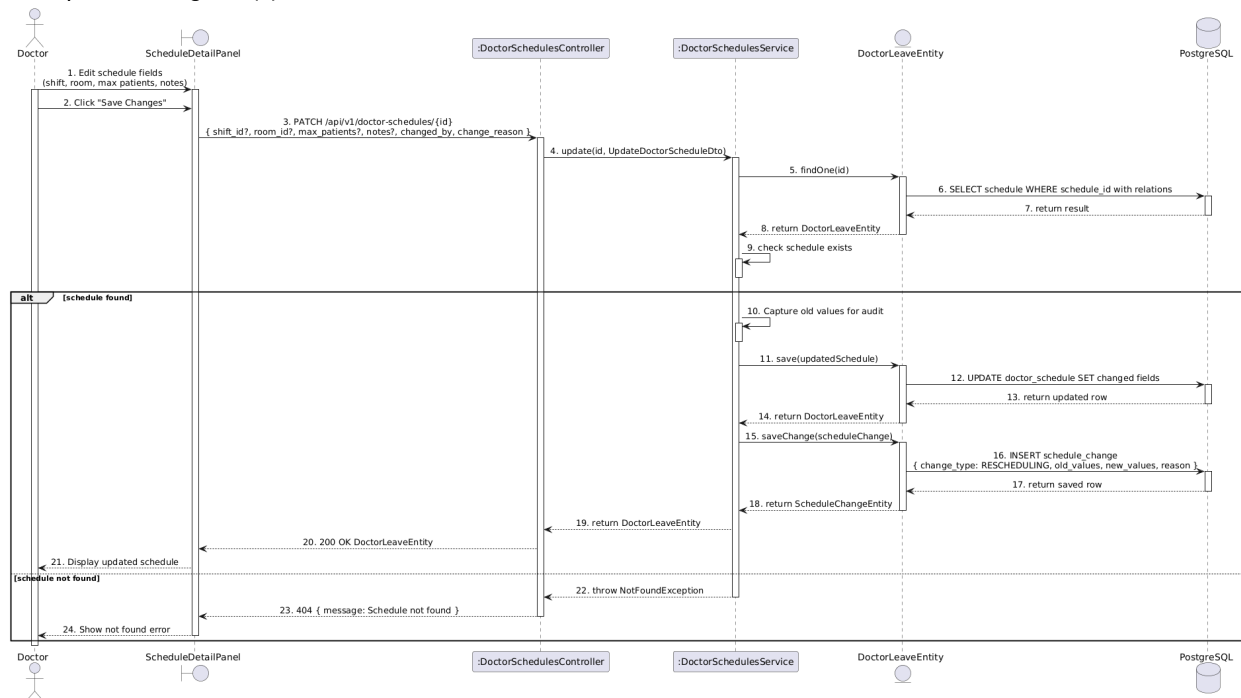


3.34 Update Personal Schedule

a. Class Diagram

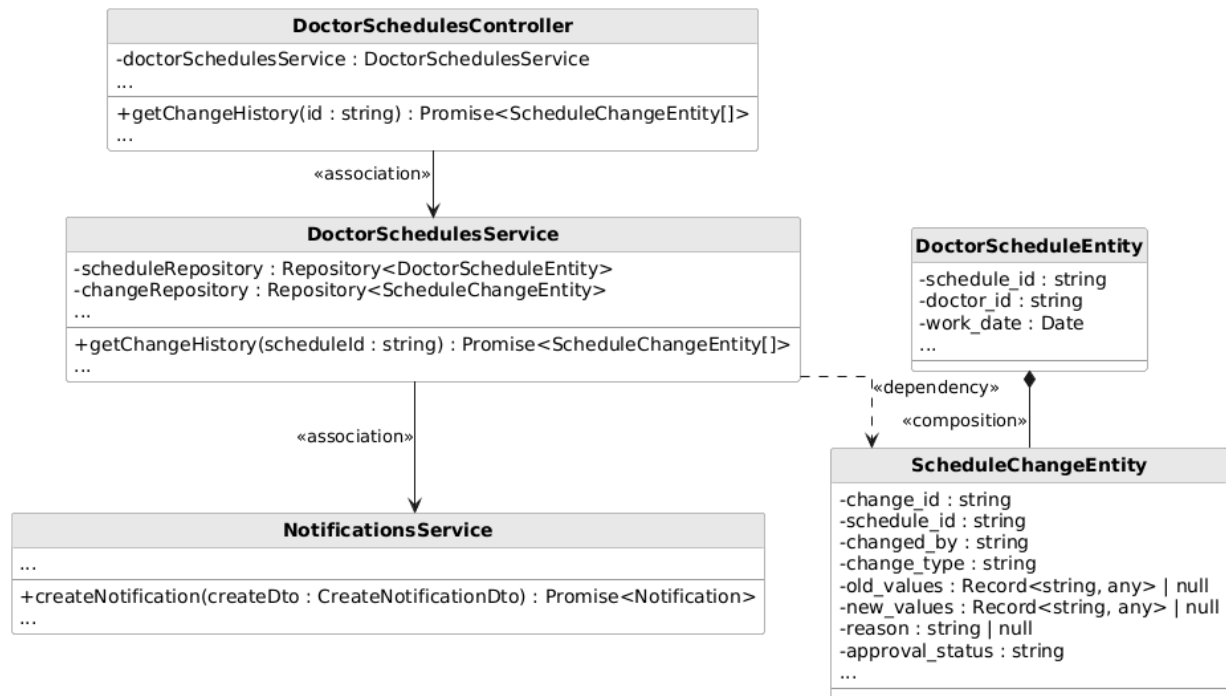


b. Sequence Diagram(s)

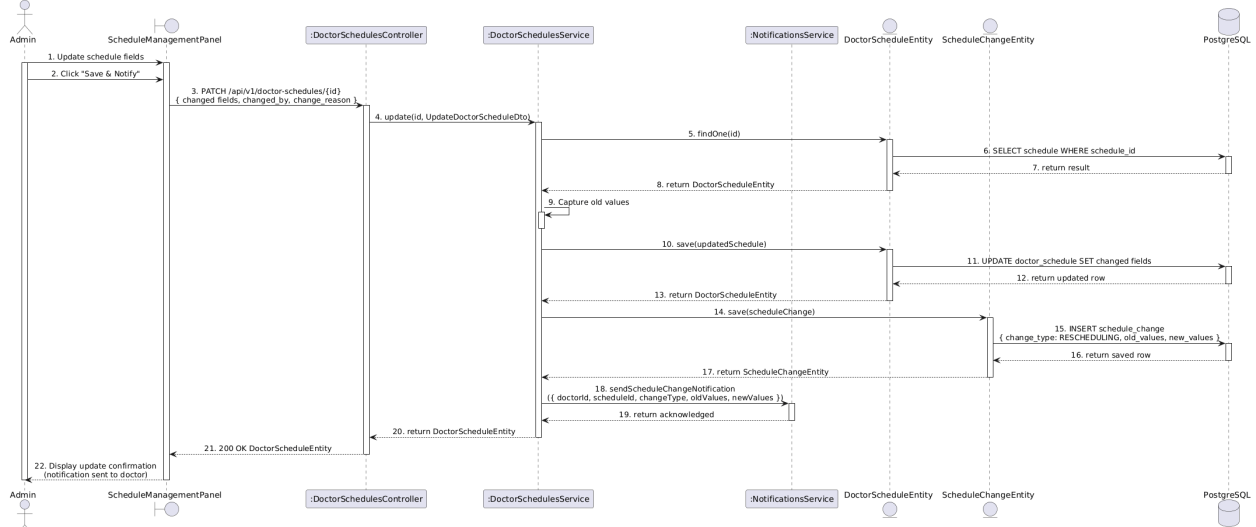


3.35 Notify Schedule Change

a. Class Diagram

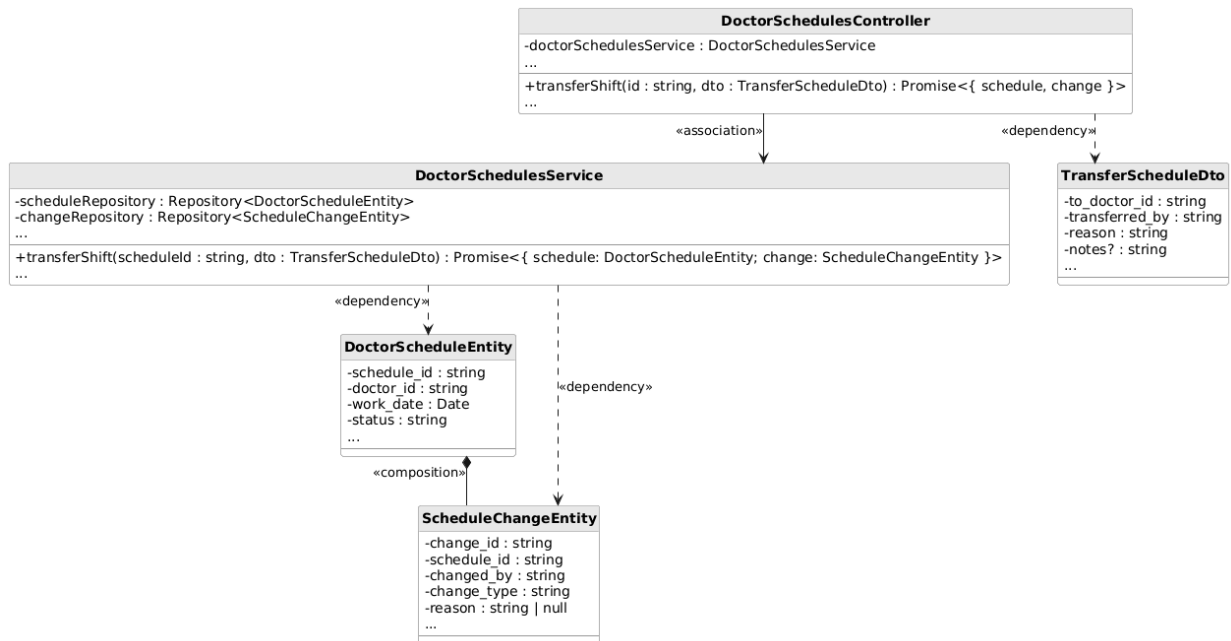


b. Sequence Diagram(s)

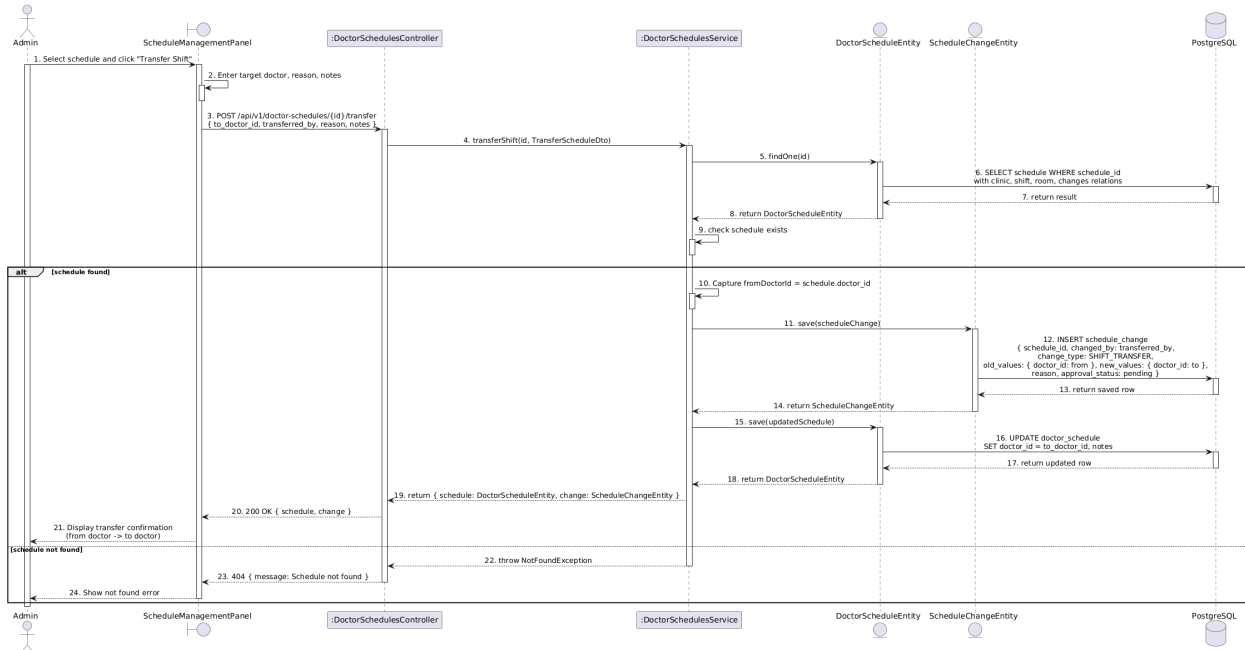


3.36 Notify Shift Transfer

a. Class Diagram

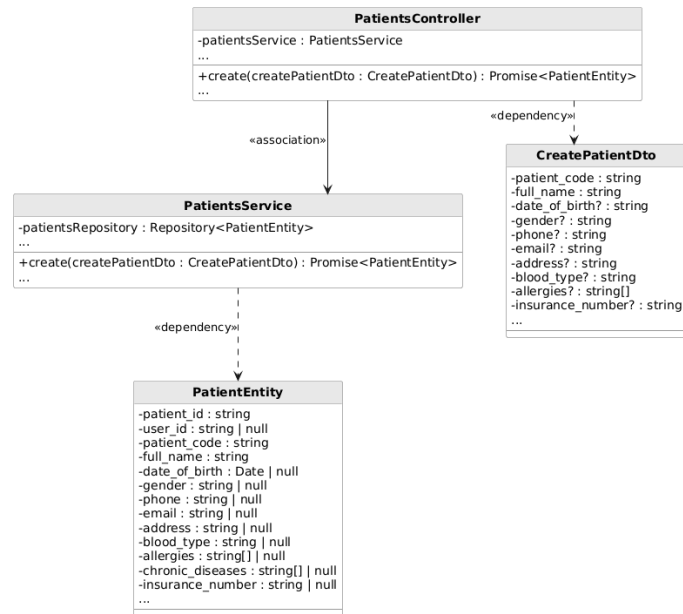


b. Sequence Diagram(s)

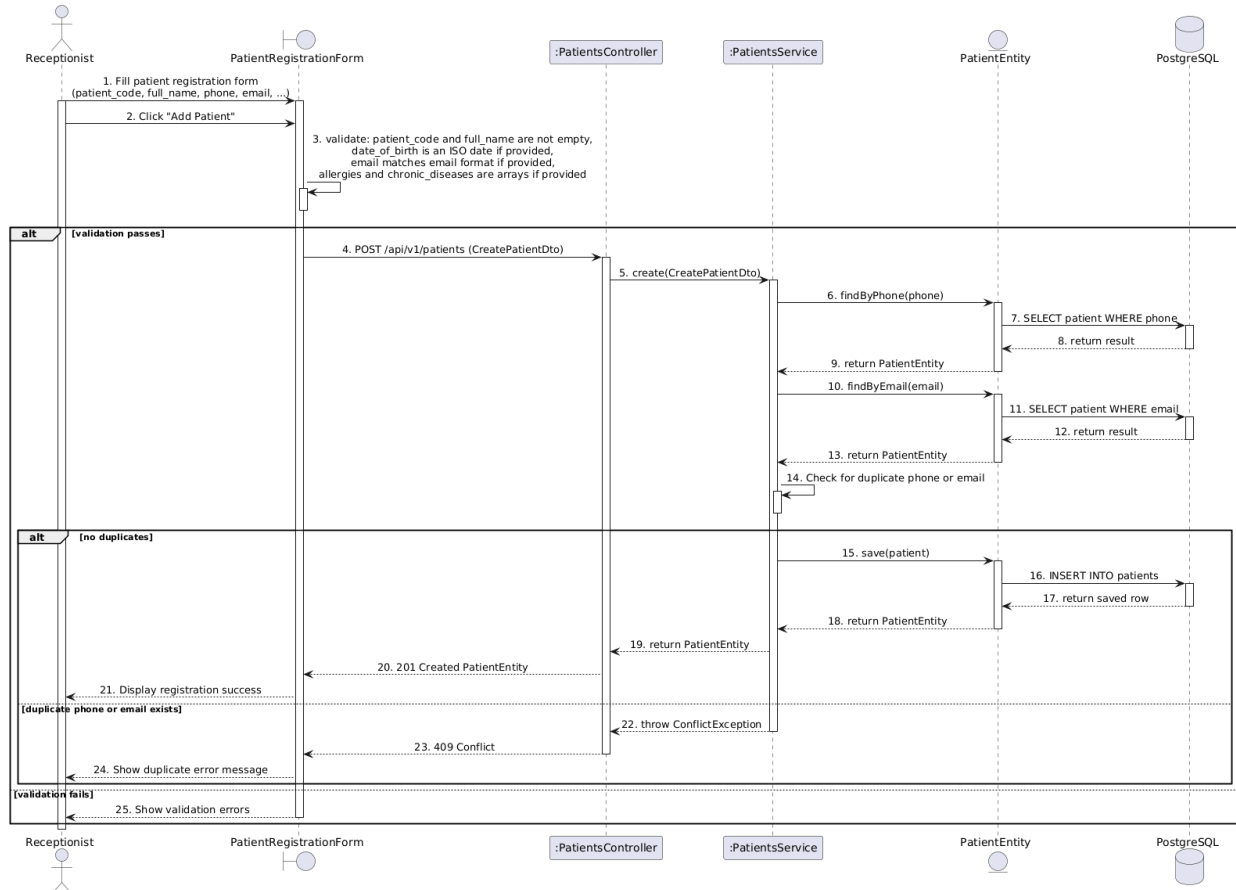


3.37 Add Patient Profile

a. Class Diagram

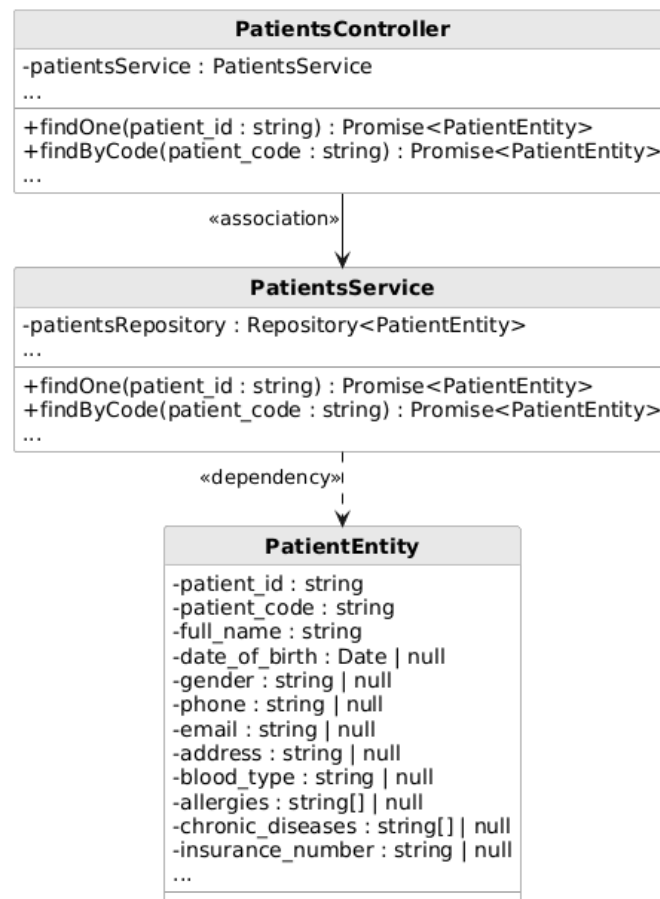


b. Sequence Diagram(s)

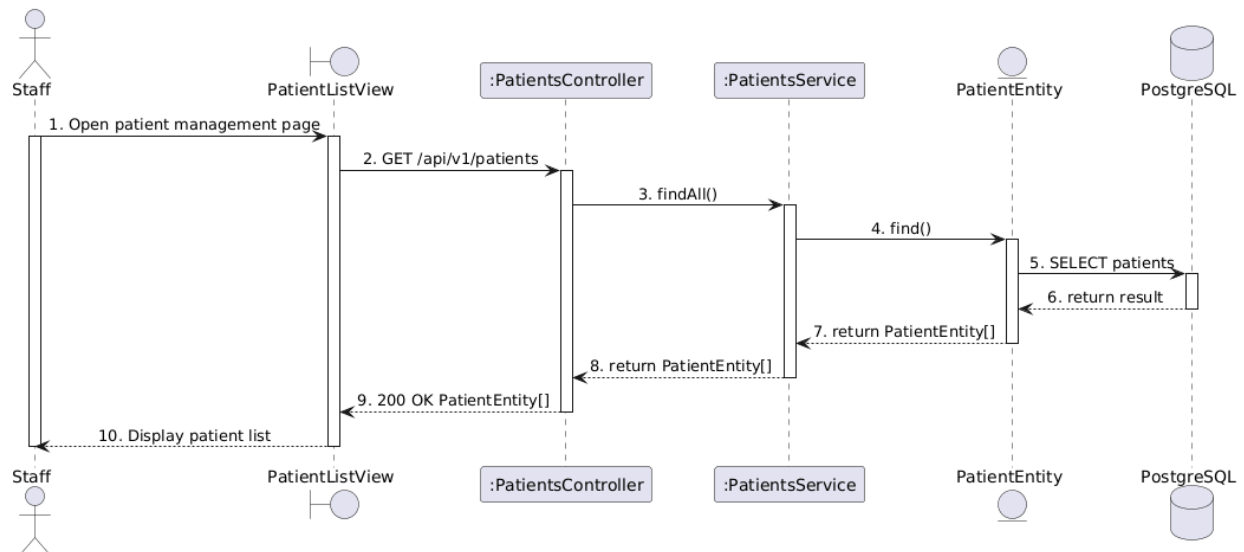


3.38 View Patient Profile

a. Class Diagram

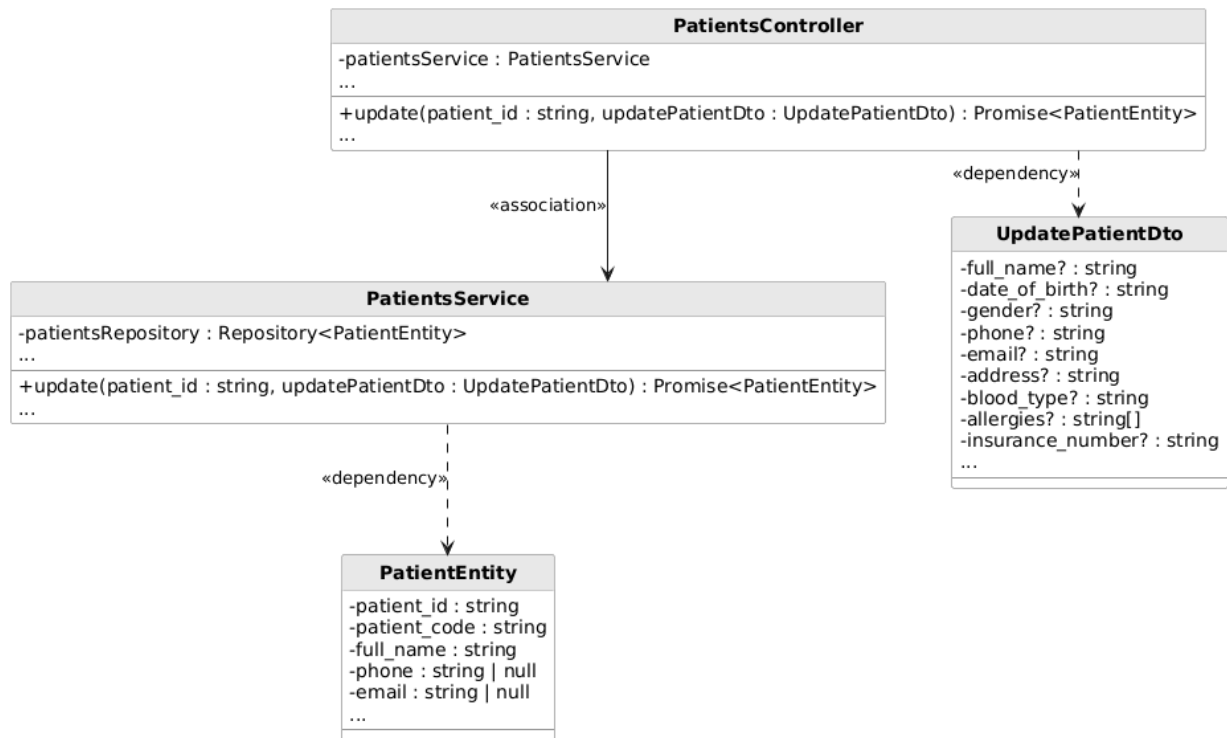


b. Sequence Diagram(s)

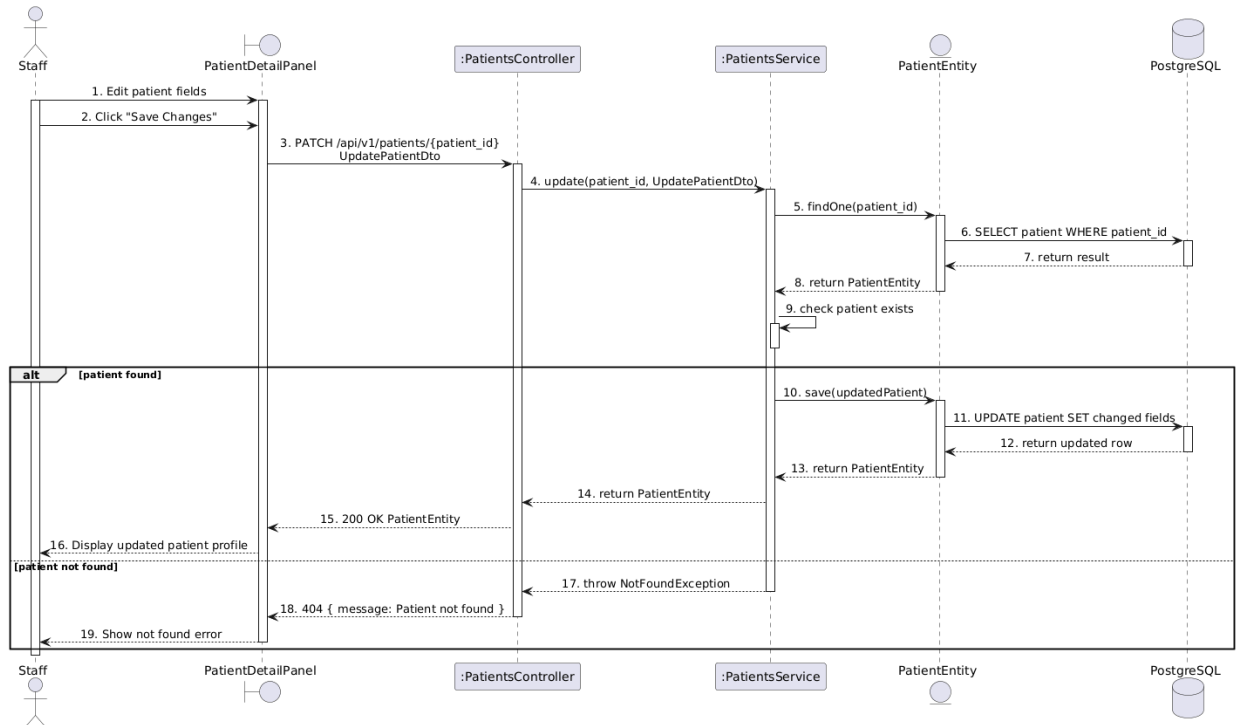


3.39 Update Patient Profile

a. Class Diagram

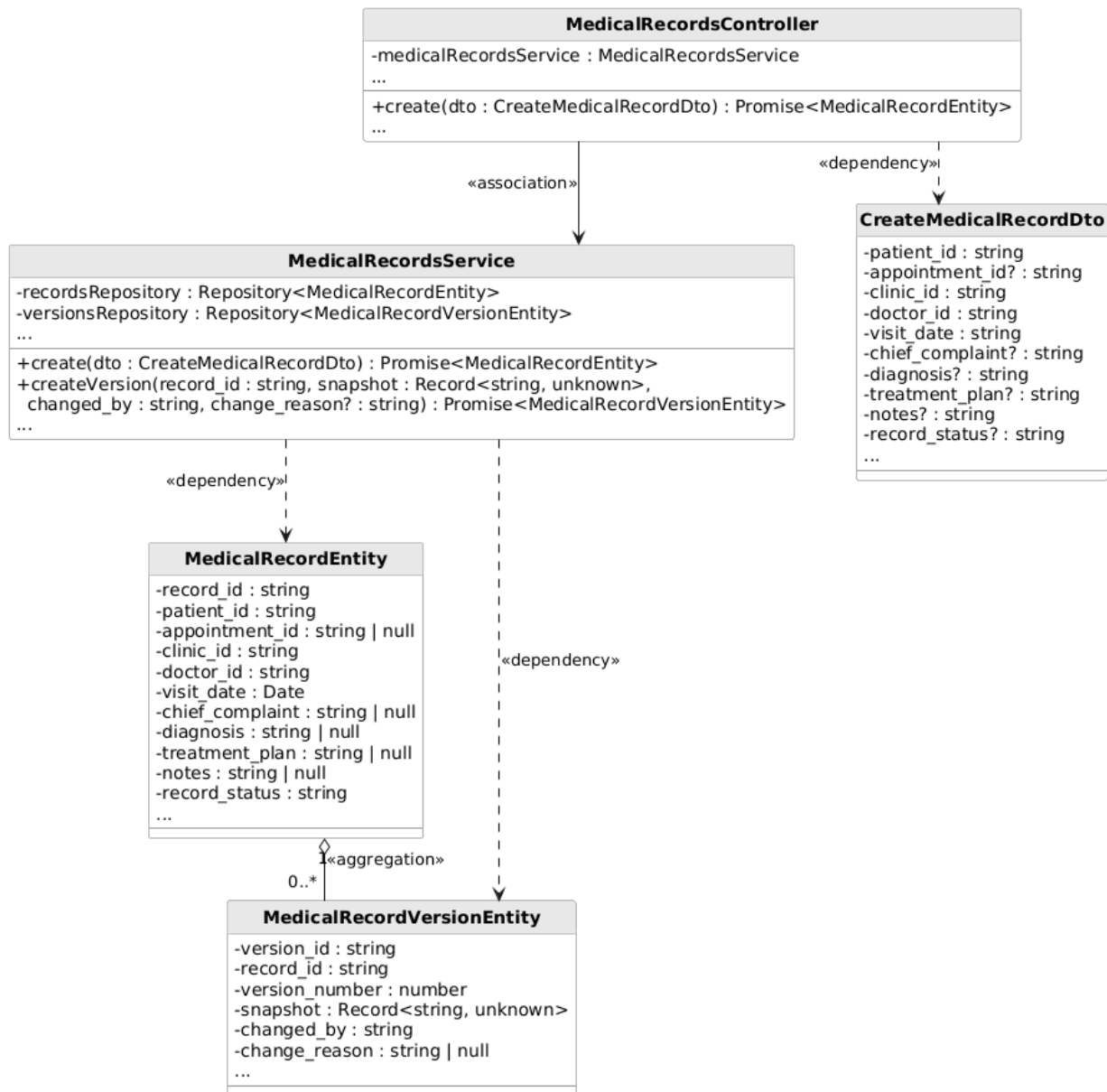


b. Sequence Diagram(s)

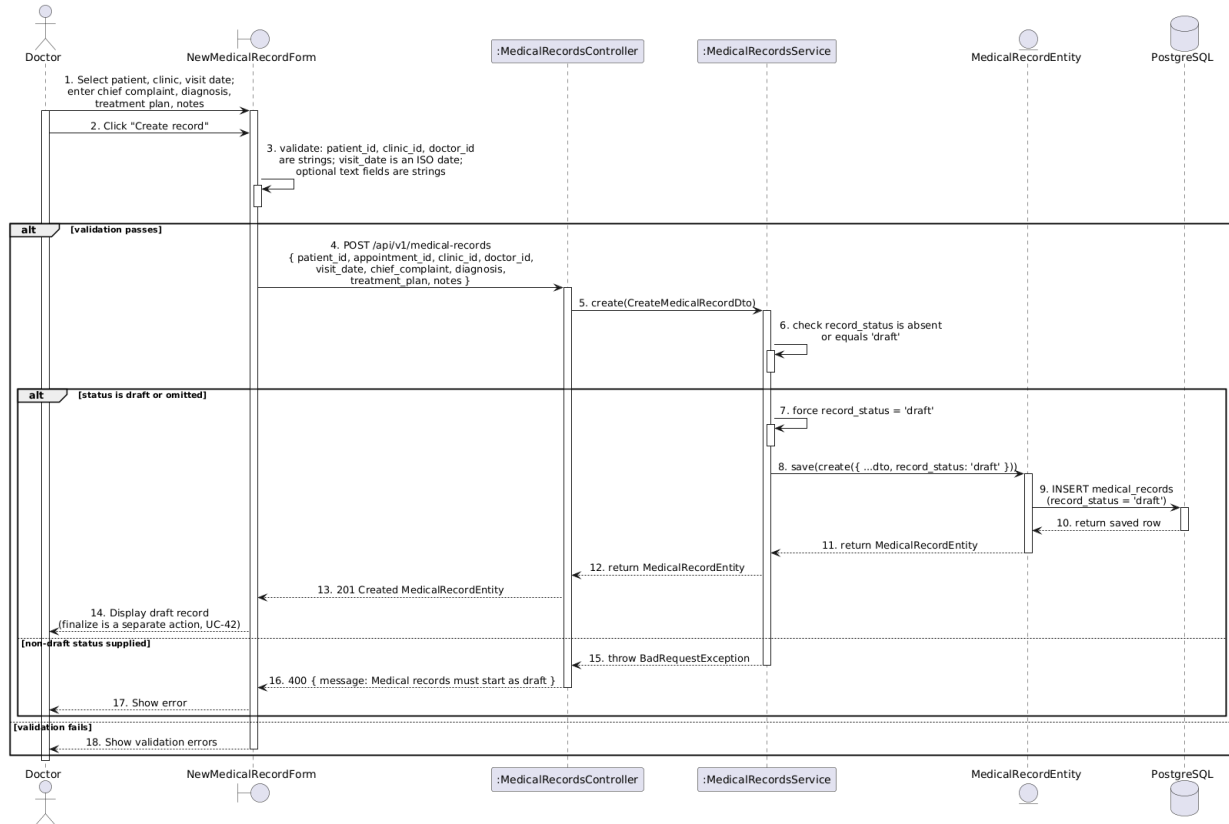


3.40 Add Medical Record

a. Class Diagram

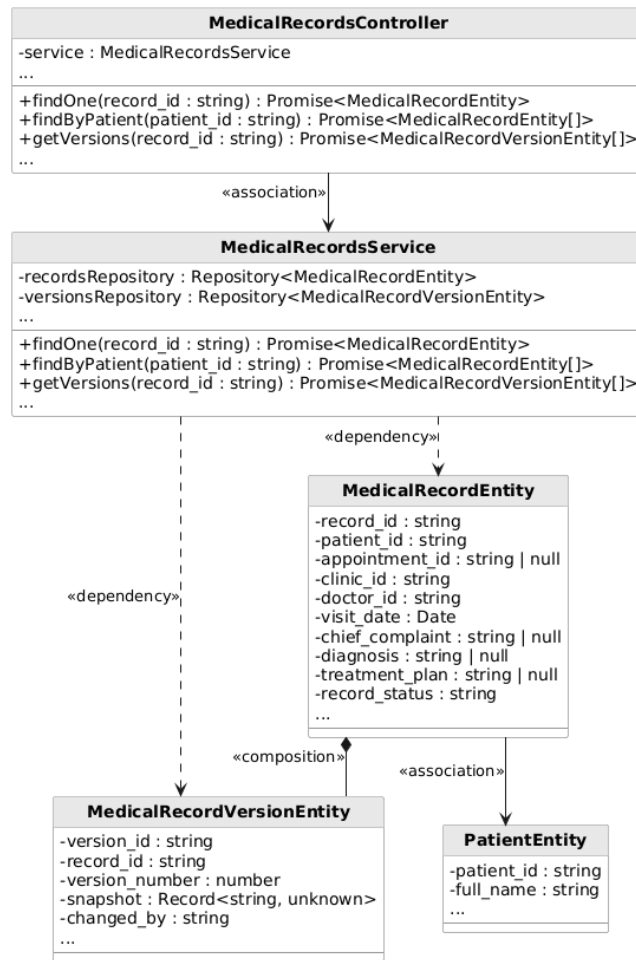


b. Sequence Diagram(s)

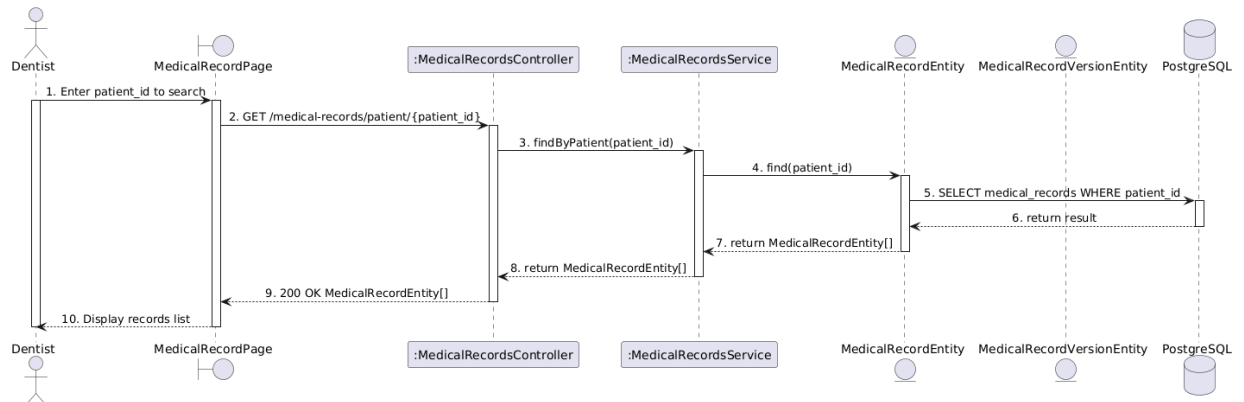


3.41 View Medical Record

a. Class Diagram

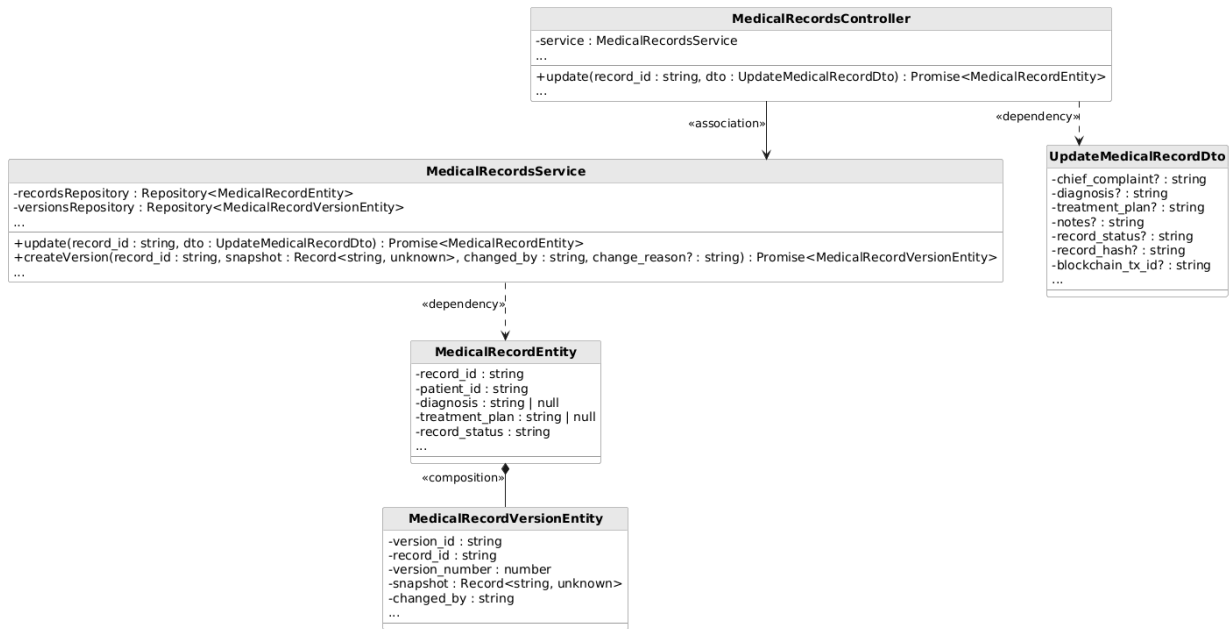


b. Sequence Diagram(s)

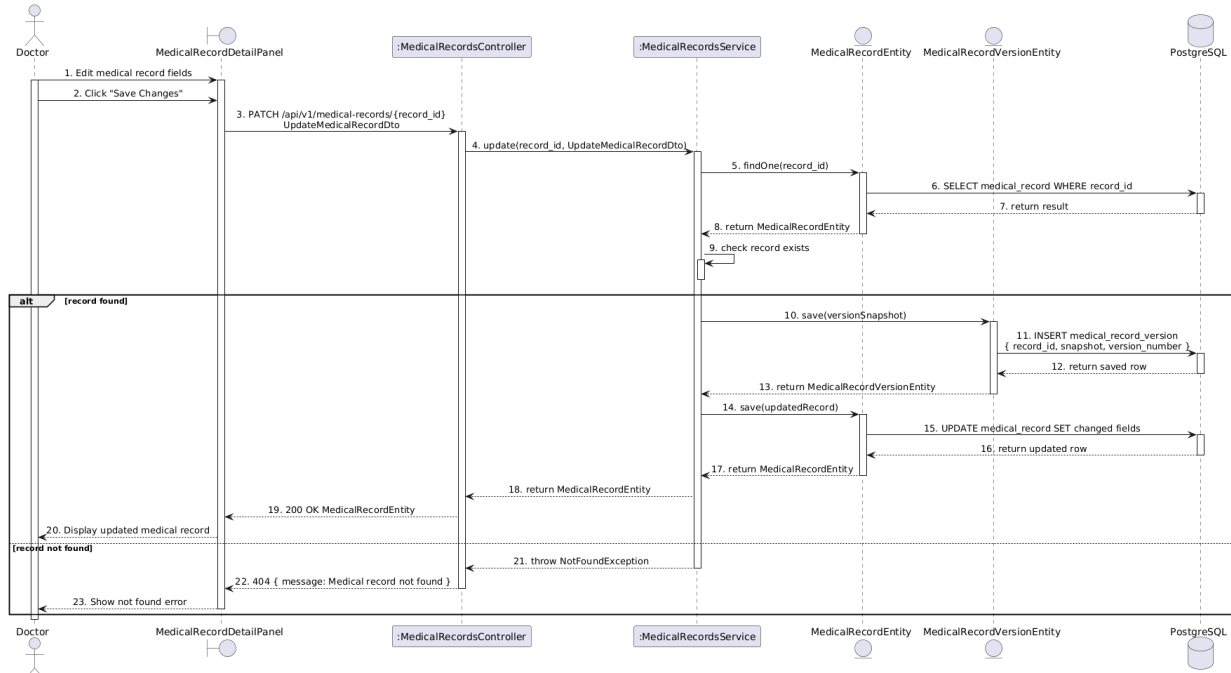


3.42 Update Medical Record

a. Class Diagram

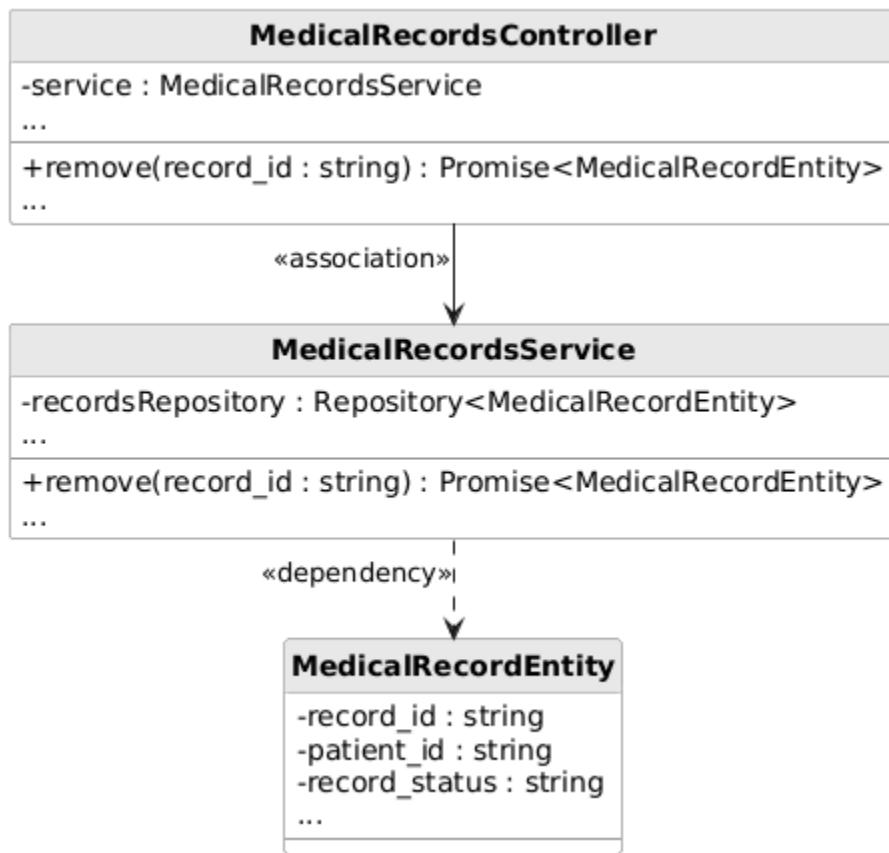


b. Sequence Diagram(s)

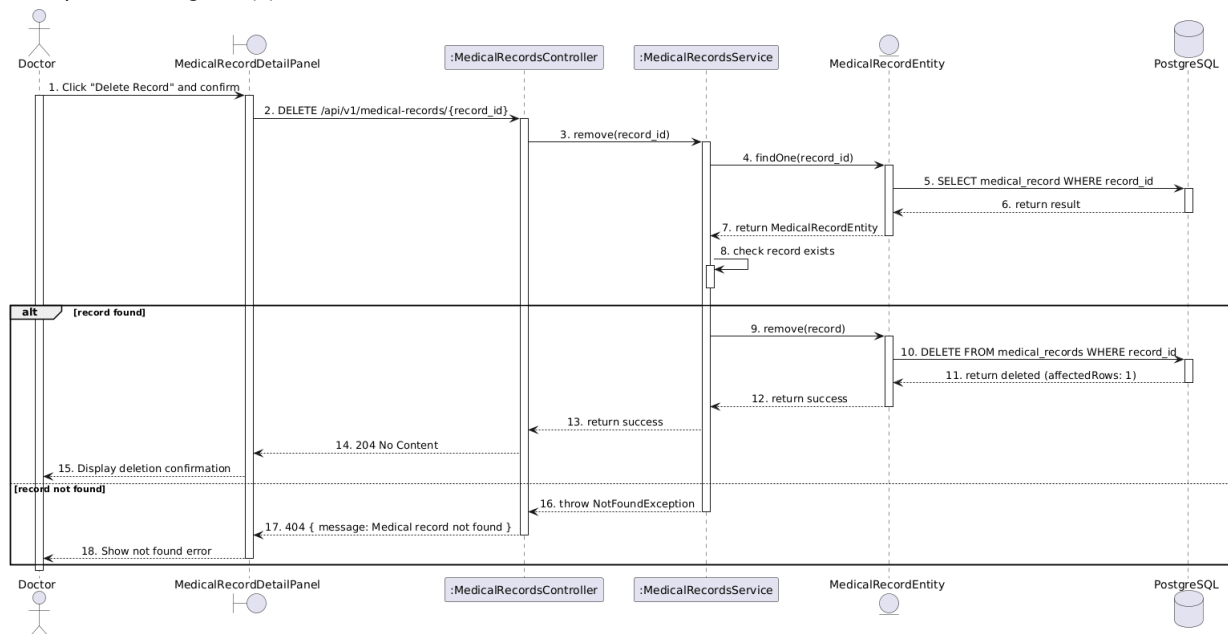


3.43 Delete Medical Record

a. Class Diagram

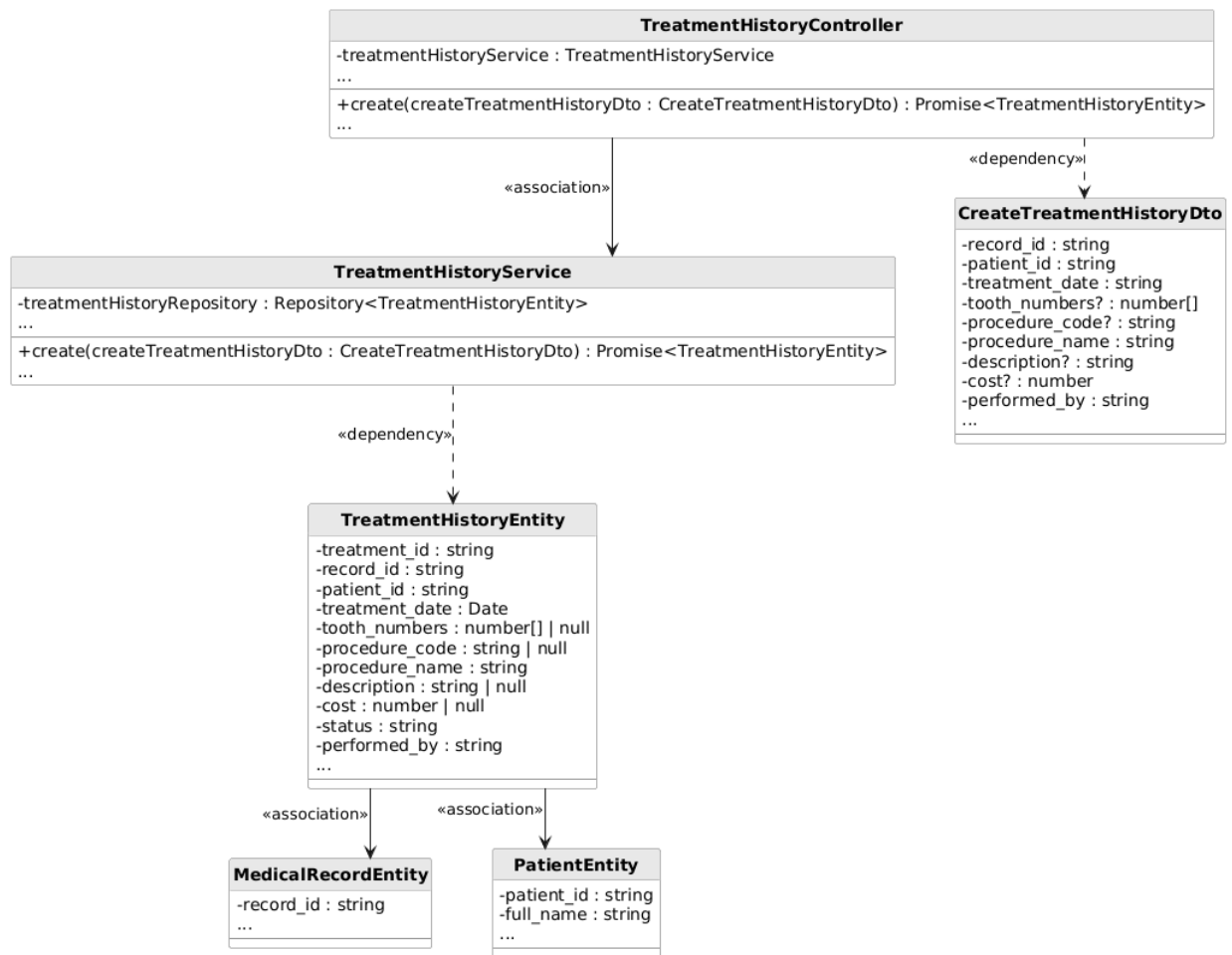


b. Sequence Diagram(s)

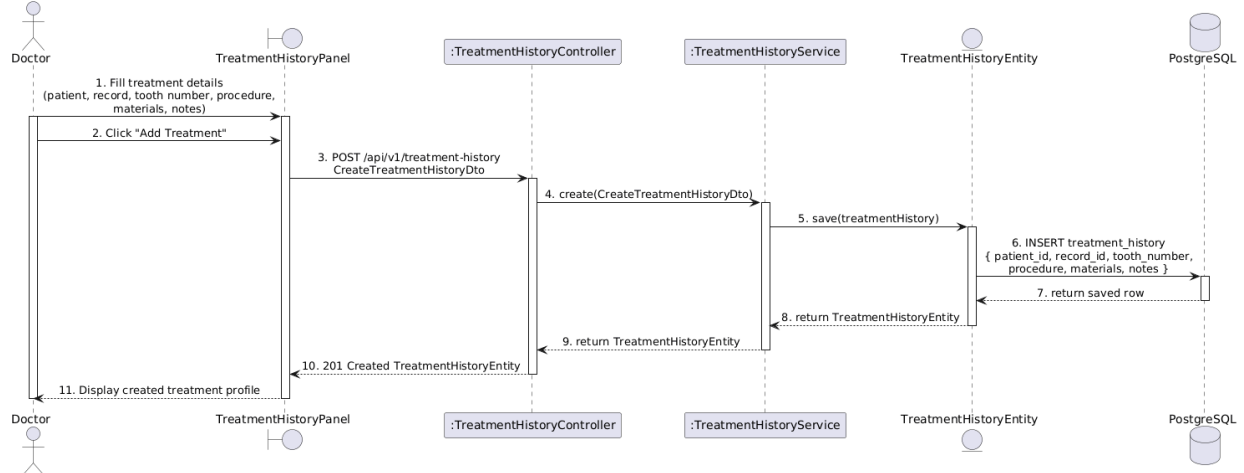


3.44 Add Treatment Profile

a. Class Diagram

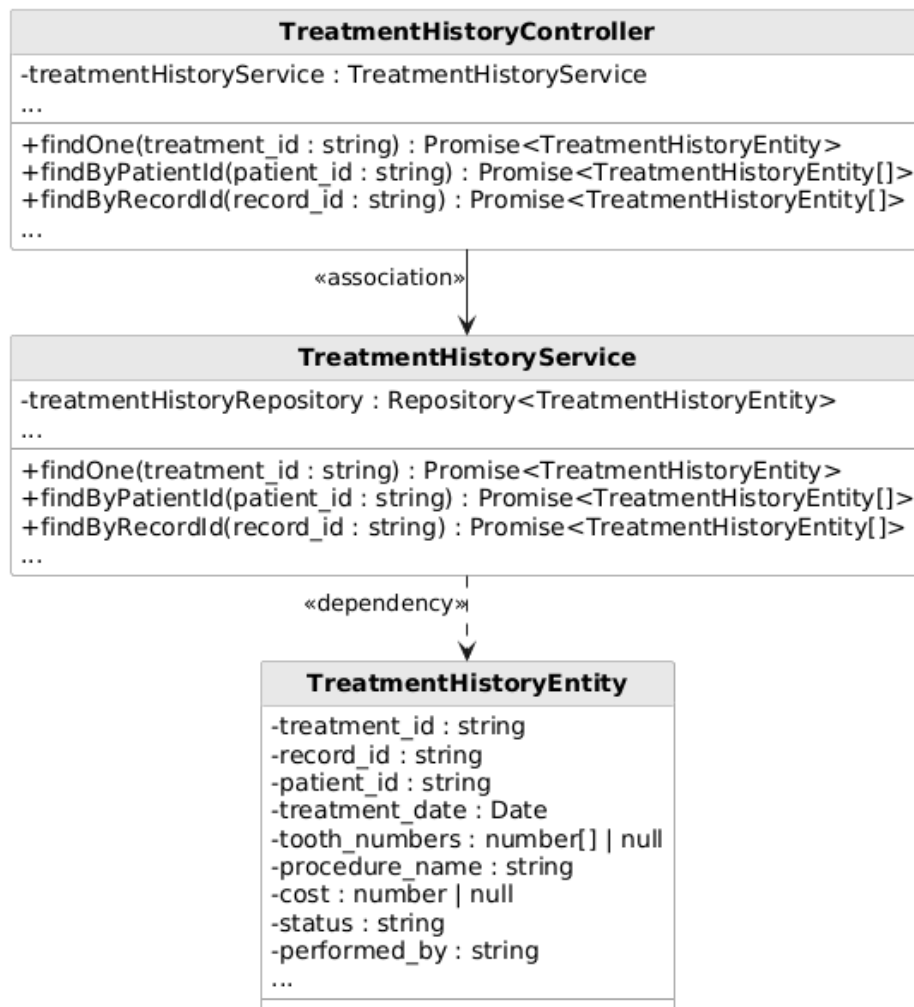


b. Sequence Diagram(s)

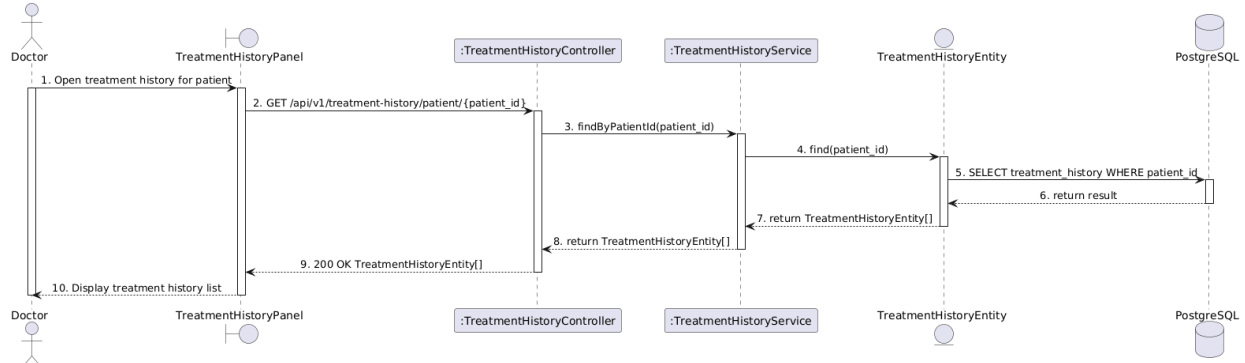


3.45 View Treatment Profile

a. Class Diagram

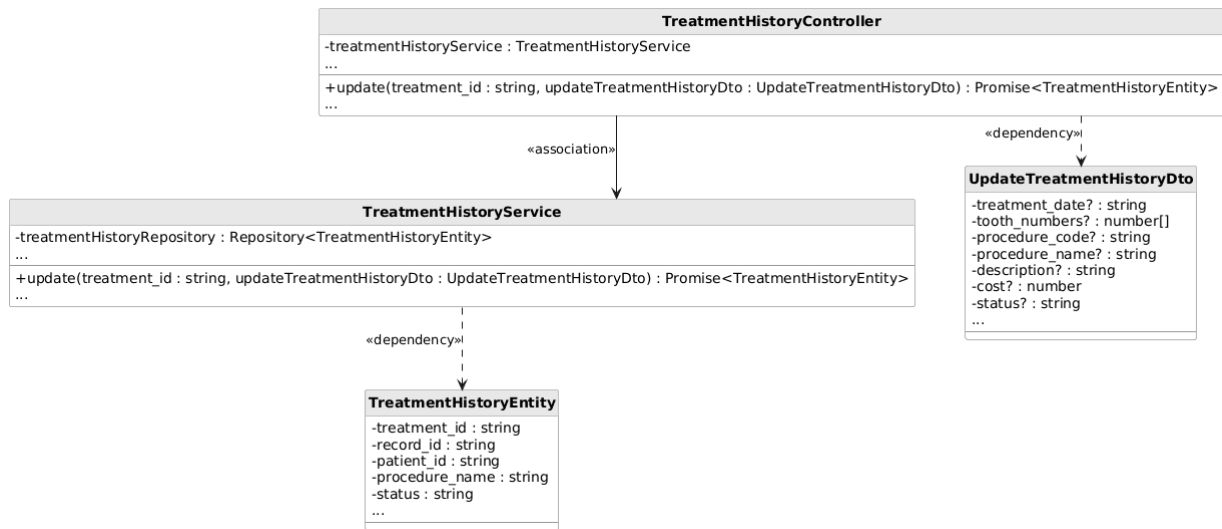


b. Sequence Diagram(s)

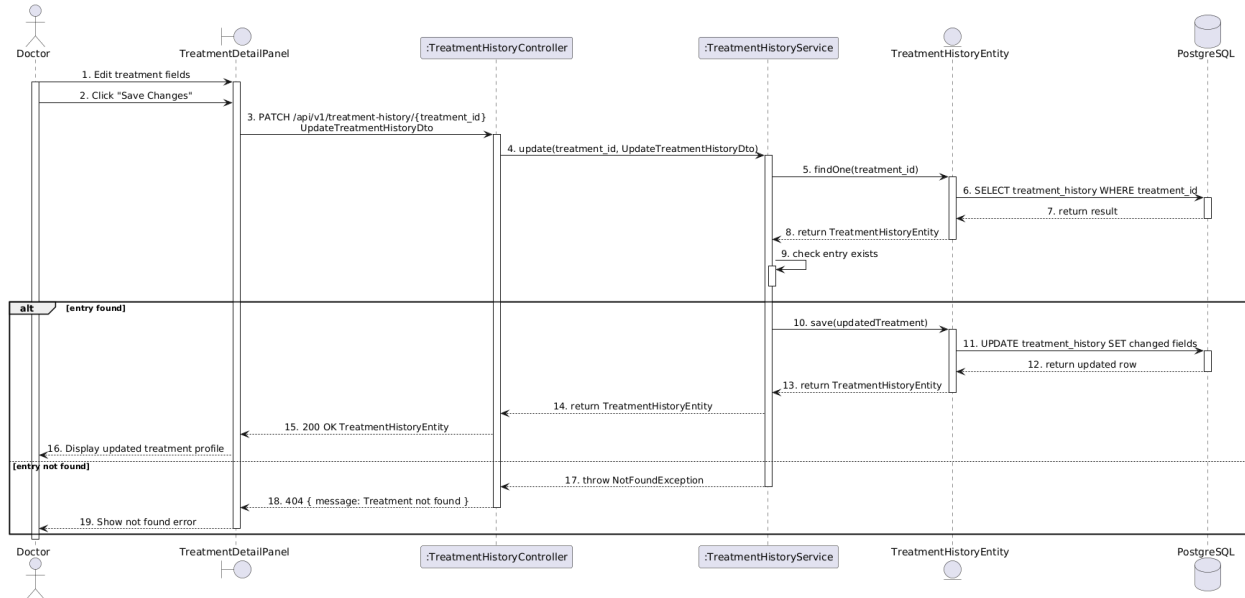


3.46 Update Treatment Profile

a. Class Diagram

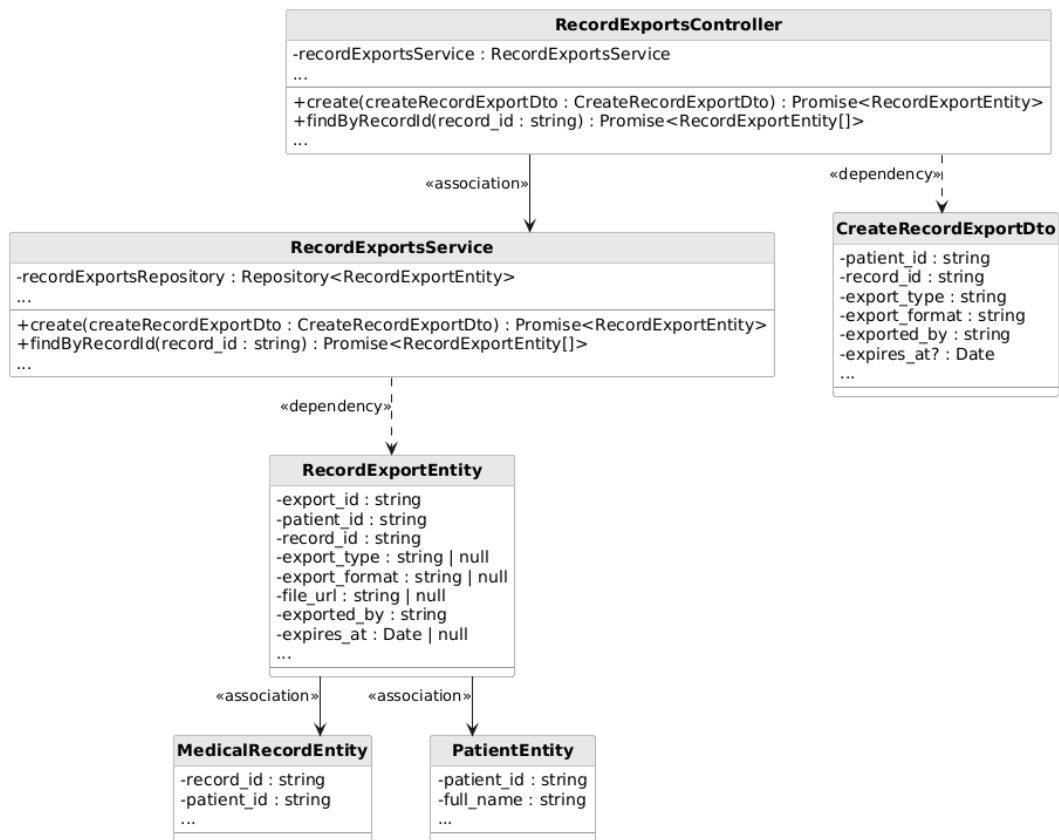


b. Sequence Diagram(s)

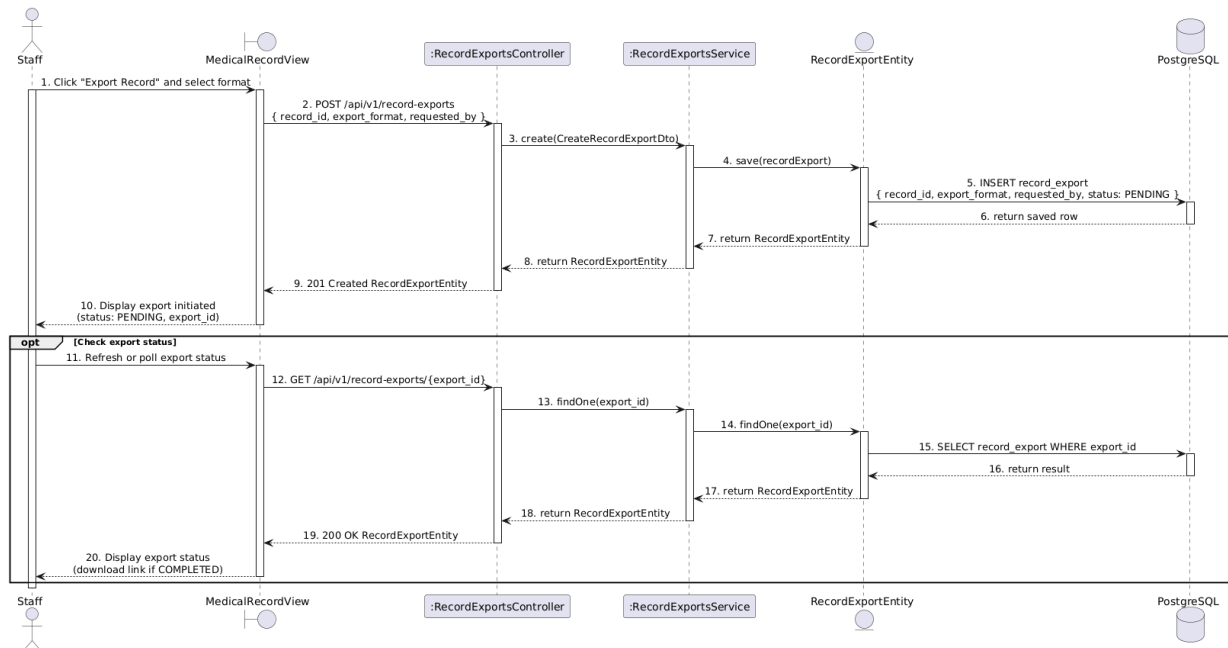


3.47 Export Medical Record

a. Class Diagram

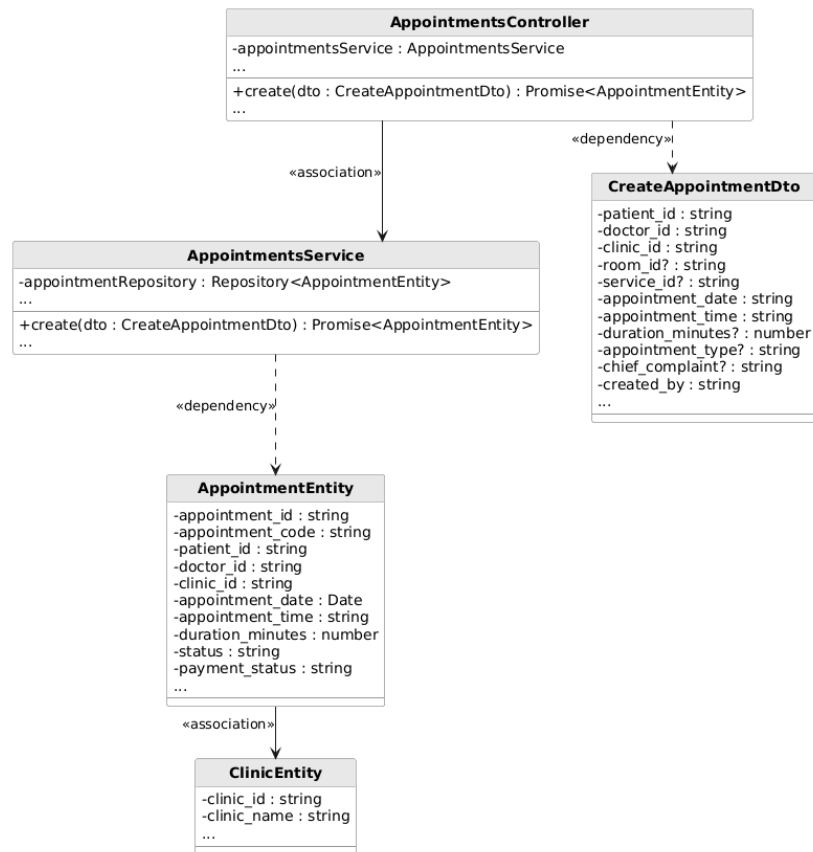


b. Sequence Diagram(s)

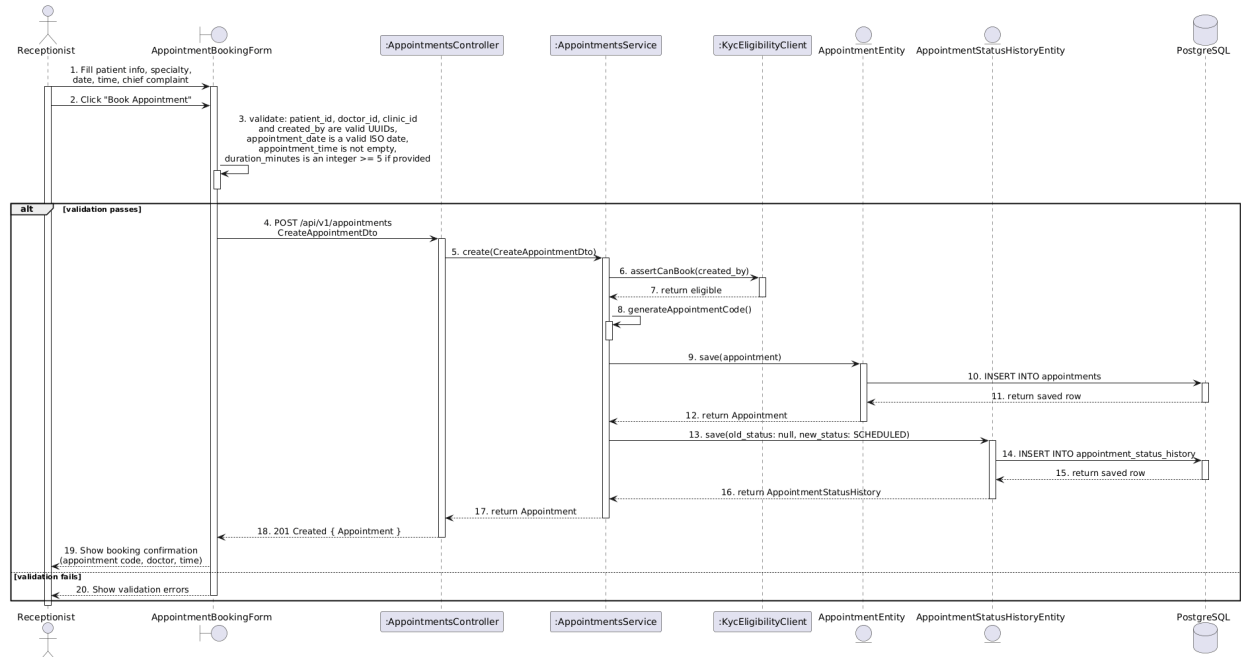


3.48 Create Appointment At Facility

a. Class Diagram

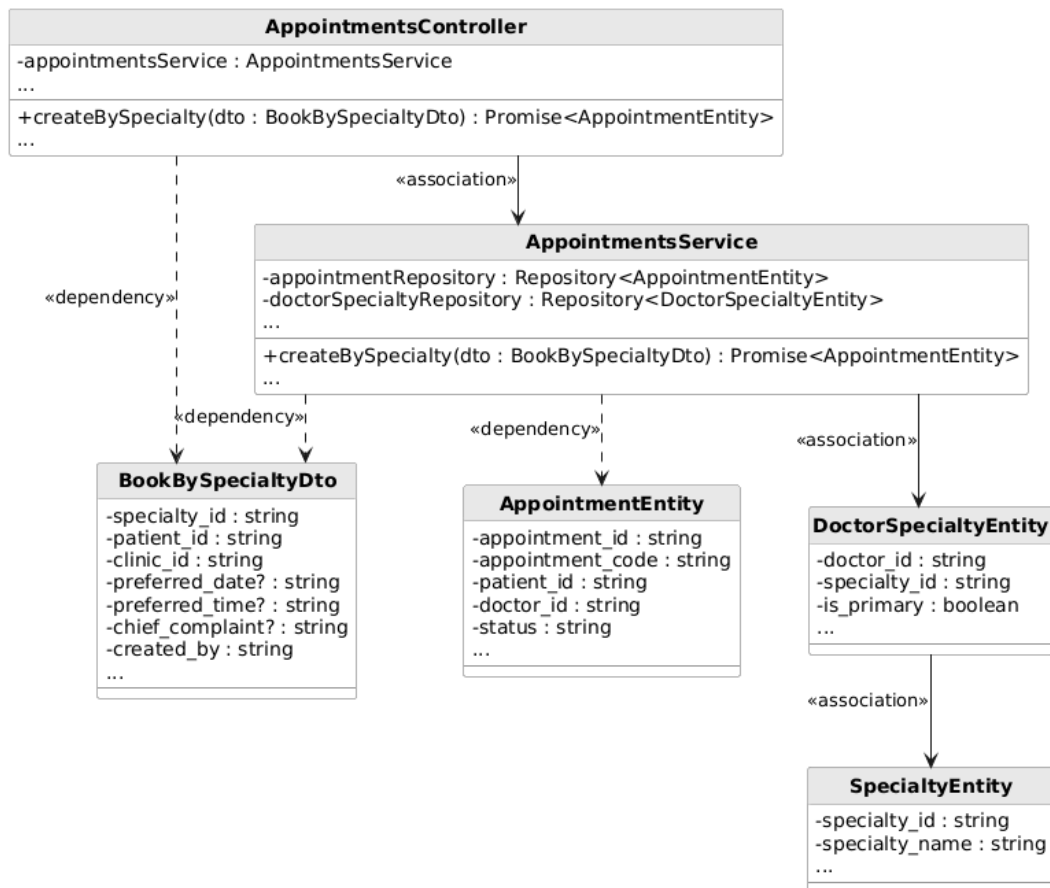


b. Sequence Diagram(s)

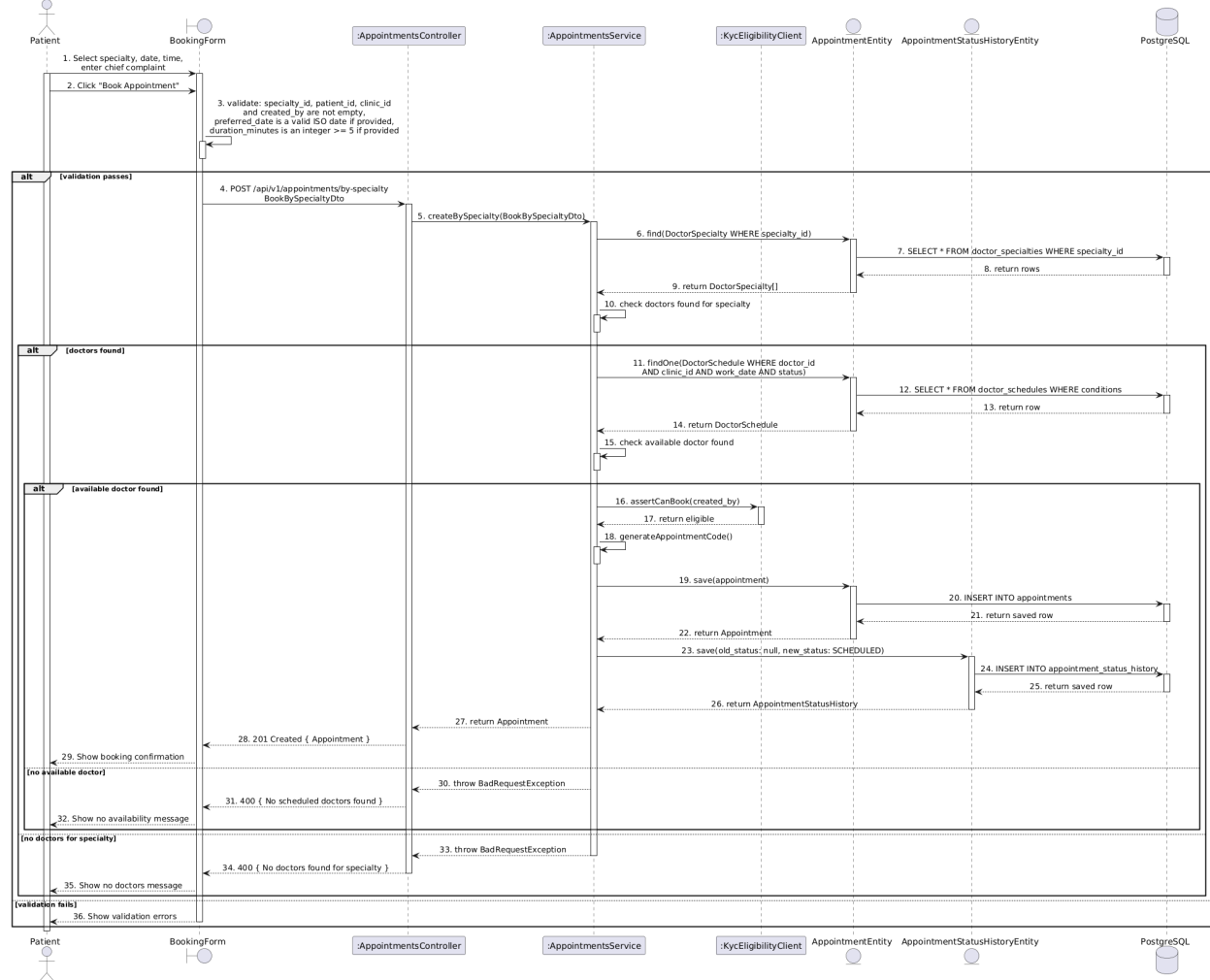


3.49 Create Appointment By Specialty

a. Class Diagram

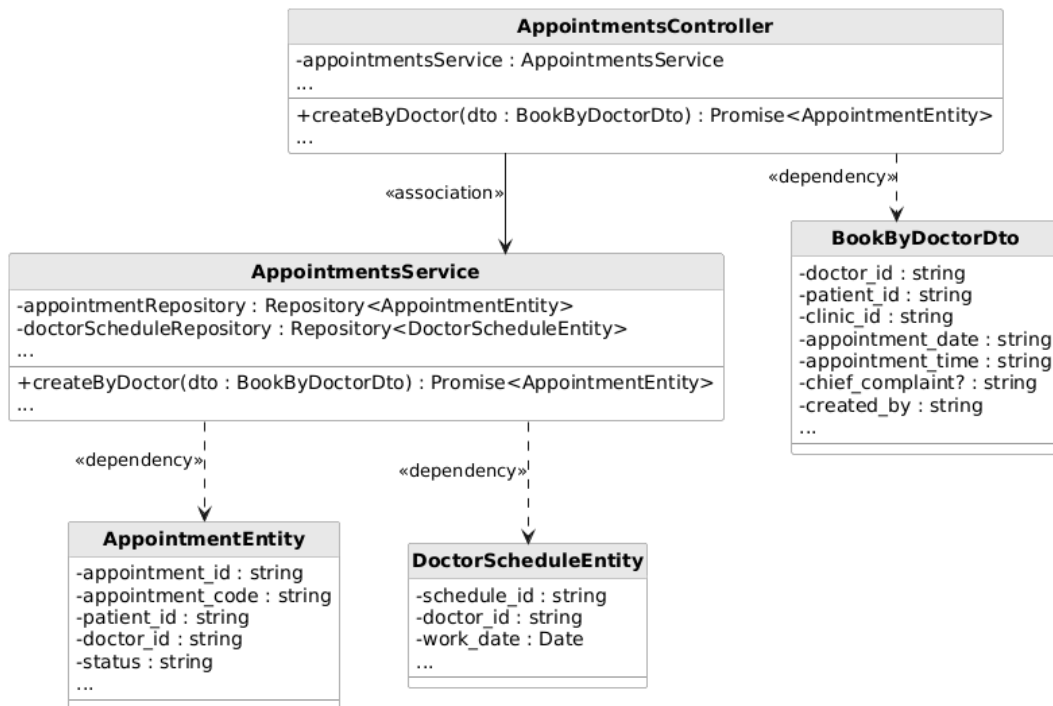


b. Sequence Diagram(s)

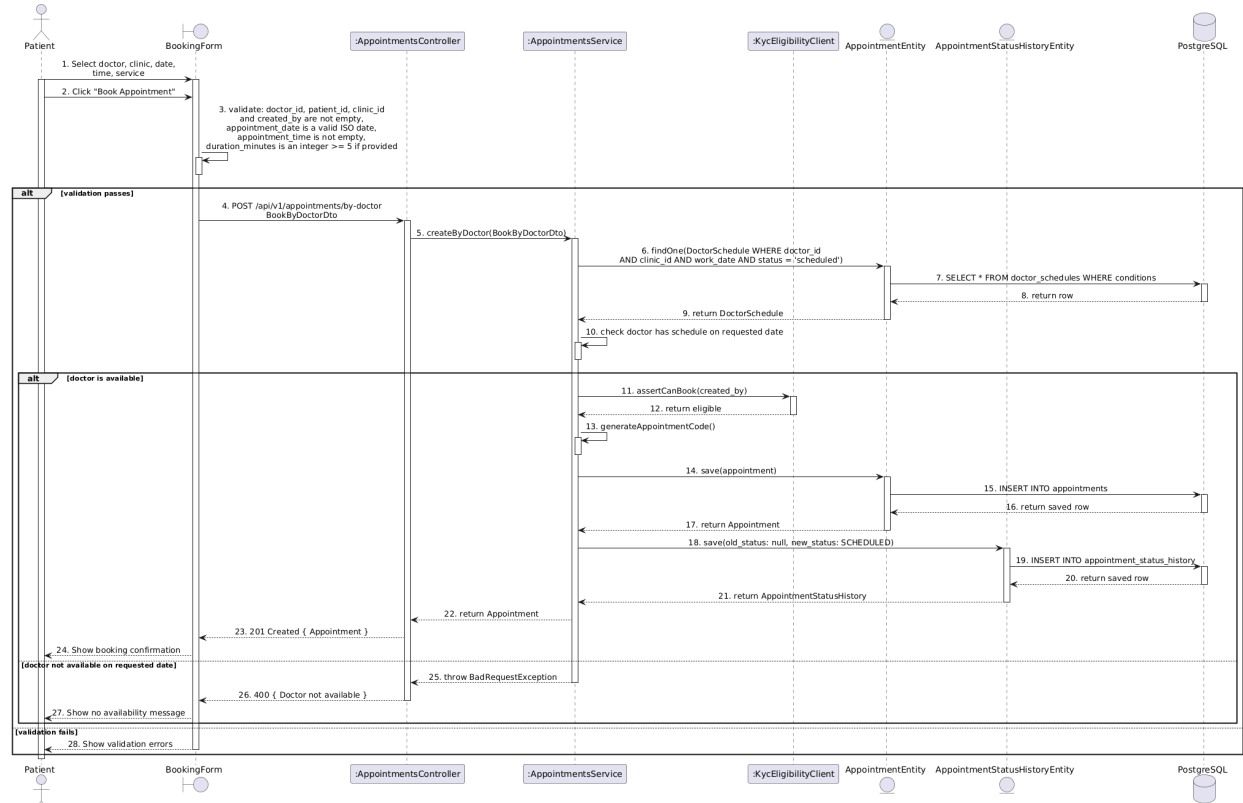


3.50 Create Appointment By Doctor

a. Class Diagram

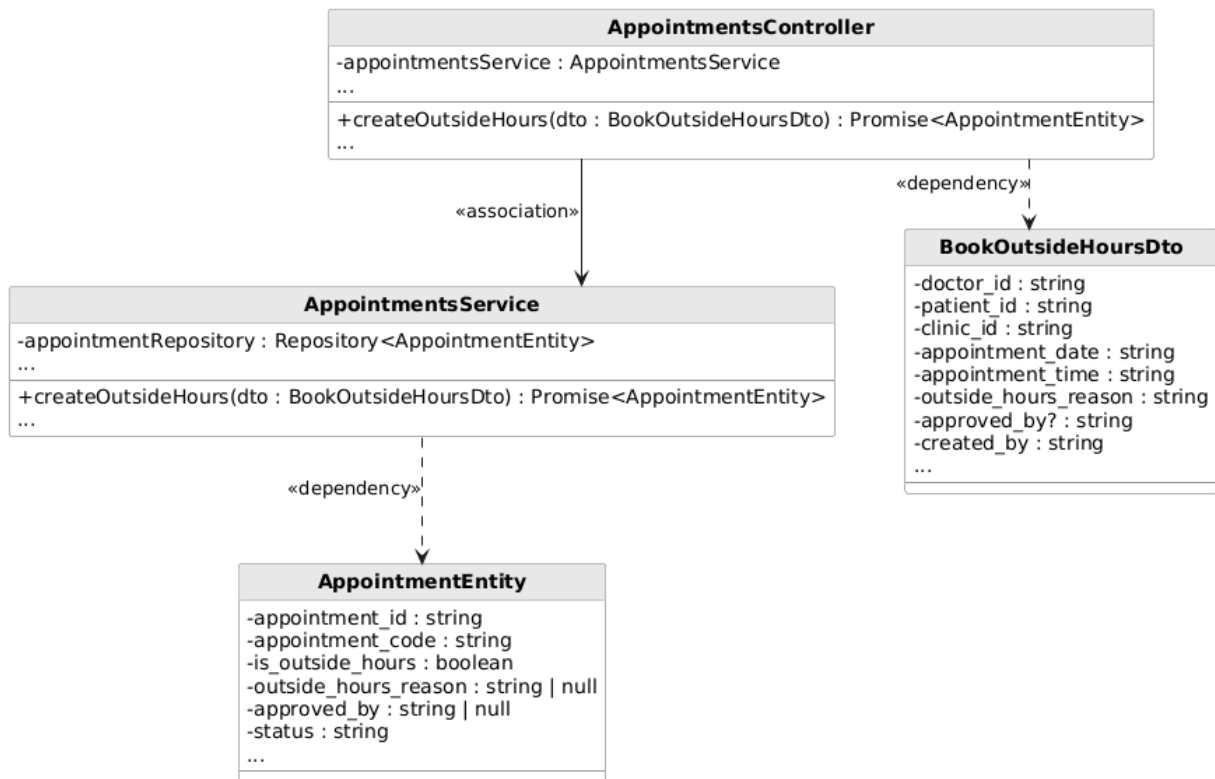


b. Sequence Diagram(s)

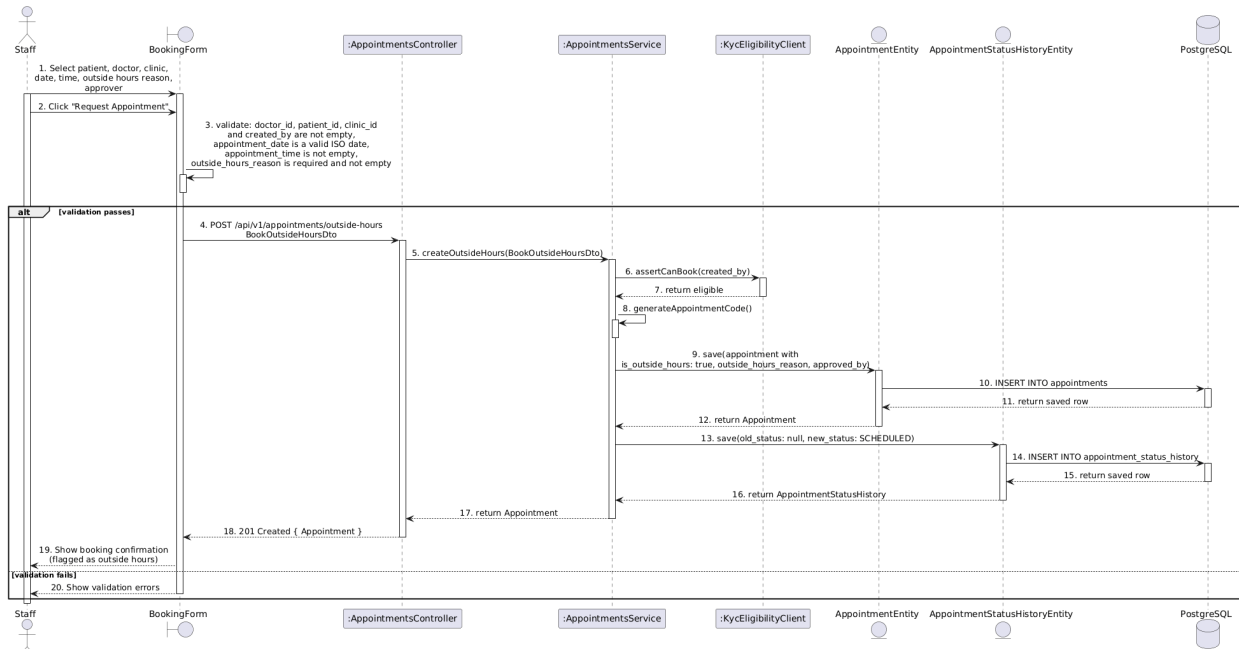


3.51 Create Appointment Outside Working Hours

a. Class Diagram

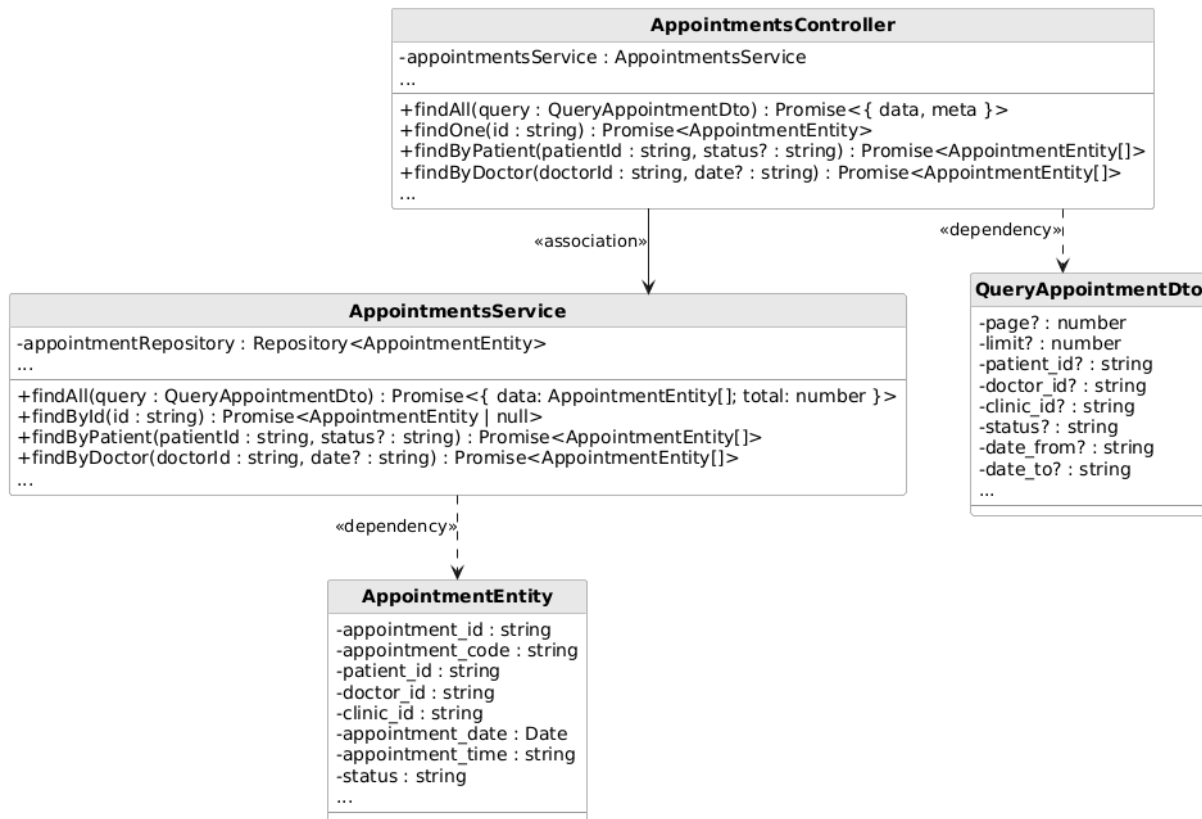


b. Sequence Diagram(s)

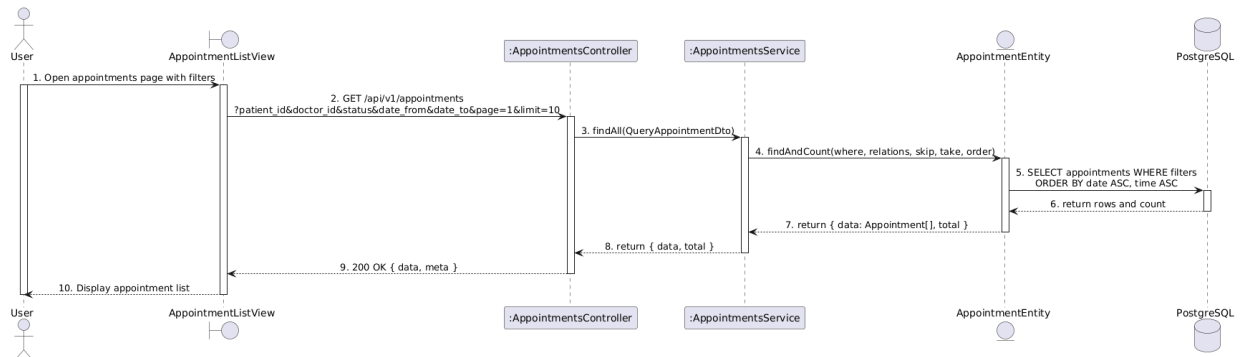


3.52 View Appointment

a. Class Diagram

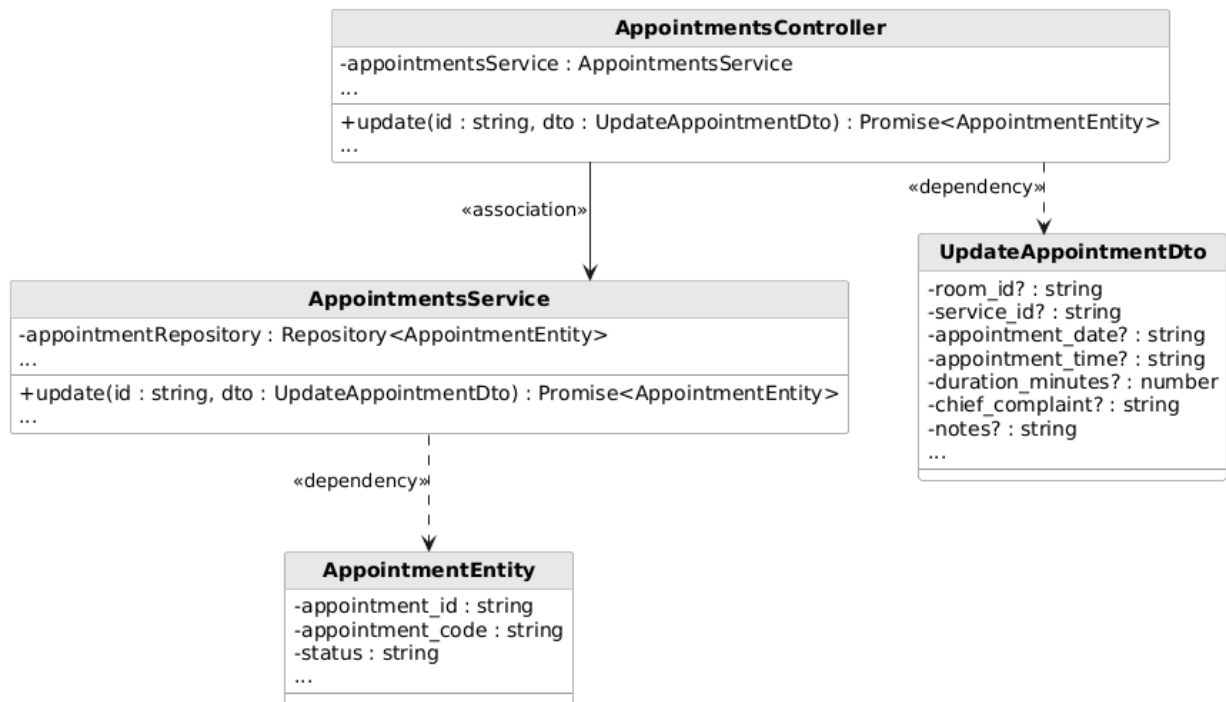


b. Sequence Diagram(s)

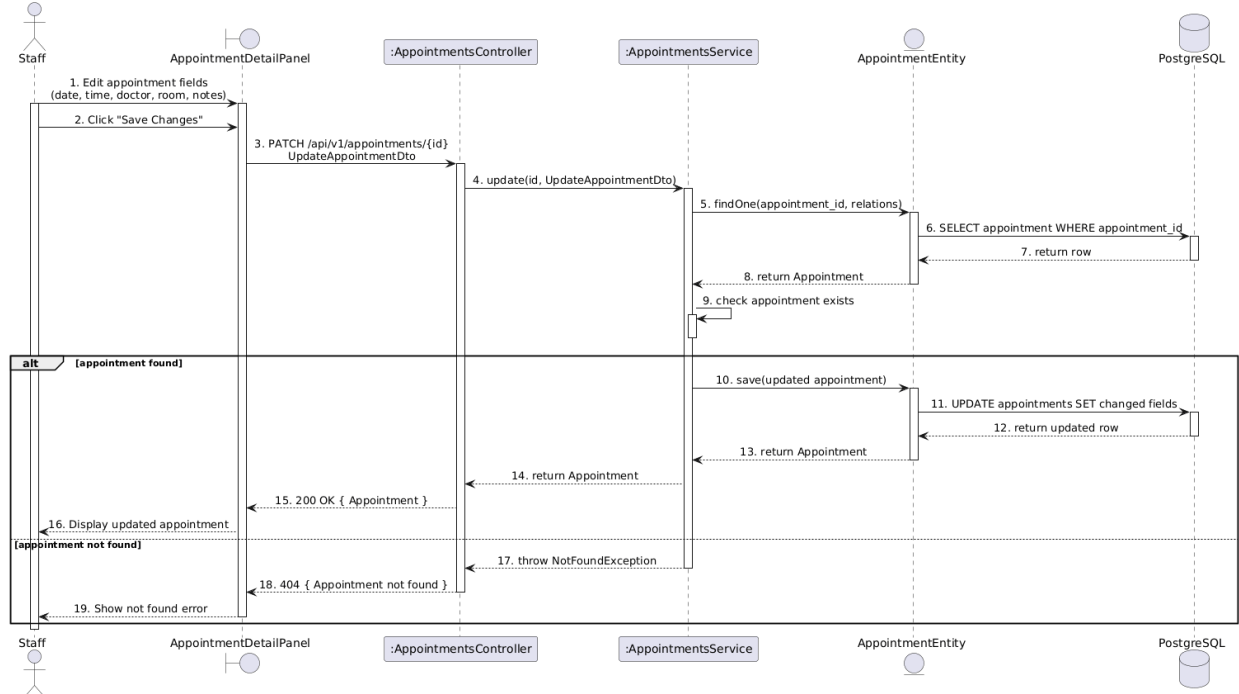


3.53 Edit Appointment

a. Class Diagram

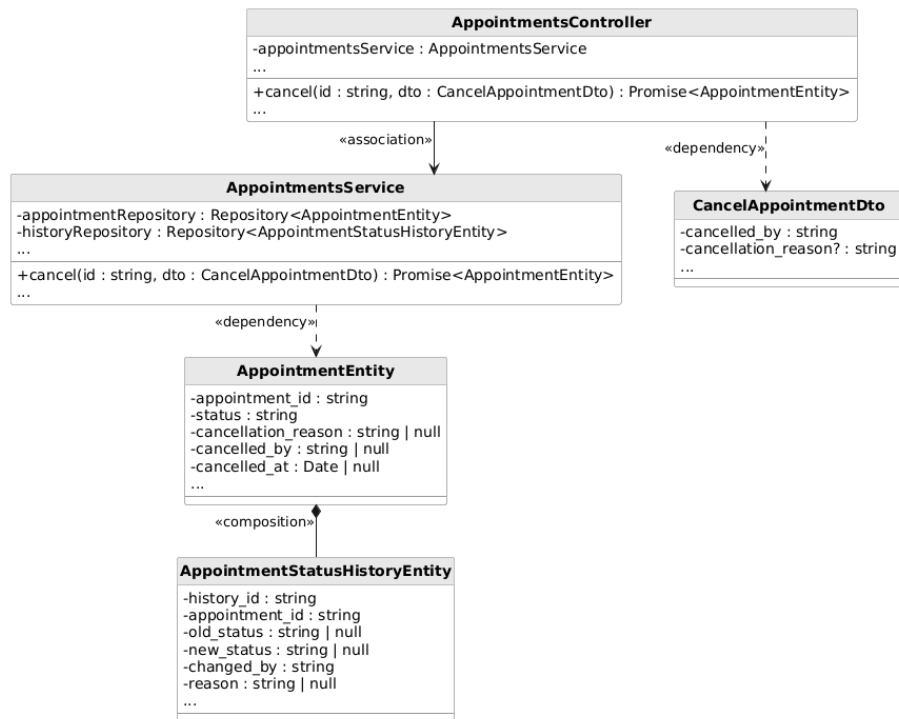


b. Sequence Diagram(s)

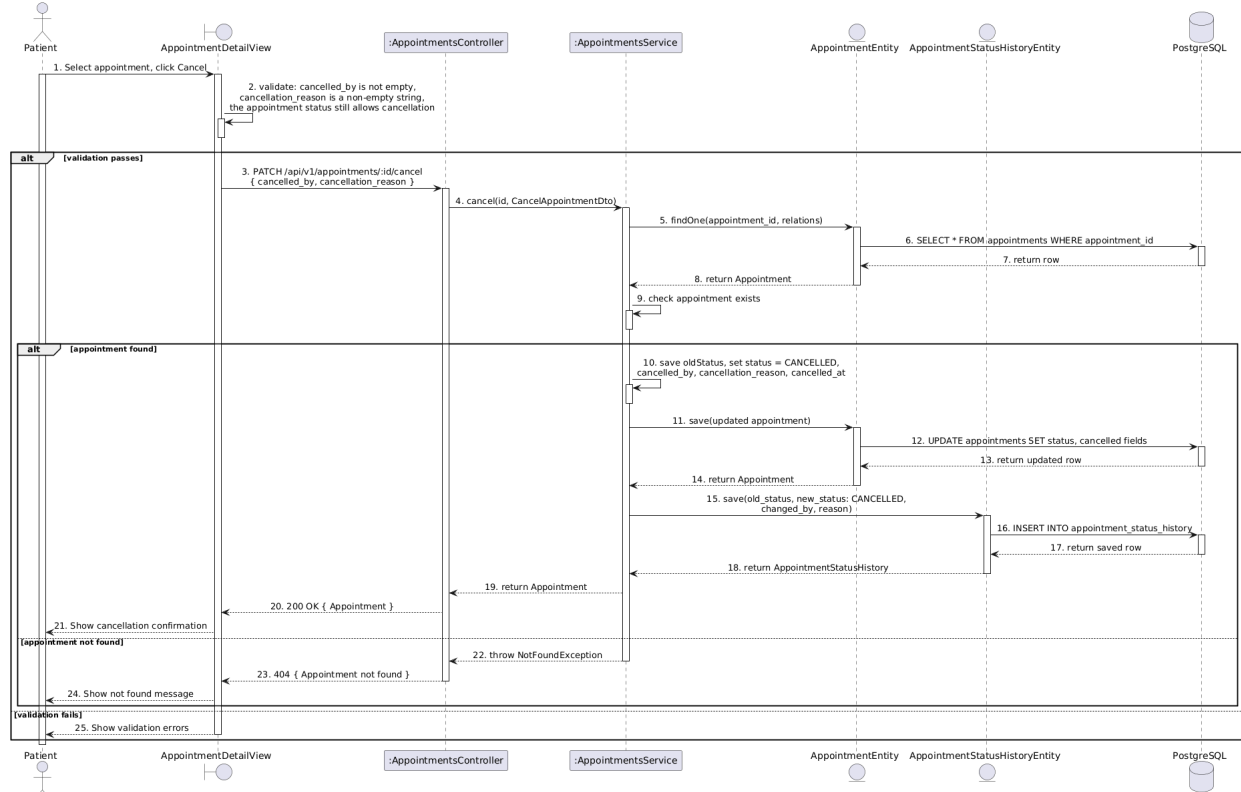


3.54 Cancel Appointment

a. Class Diagram

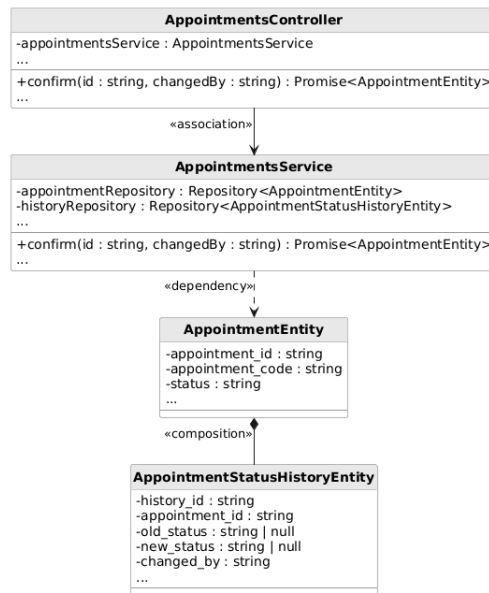


b. Sequence Diagram(s)

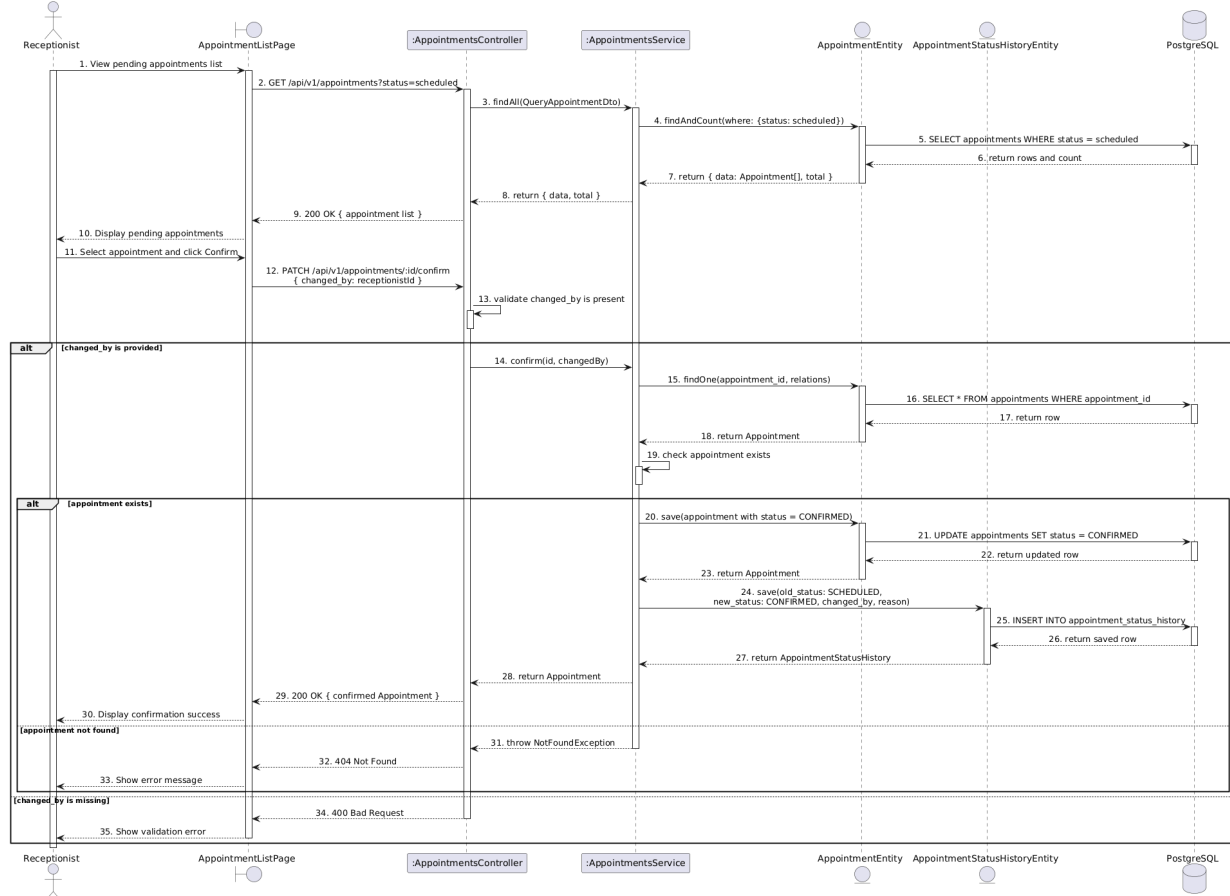


3.55 Confirm Appointment

a. Class Diagram

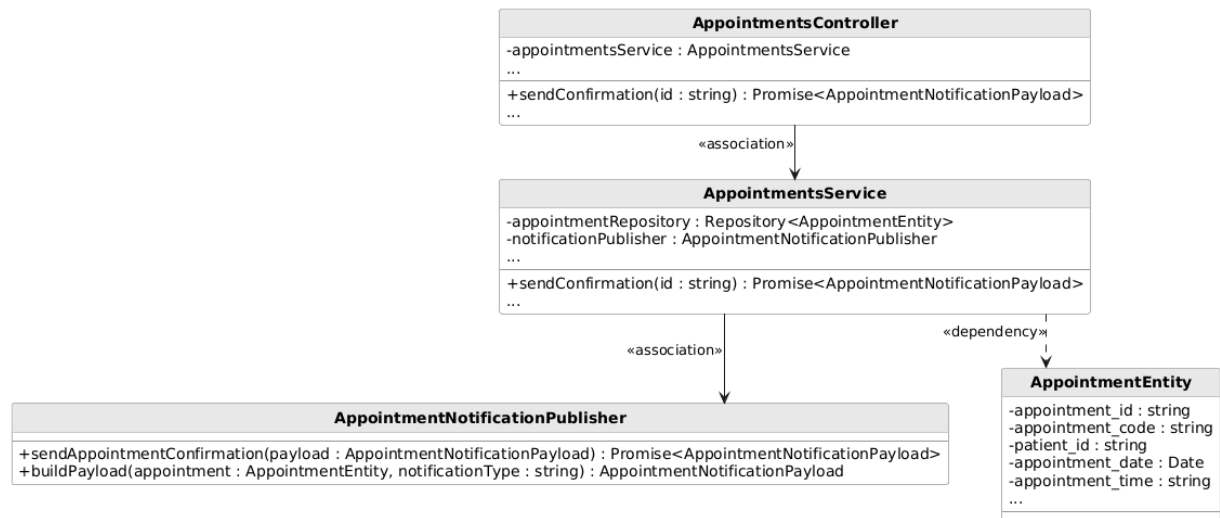


b. Sequence Diagram(s)

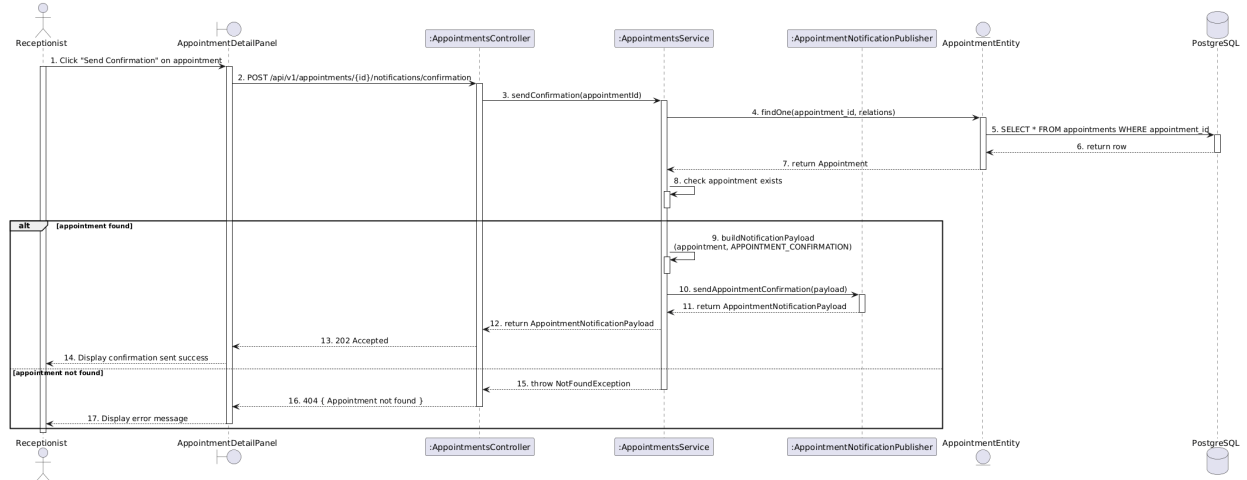


3.56 Send Appointment Confirmation

a. Class Diagram

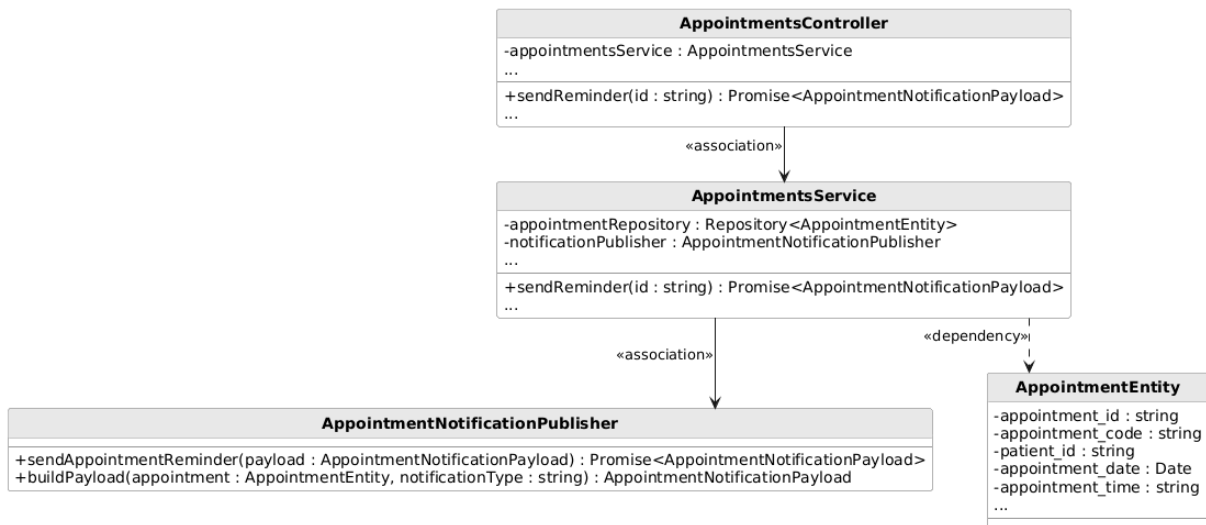


b. Sequence Diagram(s)

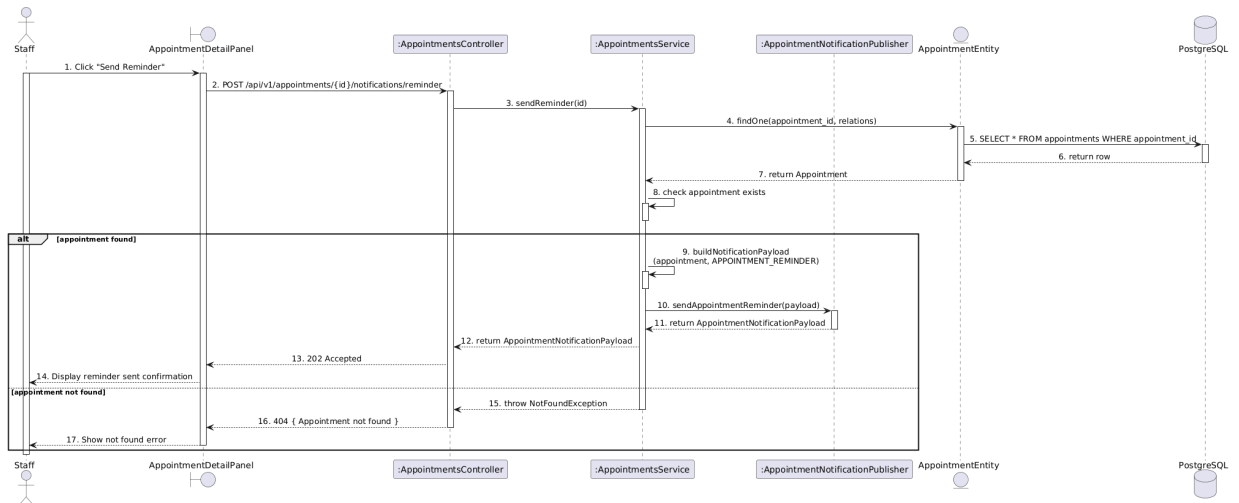


3.57 Send Appointment Reminder

a. Class Diagram

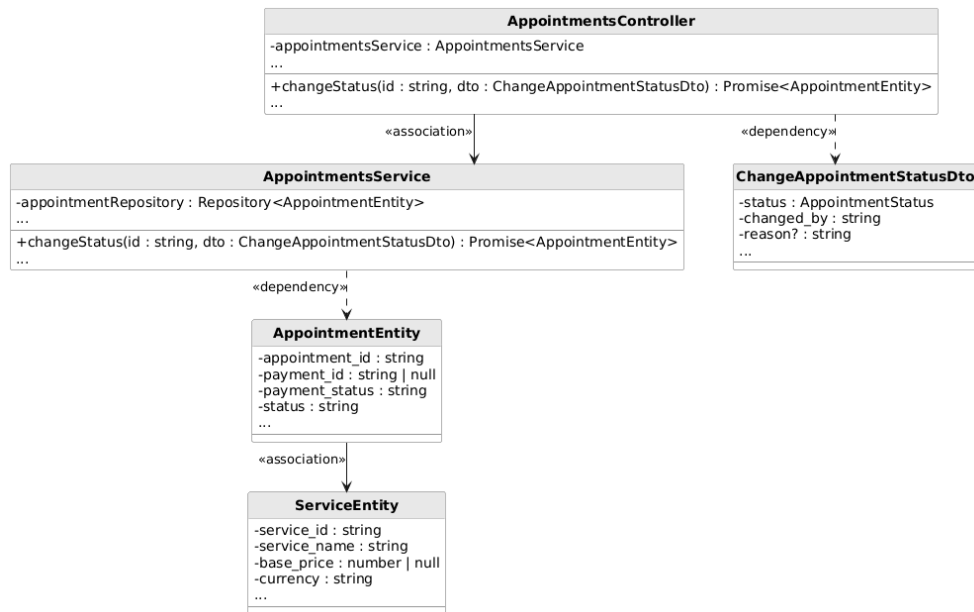


b. Sequence Diagram(s)

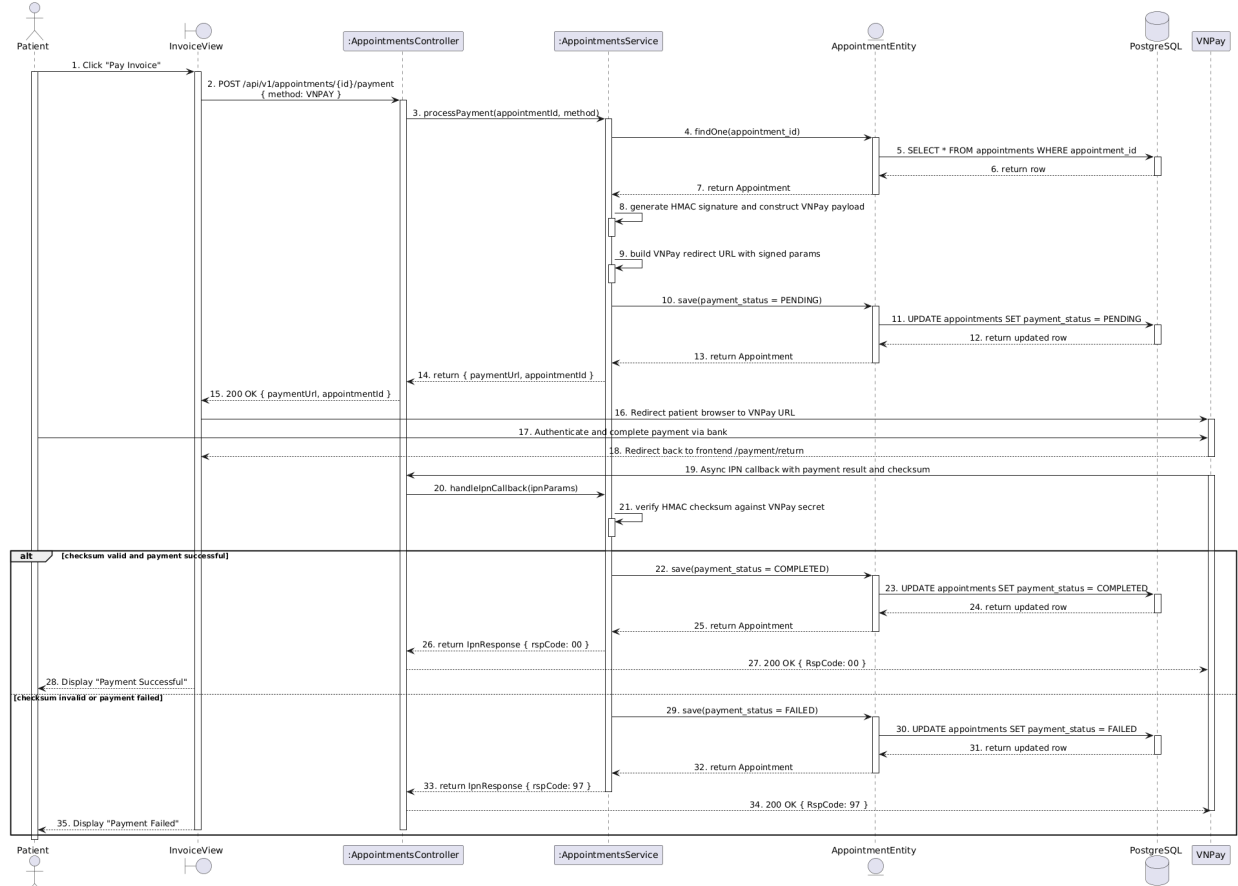


3.58 Initiate Payment

a. Class Diagram

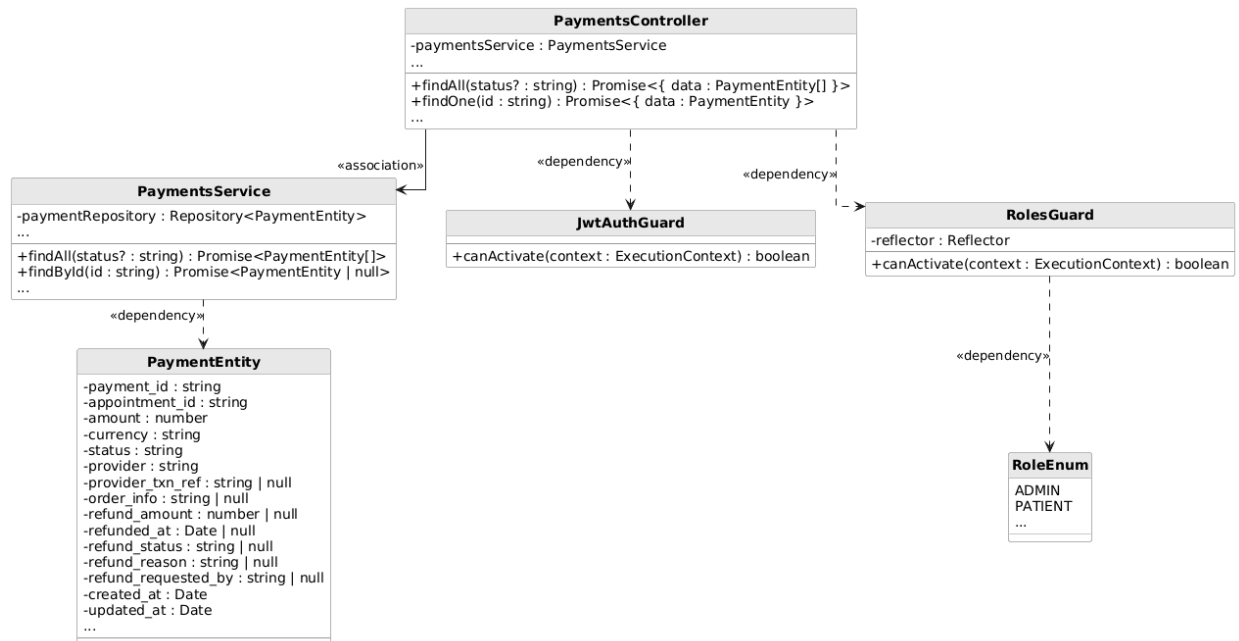


b. Sequence Diagram(s)

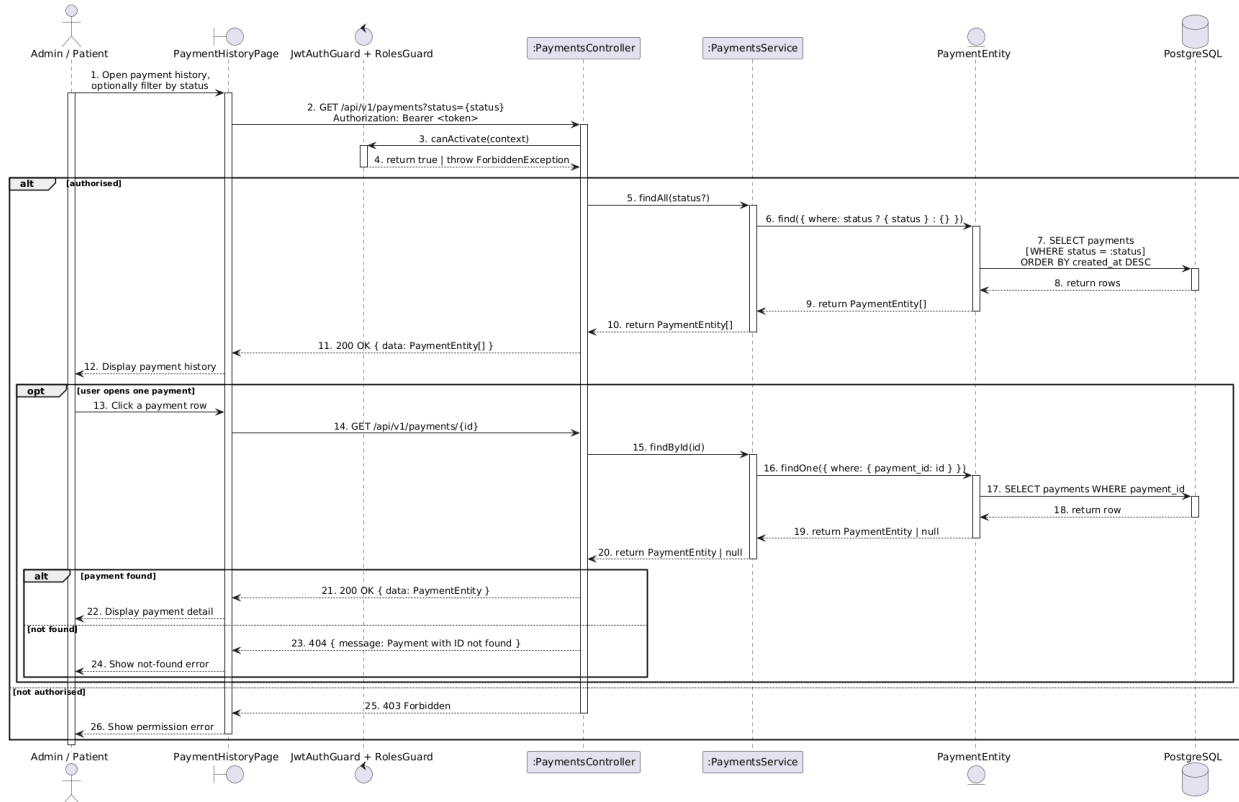


3.59 View Payment History

a. Class Diagram

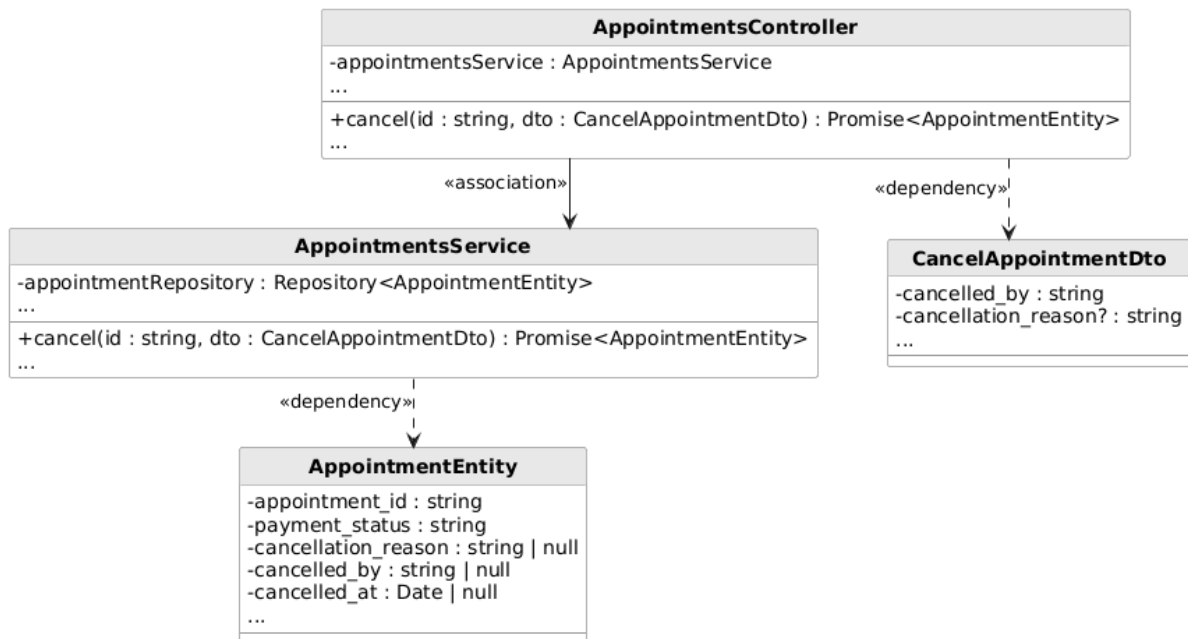


b. Sequence Diagram(s)

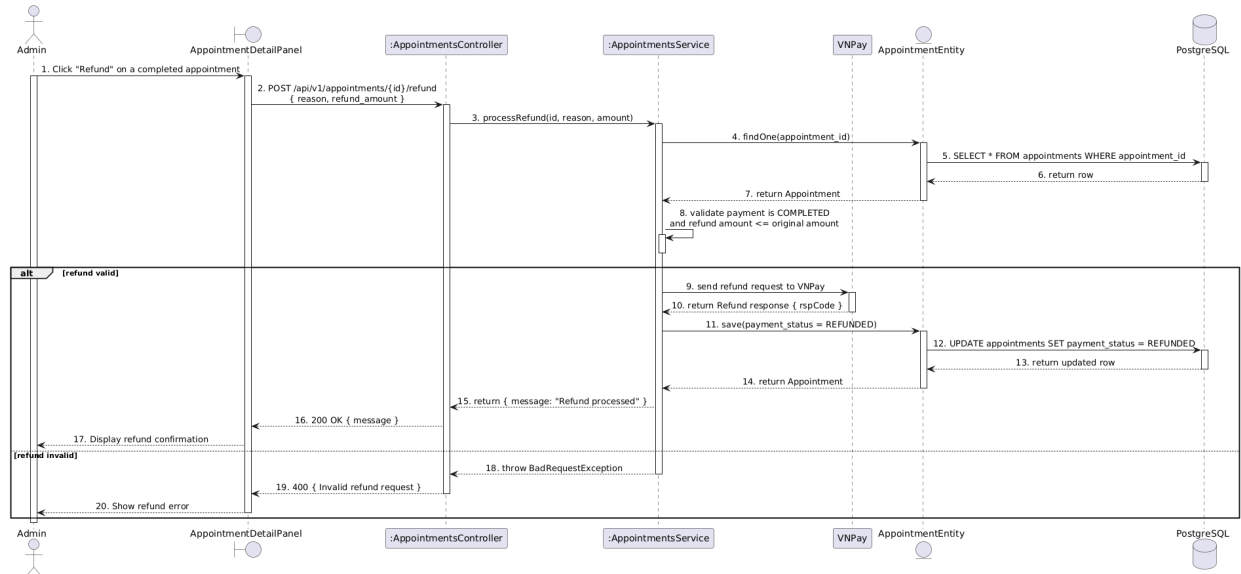


3.60 Refund Cancel Payment

a. Class Diagram

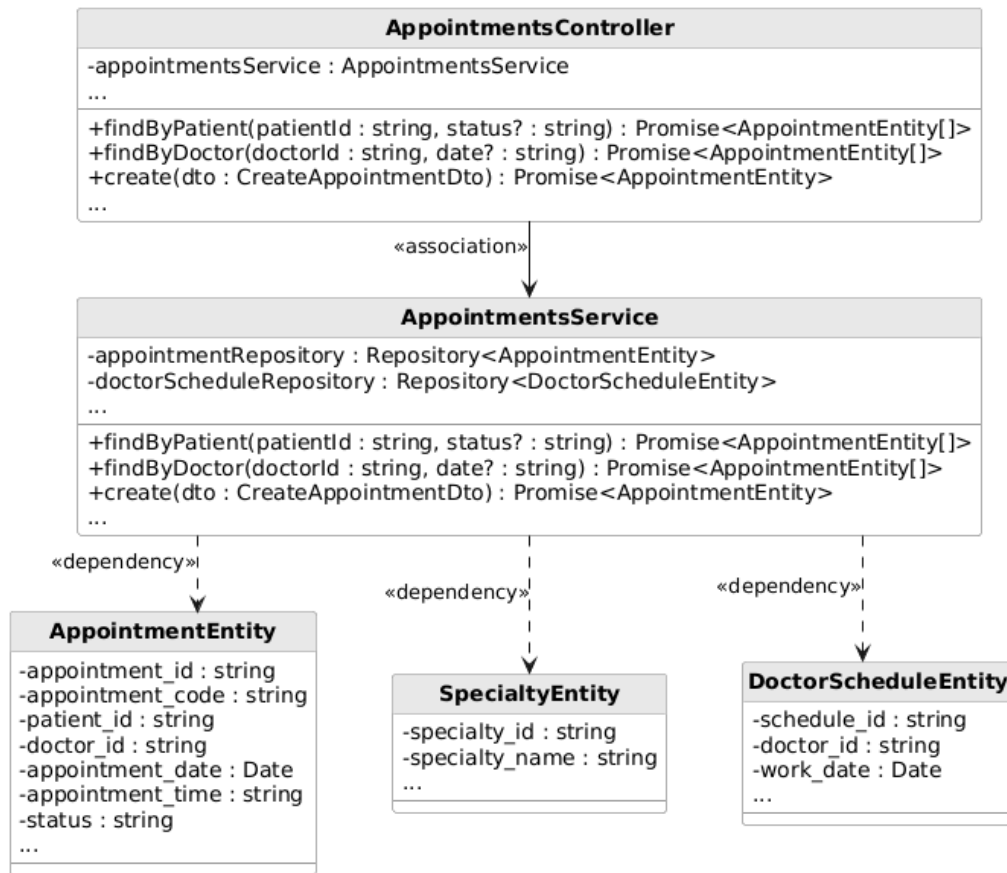


b. Sequence Diagram(s)

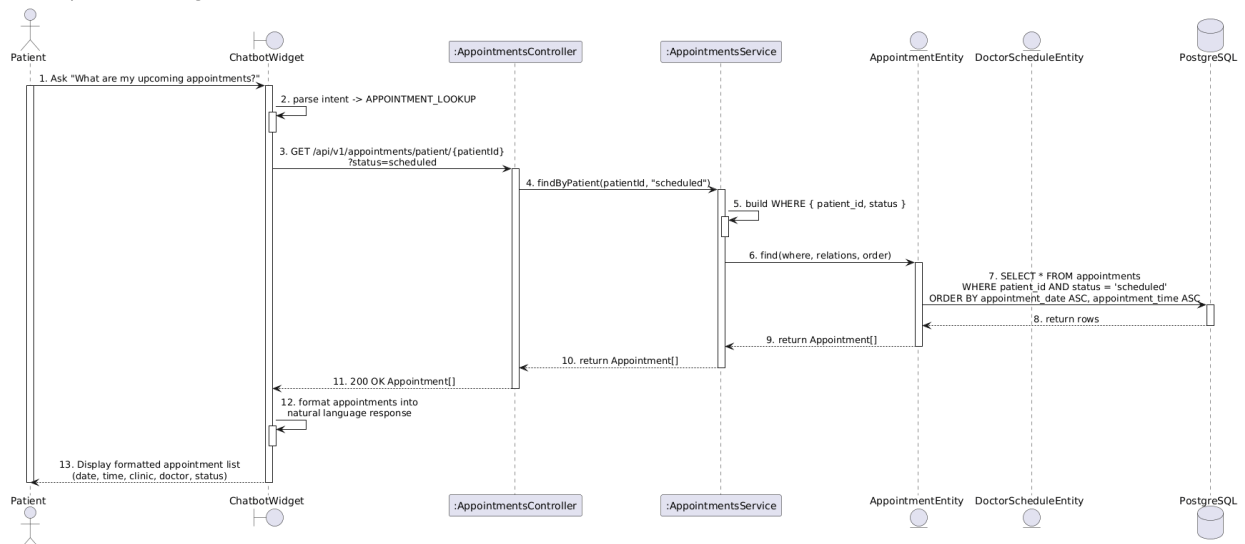


3.61 Chatbot Support For Booking

a. Class Diagram

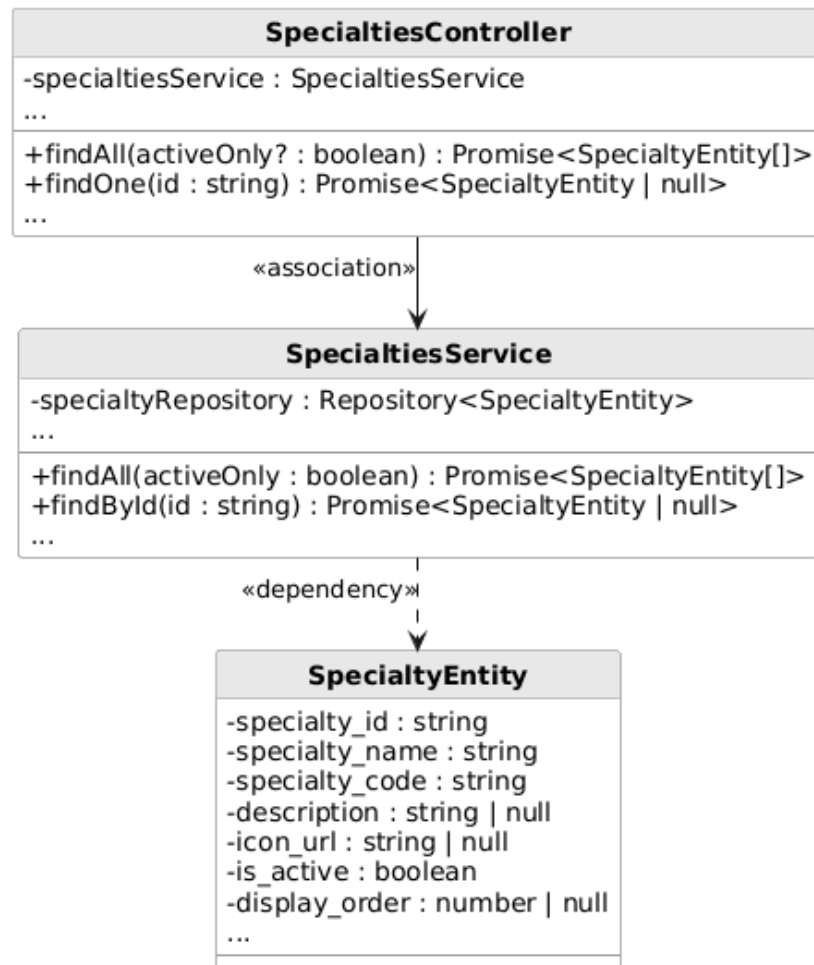


b. Sequence Diagram(s)

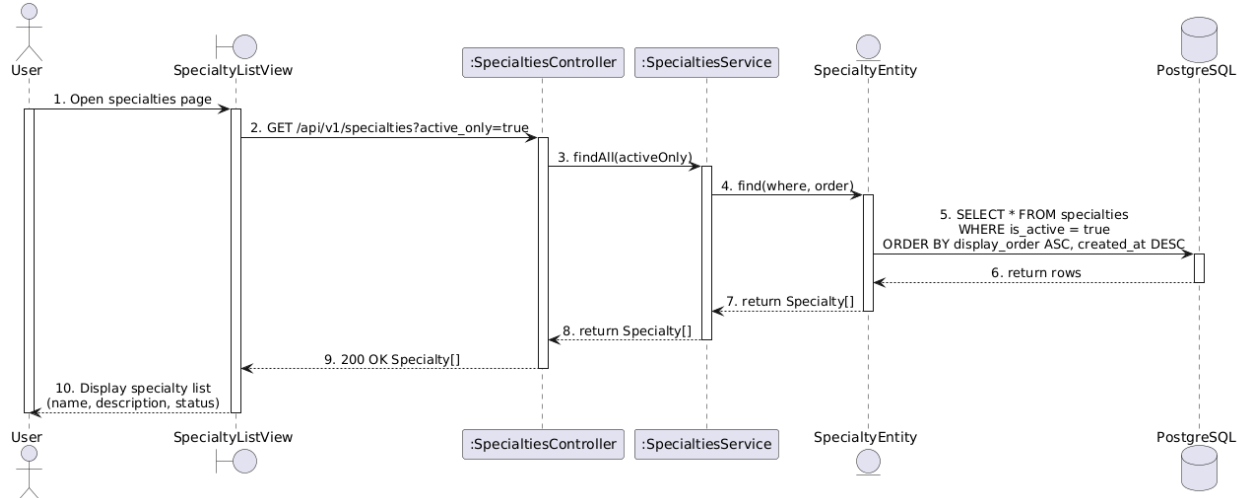


3.62 View Specialty

a. Class Diagram

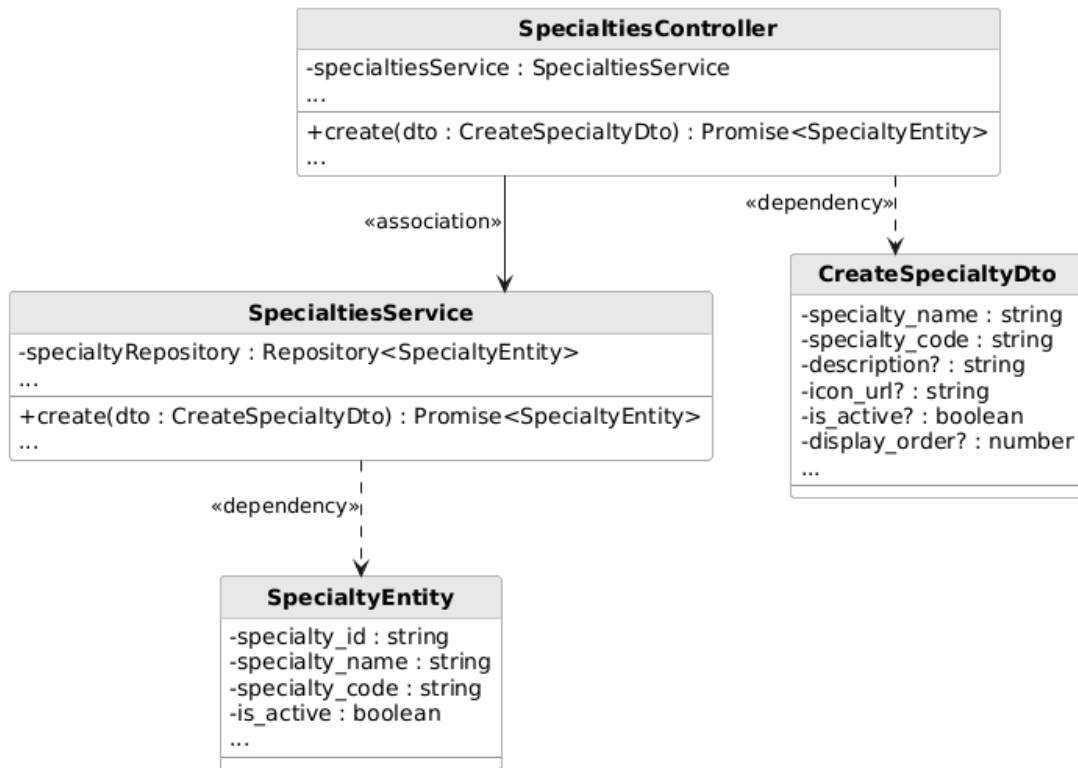


b. Sequence Diagram(s)

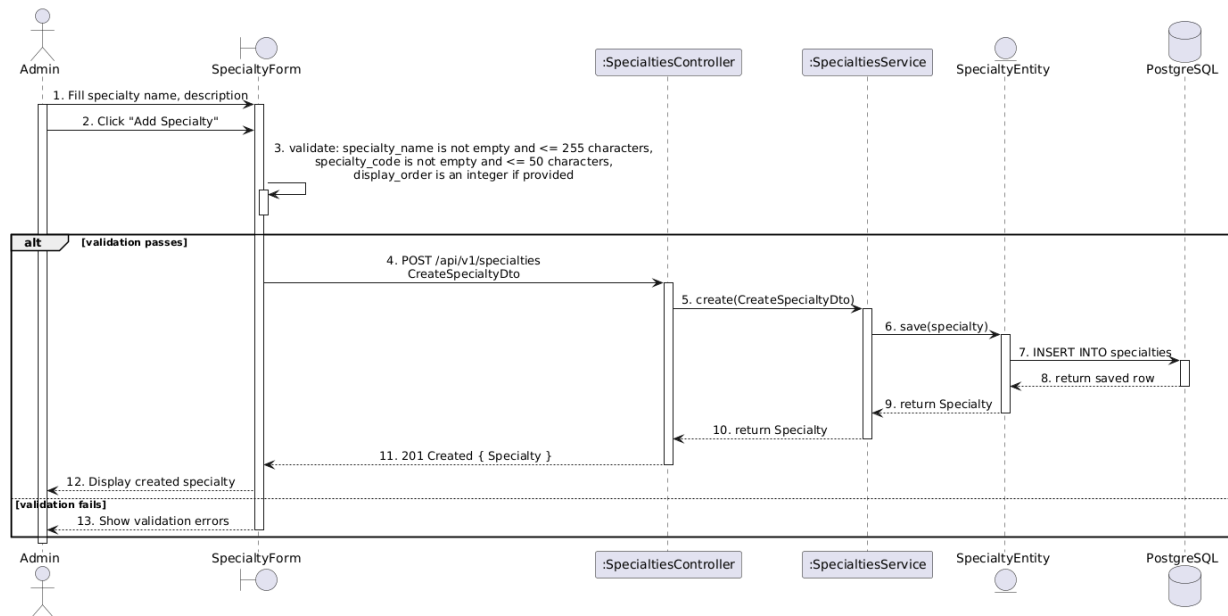


3.63 Add Specialty

a. Class Diagram

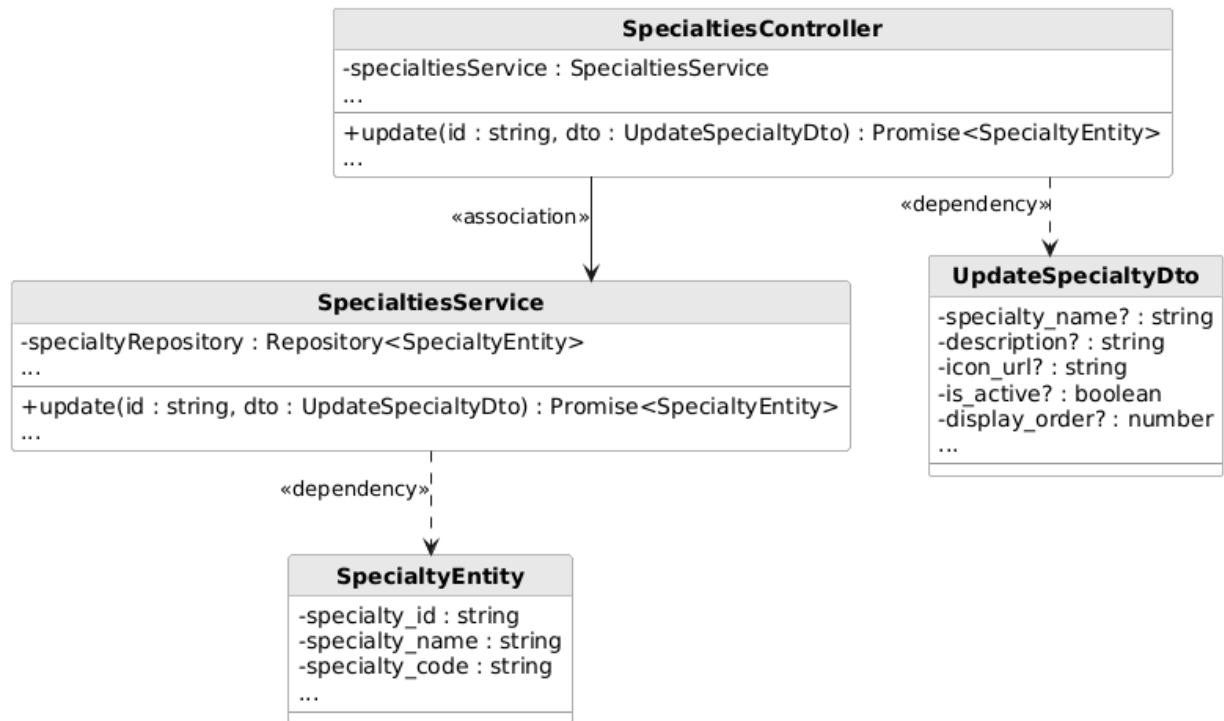


b. Sequence Diagram(s)

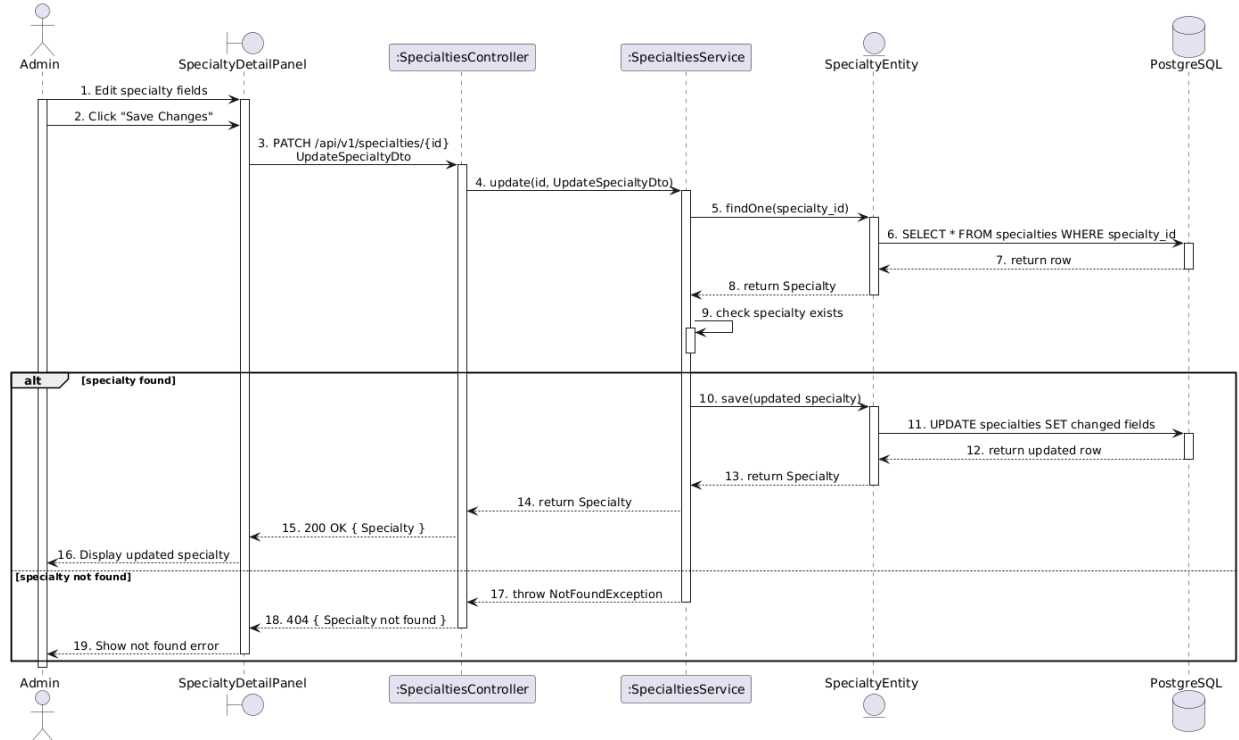


3.64 Update Specialty

a. Class Diagram

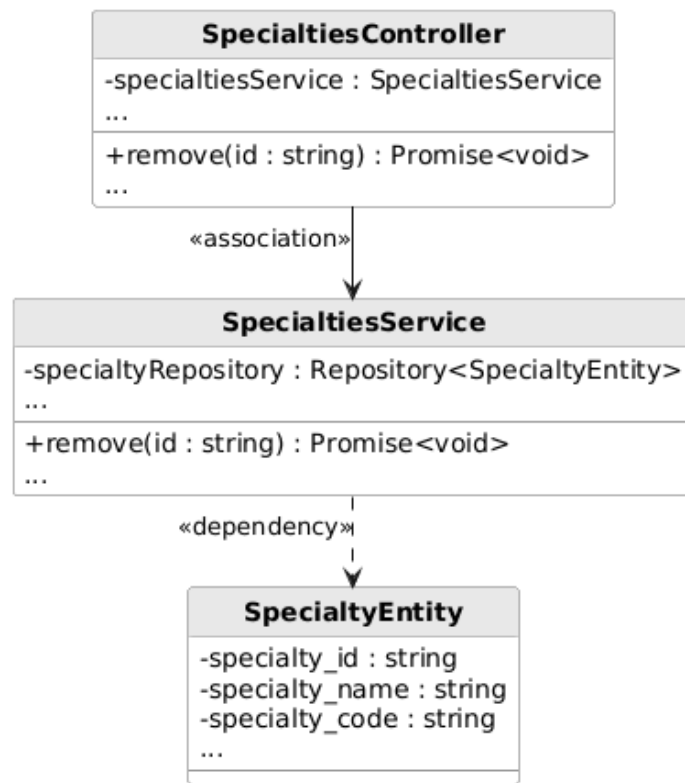


b. Sequence Diagram(s)

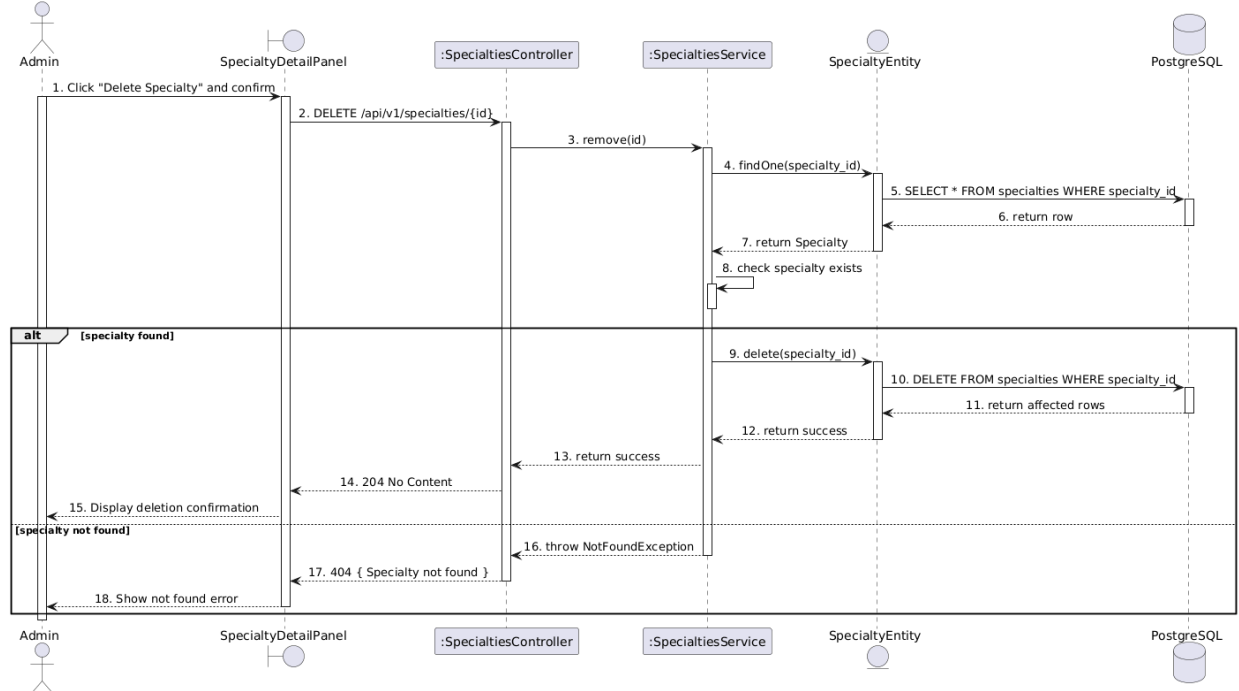


3.65 Delete Specialty

a. Class Diagram

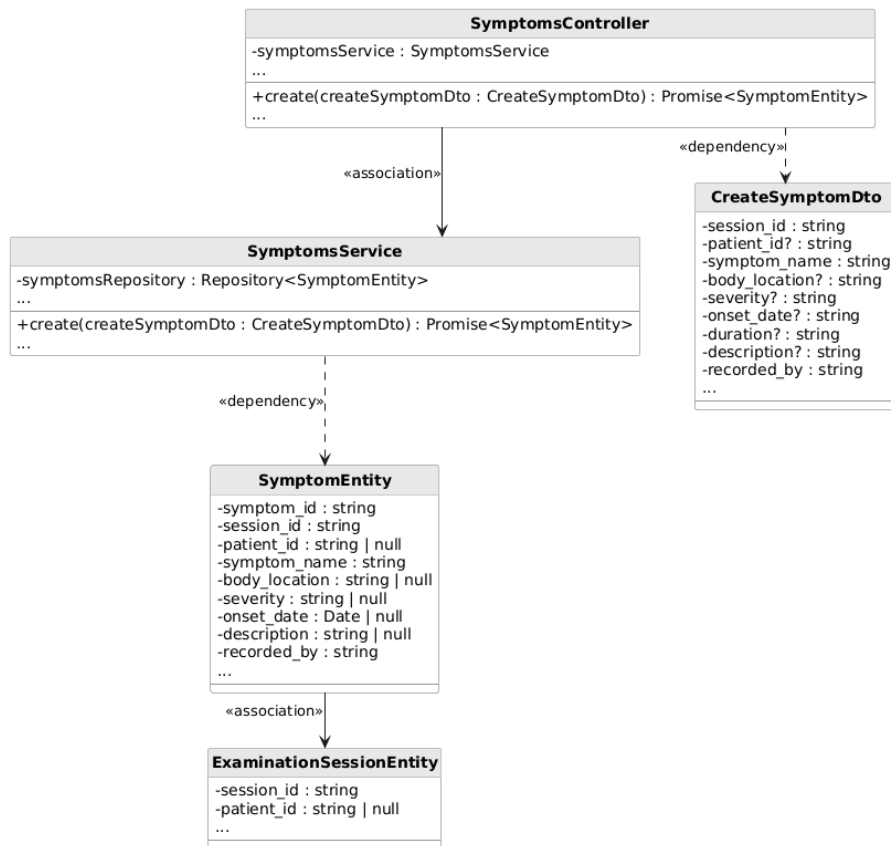


b. Sequence Diagram(s)

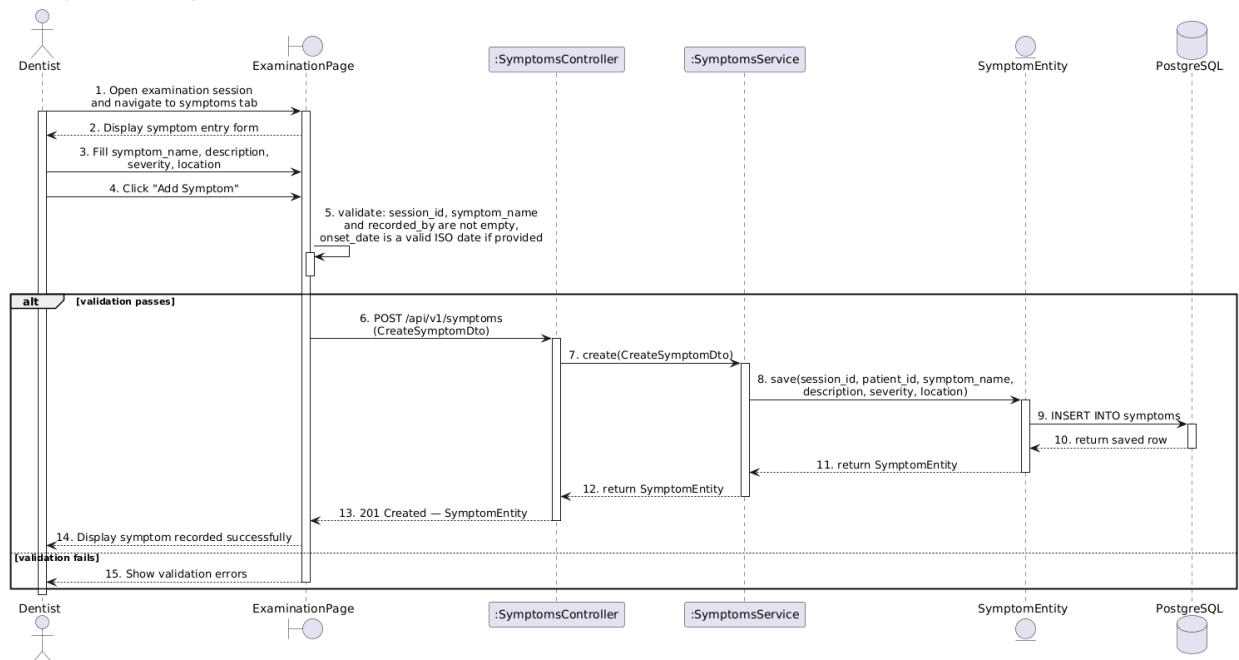


3.66 Enter Symptoms

a. Class Diagram

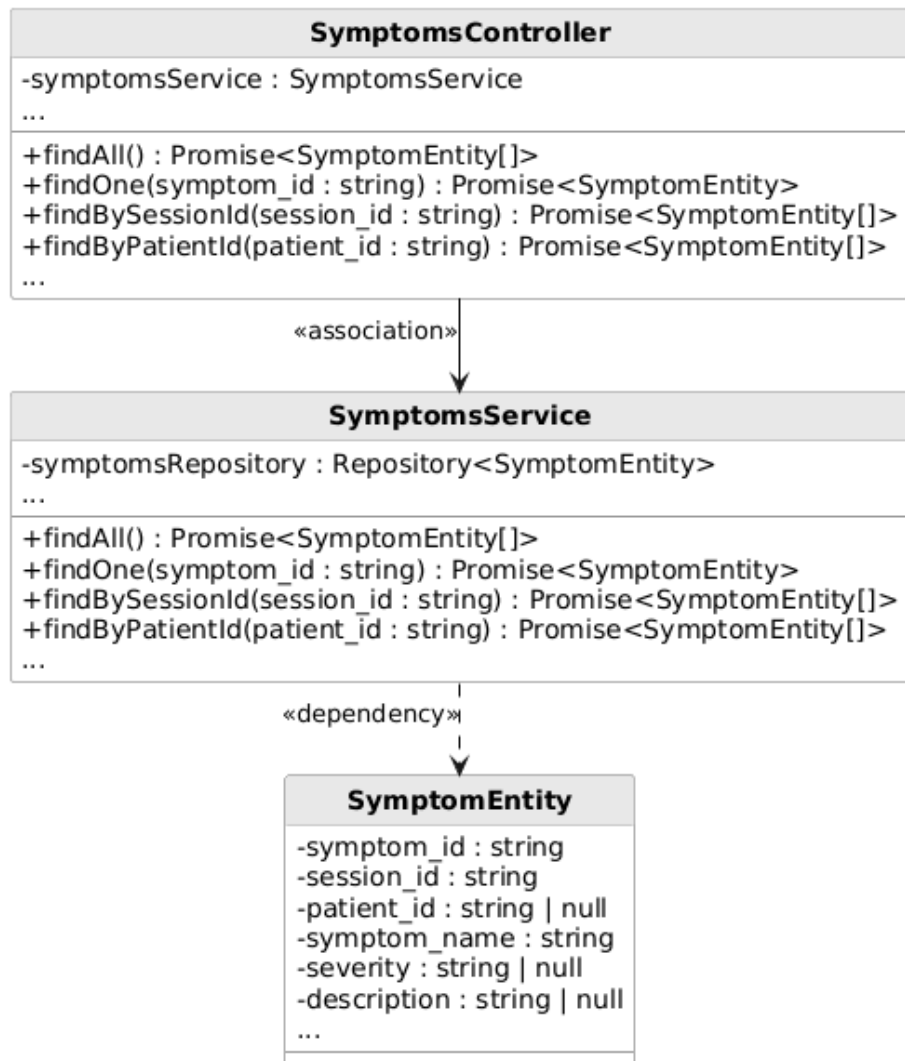


b. Sequence Diagram(s)

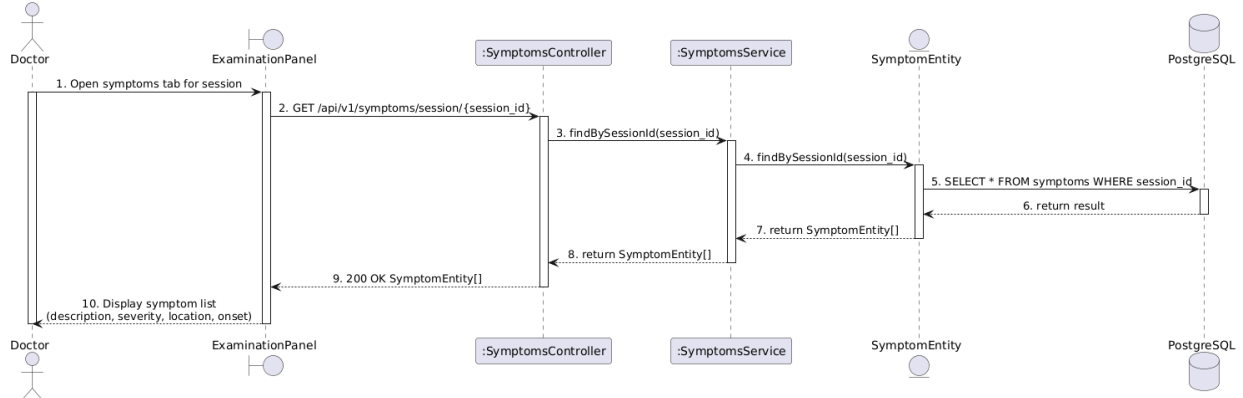


3.67 View Symptoms

a. Class Diagram

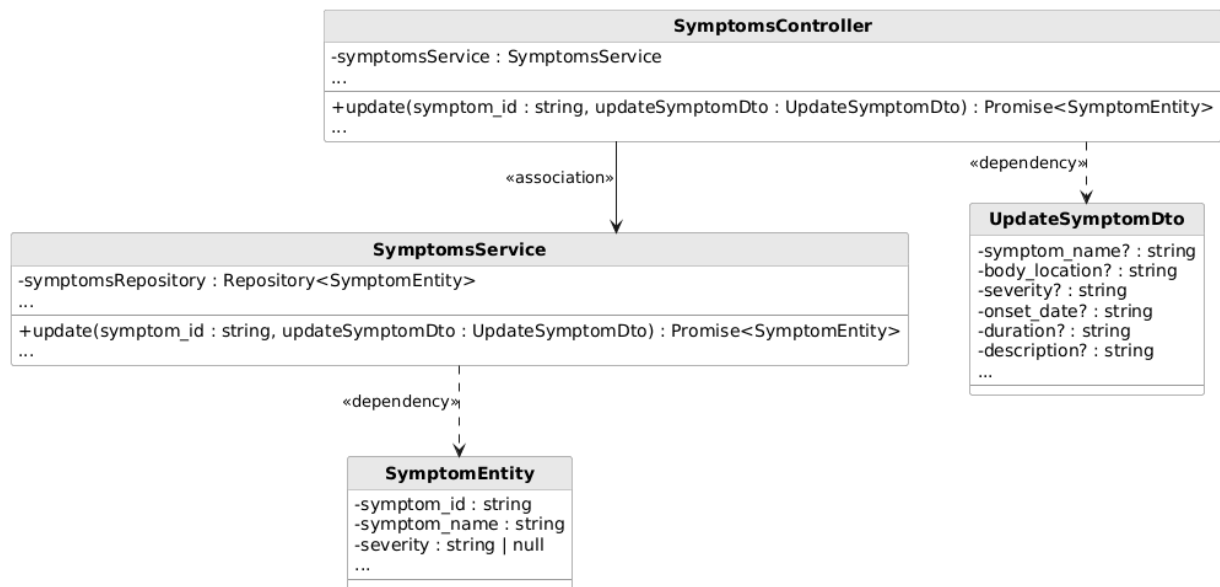


b. Sequence Diagram(s)

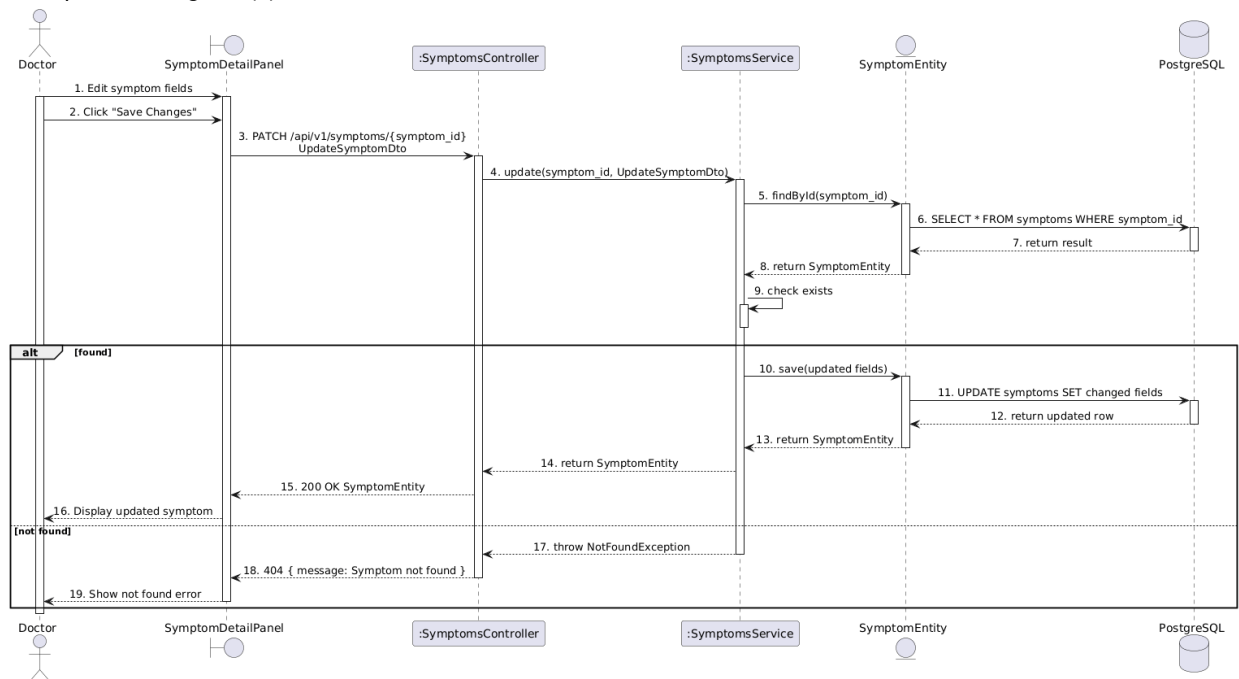


3.68 Edit Symptoms

a. Class Diagram

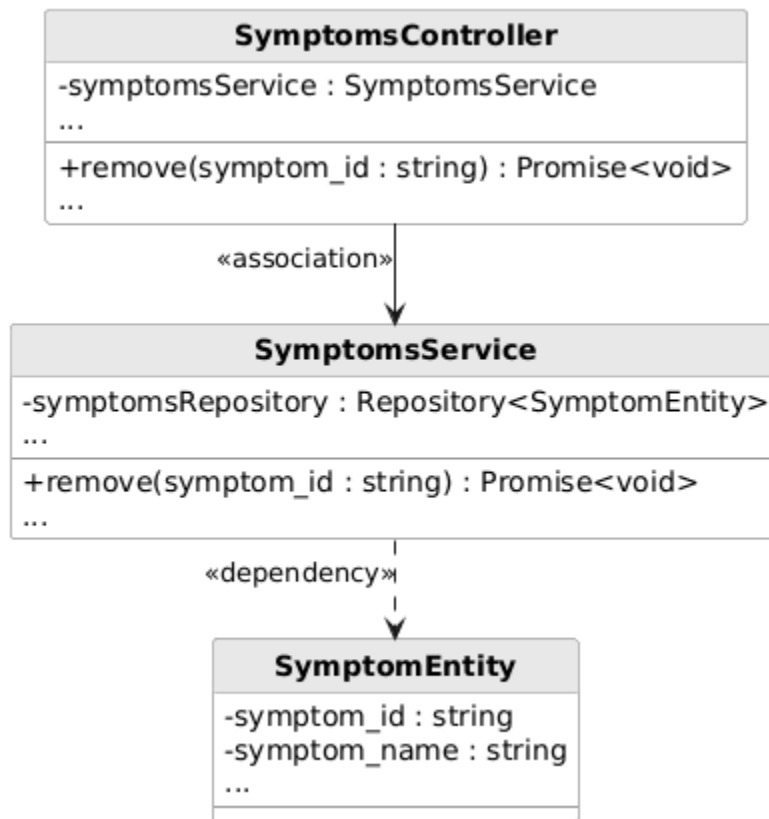


b. Sequence Diagram(s)

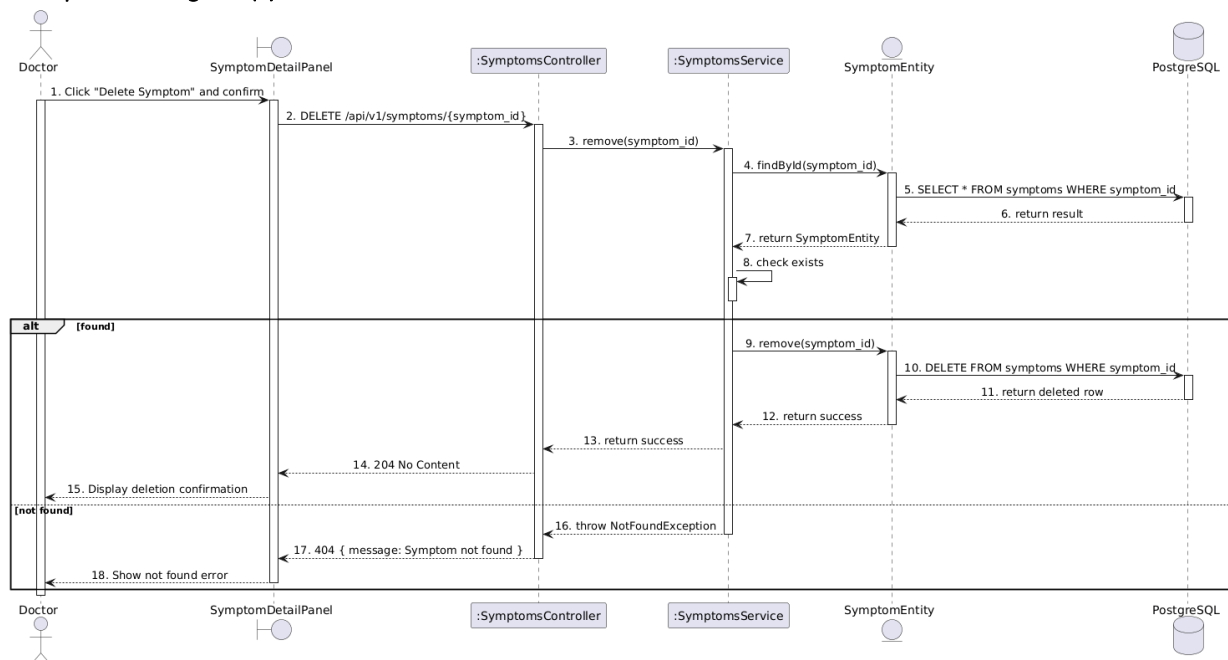


3.69 Delete Symptoms

a. Class Diagram

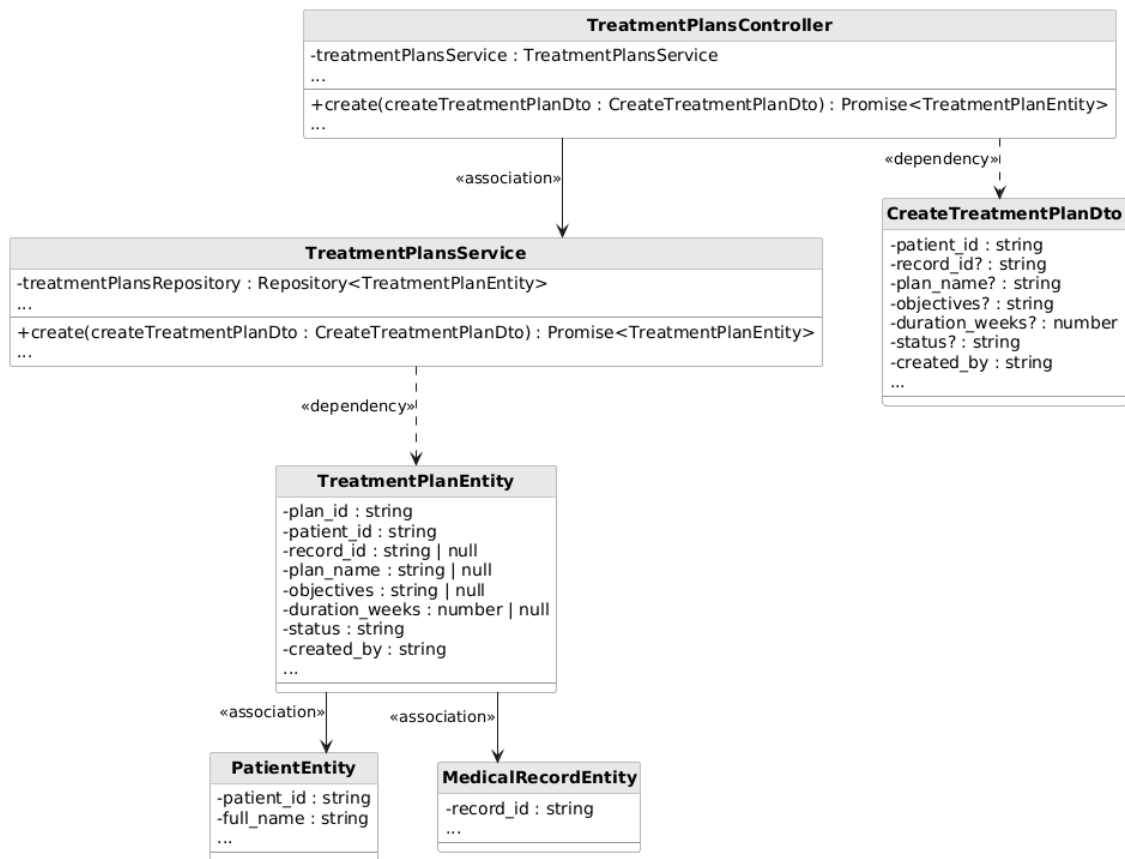


b. Sequence Diagram(s)

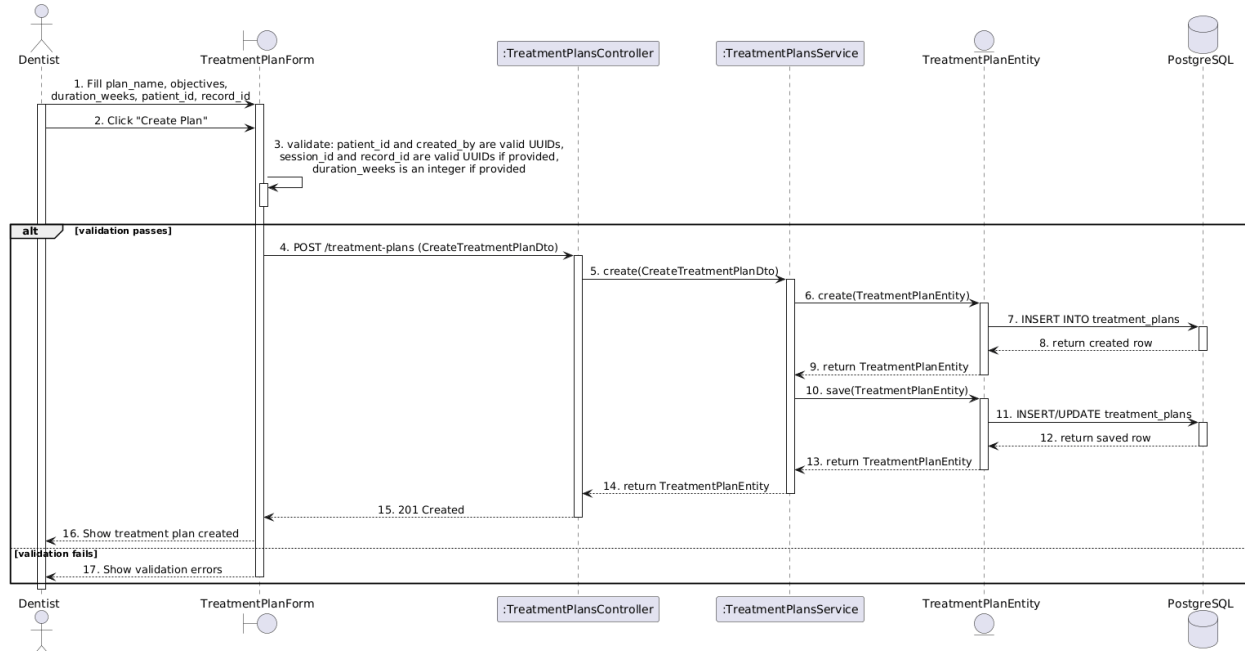


3.70 Create Treatment Plan

a. Class Diagram

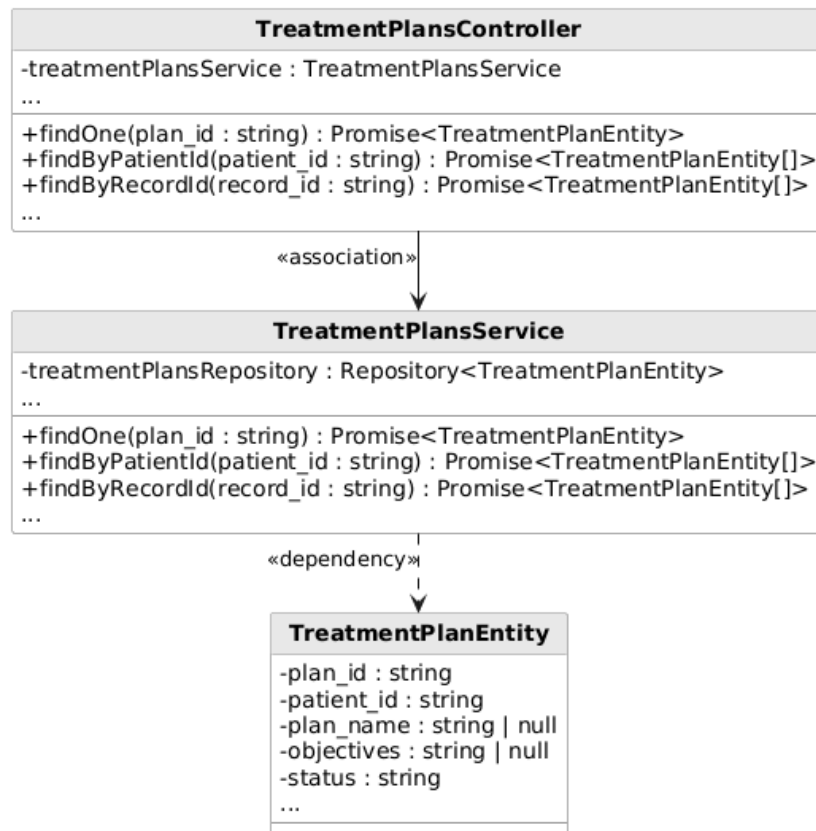


b. Sequence Diagram(s)

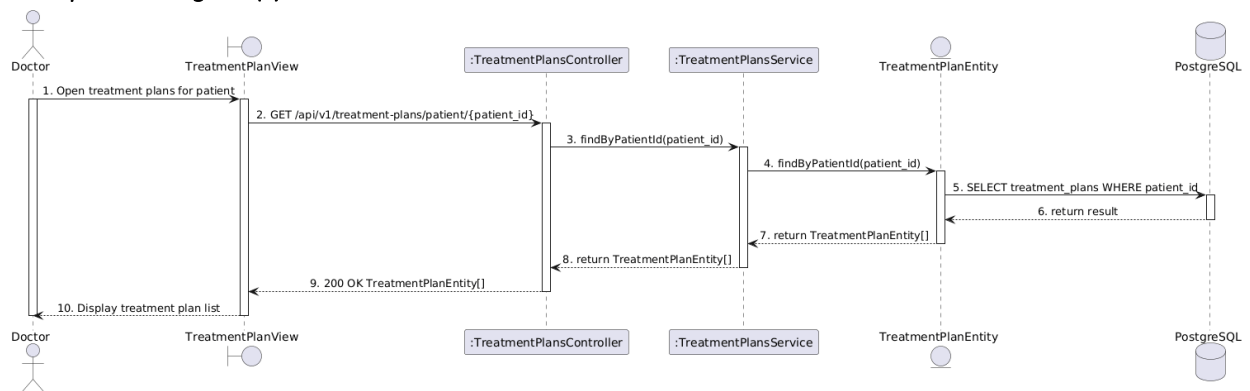


3.71 View Treatment Plan

a. Class Diagram

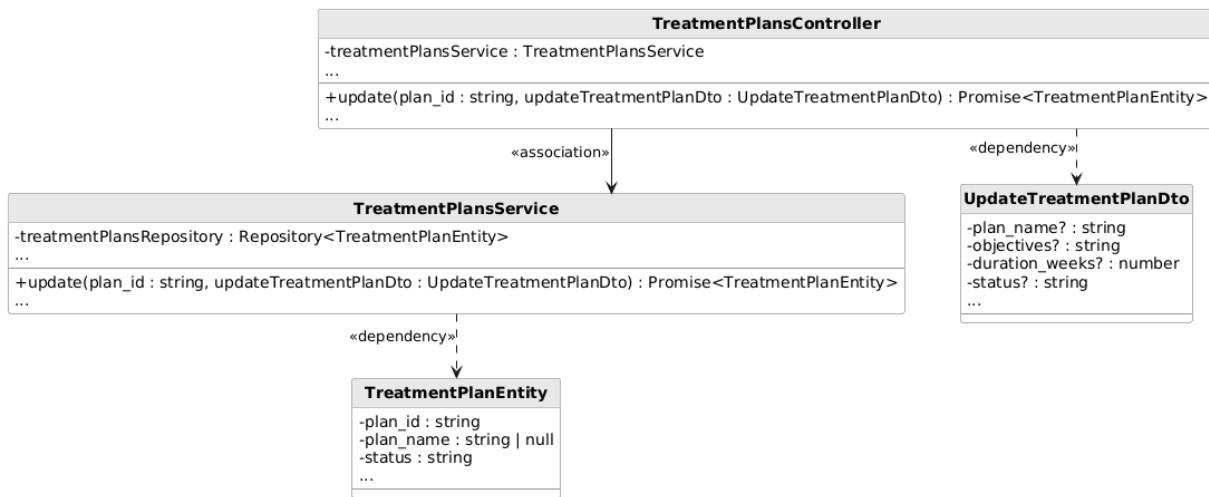


b. Sequence Diagram(s)

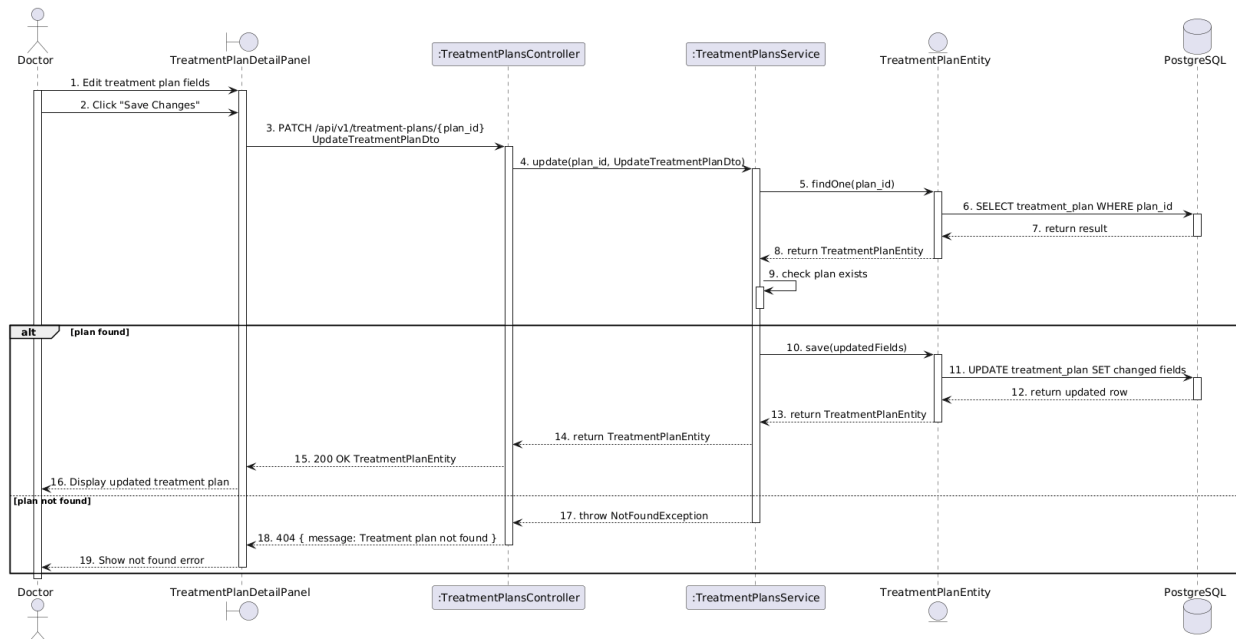


3.72 Edit Treatment Plan

a. Class Diagram

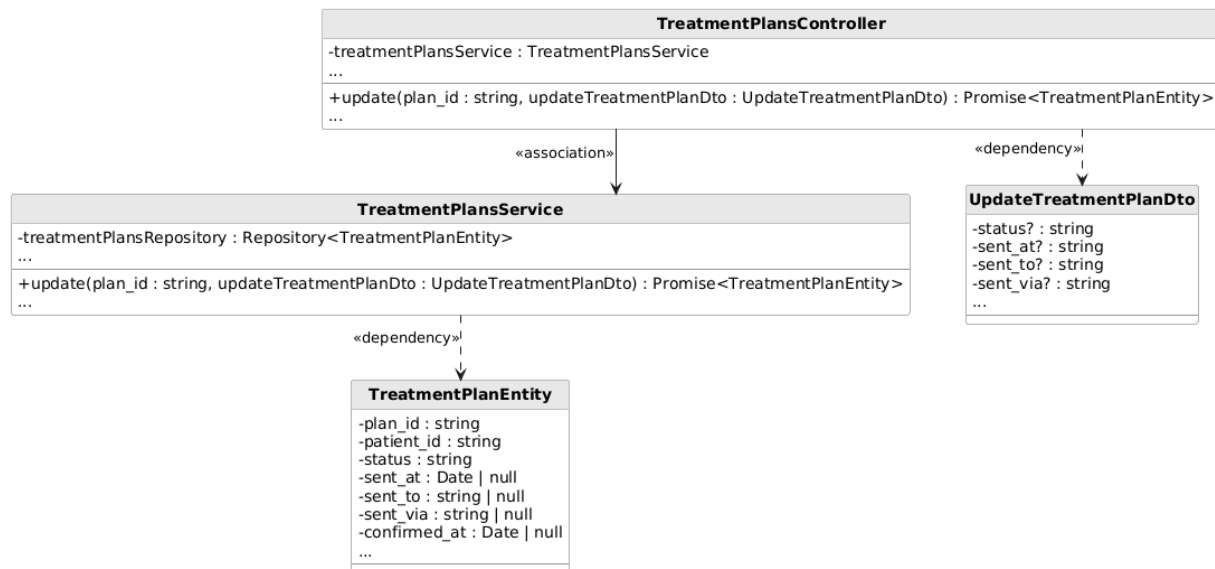


b. Sequence Diagram(s)

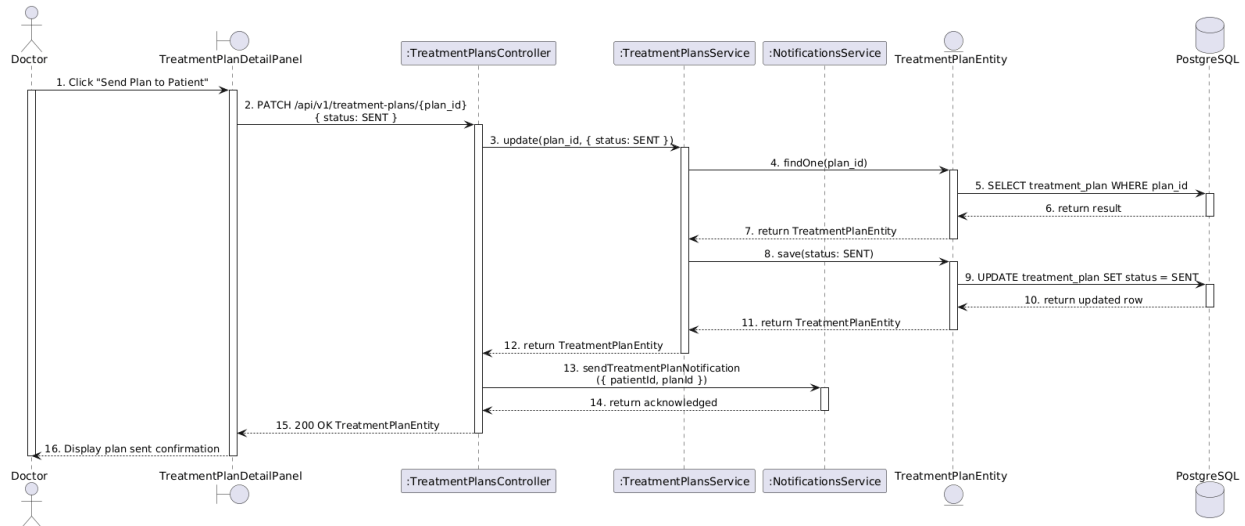


3.73 Send Treatment Plan

a. Class Diagram

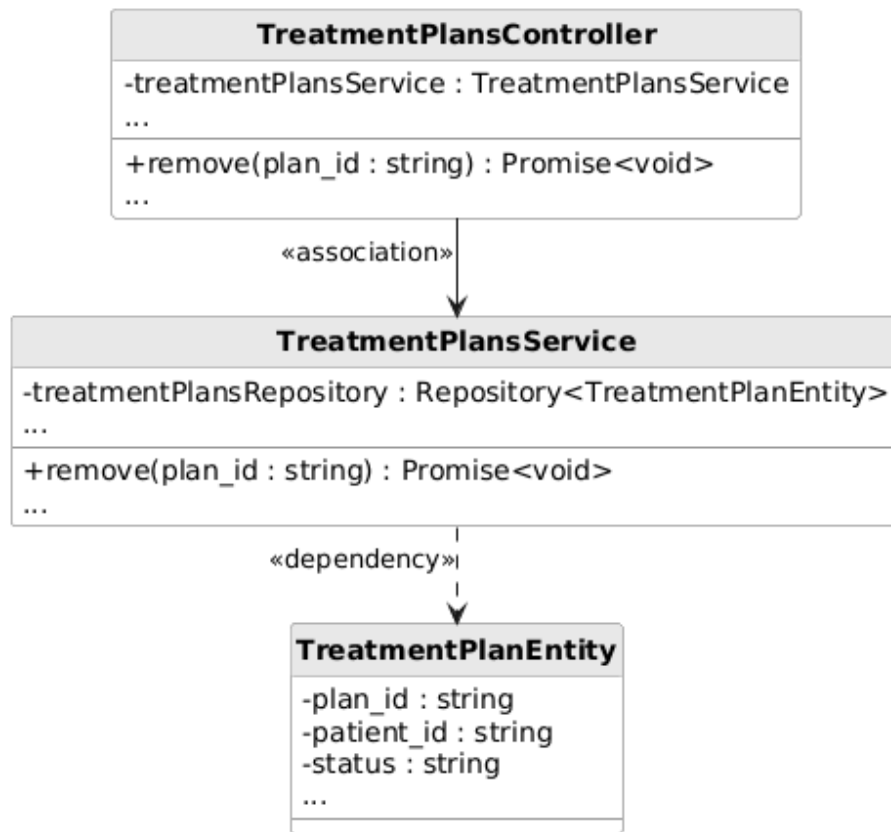


b. Sequence Diagram(s)

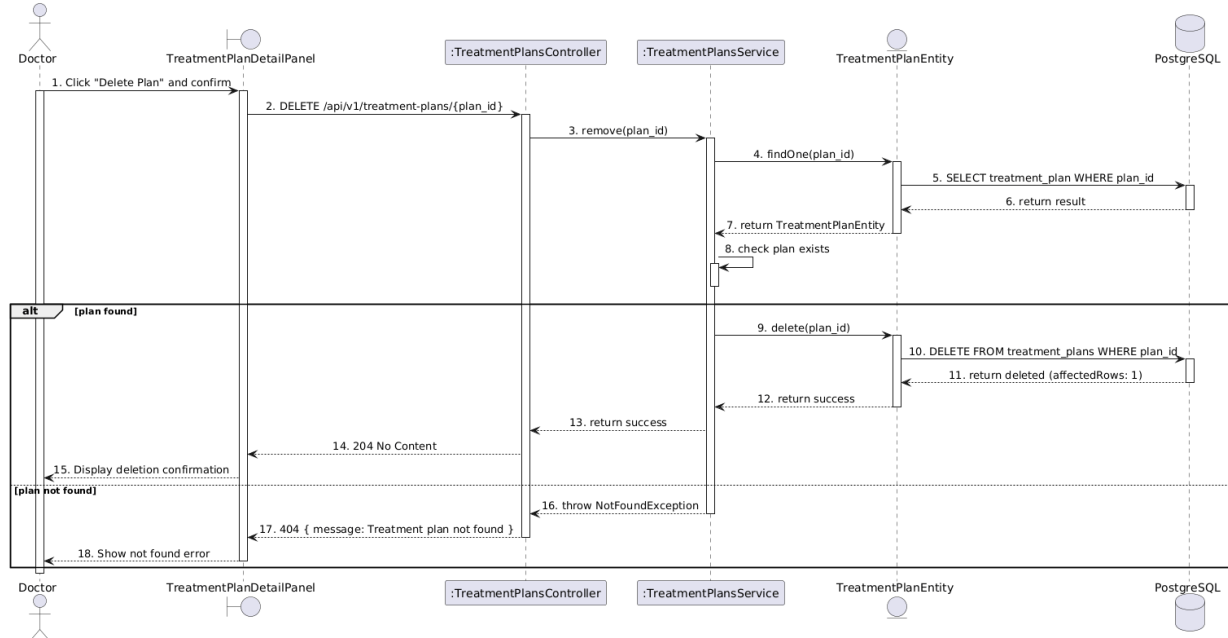


3.74 Delete Treatment Plan

a. Class Diagram

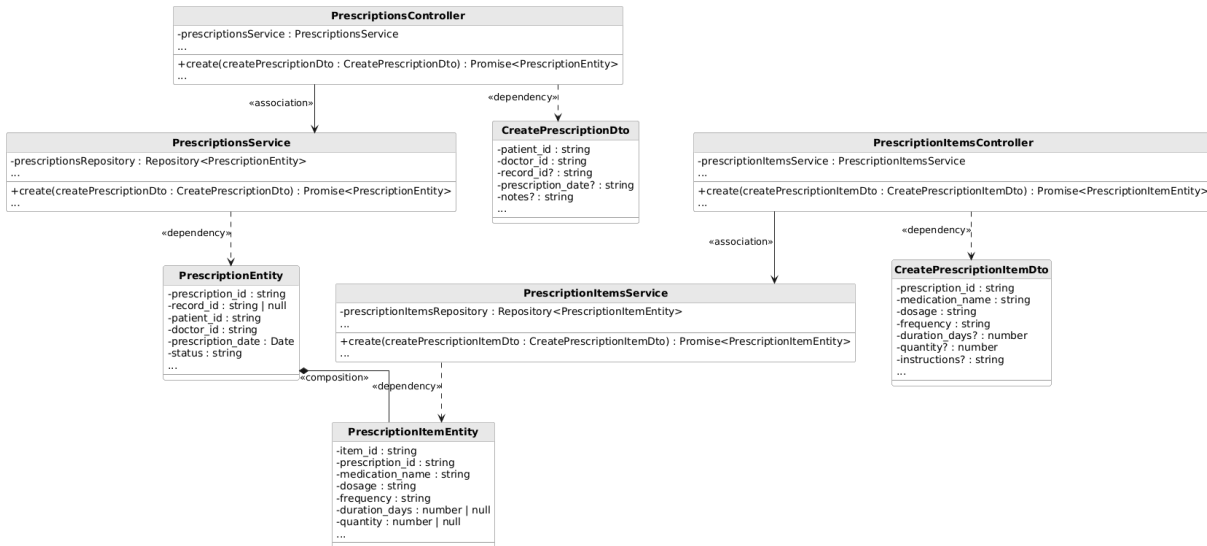


b. Sequence Diagram(s)

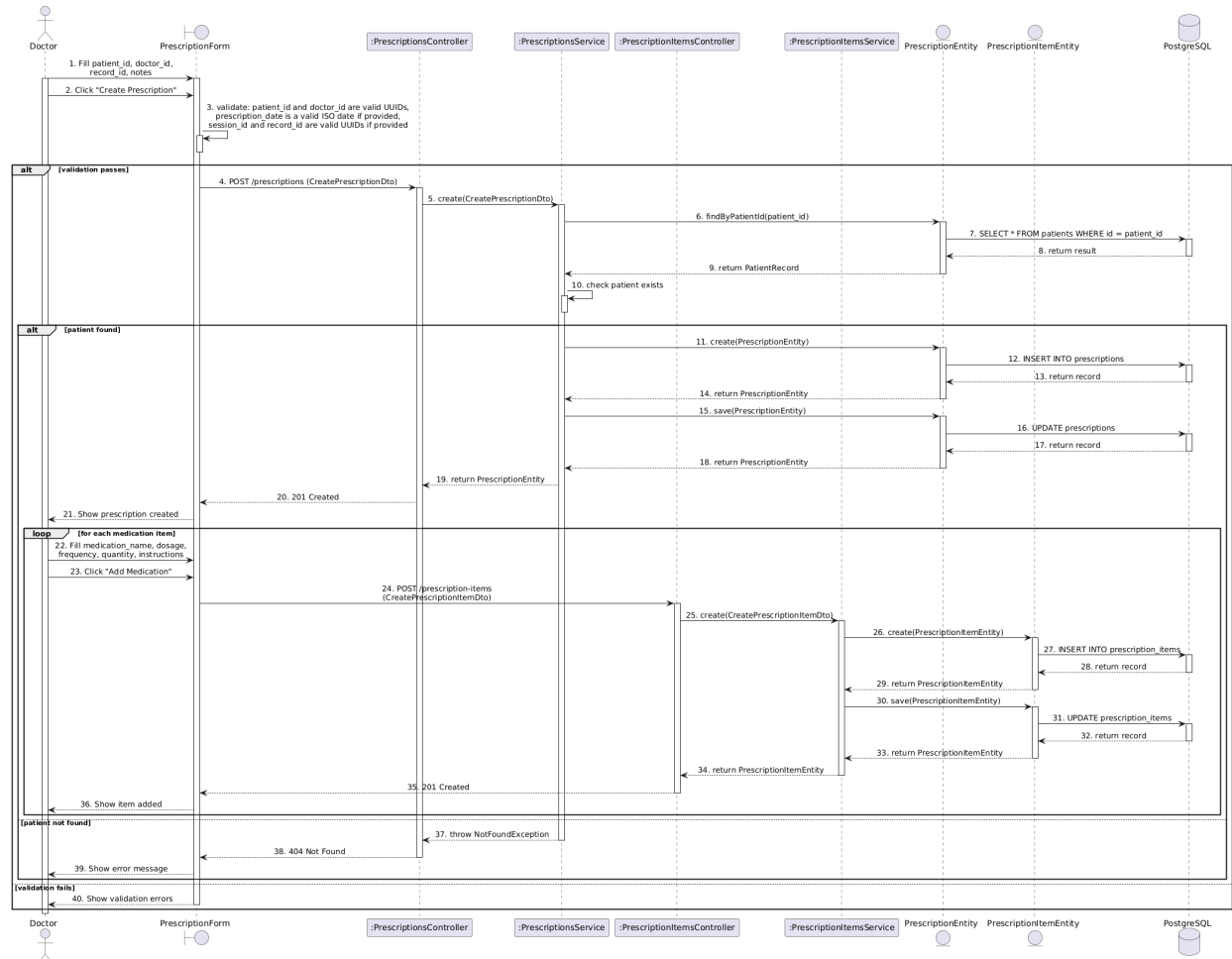


3.75 Create Electronic Prescription

a. Class Diagram

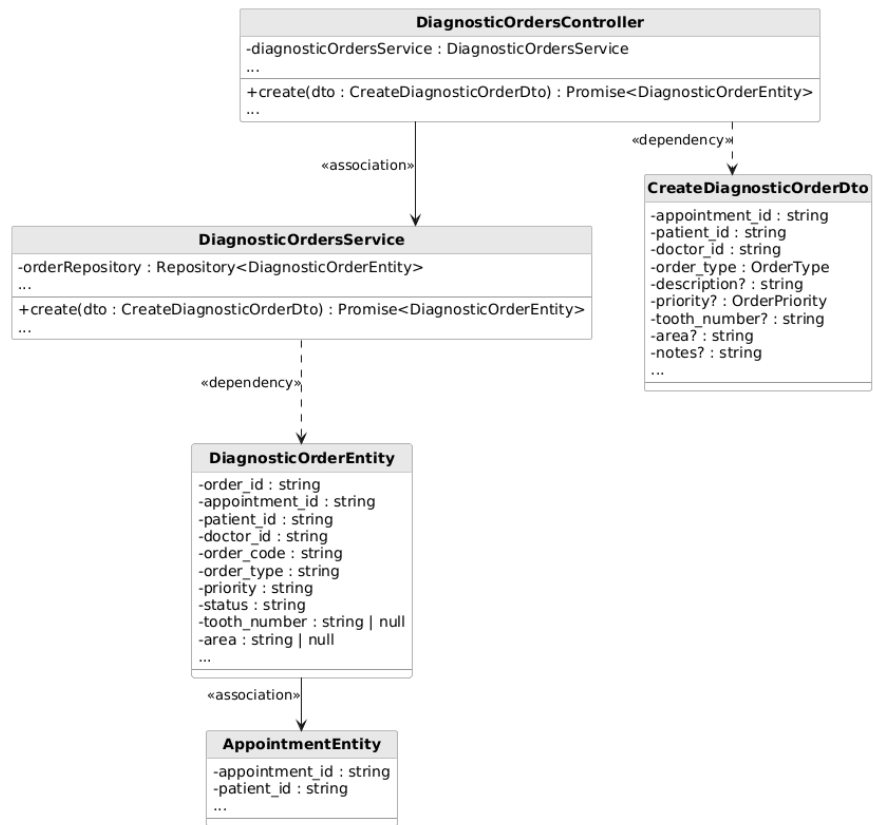


b. Sequence Diagram(s)

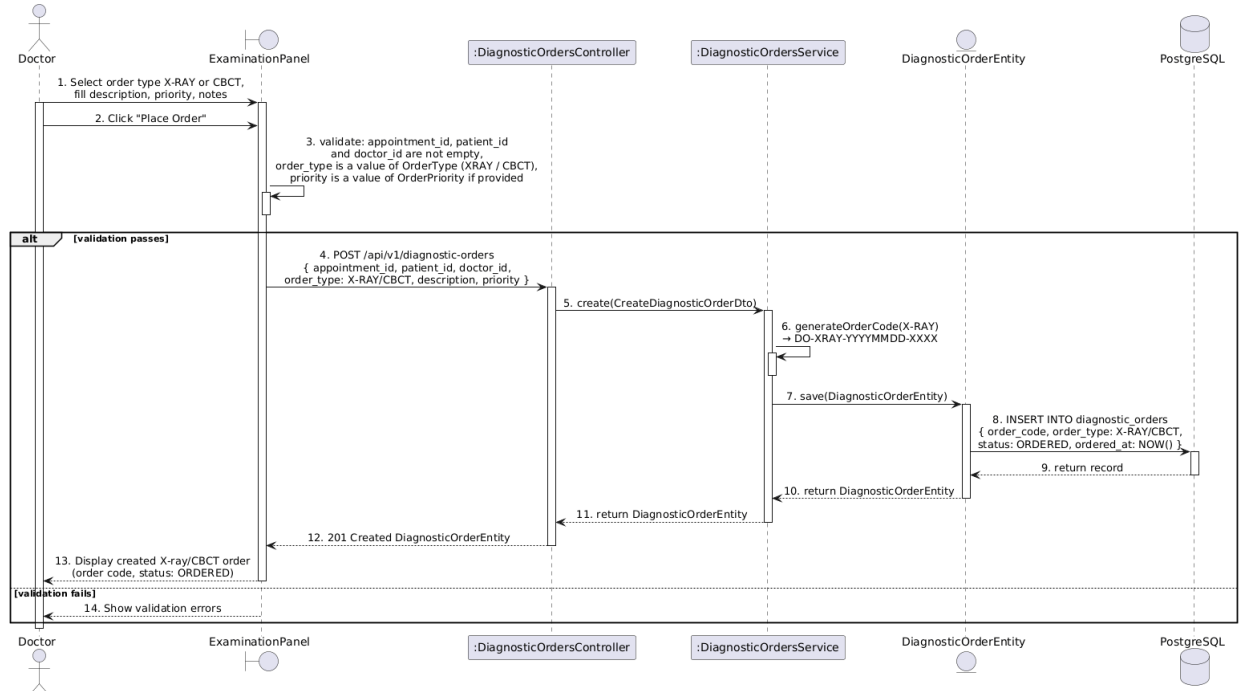


3.76 Order Xray CBCT

a. Class Diagram

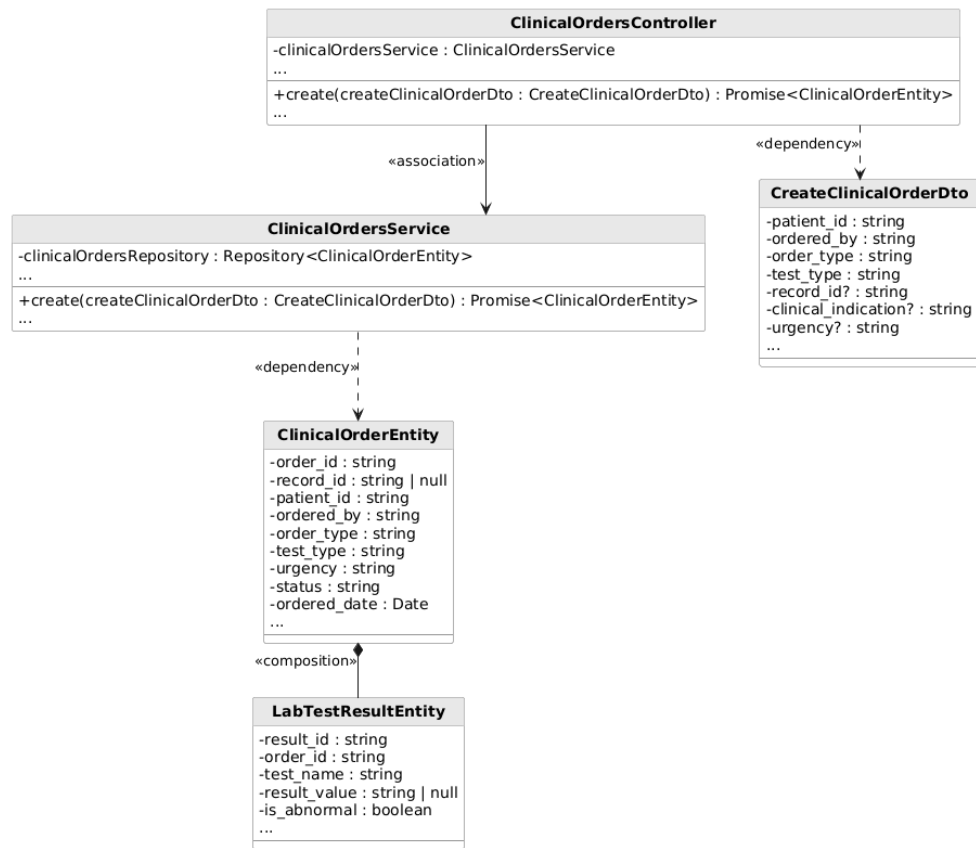


b. Sequence Diagram(s)

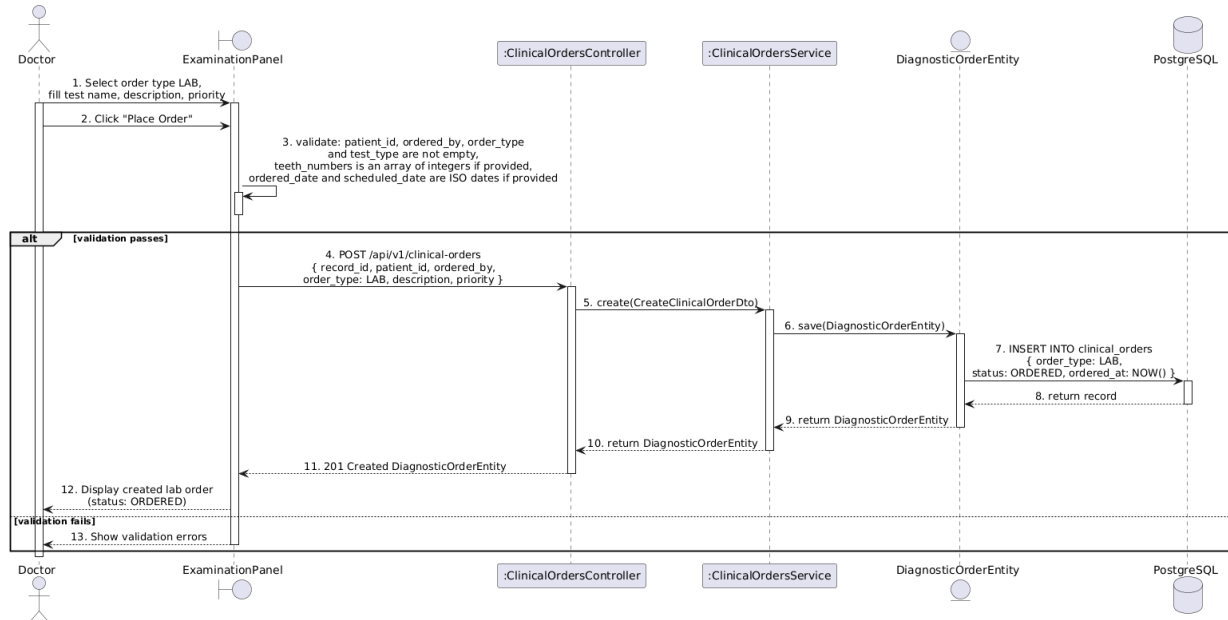


3.77 Order Laboratory Test Service

a. Class Diagram

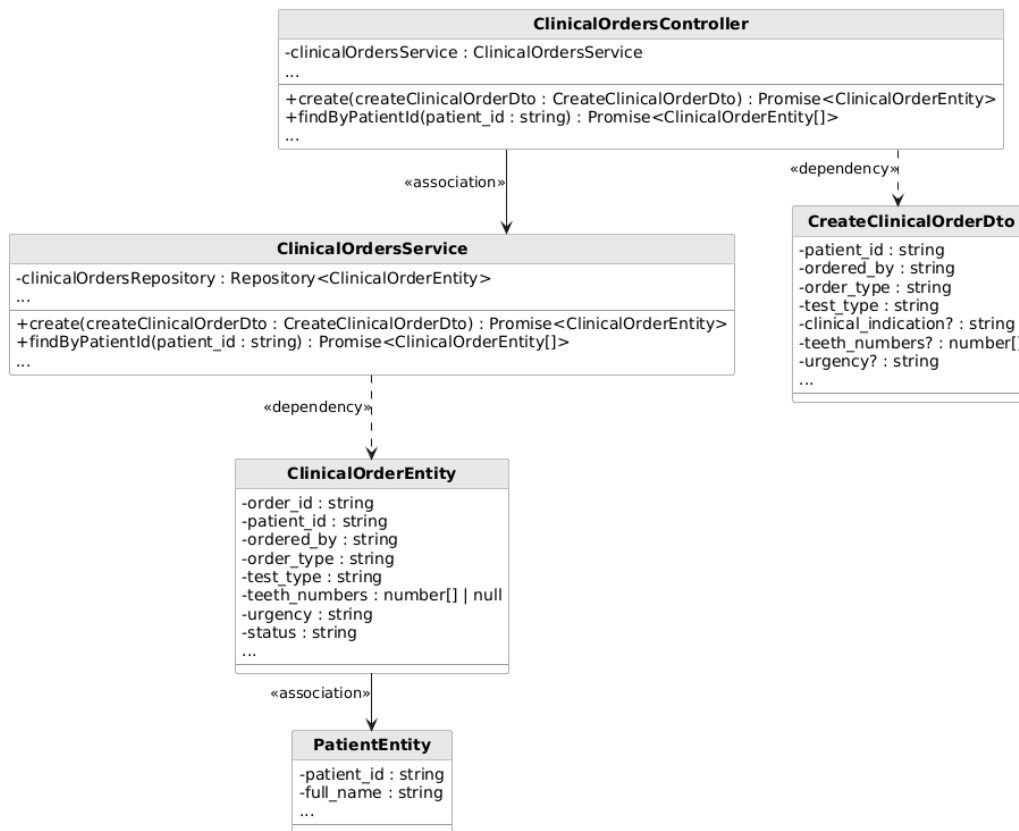


b. Sequence Diagram(s)

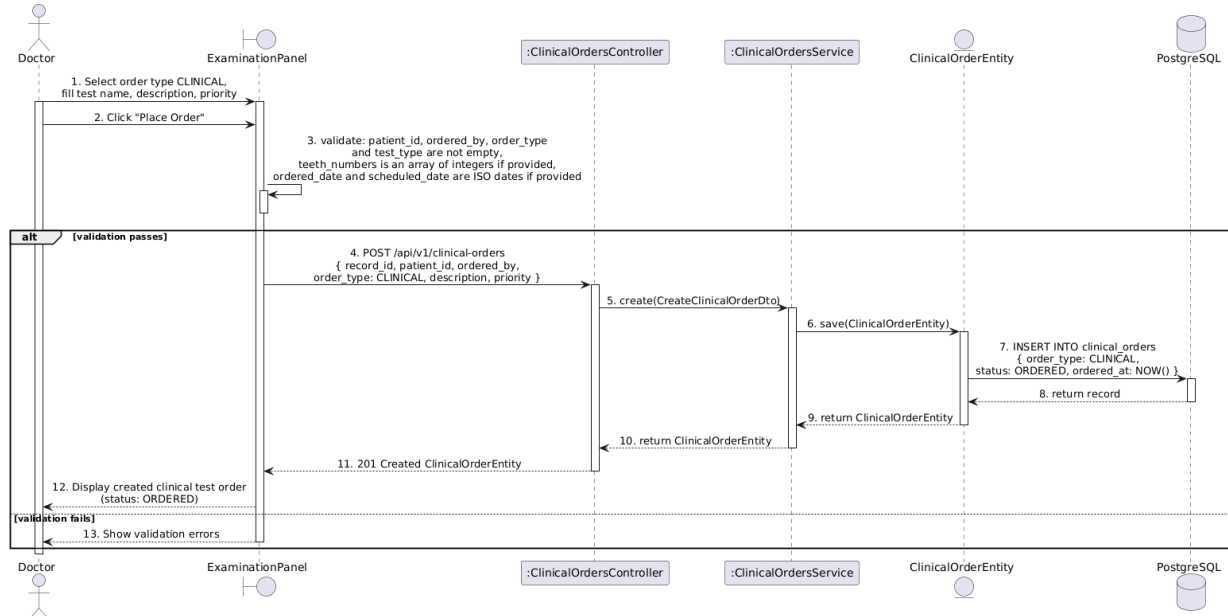


3.78 Order Clinical Test

a. Class Diagram

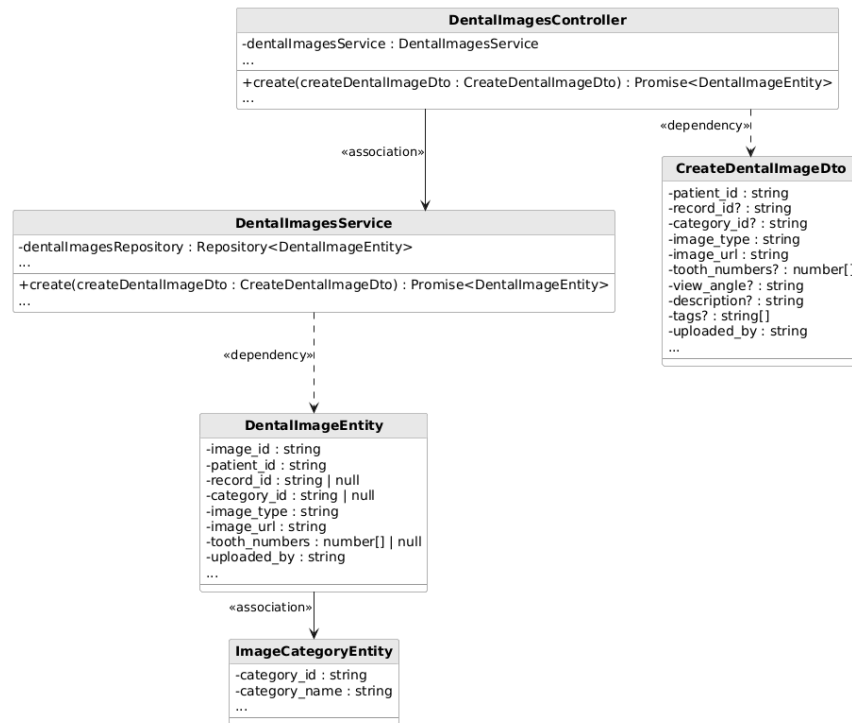


b. Sequence Diagram(s)

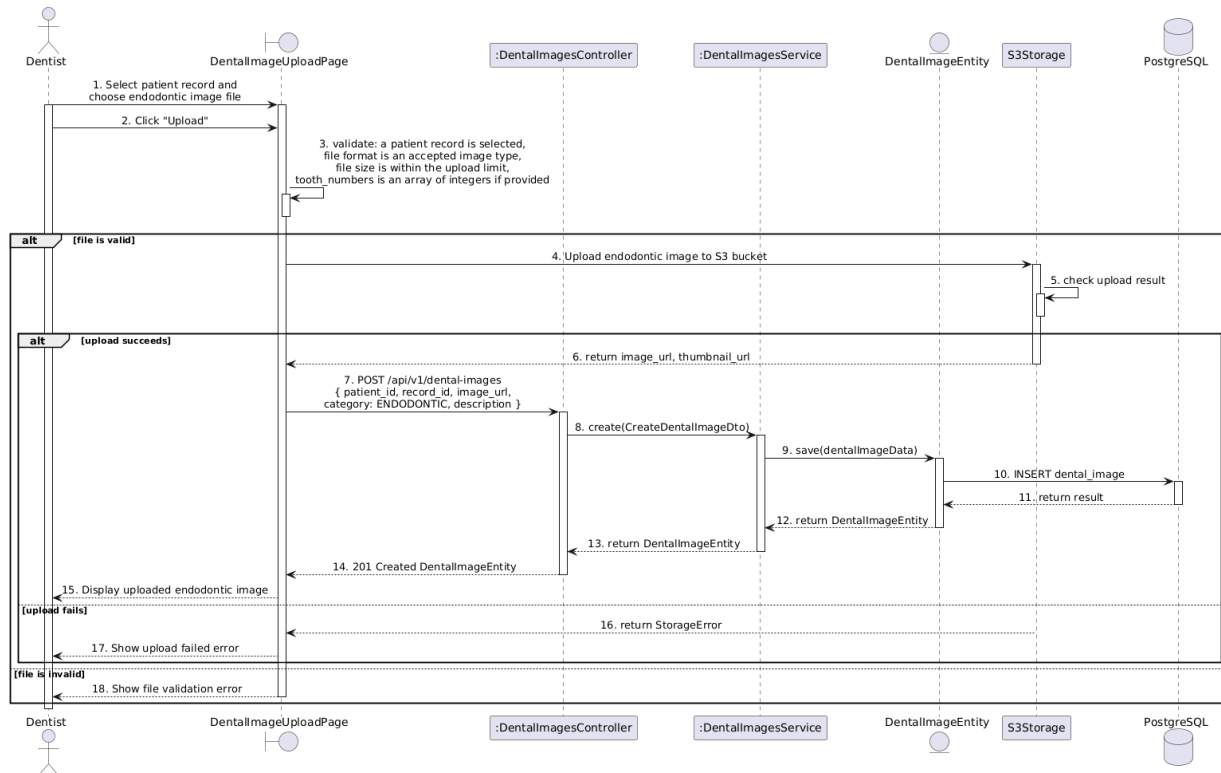


3.79 Upload Endodontic Image

a. Class Diagram

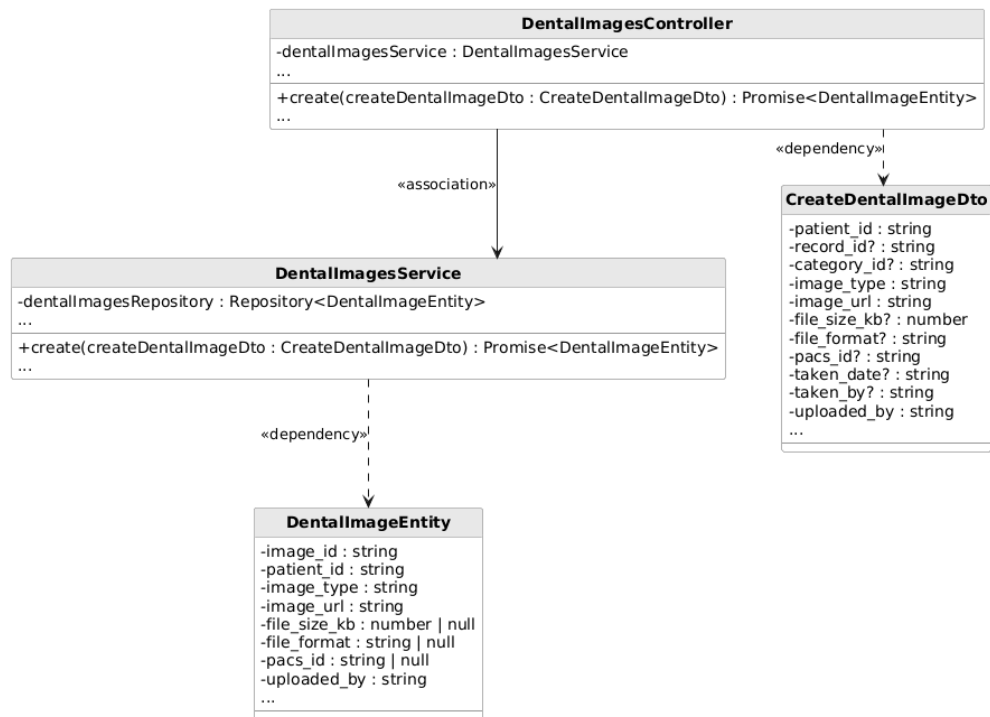


b. Sequence Diagram(s)

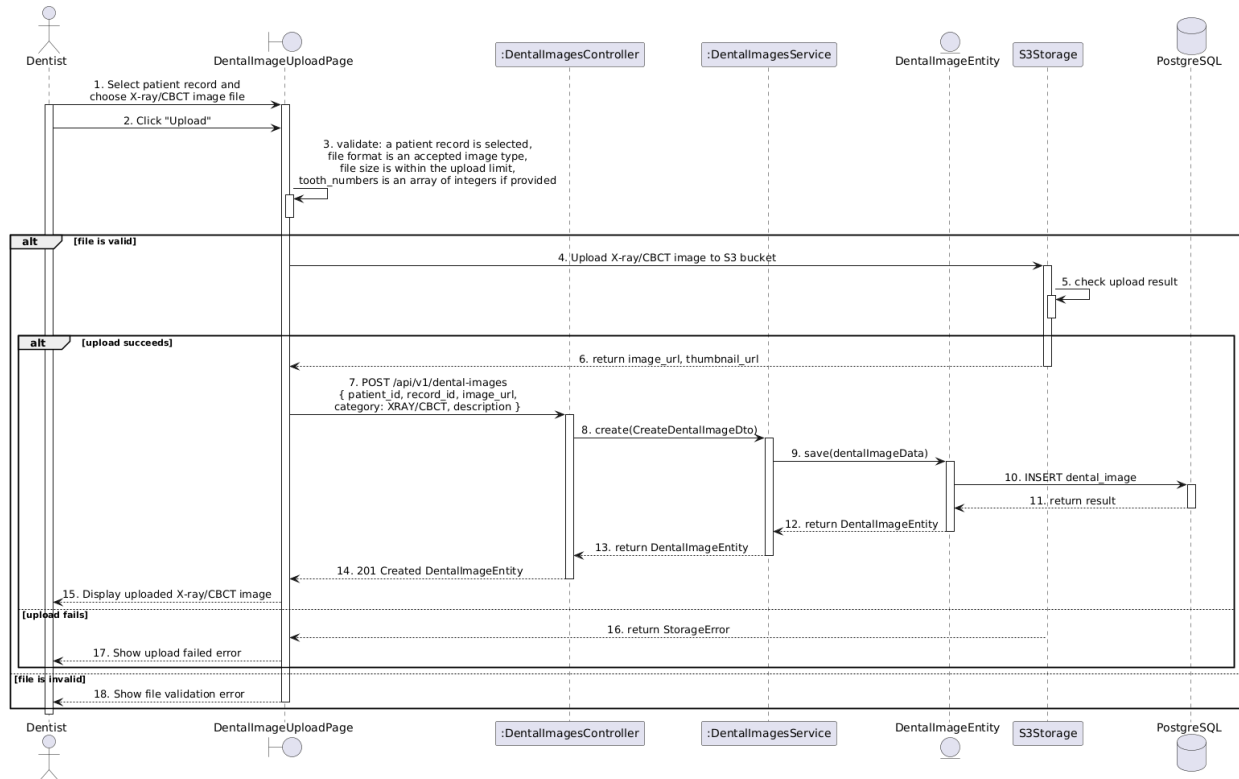


3.80 Upload Xray CBCT Image

a. Class Diagram

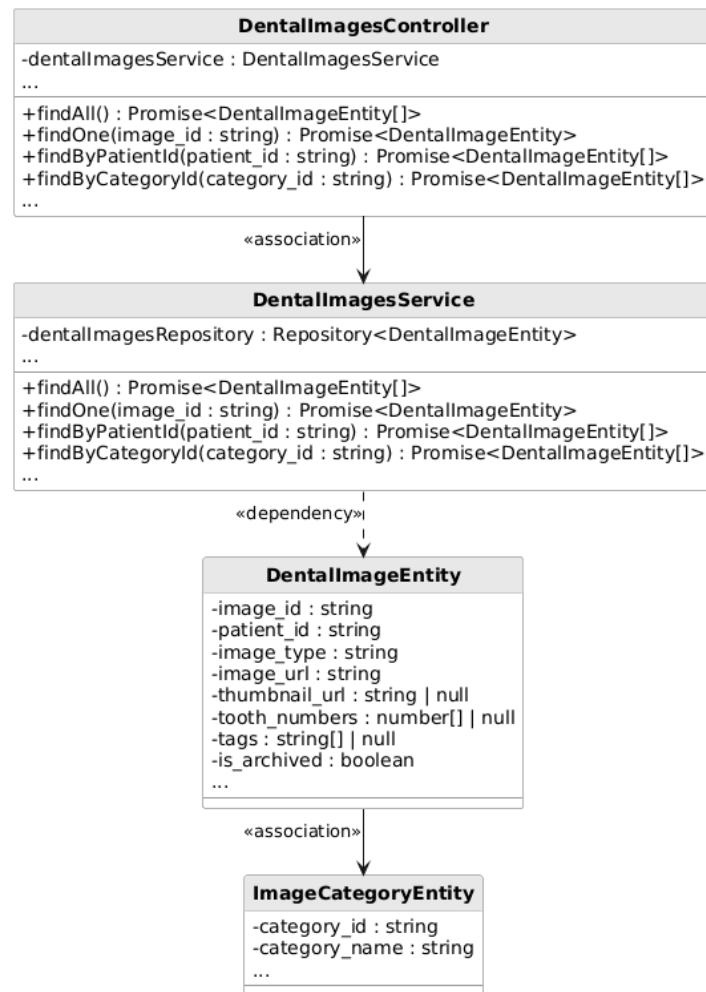


b. Sequence Diagram(s)

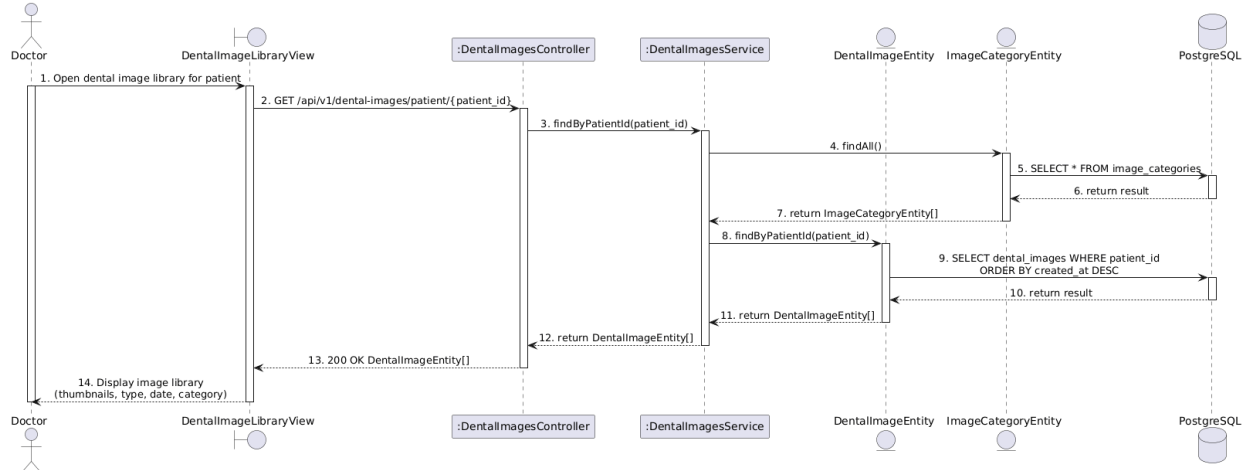


3.81 View Dental Image Library

a. Class Diagram



b. Sequence Diagram(s)

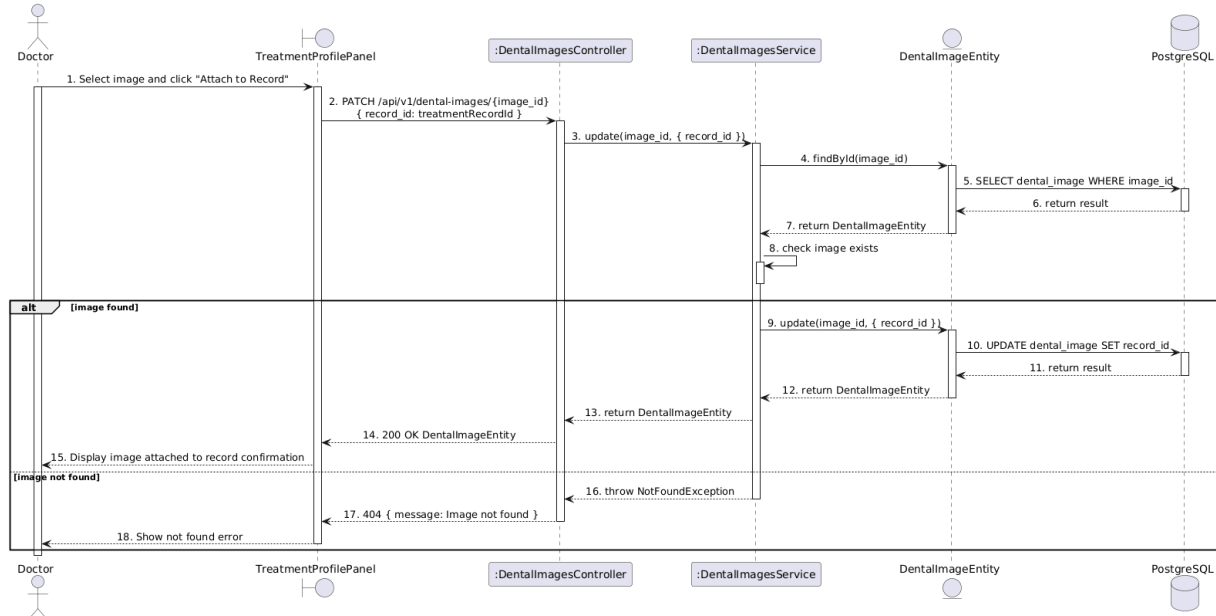


3.82 Attach Image To Treatment Profile

a. Class Diagram

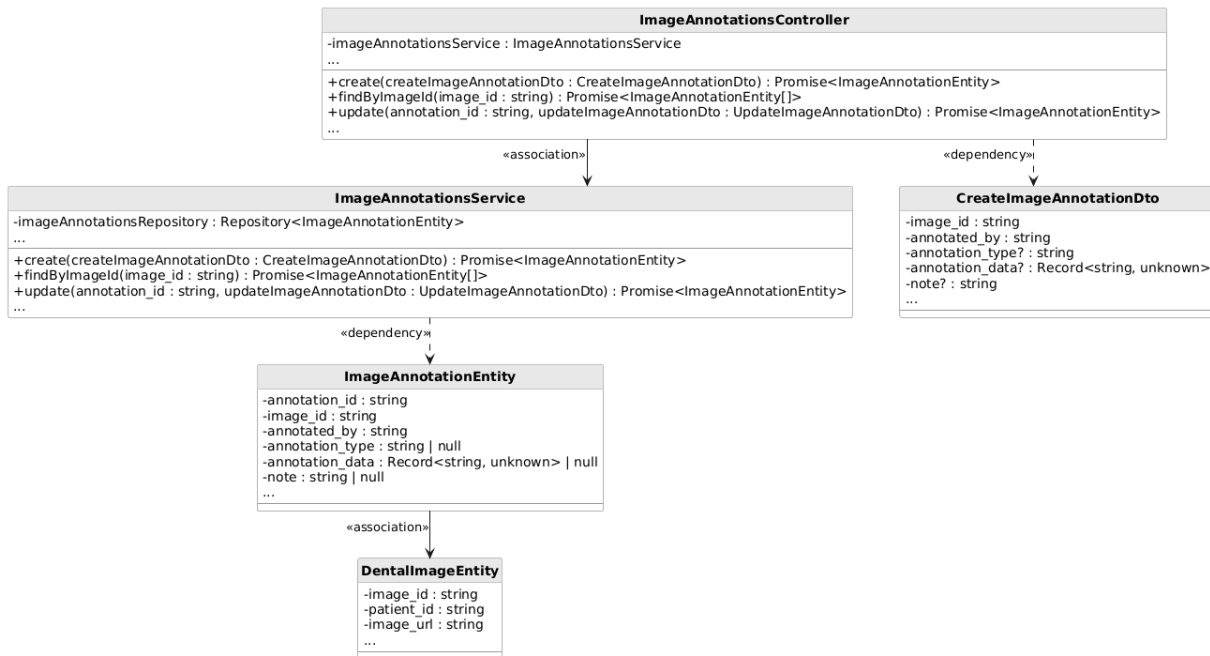


b. Sequence Diagram(s)

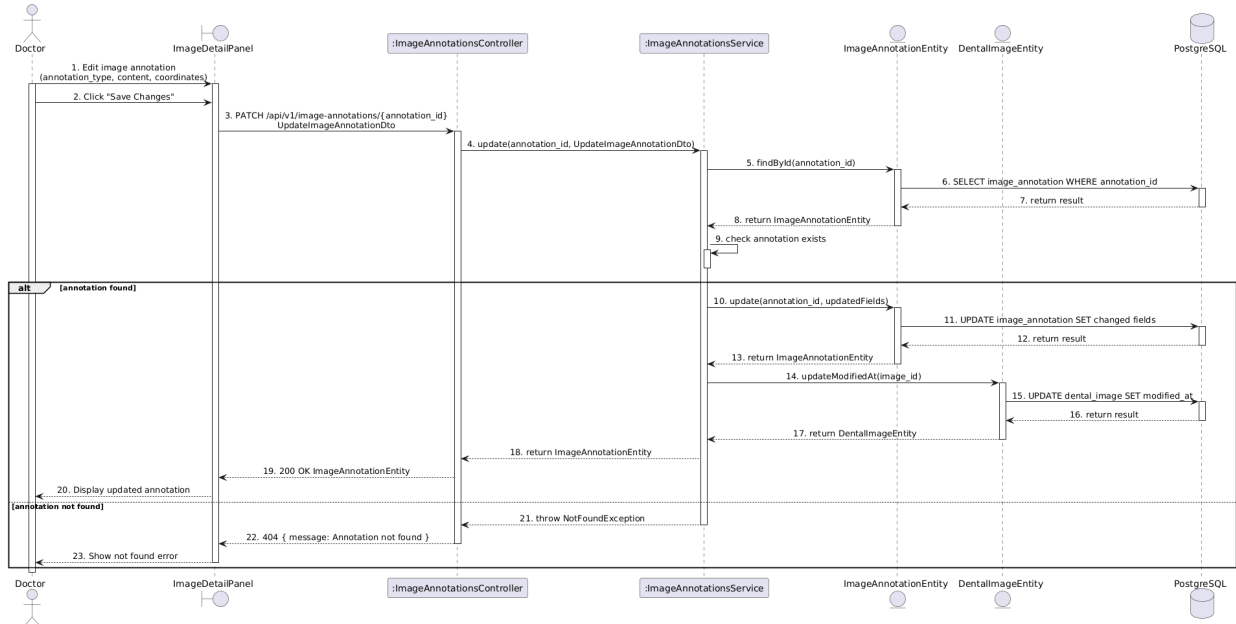


3.83 Edit Image By Treatment Profile

a. Class Diagram

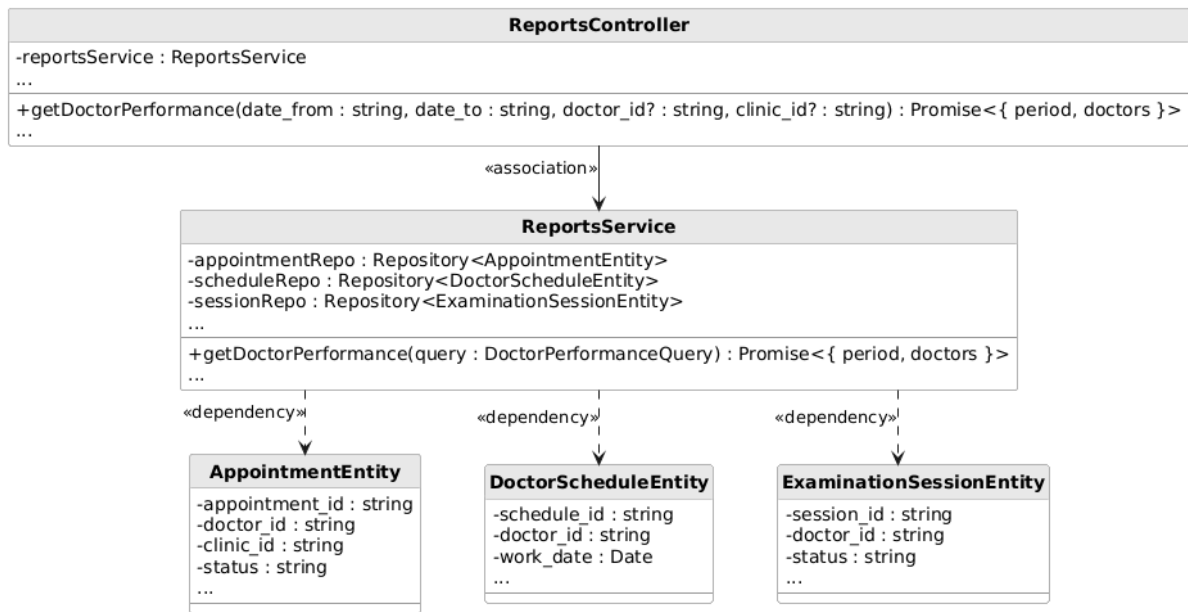


b. Sequence Diagram(s)

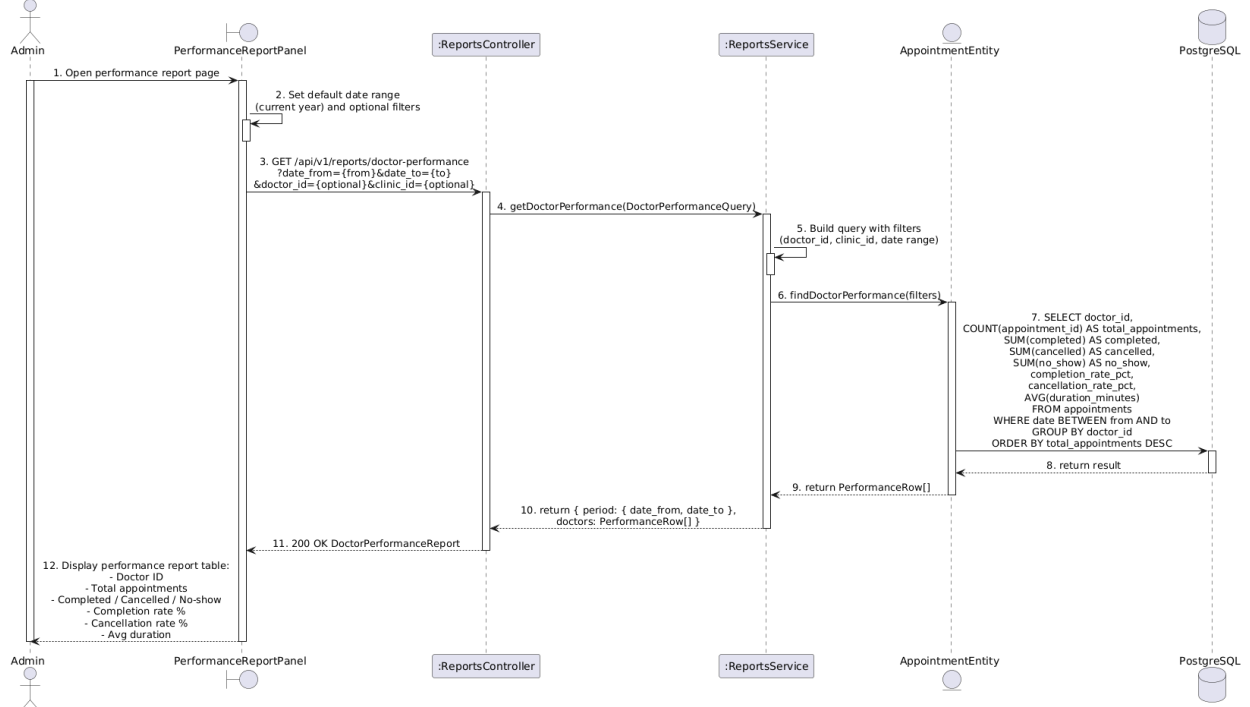


3.84 View Doctor Performance Report

a. Class Diagram

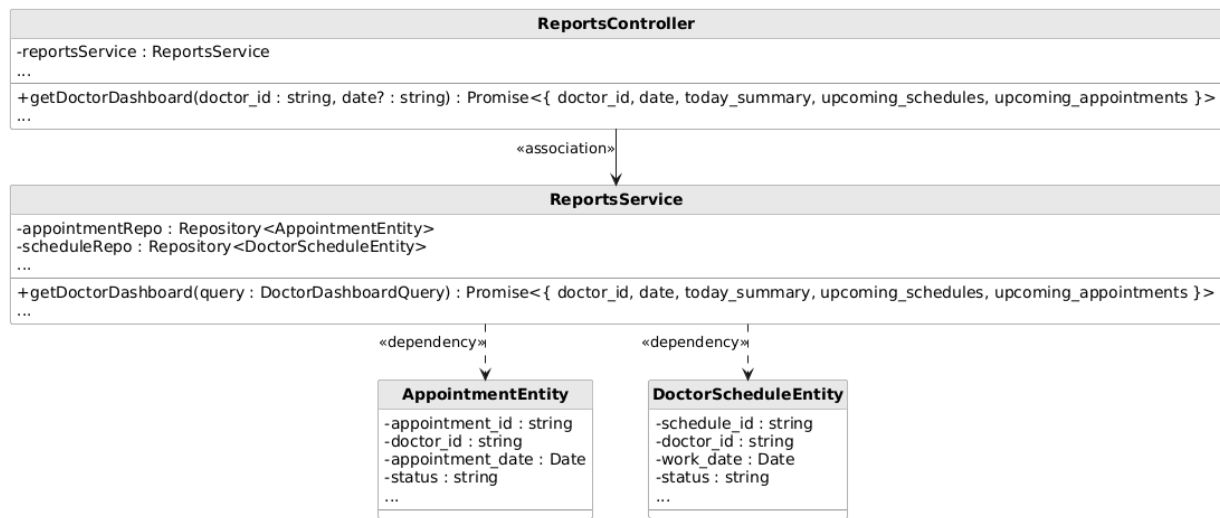


b. Sequence Diagram(s)

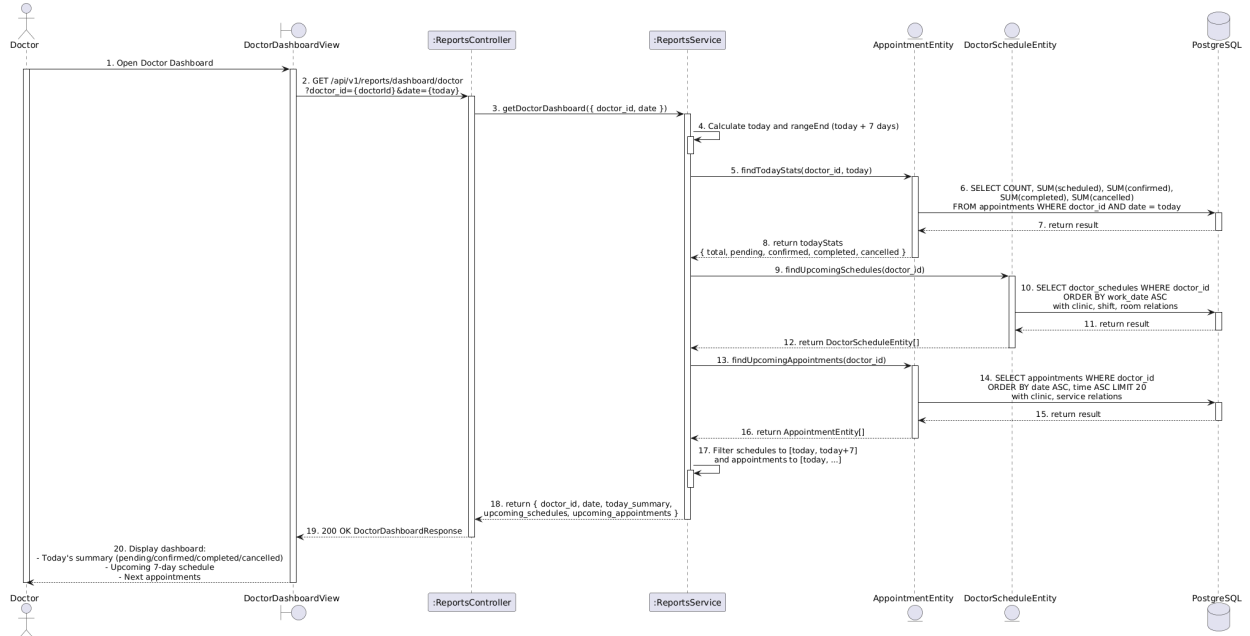


3.85 View Dashboard By Doctor

a. Class Diagram

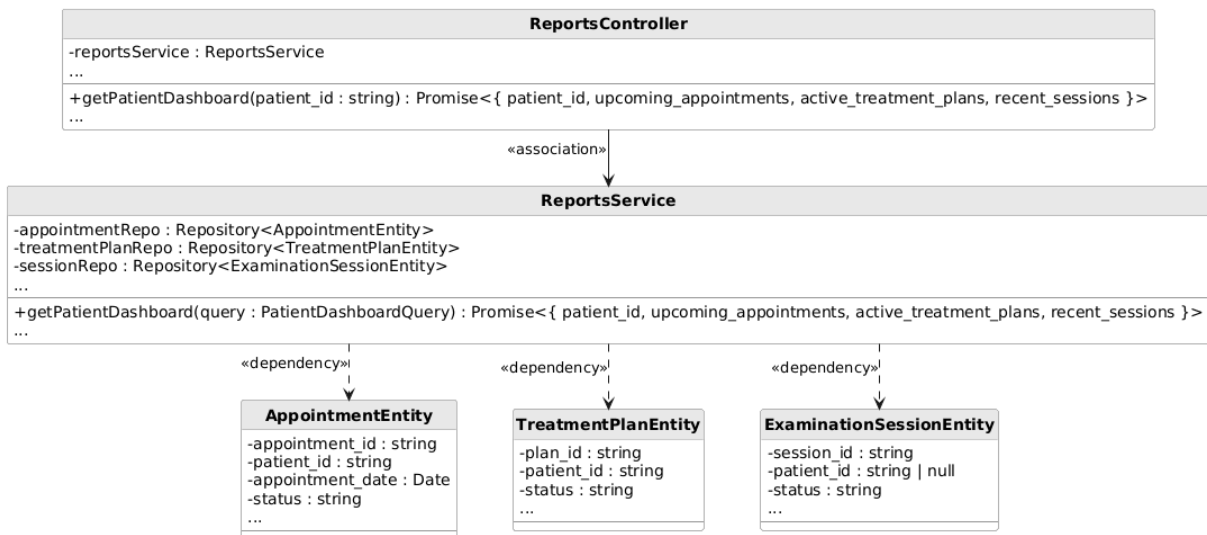


b. Sequence Diagram(s)

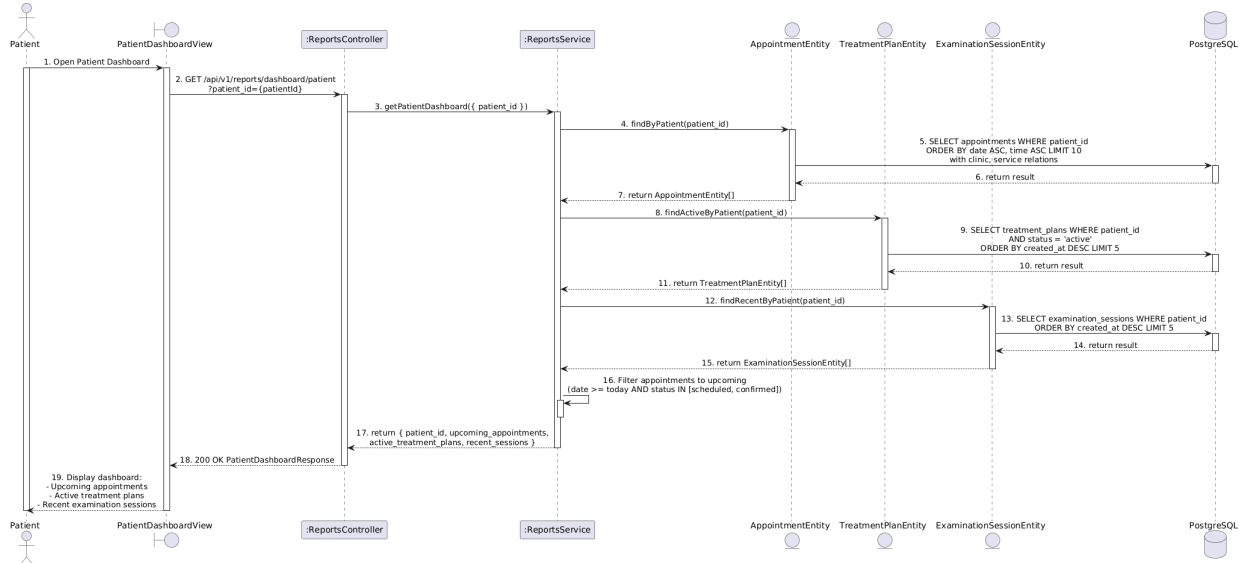


3.86 View Dashboard By Customer

a. Class Diagram

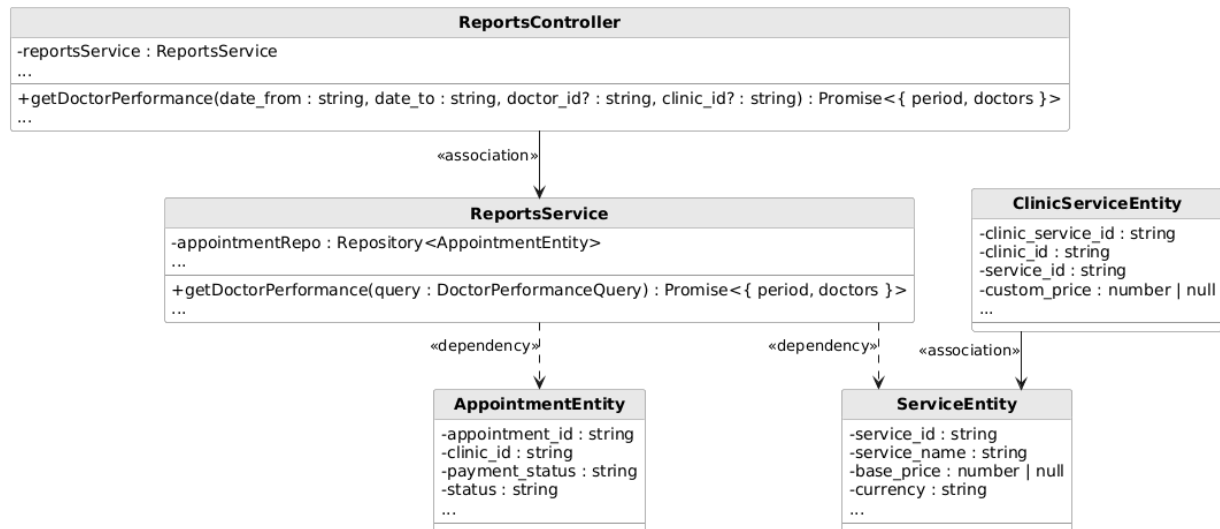


b. Sequence Diagram(s)



3.87 Revenue Financial Report

a. Class Diagram



b. Sequence Diagram(s)

